Swasti Dhagat
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Recent Advances in Bioprocess Engineering and Bioreactor Design

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Preface

In the current scenario, where the demand for innovations in bioprocess engineering is high, the book *Recent Advances in Bioprocess Engineering and Bioreactor Design* provides a comprehensive tool for the cutting-edge technologies in this discipline. This book explores all the areas of bioprocess technology, upstream and downstream processing, metabolic engineering, enzyme-catalyzed reactions, kinetics of microbial growth, bioreactor configurations, economics of bioprocesses, etc. The chapters are structured to provide a thorough understanding of each aspect, making it an invaluable resource for students, researchers, and professionals in the field.

Chapter 1 explores the stages in bioprocess technology—ranging from upstream processing, which includes inoculum development, media design, and bioreactor operation, to downstream processing, unraveling the complexities of product recovery and purification. Chapter 2 provides an insight into metabolic engineering, explaining the steps for the production of metabolites. Techniques such as pathway analysis, overexpression of key genes, and in silico metabolic engineering are meticulously discussed. Chapter 3 addresses the increasing importance of biofuels. From the types of preprocessing of lignocellulosic materials to in silico metabolic engineering, the chapter provides a comprehensive overview of strategies to enhance lipid synthesis, incorporating technological and computational advancements. Chapter 4 explores the fundamental concepts and applications of enzyme-catalyzed reactions, highlighting the versatility of enzymes in various industries. The chapter not only details the mechanisms of enzyme catalysis but also explores the emerging field of artificial enzymes, offering a glimpse into the future of biocatalysis.

From microbial growth assays to whole cell immobilization, Chap. 5 navigates through the various phases of microbial growth and the reactors employed in microbial cultivation. In-depth discussions on production kinetics and factors affecting microbial growth add depth to the understanding of this critical bioprocess aspect. Chapter 6 provides an exhaustive exploration of bioreactor configurations for fermentation processes. From traditional stirred tank bioreactors to advanced single-use bioreactors and novel photobioreactors, the chapter elucidates the design principles and control systems that drive these systems.

Chapter 7 ventures into downstream processing, unraveling the historical background and different stages involved. With a focus on integrated downstream processing unit operations, the chapter navigates through the complexities of separating and purifying diverse bioproducts, offering a comprehensive perspective on this crucial aspect of bioprocessing. Chapter 8 focuses on principles of basic and advanced techniques for the purification of enzymes. From traditional techniques like centrifugation to advanced chromatographic methods, the chapter details the isolation and extraction of enzymes, emphasizing their industrial applications. Chapter 9 delves into bioprocess strategies for the production of short-chain organic acids from microbial cells. With a focus on biosynthesis, strategies to improve production, and the role of metabolic engineering, the chapter sheds light on the challenges and prospects in this domain. The next two chapters are mainly dedicated to bioprocess strategies for plant cells. Chapter 10 explores metabolic burden analysis in in vitro culture of plants. By examining the kinetic parameters influencing cell stress, the chapter provides insights into optimizing plant cell cultures, offering a unique perspective on the intersection of plant biology and bioprocess engineering. Chapter 11 focuses on advances in bioreactor design for plant cell culture. The chapter explores the historical context, basic requirements, and various morphological forms relevant to plant cultivation in bioreactors, providing a comprehensive overview of this specialized field. Chapter 12 widens the scope to the role of nanotechnology and artificial intelligence (AI) in waste management. Covering nanomaterials, catalysts, and the application of AI across the waste management process, the chapter highlights the potential of these technologies to revolutionize the sustainability landscape. Chapter 13 focuses on the economics of fermentation processes. The chapter presents the factors affecting the cost of bioprocesses, offering insights into the estimation, optimization, and strain improvement strategies to maximize production while minimizing costs.

The book *Recent Advances in Bioprocess Engineering and Bioreactor Design* is a complete guide that connects basic principles with the latest technologies in the field. Each chapter reflects the combined knowledge of experts, offering readers a detailed insight into various aspects of bioprocess engineering. It reveals ways to innovate in the future, making it a guide for the upcoming generation of bioprocess engineers and researchers.

Raipur, Chhattisgarh, India Swasti Dhagat

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Chapter 1
Introduction to Bioprocess Technology



Fariya Khan, Shadma Andleeb Khan, Gauresh Sharma, Reena Vishvakarma, Alvina Farooqui, Mohammed Haris Siddiqui, and Archana Vimal

Abstract Bioprocess technology is the amalgamation of technology with bioprocess which is the use of any living cell (microorganism or merely its enzymes) or one of its components with defined nutritive supplements and under controlled conditions to obtain a specific desirable product that is useful for mankind. Bioprocess technology is the alteration of significant processes to create value-added products. It deals with designing and developing equipment and processes for products such as food, chemicals, feed, nutraceuticals, amino acids, polymers, or abundantly required and useful pharmaceuticals such as antibiotics, viral vaccines, etc. The use of these processes can be traced back to as old as the ancient Egyptian period where unknowingly they employed this technology for the production of beers, wines, bread, cheese, yogurts, and fermented pickles. The current perspective of bioprocess technology deals with the use of exceedingly advanced computer-operated automatic bioreactors to produce elevated quantity and quality of desired end product around which the process is curated. Future applications of bioprocess technology have significant potential since they will be used in large-scale industries, where its use is currently limited. It has been anticipated that combining bioinformatics and nanotechnology with bioprocess technology would open countless unexplored doors and lead to substantial progress in understanding complex biological systems and their underlying mechanisms and designing and screening new biologically useful components. Besides umpteen useful traits, bioprocess technology still needs to overcome a large number of hurdles and possess an advantage over other competing methods such as chemical engineering to be viable in any specific industrial context. Nevertheless, this technology holds great potential which needs to be efficiently explored to use it to the best of its capabilities.

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1.1 Introduction

Bioprocess technology is the implementation of technology to biological processes to produce an industrially important end product or enhance the quantity of a product with a limited natural production rate (Brakmann et al. 1999). It is mostly the combination of OLD and NEW biotechnology processes and involves the use of a complete living cell be it a plant, microbial, or mammalian cell, or any of its components like their organelles, or using their enzymes as biocatalysts that speeds up the entire process by bringing about some sort of physical or chemical changes in biochemical material derived from precisely designed media to produce specific desirable end products (Nandy 2016). Bioprocess technology works at the forefront of technology and biochemical reactions involving living cells with the aim of "*Enhancing life, implying technology*."

Bioprocess technology is more or less the upgradation of chemical engineering involving complex multiplexed reactions. It involves much more improvised, developed, and precisely designed processes targeted for the production of feed, food, nutraceuticals, chemicals, polymers, pharmaceuticals, and even paper from biological resources (Long et al. 2014). The development of food processing systems, the use and testing of product separation technologies, the design and operation of fermentation systems, the design of instrumentation to monitor and control biological processes, the production of biofuels, and many other application areas are just a few of the ones that are frequently associated with bioprocess technology (Singh et al. 2020). Bioprocess technology is an essential field because for some processes it is the sole practical method by which a specific desired product can be made like antibiotics and vaccines (Heinzle et al. 2007).

In all of its manifestations, bioprocess technology involves a wide range of intricate enzyme-catalyzed reactions within certain biological systems, and these reactions are highly reliant on the physical and chemical conditions that occur in their immediate surroundings (Panke and Wubbolts 2002). The only time bioprocessing will be successful is when all the necessary elements are present. This is where the bioprocess engineers come into play. Bioprocess engineers are trained to have a mindset with problem-solving approaches and inbuild skills for implementation of the process. What sets bioprocess engineers apart from other technical engineers is their comprehension of connecting the dots between engineering subjects concerning living materials (Doran 1995).

Essentially, all biotechnological phenomena or processes occur in a closed-contained system referred to as bioreactors. These processes include a large number of cells be it plant, animal, or microbe, and a specifically designed compatible medium within the bioreactor which ensures their close interactions with each other and creates a favorable environment for growth and product formation (Zhong

2011). The medium-derived substrates are transformed into primary and secondary metabolites, intra- and extracellular macromolecules, and biomass components such as proteins, carbohydrates, DNA, and RNA (Makkapati 2006). The reactions occurring are catalyzed by respective enzymes and are codependent on the physical and chemical environment of the bioreactor. A bioreactor's primary purpose is to reduce the cost of production of a desirable end product. Additionally, it must prevent cells from being released into the environment. In its most basic form bioprocess technology could be seen as inoculating microorganisms with nutrient broth and allowing the components to interact and react to produce an end product (Cooney 1983). To ensure that the bioprocess runs efficiently and can be easily and precisely replicated with the same quantities of raw materials and inoculum (the specific organism) to generate the same quantity of end product, more multifaceted and advanced processes functioning on a large scale must be in charge of the entire system.

Although the conventional types of bioprocess technology are linked to the production of feed, food, and beverages which still make up the majority of commercial bioproducts, the production of new products is emphasized from fermentations of microbial, mammalian, and plant cells (Pandey et al. 2000). Bioprocess technology curates processes specifically designed for

- The overproduction of crucial primary metabolites such as polysaccharides, acetic acid, acetone, lactic acid, amino acids, glycerol, butyl alcohol, organic acids, and vitamins.
- Generation of secondary metabolites, such as streptomycin, penicillin, cephalosporin, gibberellins, etc., which do not appear to play a crucial role in the metabolism of the producing organism.
- Production of novel recombinant products, such as therapeutic proteins, vaccines, and monoclonal antibodies.
- Synthesis of a variety of industrially relevant enzymes, including intracellular enzymes like invertase and asparaginase as well as exocellular enzymes like amylase, pectinase, and proteases, etc.

These products are all crucial to contemporary life and own substantial industrial markets today. With a few exceptions, the majority of these bioproducts cannot be manufactured profitably by other chemical methods, hence the future market expansion of these products is almost guaranteed (Langeveld et al. 2010). Moreover, by genetically modifying organisms to have greater or unique productivity and utilizing new technical advancements in processing, it is possible to achieve great cost reductions in terms of production.

No matter what organism is chosen, what medium is employed, or what product is synthesized, the mechanisms of product development in bioprocess technology are fundamentally quite similar. In each instance, cells are grown under predetermined, regulated parameters using an appropriate physical/technical confinement system (the bioreactor), proper medium composition, and environmental growth-regulating factors like aeration and temperature (Frahm et al. 2009). The biological and technological aspects of the cell and systems are both involved in the

optimization of the bioprocess. A thorough understanding of the biochemical pathways involved in product formation is necessary for the effective utilization of an organism's capacity to produce different unique products of specified quality and in vast quantities. Bioprocess technology holds great potential as a branch in the development and advancement of the industrial sector by enhancing the production of economically useful products in a cost-effective method with minimal waste generation (Gavrilescu and Chisti 2005).

1.2 Narrative from the Past

Fermentation was one of the oldest aspects of bioprocess technology where microorganisms converted sugar present in natural food into products that were useful to humans. The earliest known evidence of fermentation dates back to 10,000 BC, when the first human civilization initially appeared in what is now known as the Fertile Crescent (Middle East) as seen in (Fig. 1.1). Bioprocess was used as an art for the production of cheese in the lush Cresent between the Tigris and the Euphrates rivers in Iraq at a time when domestication just started. Even without scientific understanding, people in the past could nonetheless evaluate the needs of the people and develop tools and technology to fulfill those (Al-Hamdany and Hassan 2017). Initially, bioprocess technology evolved as a means for the production of beverages, food, and pharmaceuticals. It was used to produce bread more than 8000 years ago. Even China employed moldy soybean curd to heal skin diseases over 5000 years ago. Beer fermentation and barley malting were carried out by Egyptians as early as about 2500 BC.

The discovery of alcoholic fermentations used in brewing and winemaking is believed to have occurred between 5000 and 7000 BC by the Sumerians and Egyptians. They also developed the art of open dough fermentation used in leavened bread production. Although bioprocess technology has been used for thousands of years to produce drinks, beverages, and food, it wasn't until recently that living machinery in the form of cells was identified as the agent of the entire process. The Industrial Revolution led to the concomitant movement of people to large cities, which in turn led to a dramatic shift from food production for local communities to

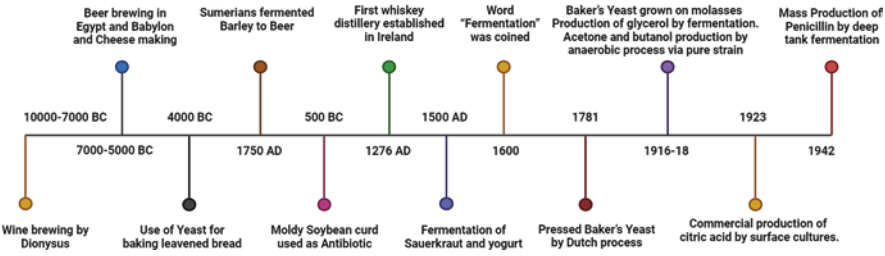


Fig. 1.1 Major breakthrough in the field of bioprocess technology

large-scale production of food, feed, beverages pharmaceuticals, etc. (Mitra and Murthy 2022). This necessitated the development of large-scale processes for commercial production of the same.

During World War II there was an increased requirement for Penicillin, therefore the government incentives encouraged pharmaceutical companies to enhance the production of penicillin. Bioprocess technology played a major role in this entire scenario. Microbiologists, industrial chemists, and chemical engineers swiftly developed techniques for recovering penicillin in an active and stable form through crystallization, lyophilization, and countercurrent extraction. They also established the viability of submerged fermentations (Quinn 2013). Henceforth, the relevance of bioprocess technology came into play. The optimization of the entire process with favorable conditions was designed by bioprocess engineers.

The nineteenth century witnessed the beginning of modern bioprocess technology as an understanding of biological systems, their constituent parts, and interactions became better with time. The first industrial-scale bioprocess operations, such as those that produced citric acid, were developed in the first part of the twentieth century. Starting with insulin, the first product manufactured using recombinant technology, the process of recombinant gene technology then resulted in a significant growth in the number of bioprocesses and their production volume (Adrio and Demain 2010). Human insulin, erythropoietin, interferons, tissue plasminogen activators, colony-stimulating factors, and monoclonal antibody-based products are some of the numerous products manufactured by the application of bioprocess technology that have received the greatest attention. Additionally, there is a significant amount of effort being invested in the development of new products for the food sector, agriculture, fine chemicals, and environmental protection (Walsh 2013).

Bioprocess technology amalgamates all the basic and applied sciences streams such as molecular biology, biochemistry, and microbiology as well as their technical engineering aspects to completely utilize the potential of different biological systems to manufacture products to fulfill the demands of the market. At present, there are numerous applications of bioprocess technology in terms of commercial biotechnology including the manufacturing of pharmaceuticals and the process of drug discovery. Albeit, the biotechnological processes have seen quick and remarkable advancement in recent years, it has been around since the dawn of time, making it one of the oldest technologies even before the field of microbiology was discovered (Schaechter et al. 2004).

The discovery of the field of biotechnology which operates via the utilization of bioprocess technology, and the advancement of these processes offers technical solutions to a wide range of issues in the food, health, energy, agriculture, and many other industrial sectors. Therefore, the interest in bioprocess technology has grown among academics, industries, governments, non-governmental organizations, and communities where all these applications have a favorable economic, social, or environmental impact (Barragán-Ocaña et al. 2020). However, the options proposed by the bioeconomy to foster economic growth should not be restricted to technical improvement in and of itself, but also consider other factors of importance.

1.3 Bioprocess Technology and Its Stages

Bioprocess technology entails the optimization, design, operation, and control, of biochemical processes involving various biological pathways or reactions mediated by living cells of plants, animals, and microorganisms or enzymes under controlled conditions to effectively carry out biotransformation of raw materials into a variety of products at the necessary scales. Therefore, the basic principle of bioprocess technology deals with understanding the basic mechanisms of the cell or the biocatalyst involved which in turn guarantees the unswerving quality and reliability of the product irrespective of the scale (Moser 2012). It also takes into consideration the impact of product manufacturing and usage on the environment and tackles the pressure input by the consumer market through the development of processes in response to the same.

The product developed by bioprocess technology may either have direct applications as feed, food, medicine, beverages, and industrial chemicals, or it may have indirect applications through bioprocesses that do not directly manufacture products, such as the detoxification of industrial wastes or the treatment of manufacturing effluents. Microorganisms have a history of being used for the production of food in the form of fermented products and since then bioprocess technology has been utilized to manufacture a wide range of desired end products ranging from economical and affordable to expensive items such as vaccines, antibiotics, therapeutic proteins, etc.

Also, to re-engineer microorganisms to "over-produce" end products and intermediates of primary metabolism, new and powerful approaches for metabolic engineering have been developed as a result of breakthroughs in genomics, proteomics, and metabolomics (Sweetlove et al. 2017). These developments are now making it easier for the development of cost-effective bioprocesses for the production of bulk chemicals and feedstocks for the chemical industry, especially in light of the rising price of petroleum and the desire for eco-friendly alternatives.

Consequently, bioprocess technology serves as a foundation or building block for the biotechnology sector or indeed is the backbone of these industries that serves as a catalyst in the transformation of research-based knowledge into product-based output and has two counterparts (Fig. 1.2):

- Upstream processing
- Downstream processing

The upstream process of bioprocess technology is the initial planning which includes steps such as identifying and isolating the desired organism to be used, optimizing the bioreactor conditions favorable for its growth, and then designing the media for maximum production of the desired end product (Lindskog 2018). Particularly in the case of biopharmaceuticals, the upstream procedure entails isolating the cell line to be used, culturing those cells essential for the end product, and then harvesting these cells.

The bioprocess itself, which is typically carried out in enormous tanks known as fermenters or bioreactors, is a part of the upstream processing. To facilitate cell

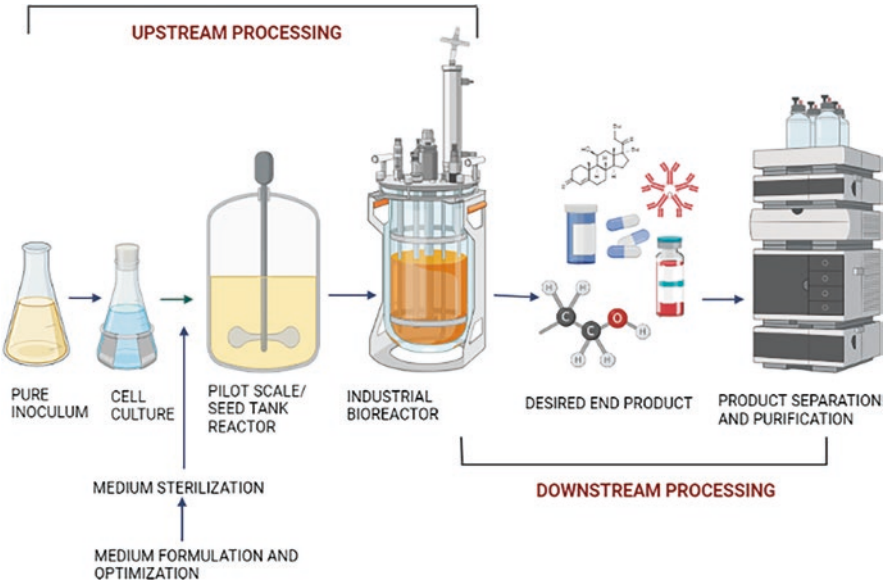


Fig. 1.2 Pictorial representation of bioprocess technology

growth and product synthesis under the best possible conditions, the tank is typically fitted with advanced sets of monitoring and control devices in addition to mechanical parts that create the desired optimum conditions inside the bioreactor, such as aeration, cooling, agitation, etc. Numerous variations of engineering technologies can be used to manipulate the internal bioprocess reactions in the bioreactor (Clapp et al. 2018).

The second stage of bioprocess technology, i.e., downstream processing includes isolation and purification of the end product with acceptable quality and adequate quantity. The various steps of downstream processing, which come after the fermentation process, require appropriate procedures and methods for recovering, purifying, and identifying the desired bioprocess result (Chisti 1998). For downstream processing, a wide range of techniques, including chromatography, filtration, and centrifugation, may be used. These procedures change depending on the final product's desired grade, as well as its chemical and physical characteristics.

Both of these stages constitute the entire bioprocess technology with intermediate steps in each playing very crucial steps important for the efficient implementation of the complete process (Narayanan et al. 2020).

1.3.1 Upstream Processing

The initial step of bioprocess technology is called upstream processing which deals with the selection of the desired strain of the cell, isolating the same from its pure culture along with designing of medium with all the necessary components such as

essential chemicals, growth factors, inhibitors, precursors, etc., followed by optimizing the conditions and factors playing an important role in growth and maintenance of the reactions such as temperature, pH, along with favoring the production of desired end product (Berovic 2009). It also implements different streams like genetic engineering for inoculum improvement.

The crucial components of upstream processing include:

- The selection and production of an active and pure inoculum of desired cell culture in adequate quality and quantity.
- The designing and optimization of appropriate media that will be used for culturing the organism during the development of the inoculum and in the fermenter or bioreactor with precise separation of particulate and inhibitory substances from the medium.
- Bioreactor and mode of operation of bioreactor.
- Sterilization of all the physical components involved in the process including bioreactor, vessels, ancillary equipment, and most importantly the medium.
- The optimization of favorable conditions is necessary for the growth of cells in the bioreactor.

All the steps mentioned are very crucial and each holds its utmost importance for a successful completion of the entire process. The initial step where a particular strain is selected depends upon the capability of the cells to synthesize our desired end product having the desired economical value (Jain and Kumar 2008). Strain improvement techniques are then implemented to increase the efficiency of production and sometimes to improve the quality of the product.

1.3.1.1 Inoculum Development

The term "inoculum development" describes the increase in viable cell count from a dormant stock stage to a final productive exponential state. The stock culture used for inoculum development should possess some desirable characteristics such as it should be free from any form of contamination. Also, it should be in an active state to skip the transition from dormant to active state. It should be in adequate amounts to be used as seed culture. Last but most importantly, it should support the maximum production rate of the desired product and that can only be ensured when the media is designed precisely and accurately and is complementary to the needs and counteracts any hindrance caused by intermediate products that affect the quality, quantity or rate of the desired end product (Roussos et al. 1991).

1.3.1.2 Media Design and Optimization

All types of cells require certain elements such as carbon, phosphorus nitrogen, minerals (possibly vitamins), and water for growth along with the presence of sufficient amount of dissolved oxygen (if aerobic). Therefore, the most crucial need for

the optimal creation of the desired product, whether cell biomass or a particular metabolite, is the formulation of a proper culture medium with a sufficient amount of nutrients to support growth and production. Precursors, inhibitor compounds, inducers, or antifoam agents are often a part of media design and optimization to target the production of desired products (Nurfarahin et al. 2018). Optimum environmental conditions, i.e., pH, temperature, humidity, etc. inside the bioreactor and media formulations severely affect the biochemistry, morphology, and physiology which in turn affects the production rate, yield, and other properties.

1.3.1.3 Bioreactor and Its Mode of Operation

The heart of bioprocess or fermentation technology is the bioreactor. A bioreactor sometimes also referred to as a "Fermenter" is a large vessel that provides a controlled and optimized environment for the growth of microorganisms, plant, or animal cells to yield a specific end product. The bioreactor should be capable of being operated aseptically for several days and should be reliable in long-term operation and fulfill the containment regulations (Allman 2018). Besides it should favor the maximum growth rate of the cells along with maintenance of proper aeration and agitation. A bioreactor should be advanced with automated sensors and controllers along with being economical i.e., it should consume less power. There are different types of bioreactors available each having their own sets of advantages and disadvantages (Spier et al. 2011).

These are made to facilitate the optimal environment that supports biosynthesis and production for industrially significant cultures and provides flexibility to manipulate them for all tasks related to using the bioreactor. These biovessels need to be sturdy enough to withstand the weight of a significant amount of an agitated medium. The bioreactor should not corrode the end product, degrade the quality or make the growing medium hazardous (Tekere et al. 2019). This requires careful design of each component of the vessel as well as any opening ports that are a part of the process.

Novel and advanced bioreactor types and modes of operation have been employed in response to the rising demand for efficient and economical ways to produce significant amounts of critical biological components from living cells and enzymes (Wang and Zhong 2007). Currently, bioreactors are utilized for small- to large-scale manufacture of several diverse items from isolated enzyme systems of microbial, animal, and plant cells. Biotechnologists always keep in consideration the medium costs, downstream processing expenses, and technical benefits when selecting a bioreactor type and its method of operation (Felo et al. 2013).

Cells vary in terms of their growth pattern and mode of synthesis or production. Different cells are therefore cultivated in batch, fed-batch, or continuous cultures to produce the necessary end product. Continuous fermentations are less prevalent than batch and fed-batch fermentations. The type of product being produced determines the mode of operation of the bioreactor each having its own set of advantages and disadvantages (Büchs 2001).

1.3.1.4 Sterilization of the Components

Sterilization is the elimination of all sorts of contamination, i.e., unwanted microorganisms in all forms be it active or as spore by either killing or physical removal to facilitate an unhampered reaction environment. One of the most important elements of upstream processing is the sterilization of fermentation media, gases from the inlet and outlet, and the maintenance of aseptic conditions during bioprocessing (Berenjian 2019). There are different kinds of sterilization methods depending upon the requirement namely, physical and chemical. Filtration, radiation, ultrasonic treatment, chemical treatment, or heat are mostly used to sterilize media. Steam, however, is almost always employed for the disinfection of fermentation medium due to practical considerations (Silindir and Özer 2009). The main exception is the sterilization of media for animal-cell culture, for which filtration is the preferred procedure since the media are soluble and include components that are heat labile (Whitford et al. 2018).

1.3.1.5 Optimization of Favorable Conditions

The ultimate goal of bioprocess technology in all forms is to guarantee that every component of the system is exposed to the same circumstances with identical conditions. The cells are suspended in the nutritional medium of the bioreactor, which contains the building blocks essential for cell growth and the production of vital products (Ge et al. 2023). Each cell must get all necessary nutrients, including dissolved oxygen, and carbon and nitrogen sources, with micronutrients, growth factors precursors, etc. along with efficient elimination of waste products.

The range of dissolved nutrients also plays a critical role, as low levels will slow the pace of an organism's metabolism, while high quantities might be toxic, hence the concentration of nutrients around must be kept within a specific range. In biotechnological processes, these conditions must be given on a micro-scale to ensure that each cell is equally catered for. Biological reactions function most effectively within optimal ranges of environmental parameters (Liu 2020). It is quite challenging to create these circumstances in a large scenario when the enormous scale of multiple bioreactor systems is taken into account. This is where the abilities of the microbiologist and the process or biochemical engineer must meet. The bioprocess parameters also have a significant impact on the effectiveness of sedimentation, filterability, and homogenization later in the downstream processing stage (Gerzon et al. 2022).

Only when all the specific growth-related factors are optimized and implemented accurately then only, bioprocessing would be successful. To comprehend and control a bioprocess, it needs an analysis of its state over a brief period as well as an understanding of how the cells react to several quantifiable external factors. For processes to be optimized, feedback control must be precise and quick (Sonnleitner 2006). Therefore, computer has become a crucial component of the majority of

bioreactor systems in the present. The chemical components of the bioprocess cannot currently be analyzed online due to a lack of effective sensor probes. Data specific to growth and production along with internal factors, should be monitored, recorded, and utilized to enhance and optimize the process efficiently.

1.3.2 Downstream Processing

The isolation and purification of the desired end product in an acceptable quality to be used for the intended purpose is referred to as downstream processing. The desired end product(s) are present usually in a diluted aqueous solution containing exhausted media, unused media components, cells, undesired products, toxins, etc., and the final stage of bioprocess technology, i.e., downstream processing deals with the isolation of desired end product and acquiring the level of purity needed (Kalyanpur 2002). After the reaction has been completed, the desired product must be recovered. This will, at the very least, entail separating the cells from the broth, but it may also involve purifying the metabolites with or without cell disruption (Fig. 1.3). Downstream processing is the term used to describe such procedures. Finding highly selective recovery methods that leave as little residual product in the medium as possible will require research, which will reduce the labor involved in downstream processing (Flickinger 2013).

In the realm of bioprocess technology, a diverse range of substances, such as monoclonal antibodies, whole cells, organic acids, amino acids, enzymes, vitamins, solvents, vaccines, and therapeutic proteins, are routinely extracted and refined from the fermentation broth during downstream processing. These end products exhibit considerable differences in both their molecular size and chemical intricacy, which mandates the utilization of a wide array of separation methods to isolate and

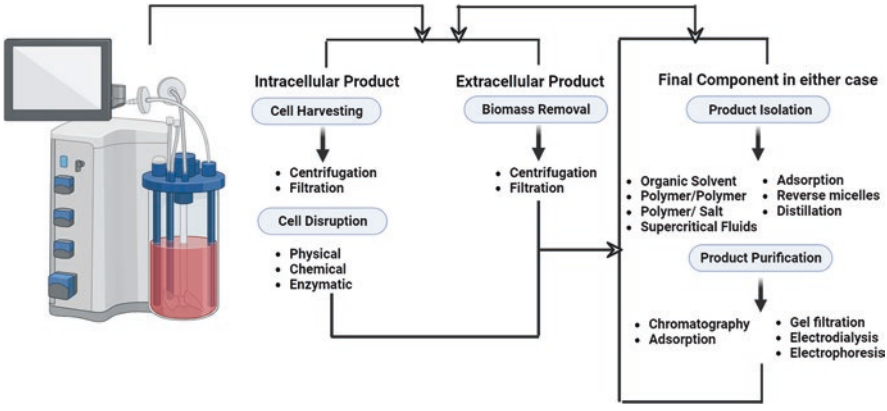


Fig. 1.3 Steps involved in downstream processing

purify the desired compounds (Haldar and Purkait 2020). While many of these products possess robust and stable structures, some are characterized by delicate and labile configurations, necessitating a cautious approach during their processing.

The successful utilization of isolated end products for commercial purposes hinges on the careful design and efficient execution of downstream processing. It is crucial to minimize any loss, no matter how small, of the desired product during these operations. Downstream processing comprises multiple stages, with the primary focus being on the initial separation of the fermentation broth into liquid and solid phases. Subsequently, the concentration and purification of the target product become the central objectives in this process (Wheelwright 1989).

After the removal of broth from the bioreactor, it is usually unstable and is vulnerable to microbial contamination, therefore it requires immediate downstream processing. The chemical and physical characteristics of the products also profoundly influence downstream processing. For the initial pre-treatment or conditioning of the broth, cells or solids may be flocculated, or cells may be broken down to release intracellular products like enzymes. The final level of processing depends on the product's intended use and the required quality when separating the components in the liquid phase of the broth through settling, filtration, centrifugation, or electrokinetic separation (Jornitz 2019). The product is separated into its liquid and solid components (such as microbial biomass), and the broth is then disposed of via conventional sewage treatment.

The isolation of the final product makes use of the most advanced and efficient chemical and biological techniques. The most effective extraction methods will be determined by properties such as molecular size or centrifugation, and they play a huge role in the overall economics of the production and quality of the final product (Chemat et al. 2020). The purity of the finished product depends on the kind of product and the needed level of purity. Gel-filtration chromatography, evaporation and distillation, membrane filtration, HPLC, GC-MS, precipitation, adsorption, affinity, and ion exchange chromatography are some of the techniques currently in use.

The final products arising after downstream purification processes ought to be sufficiently stable for commercial distribution. Most products are effectively stable when they are dried in some way. In actuality, this is accomplished through spray-drying, freeze-drying, or fluidized-bed drying. The technique of choice depends on the desired end product and the cost to make the process economical (Pisano et al. 2023). Polysaccharides, organic acids, enzymes, amino acids, antibiotics, single-cell proteins, and many other products are available in dry form. Many items must be sold as liquid formulations because they are difficult to supply in dried form. When the product is proteinaceous it is better to take precautions to prevent denaturation as well as microbiological contamination and product degradation during downstream processing (Bergemann et al. 2007).

Downstream processing continues to be one of the most difficult and time-consuming aspects of bioprocess technology. The majority of high-value biotechnological products are characterized by their purity and stability and therefore

downstream processing should be designed very precisely with a smart selection of isolation and purification techniques. The downstream processing accounts for almost 60% of the total production cost (Grote et al. 2012).

1.4 Scale Up in Bioprocess Technology: *It's All in the Volume*

The majority of bioprocesses are initially performed at a laboratory scale and then moved to a large scale. The eventual commercial viability of the process depends on its capacity to scale up, first from the laboratory to the level of pilot scale and subsequently to industrial-scale production for commercial use. Successful process scale-up must take into account a variety of practical and financial constraints. Typically, laboratory scale culture is used to identify some of the regulating factors before being scaled up to the pilot level. A pilot plant is a laboratory at a large scale that is built with flexibility for accommodating equipment and adaptability in process operation. The total volume of pilot plant bioreactors ranges from 100 to 10,000 L, and the larger ones occasionally serve as production units (Henry and Gilbert 1973). Volumes of full-scale industrial bioreactors range from 20,000 to 400,000 L. High capital expenditures and investments are needed for operations such as mixing and aeration, along with advanced monitoring and control equipment, and stringent sterility maintenance in the management of scale-up.

A reliable bioprocess scale-up is influenced by several different parameters each playing a crucial role. The reaction volume essentially has no impact on kinetics or thermodynamics. Nevertheless, the scale has a significant impact on mass transfer in a bioprocess. Just as the agitation speed and gassing rate are affected by the bioreactor's geometry and impeller, so are the oxygen intake, substrate, and the addition of chemicals for pH regulation. An efficient scale-up can be helped by standardized bioreactor formats, such as height-to-diameter ratios (Garcia-Fernandez et al. 2020).

Microtiter plates and traditional shake flasks are still frequently employed in the early stages of bioprocess technology development due to their low cost and ease of handling. These instruments, however, are handled manually have one or no instrumentation, and provide only a few monitoring and control options. Furthermore, in larger-scale bioreactors, shaken cultures entirely vary from stirred-tank cultures in terms of hydrodynamics. These early-stage development techniques do not mimic or accurately reflect the conditions present inside advanced large-scale, well-equipped stirred-tank bioreactors, thus making them unsuitable for the design and development of industrial production methods (Lim 2005). Miniaturized stirred-tank bioreactors, which are comparable to large-scale production reactors in terms of design and fluid dynamics, and mimic the conditions more efficiently offer a more reliable scale-up technique.

Scale-up criteria are often based on engineering characteristics and factors such as vessel and impeller design, tip speed, mixing duration, oxygen transfer rate,

volumetric mass transfer coefficient (kLa), or power number are taken into consideration. The basic approach is to maintain a particular process parameter constant and manipulate other factors throughout the scale-up procedure (Zlokarnik 2006).

In bioprocess technology, for cells that require aerobic conditions, the transfer of oxygen is often a rate-limiting step because it is a critical step and is required for the growth of cells in the culture. Utilizing equipment with identical oxygen transfer rate capabilities across all scales is crucial for an efficient scale-up. The kLa, which may also be used as a scale-up criterion, can be used to describe oxygen transfer inside a bioreactor. A common scale-up technique is to keep the impeller's power consumption per liquid volume (P/V) constant (Labík et al. 2018). This is achieved by altering the impeller's size and shape to correspond with the various working volumes. The impeller power number (Np), also known as the Newton number (Ne), is a dimensionless number connected to various types of impellers, and it is used to determine the P/V value.

The dimension analysis and similarity criteria are the basic foundations of traditional scale-up and usually, the biological characteristics of the cells are not taken into account and are ignored. The intricate connection between cellular machinery and the extracellular environment is addressed by the multi-scale analysis, on the contrary, and at all scales (Prothero 2002). It takes into account a wide range of influencing and interacting elements of the intricate biological cultivation system.

For successful implementation of multi-scale analysis, a thorough knowledge of the metabolism and activity of the cells, as well as the technical capacity for extensive measurement and the incorporation of real-time analysis is required (Hartmann et al. 2022). The multi-scale analysis is an effective method for creating scalable and reliable bioprocesses due to the expanding opportunities for accurate online monitoring, analysis, and control of bioprocess changes.

Even though the designs of bioreactors of various sizes are similar, it is impossible to anticipate identical process performance at every scale (Fig. 1.4). Different gradients may develop in large-scale bioreactors as a result of inadequate mixing and mass transfer restrictions. A heterogeneous culture environment is created due

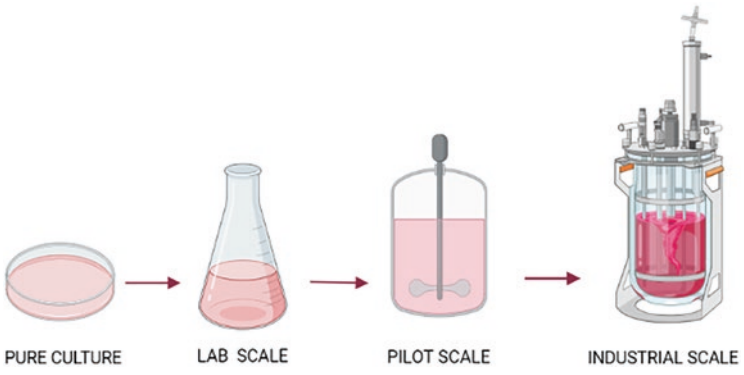


Fig. 1.4 A simple representation of scale-up

to dissolved oxygen, pH changes, or substrate gradients. Large-scale bioreactor heterogeneities could cause a variety of physiological reactions in the cultured cells and organisms which is highly undesirable (Delvigne et al. 2014). In comparison to lab-scale bioreactors, growth and productivity can be altered, which could lead to decreased biomass and product yields or higher by-product generation. Therefore, for a successful scale-up, it is essential to comprehend how the cells perform in a dynamic environment.

An innovative tool for replicating the heterogeneities in industrial-scale bioreactors is the two-compartment bioreactor. Even in the process development stage of a laboratory scale, this type of scale-down method facilitates anticipating probable quality loss during scale-up. The plan is to create a two-compartment bioreactor system out of two separate bioreactors. The upper and lower gradient environments are represented by the culture environments that are unique to each of the single bioreactors and have their parameter values (Nadal-Rey et al. 2021). Cells are constantly switched from one compartment to another during culture. The outcomes of the two-compartment cultivation can shed light on the bioprocess phenomenon that might have first surfaced during the actual industrial production process using a conventional method.

There are many brand-new prospects due to the speedy advancement of computer technology and simulation techniques. The characteristics and behavior of cultivated cells, as well as the entire bioprocess, can be understood, predicted, and optimized using sophisticated mathematical models. These integrative models link cell kinetics and fluid dynamics (Wang et al. 2015). They can be used to model physiological reactions and hydrodynamic effects at different scales and types of bioreactors.

Each bioprocess is a sophisticated system. The living cell or organism that is being cultured interacts constantly with other cells, itself, and the environment where it is being cultivated. Every modification of the cultivation conditions sets off reactions and internal controls in the cells. Additionally, when bioreactor size increases, heterogeneities appear, and the cells must adjust to a fast-changing environment (Neubauer and Junne 2010). The use of fully scalable bioreactor technology in conjunction with scale-up tactics that take into account the aforementioned difficulties can provide effective methods for scaling up bioprocesses from the early stages of development to final production.

The cost to develop the entire production in the lab-scale process is typically less than the cost to scale up a bioprocess technology to a manufacturing scale. This can range from US $100 million to US $1 billion, taking into account manufacturing plant development and startup costs as well as intermediate process validation (on pilot and demo sizes). Typically, it takes 3–10 years to go from lab scale to industrial scale (Marques et al. 2010). When financial risk is this high, a decline in process performance during scale-up will be expensive and disruptive, and it could even result in project failure. Even slight (5–10%) underperformance and/or delays during scale-up can significantly lower investor returns and erode stakeholder and consumer confidence, excluding collapse. Therefore, it is important to do it right and to get it right the first time when scaling up microbial operations.

1.5 Bioprocess 2.0: Insight into the BioFuture

Bioinformatics and nanotechnology are gaining momentum worldwide and has become vital tool for the bioprocessing industry in meeting the anticipated ever-increasing worldwide demand for resources that will result from tremendous population growth. The combination of nanotechnology and bioinformatics, known as nanoinformatics, may enhance the manufacturing process and their optimization and result in products with improved properties and novel functions for bioprocessing industries (Xia et al. 2013). Nanosensors that are combined applications of nanotechnology and bioinformatics serves with real-time quality assurance by tracking cell growth pattern and characteristics, toxins generation, and any chances of contamination throughout the bioprocess through data capturing for enhanced optimization, automatic control of functions, and data storage and documentation.

The feature to manufacture desired products like lifesaving drugs that can transform healthcare all around the world for patients makes bioprocess technology the backbone of the biotechnology and pharmaceutical industries. For the same reason, the idea of developing novel technologies that can enhance these processes is of great interest (Hole et al. 2021). Bioprocess technology engineers must have a competitive spirit and should be innovative to walk along with engineers from different fields. The industries dealing with bioprocess technology embraced the future by incorporating intelligent automation using artificial intelligence (AI) and machine learning. Robotics, deep learning, big data analytics, artificial intelligence, and machine learning are just a few examples of the wide spectrum of technical breakthroughs that are referred to as "intelligent automation" in manufacturing settings in industrial aspects (Jha et al. 2021).

It is now more convenient than ever before to assemble data from multiple sources due to the advent of AI solutions in the bioprocessing industries, which enables multiple stakeholders within one establishment or across several organizations to use this knowledge to optimize and improve their operations and the quality of their products (Davenport and Kalakota 2019). To comprehend and trace crucial process parameters, such as substrate utilization rate, mixing efficiency, and cell growth rates at various stages of growth in cell culture, bioprocess technology experts have started implementing AI techniques such as machine learning algorithms (e.g., multivariate data analysis, or more complicated deep neural networks), text mining, pattern recognition, natural language processing (NLP) along with other signal processing approaches.

By implementing the use of nanosensors, artificial intelligence, and machine learning, bioprocesses can be monitored and controlled more efficiently. AI-based technologies are being utilized to support process monitoring operations and optimization at various stages of a bioprocess technology, including both upstream like cell culture, growth kinetics, and synthesis of products at larger scale and downstream processing where product purification is involved (Gerzon et al. 2022).

By delivering more precise data that would have been challenging to gather without these technologies, their application enables companies to benefit in terms of

better optimization, enhanced quality, efficient yield, large productivity, cost reductions, and greater safety (Javaid et al. 2022). They can be used in a variety of ways to maintain ideal conditions for growing cells at all stages of the process, ranging from cultivation to product harvesting and purification that require precise control to attain a high level of purity like chromatography systems, etc.

The core of bioprocess technology is cell culture techniques. These are either based on kinetic equations of growth or experimental data that take into account several variables like concentrations of various medium components in media design, time, and temperature. To maximize output and produce more biomass per unit volume at the lowest possible metabolic cost while using the least amount of energy and resources, all these factors must be tuned throughout cell culture (Sun et al. 2018). By reducing the time and expense involved with experimental processes, machine learning algorithms can be utilized to create models that aid decision-making during new product research and development. These models provide a better understanding of the effect of factors such as oxygen concentrations, pH levels, and mixing conditions on cell growth kinetics and production (Vora et al. 2023).

While there has been progress in modeling different aspects involved in downstream processing, there is still a considerable distance to cover before achieving comprehensive success at every stage. The journey involves the development of more precise models rooted in scientific principles rather than solely relying on trial-and-error approaches or statistical correlations derived from observed variables over time. Successful model development requires data derived from experimental work on actual samples as well as with the help of computational methods like deep learning through neural networks & machine learning algorithms (Sarker 2021).

The bioprocess industry as a whole has widely acknowledged the need for digitalization due to its potential advantages including improvements in efficiency with reduced time, delays, resources, etc., and enhanced productivity, leading to increased revenue streams because of better decision-making abilities (Pandey et al. 2023). The ideal balance between robustness, process design, and product quality is still something that bioprocess engineers are still working to achieve while avoiding risks related to regulatory compliance. When integrated into process development and manufacturing facilities, AI models can assist in achieving their goal by giving a more data-driven strategy for bioprocess optimization and control (Rathore et al. 2023).

One of the main advantages of using an advanced process control system is the capacity to detect, identify, or analyze anomalies through real-time process monitoring combined with anomaly detection, classification, or predictive analytics. For enhanced real-time monitoring of bioprocesses, it is necessary to integrate process analytical sensor technologies, spectroscopic techniques, data management, visualization, and sophisticated data analytics (Clementschitsch and Bayer 2006). Advanced process control can be used to thoroughly monitor and anticipate the complete bioprocess production stage (and not single unit operation), followed by advanced data-driven modeling based on artificial intelligence technologies.

Ingenious bioprocess technology development based on continuous production could be achieved by automating the control of large-scale reactors using machine learning models trained on data sets from benchtop bioreactors (Mowbray et al. 2021).

1.6 Roadblocks in the Way: Challenging but Possible

The difficulty in bioprocess engineering is in determining the demands of the sector, fostering knowledge transfer, and training engineers to meet the diverse markets, needs, and products that are being covered by breakthroughs in bioprocess technology and biotechnology (Rendón-Castrillón et al. 2023). A big portion of this issue is developing the enabling technologies for industry to fully utilize the potential of contemporary biology and chemistry through synthesis and innovation.

Although additional effort is still required to advance toward an integrated sustainability framework, the scientific and technical trajectory demonstrates how sustainability and bioprocesses are subjects of considerable interest and are constantly expanding (Liu 2020). In this technological field, microbiology and enzymology are frequently prevalent, but new opportunities are opening up in response to the increased demand for sustainable solutions to support economic growth and industrial development, particularly basic science projects, which should be thoroughly investigated and exploited (Barragán-Ocaña et al. 2020). Emerging technologies and products of the future include innovative bioprocesses like biorefining, bioethanol production, consolidated bioprocessing, microalgae utilization, lignocellulose conversion, biocatalysts, and biohydrogen production.

Specifically, designed equipment that complies with regulatory, biological, and economic requirements, the incorporation of manufacturing processes into environmentally acceptable and financially viable process concepts, rapid purification and monitoring of purification processes to produce products of high-quality, high-purity, and consistent output are some of the challenges that bioprocess technology must overcome (Rader 2018).

Although biotechnology has made unheard-of advancements, it is also true that bioprocess technology design needs to be tailored toward sustainability standards, where social effects must be given priority. Legislative, normative, and ethical considerations, resource optimization and conservation, technical and risk assessment, biosecurity, and intellectual property are additional factors to be taken into account to ensure the successful development of future bioprocess technology applications for the benefit of economic development, environmental protection, and social welfare (Singh et al. 2012).

It is also necessary to improve conditions for culturing cells and producing desirable products, as well as bioreactor design. Submerged and solid-substrate fermentations provide significant potential for bioprocess engineering-based

biomanufacturing and process optimization (Molina et al. 2023). The economics of manufacturing specialty bioproducts could be improved by new bioreactor designs, membrane separation technology, and processing aids.

Process productivity has seen a significant improvement during the past 20 years. Process control and monitoring developments have improved affordability by lowering development costs. Stochastic and mechanistic model-based controllers are presently becoming well-liked options for bioprocess control, inspired by the most recent regulatory requirements within the R&D framework (Nikita et al. 2023). It has been found that using simulations and sophisticated statistics produces results at a lower cost and in less time than the conventional experimental approach. The introduction of neural network (NN)-based controllers and the development of process control schemes ranging from cascade to adaptive to hybrid are the results of this. Recently, efforts have been undertaken to develop controllers based on holistic process models that provide a digital representation of the entire bioprocess (Rathore et al. 2021).

The ideal balance between robustness, process design, and product quality is still something that bioprocess engineers are still working to achieve while avoiding risks related to regulatory compliance. When integrated into process development and manufacturing facilities, AI models can assist companies in achieving their goals by giving a more data-driven strategy for bioprocess optimization and control (Borgosz and Dikicioglu 2024). Additionally, manufacturing automation would offer significant potential to overcome many of the obstacles to commercial success (Tung et al. 2019).

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Chapter 2
Understanding the Metabolic Pathways to Increase the Product Yield



Surender Jangra, Soumita Paul, and Kongara Hanumantha Rao

Abstract Synthesis of biochemical products at mass scale is highly desirable in industries. Metabolic control of enzymatic activities plays an important role in the production of products commercially. Metabolic pathways are regulated by activation or deactivation of allosteric enzymes. For this, the removal of inhibitors and addition of activators for allosteric enzymes is highly required. Moreover, mutant versions of enzymes by recombinant technology could serve as a promising approach for the synthesis of desirable products at mass scale. In this chapter, we discussed all the possibilities of optimization and standardization of biotechnological approaches that would provide new insights understanding basic principle which can be followed to attain better yields.

Keywords Metabolites · Metabolic flux analysis · Optimization · Gene regulation

2.1 Introduction

Metabolites are generally organic molecules that are intermediate or by-product of various metabolic reactions within the living organism. In the context of metabolism, metabolites can be classified into two main categories: primary and secondary metabolites. Primary metabolites are small molecules that are produced in the

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exponential phase of the growth (Sanchez and Demain 2008) and found in every living cell capable of dividing (Thirumurugan et al. 2018). They are typically essential for cellular functions and present in relatively high concentrations. They play crucial roles in growth, development, reproduction, and survival of an organism. These metabolites are typically synthesized through primary metabolic pathways, which are conserved across various species, indicating their fundamental importance in biological systems (Pickens et al. 2011). Examples of few commercially produced primary metabolites are ethanol, lactic acid, citric acid, acetic acid, amino acids, nucleotides, vitamins, organic acids, polyols, glycerol, etc. (Sanchez and Demain 2008). They are produced on a large scale to meet the demand of various industries and serve as essential components for numerous products and applications. In industries microbial fermentation is preferred over chemical synthesis of these metabolites as it is economically competitive and produces specific isomeric form of metabolite (Sanchez and Demain 2008). Secondary metabolites are molecules that are not directly involved in the primary metabolic pathways required for growth and development. Instead, they often have specialized roles, such as defense against pathogens or attracting pollinators. Secondary metabolites are generally produced in lower quantities compared to primary metabolites and are synthesized during the end or near the stationary phase of growth (Calvo et al. 2002). These molecules are found only incidentally and are not of paramount significance to organism's life (Thirumurugan et al. 2018). Many secondary metabolites have commercial value due to their medicinal, agricultural, or industrial applications. Metabolites such as antibiotics, alkaloids, flavonoids, terpenoids, polyketides, steroids, tannins, resins, saponins, pigments, etc. are few examples of industrially produced secondary metabolites (Chandran et al. 2020). The production of secondary metabolites often involves microbial fermentation or plant cell and tissue culture approaches to meet the demand for these valuable compounds (Hussain et al. 2012).

Secondary metabolites are produced by plants, bacteria, fungi, and many marine organisms such as snails, corals, sponges, and tunicates. Among them plants are major sources of secondary metabolites. Approximately, 80% of the secondary metabolites are extracted from plants only (Bérdy 2005). Plant secondary metabolites are bioactive compounds that have various applications in the food industry, including flavorings, colors, and health-promoting properties. Metabolites with antioxidative properties can be used as preservatives or anti-browning agents. Traditionally, these compounds have been obtained from plants themselves by different phytochemical procedures such as solvent, steam, and supercritical extraction (Thirumurugan et al. 2018). But the direct extraction of these compounds from plant materials is often challenging due to the following reasons: (1) chemically related complex mixture of secondary metabolites are often found in plants; (2) desired metabolite may not be produced all times, and in every tissue; (3) metabolite conc. is affected by variation in season and geographical location (Kallscheuer et al. 2019). However, with advancements in genetic engineering and synthetic biology, it is now possible to produce plant secondary metabolites using genetically

engineered microbes. These microbes are preferred for the production of plant metabolites as they have higher growth rate, can be grown easily on cheap media to produce high biomass.

2.2 Steps for Production of Metabolites

The production of metabolites involves several steps, including strain selection, genetic engineering, fermentation, and downstream processing (Fig. 2.1).

2.2.1 Strain Selection

Identify and select a suitable microbial strain capable of producing the desired metabolite. This can be done through screening natural isolates or using genetically modified organisms (GMOs) (Ajikumar et al. 2010; Nielsen et al. 2014).

2.2.2 Genetic Engineering

Modify the microbial strain's genetic makeup to enhance metabolite production. This can involve the introduction of genes, deletion of competing pathways, or overexpression of key enzymes (Pickens et al. 2011; Keasling 2010).

2.2.3 Fermentation

Cultivate the genetically modified strain in a controlled fermentation process to produce the metabolite of interest. This involves optimizing culture conditions, such as temperature, pH, oxygen levels, and nutrient availability (Zhang et al. 2015).

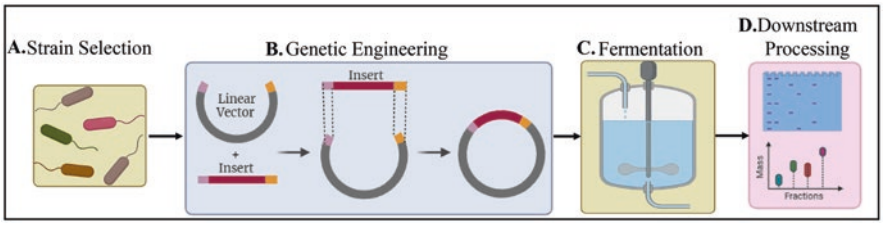


Fig. 2.1 A flowchart consisting of different steps of metabolite production. Four steps of metabolite production are depicted here. (**a**) Strain selection for desired metabolite production; (**b**) Genetic Engineering to modify the strain; (**c**) Fermentation of the genetically modified strain; (**d**) Downstream processing consists of extraction and purification of enzymes

2.2.4 Downstream Processing

Extract and purify the metabolite from the fermentation broth or cell biomass. This may involve techniques such as filtration, centrifugation, chromatography, crystallization, and drying (Su et al. 2018).

It is important to note that the specific steps and techniques can vary depending on the metabolite of interest, the host organism, and the overall production strategy.

2.3 Enhance Product Yield

Increasing yield in metabolic pathways involves optimizing the efficiency of biochemical reactions and manipulating cellular processes to enhance the production of desired compounds. Here are some key strategies and considerations for understanding and enhancing metabolic pathways to increase yield:

2.3.1 Pathway Analysis

Begin by analyzing the existing metabolic pathway(s) involved in the production of the desired compound(s). This includes understanding the enzymes, intermediates, and regulatory mechanisms involved. Pathway analysis is a crucial step in understanding the metabolic pathway and its components, which can guide efforts to increase yield. Here are some key considerations for conducting pathway analysis to enhance yield:

2.3.1.1 Pathway Mapping

Map out the metabolic pathway of interest, identifying the enzymes, intermediates, and reactions involved. This can be done using existing pathway databases, literature resources, or computational tools for pathway reconstruction (Büchel et al. 2013; Kanehisa et al. 2017).

2.3.1.2 Metabolic Flux Analysis

Utilize experimental techniques such as metabolic flux analysis (MFA) to measure and analyze the actual metabolic fluxes within the pathway. MFA provides insights into the actual flux distribution and can guide efforts to optimize flux through key steps for increased yield (Wiechert 2001).

2.3.1.3 Enzyme Kinetics and Regulation

Analyze the kinetic properties of the enzymes involved in the pathway, including their substrate specificity, affinity, and maximum catalytic rate. Investigate the regulatory mechanisms, such as allosteric regulation or feedback inhibition that control enzyme activity within the pathway (Bailey and Ollis 1986).

2.3.1.4 Genetic Regulation

Study the genetic regulation of the pathway, including the transcriptional control of key genes encoding enzymes and regulatory factors. Identify transcription factors or regulatory elements that influence the expression of these genes and understand their impact on pathway yield (Hahn-Hägerdal et al. 2007).

2.3.1.5 Omics Approaches

Utilize omics technologies such as genomics, transcriptomics, proteomics, and metabolomics to gain a comprehensive understanding of the pathway components, their expression levels, and metabolic intermediates. This can reveal correlations between gene expression, enzyme activity, and pathway yield (Ramon and Ciruela 2018).

2.3.1.6 Perturbation Studies

Conduct perturbation studies, such as gene knockouts or knockdowns, to evaluate the impact of specific genes or enzymes on pathway yield. Perturbation experiments can provide insights into the importance of individual components and guide strategies for pathway optimization (Ranganathan et al. 2012).

2.3.1.7 Modeling and Simulation

Develop mathematical models or computational simulations of the pathway to predict the effects of genetic modifications or environmental changes on yield. These models can aid in the design of optimal interventions and guide experimentation for yield improvement (Nielsen and Oliver 2005).

2.3.1.8 Comparative Analysis

Compare the pathway of interest with related pathways in other organisms or organisms with higher yield. Identify key differences in pathway components, regulation,

or enzyme properties that may contribute to higher yield and explore strategies to implement these differences in the target organism (Ajikumar et al. 2010).

Pathway analysis should be an iterative process, with insights gained from analysis informing experimental interventions and vice versa. Combining computational tools, experimental techniques, and systems biology approaches can provide a comprehensive understanding of the pathway and guide strategies for increasing yield.

2.3.2 Limiting Steps and Rate-Limiting Enzymes

Identify the rate-limiting steps and enzymes in the pathway. These are the steps that have the most significant impact on the overall pathway yield. Targeting these steps for optimization can lead to substantial improvements. Identifying and addressing the limiting steps and rate-limiting enzymes in a metabolic pathway are critical for increasing yield. By focusing on these bottleneck processes, you can enhance the flux through the pathway and ultimately improve the production of the desired compounds. Here are key steps to consider when addressing limiting steps and rate-limiting enzymes:

2.3.2.1 Enzyme Kinetics

Study the kinetic properties of enzymes involved in the pathway to understand their catalytic rates and substrate affinities. Identify enzymes with low turnover rates or high substrate K_m values, as they may be potential rate-limiting enzymes (Cornish-Bowden et al. 2002).

2.3.2.2 Metabolite Availability

Assess the availability of substrates, co-factors, and precursors for the rate-limiting enzymes. Limited availability of these molecules can hamper the enzymatic activity and lead to reduced pathway yield. Optimize the supply of these metabolites through genetic engineering, precursor supplementation, or altered environmental conditions (Siedler et al. 2014).

2.3.2.3 Genetic Regulation

Investigate the genetic regulation of the rate-limiting enzymes. Transcriptional control, post-transcriptional modifications, or feedback inhibition mechanisms can influence enzyme expression and activity. Identify regulatory elements, such as transcription factors, and modify their activity to enhance enzyme expression (Raman et al. 2009).

2.3.2.4 Enzyme Engineering

Apply enzyme engineering techniques to enhance the activity and stability of rate-limiting enzymes. Rational design or directed evolution approaches can be employed to optimize enzyme performance. Focus on improving kinetic parameters, such as k_{cat} (catalytic rate constant) and K_m (substrate affinity), to increase enzymatic efficiency (Arnold et al. 1996).

2.3.2.5 Co-Factor Optimization

Evaluate the availability and regeneration of co-factors required by rate-limiting enzymes. Insufficient co-factor levels can limit enzyme activity and pathway flux. Modify co-factor specificity, enhance co-factor biosynthesis pathways, or supplement exogenous co-factors to improve their availability (Li et al. 2018).

2.3.2.6 Allosteric Regulation

Investigate the allosteric regulation of rate-limiting enzymes. Allosteric activators or inhibitors can modulate enzyme activity and flux through the pathway. Identify potential allosteric effectors and engineer the enzymes to respond favorably to these effectors (Ke and Yadav 2018).

2.3.2.7 Expression Level Optimization

Increase the expression levels of rate-limiting enzymes by overexpressing their genes. Employ genetic engineering techniques to enhance gene expression using strong promoters, optimized codon usage, or gene copy number amplification.

2.3.2.8 Protein Stability Enhancement

Improve the stability of rate-limiting enzymes to prevent their degradation or denaturation. Utilize protein engineering strategies, such as site-directed mutagenesis, to enhance protein stability and prolong enzyme half-life (Lehmann et al. 2000).

2.3.2.9 Pathway Engineering

Consider pathway engineering strategies to bypass or alleviate the bottleneck steps altogether. This may involve introducing alternative pathways, eliminating competing pathways, or rewiring the metabolic network to redirect flux toward the desired compound (Nielsen and Keasling 2016).

2.3.2.10 Iterative Optimization

Continuously assess the effects of modifications and interventions on the pathway yield. Combine different strategies and iterate through the optimization process to achieve the desired increase in yield. Experimentally validate the effects of the modifications and fine-tune the interventions based on the results (Liu and Redden 2015).

By focusing on the limiting steps and rate-limiting enzymes, and implementing appropriate interventions, it might be possible to increase the yield of a metabolic pathway (Moreno-Sánchez et al. 2008). The specific strategies depend on the characteristics of the pathway, the host organism, and the desired compound (Fig. 2.2).

2.3.3 Overexpression of Key Genes

Increase the expression of genes encoding enzymes that play crucial roles in the pathway. This can be achieved through various genetic engineering techniques, such as gene amplification, promoter engineering, or using strong expression systems. Overexpression of key genes involved in a metabolic pathway is a common strategy to increase the yield of desired compounds. By increasing the expression levels of enzymes or regulatory factors, you can enhance the flux through the pathway and ultimately boost the production of the target metabolite. Here is an overview of the steps involved in overexpressing key genes:

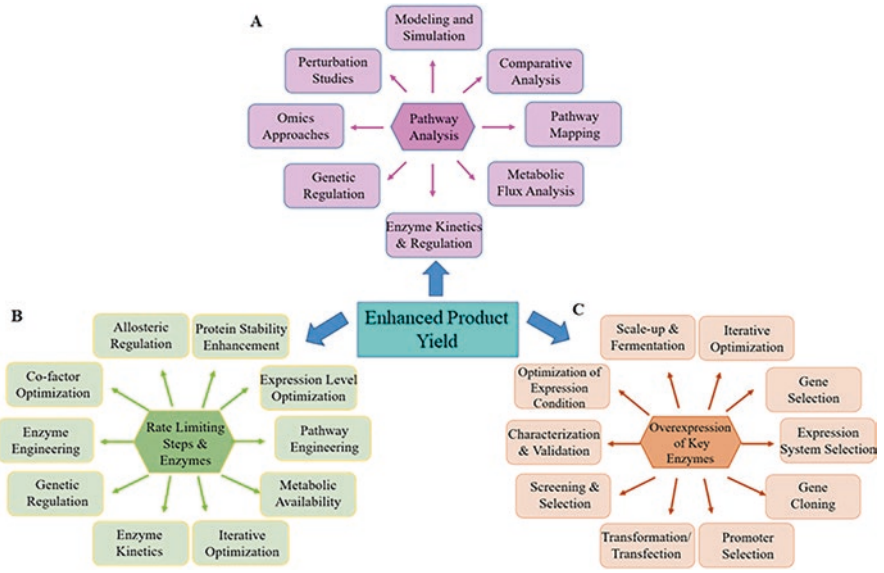


Fig. 2.2 A diagram to understand the metabolic pathways to increase the yield of metabolite production. (**a**) Metabolic pathway analysis and key considerations for this analysis; (**b**) Rate-limiting steps and enzymes in the pathway and major key points; (**c**) Overexpression of key genes and an overview of the steps involved in this process

2.3.3.1 Gene Selection

Identify the critical genes within the metabolic pathway that have the most significant impact on the yield of the desired compound. These genes may encode enzymes involved in rate-limiting steps or regulatory factors that control the pathway flux (Ro et al. 2006).

2.3.3.2 Expression System Selection

Choose an appropriate expression system for overexpressing the target genes. Common options include plasmids for bacterial expression, viral vectors for mammalian cell expression, or inducible promoters for controlled gene expression.

2.3.3.3 Gene Cloning

Clone the target genes into an appropriate expression vector using molecular cloning techniques. Ensure that the genes are correctly inserted into the vector and that the resulting construct is suitable for expression in the desired host organism (Sambrook and Russell 2001).

2.3.3.4 Promoter Selection

Select a strong and compatible promoter to drive the expression of the target genes. The promoter should be compatible with the expression system and provide high transcriptional activity to ensure robust gene expression (Studier and Moffatt 1986).

2.3.3.5 Optimization of Expression Conditions

Optimize the growth conditions for the host organism to achieve maximum gene expression. This includes optimizing temperature, media composition, inducer concentration (if applicable), and other factors that can influence gene expression levels.

2.3.3.6 Transformation/Transfection

Introduce the expression construct into the host organism using appropriate transformation or transfection techniques. This may involve electroporation, chemical transformation, viral transduction, or other methods specific to the host organism (Chong et al. 2021).

2.3.3.7 Screening and Selection

Identify transformed or transfected cells that have successfully incorporated and expressed the target genes. This can be done using selectable markers present in the expression vector or through other screening methods specific to the target genes or metabolites.

2.3.3.8 Characterization and Validation

Characterize the overexpression strains to confirm increased gene expression and assess the impact on the metabolic pathway. This can involve various techniques such as quantitative PCR, enzyme activity assays, metabolite profiling, or other analytical methods.

2.3.3.9 Scale-Up and Fermentation

Once validated, scale up the production process to larger volumes, such as bioreactors, to enable the production of the desired compound at an industrial scale. Optimize the fermentation conditions to maximize the yield and productivity of the pathway.

2.3.3.10 Iterative Optimization

Monitor the production process and evaluate the results. If the desired yield is not achieved, iterate the process by fine-tuning the expression levels, optimizing growth conditions, or considering other strategies such as co-expression of additional genes or pathway engineering.

It is important to note that overexpression of genes should be carefully balanced to avoid potential issues such as metabolic burden, toxicity, or imbalances in cellular resources. Optimizing the expression levels of target genes within the context of the overall cellular metabolism is crucial for achieving the desired increase in pathway yield.

2.4 Conclusions and Future Perspectives

In vitro metabolite production has a significant role in commercial purposes. In industries, in-vitro production can be achieved by bioprocess engineering and their main focus is to increase the yield of metabolite production. Accumulated evidence revealed that the different steps of metabolite production need to be optimized to get the maximum yield. In this chapter, our detailed study can motivate others to find

out the strategies and key consideration for metabolic pathway analysis to improve their work. The detailed study on rate-limiting step and overexpression of key genes revealed the prospect and constraints of each step along with focusing on the step which need to be considered at the time of metabolite production. Bioprocess engineering is a common strategy used in industries in daily manner. This process need to be optimized more to get better production. Artificial intelligence (AI) has emerged now into the science field to improve the quality of research. In future, there is a possibility to use AI in bioprocess engineering, that can improve the pathway by good model development. This can enhance the yield of the product as well as we can expect a better product.

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Chapter 3
Metabolic Engineering of Lipid Biosynthesis Pathway to Enhance the Oil Content in Microalgae

Zainab Mahmood, Mohit Nigam, and Lalit Kumar Singh

Abstract Today, a large range of software programmes are used in the field of metabolic engineering. Such applications were shaped to provide support for a variety of experimental and analytical practices. Bioinformatics is used all over the metabolic engineering approach to separate as well as evaluate useful data derived from enormous data volumes. In the current state of renewable energy, which is fixated on the expansion of alternative and sustainable sources of energy, microalgae are at the top of the list as the most capable feedstocks for the manufacturing of biofuels. However, the complexity of microalgal culture, which involves collecting, concentrating, drying, and extraction of lipids, is the main barrier to the switch to algae-based biofuel production. Triacylglycerols (TAGs), the main building blocks for lipid synthesis, are stored by a number of green microalgae. It has been discovered that novel metabolically important genes exist in oleaginous microalga like *Chlamydomonas reinhardtii*, etc., which would improve lipid production in these microalgae. An interactive perspective on enhancing lipid synthesis, which ultimately enhances oil production, is provided by several additional components of metabolic engineering employing OptFlux and effective bioprocess design. The main goal of this research is to use metabolic engineering and reaction engineering techniques to examine the ability of microalgae to generate increased bio-oil.

Keywords Metabolic engineering · Microalgae · Biofuel · Triacylglycerols · *Chlamydomonas reinhardtii*

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3.1 Introduction

Generally, the energy demand increases with an increase in population, and the energy consumption per year grows rapidly with upgraded living standards. It is expected that the world's approximately eight billion population will escalate by 2030 (Leung et al. 2010). With the rise in CO_2 and several other greenhouse gas (GHG) discharges, global warming problem, as well as severe environment change, a few interests have emerged and shifted the concern to utilize alternative sources of energy from renewable resources instead of fossil fuels (Witsuthammakul and Sooknoi 2012). Oleaginous microorganisms mostly produce triacylglycerols (TAGs), some of which are free fatty acids (FAs). Lipids have high carbon-to-heteroatom ratios, which make them very useful feedstocks for the preparation of renewable fuels. Through the transesterification of TAGs (lipids) and hydrotreatment, biofuels are produced (Santacesaria et al. 2012). The hydrolysis of TAGs converts lipids into oleochemicals when influenced by water, pressure, and temperature (Tan et al. 2013). Essential fatty acids (EFAs) such as oleic acid (OA), gamma-linoleic acid (GLA), eicosapentaenoic acid (EPA), arachidonic acid (ARA), and docosahexaenoic acid (DHA), are very important lipids produced by microbes in food applications with a variety of by-products. Corma et al. (2007) stated that glycerol is one of the important by-products, that is produced during the production of lipids and have variety of uses.

Microorganisms produce microbial lipids such as triacylglycerols, glycolipids, phospholipids, and sterol esters (Bouchedja et al. 2017). In addition to these lipids, TAGs containing non-polar fatty acids make up the majority of the internal structure of microbial cells (Kumar et al. 2015). The lipids (TAGs) formation occurs at the cytoplasm of the cell and it begins at the stationary growth phase of the growth curve, when glucose (hexoses), glycerol, or polysaccharides are used as the only carbon source. TAGs can be produced in enormous amounts at the cytosol and can be preserved for future uses known as a "preserve component." It can perform a key role in the generation of energy reserves full of preserved lipids. (Dourou et al. 2017; Thakur et al. 2022; Nigam et al. 2018, 2021).

3.1.1 Lignocellulosic Material Vs. Algal Biomass

Cellulose, hemicellulose, and lignin are the main components of lignocelluloses with minor quantities of pectin, protein, extractives (dissolved nonstructural elements as well as waxes), and ash particles serve as the primary structural component for plant cell walls (Jørgensen et al. 2007). Its compositional contents may change with other factors such as the various phases of development or age, as well as across plant species, with softwoods being high in hemicellulose whereas hardwoods in cellulose (Sun and Cheng 2002).

The chief structural constituent of plant and algal cell walls is made up of cellulose, and it has a fibrous structure made of subunits of D-glucose, joined together via beta-1,4-glycosidic linkages. Cellobiose, a repeating unit created by hydrogen and Van der Waals bonds in cellulose chains, makes the microfibrils which are cellulose in an organized form. Biomass can have cellulose in both crystalline as well as amorphous arrangements. Much of the cellulose content is present in its crystalline state, whereas a minor, disorganized component of the cellulose chain manifests as amorphous cellulose (Béguin and Aubert 1994).

Hemicellulose differs from cellulose in that it has small lateral chain branches with various sugars as opposed to cellulose's long, continuous chain. The backbone of hemicellulose is made up of two homopolymers and two heteropolymers, and its short arms are sometimes connected with -1,3-glycosidic links and other times with beta-1,4-glycosidic linkages. Hemicelluloses include polymers that are simpler to hydrolyse than cellulose and that do not combine even when co-crystallized with chains of cellulose (Kuhad et al. 2006). In the main cell wall, lignin is considered to be a conglomerate substance that has an interconnect polymer of phenolic monomers that offers rigidity, structural support, and defence against microbial infection. Lignin is a heterogeneous aromatic polymer mainly constituted by three precursors: *p*-coumaryl alcohol, coniferyl alcohol, and sinapyl alcohol known as lignin monomers joined together by aryl-aryl ether, alkyl-aryl, and alkyl-alkyl linkages. This structural configuration gives lignin its key properties. Small plants have low levels of lignin, while softwood has the highest levels (Jørgensen et al. 2007).

The algal biomass, on the other hand, mostly comprises of lipids, proteins, and carbohydrates (Li et al. 2017)**.** The microalgae-based carbohydrates primarily consist of cellulose and starch without lignin, making them a convenient source of carbon for the fermentation sector (Yen et al. 2013). Other beneficial elements of microalgal biomass include proteins, long-chain fatty acids, pigments, and carbohydrates.

3.1.2 The Need for Pre-processing of Lignocellulosic Materials and Its Effect on Biofuel Production Cost

Researchers recognized the beneficial effects of pre-processing lignocellulosic materials long ago; its main aim is to take out the lignin and hemicelluloses and improve the permeability of lignocellulosic biomass and the lucidity of disaccharide. Pretreatment undergoes the successive prerequisites:

- Highly improved sugar formation or afterwards the capability to form carbohydrates by hydrolysis,
- Avoid adulteration, deprivation of carbohydrates,
- Invalidate the by-product generation which can cause repression of successive hydrolysis, and
- It should be cheap.

Pretreatment may be classified as either physical, physicochemical, biological, electrical, or a blend of any of them.

3.1.3 Types of Pre-processing of Lignocellulosic Constituents

3.1.3.1 Physical Pretreatment

- **Mechanical Pulverization**: To rationalize the crystallinity of cellulose, lignocellulosic materials are pulverized using a combination of chipping, grinding, and/or milling, typically in sizes between 10 and 30 mm (by chipping) and 0.2–2 mm (by milling or grinding) (Sun and Cheng 2002). The power required for mechanically pulverizing agricultural materials depends on the characteristics of the biomass and the size of the particles. According to Cadoche and López (1989), who studied the energy demand for hard wood's pulverization ratio and final particle size, the energy consumed is often more than the theoretical energy content usable in the biomass.
- **Pyrolysis**: This method has been used to pre-treat lignocellulosic material. The treated biomass cellulose rapidly breaks down into gaseous components and char at 300 °C (Kilzer and Broido 1965). When materials from pyrolysis pretreatment are hydrolysed with an elemental acid, such as 1N H_2SO_4, at 97 °C for 2.5 h, the cellulose is converted to 80–85%reducing sugars with more than 50% of the glucose in it (Shafizadeh and Bradbury 1979). This process is accelerated when the materials are transferred in the presence of oxygen.

3.1.3.2 Physicochemical Pretreatment

- **Steam Explosion**: The steam explosion is most often used technique for pre-processing lignocellulosic materials. A few seconds to a few minutes are spent treating biomass with high-pressure saturated steam at a pressure of 0.69–4.83 MPa (at a temperature of 160–260 °C), and this steam is then quickly decreased to air pressure, causing the materials to undergo a violent decompression (Sun and Cheng 2002). The primary determinants of steam explosion are chip size, moisture content, temperature, and residence duration. Hemicellulose hydrolysis works best at high temperatures (270 °C) and short intervals of time, such as for 1 min, or low temperatures (190 °C) and longer intervals of time, such as for 10 min (Duff and Murray 1996). Its advantage over the other process includes requiring less energy or being more affordable.
- **Ammonia Fibre Explosion (AFEX)**: It is a physiochemical pretreatment method that involves exposing biomass to liquid form of ammonia at high temperatures and pressures for a time period that is predetermined before the pres-

sure abruptly lowers, simulating a steam explosion (Gollapalli et al. 2002). In a typical AFEX process, 30 min of residence time at 90 °C and 1–2 kg of liquid ammonia per kg of dry biomass are employed. According to Alizadeh et al. (2005), it may increase and improve the rate of fermentation in a variety of grasses and crops, including wheat straw and lucerne.

- **Carbon Dioxide Explosion**: To improve the pretreatment methods for lignocellulosic biomass, supercritical CO_2 explosion sought to create a method that was less expensive than AFEX and had a lower temperature than steam explosion. A supercritical fluid is a fluid with a liquid-like density that is pushed to a temperature above the critical bond in a gaseous state. It was proposed that CO_2 aids in cellulose and hemicellulose hydrolysis. The cellulose structure is damaged, and a volatile CO_2 pressure is produced, which increases the surface area of the hydrolysable substrate. According to Dale and Moreira (1982), pre-treating lucerne at 5.62 MPa with 4 kg of CO_2 fibre resulted in the release of 75% of the potential glucose in 24 h, as opposed to enzymatic hydrolysis.

3.1.3.3 Chemical Pretreatment

- **Ozonolysis**: This process uses ozone to reduce the lignin concentration of lignocellulosic biomass. When compared to other chemical processes, ozonation improves the digestibility of hydrolysed materials and leaves behind less hazardous waste. Many lignocellulosic materials, including pine, bagasse, sawdust, wheat straw, etc., are subjected to ozonolysis in order to remove or decrease lignin; unlike hemicelluloses, cellulose is unaffected (Ben-Ghedalia et al. 1983). Ozone therapy have several benefits like all reactions take place at normal pressure and temperature. The disadvantage of ozone treatment is that it uses a huge quantity of ozone, and it might make this procedure costly (Quesada et al. 1999). Ozone treatment is also designed to reduce environmental pollution and can be readily degraded by an increase in temperature.
- **Acid Hydrolysis**: Concentrated acids like H_2SO_4 and HCl are being used to treat biomass of lignocellulosic material because they facilitate enzyme-based hydrolysis and enable the discharge of fermentable sugars from the material. Concentrated acids are toxic, corrosive, and dangerous, but they are also very effective agents for hydrolysing cellulose and need reactors which are impervious to weathering, hence making the operations costly (von Sivers and Zacchi 1995). However, to make the process cost-effective after hydrolysis, the concentrated acid should be recovered. For making the pretreatment of lignocellulosic materials commercially viable and cost-effective, 4% w/v sulphuric acid has been utilized. Furfural has been produced from cellulosic materials for commercial use (Zeitsch 2000).
- **Alkaline Hydrolysis**: Depending on the amount of lignin in the biomass, different bases may be utilized in the pretreatment of lignocellulosic biomass. Alkaline

hydrolysis uses low temperatures and pressures compared to other pretreatment techniques. This alkaline hydrolysis follows moderate conditions, even though they take several hours or days for pretreatment process to complete instead of only a few moments. In comparison to acid hydrolysis, it results in less degradation of cellulosic sugars and may be recovered as a variety of caustic salts. Sodium hydroxide is the alkaline agent that is used the most for pretreatment, followed by sodium, calcium, potassium, and ammonium hydroxides (MacDonald et al. 1983). Although slaked lime, or calcium hydroxide, is an effective pretreatment agent and more cost-effective per kg than hydroxide, kiln technology, which involves balancing the process with inexpensive carbon dioxide, is used to recover the calcium from the insoluble aqueous solution of calcium carbonate. Dilute NaOH produces inflammation or expands the inner surface area of lignocellulosic material as well as weakens the connections formed between lignin and carbohydrates. As a result, the extent of lignocellulose polymerization and crystallinity decreases, which lead to disruption of the lignin structure. According to Bjerre et al. (1996), pretreatment with NaOH may effectively reduce the lignin content of softwoods by up to 26% and that of hardwoods by 14–45%.

- **Oxidative Delignification:** A peroxidase enzyme and H_2O_2 work together to catalyse the biodegradation of lignin. The sensitivity of cane bagasse to enzymatic hydrolysis is significantly increased by pre-treating the bagasse with H_2O_2. By using 2% H_2O_2 at 30 °C for 8 h, hemicellulose and about 50% of the lignin are disintegrated. By saccharifying cellulose for 24 h at 45 °C, approximately 95% of glucose is produced (Azzam 1989). Bjerre et al. (1996) employed alkaline hydrolysis with wet oxidation of wheat straw, converting 85% of the cellulose into glucose at 170 °C for 5–10 min. The presence of microbial inhibitors such as furfural and hydroxymethyl furfural was not seen; however, the combination of wet oxidation and base addition increases the polysaccharide's sensitivity to enzymatic hydrolysis and readily oxidizes lignin (Bjerre et al. 1996).
- **Organosolv Process**: The processing of lignocellulosic biomass piques attention with the Organosolv process. The breakdown of linkages between the lignin and hemicelluloses occurs during this procedure, which combines inorganic acids like HClor H_2SO_4 with aqueous organic or organic solvent amalgams like methanol, ethanol, ethylene glycol, acetone, triethylene glycol, and tetrahydrofurfuryl alcohol (Chum et al. 1988). In the Organosolv method, several organic acids, including oxalic, acetylsalicylic, and salicylic acids, are used as stimulants (Sarkanen 1980). Initiated by organic solvents, it simultaneously includes lignocellulosic hydrolysis and delignification of biomass and primarily dilutes aqueous acid solutions (Thring et al. 1990). In biorefineries, a combination of softwood pulps and remaining lignin ranging from 6.4% to 27.4% (w/w) is included using a lignol process based on aqueous ethanol organosolvation mining (Pan et al. 2005).

3.1.3.4 Pulsed Electric Field Pretreatment

The sample is put between two electrodes with a brief break from high voltage during the PEF pretreatment (pulsed electric field), which has a significant impact on the structure of plant tissue (Lynch and Davey 2012). When the cell membrane is exposed to high-intensity external voltage, the cell wall, tissue, and its local structure break down quickly. When an electric field is supplied as the square waves or in the form of exponential decay, the permeability of the aggregation rises abruptly, and in certain instances, the plant tissues are physically disrupted (Zimmermann and Neil 1996).

3.1.3.5 Biological Pretreatment

Biological pretreatment (Okano et al. 2005) involves eco-friendly microorganisms to efficiently break down lignocellulosic material, reducing energy needs compared to conventional methods. Conventional pretreatment methods for lignocellulosic materials demand substantial energy and advanced equipment due to their complexity. Alternatively, eco-friendly approaches utilize microorganisms such as bacteria, fungi, yeast, and microalgae. These organisms facilitate the efficient breakdown and removal of lignin from biomass, presenting a sustainable solution that avoids the high energy demands associated with traditional methods, even when faced with challenges like uneconomical lignin removal (Okano et al. 2005). In essence, employing environmentally friendly variants enables effective lignocellulosic processing with reduced energy requirements.

3.1.4 Features of Microbial Lipids

Numerous articles and publications on the creation of microbial lipids have recently been published in international journals (Souza et al. 2017). Scientists are becoming more and more interested in the synthesis of lipids, from the simplest class, which includes bacteria, to the most complicated class, which includes fungus, due to economic and technological considerations. Microbial oils are consumed less often than regular oils and fats due to the high expense of manufacturing and the maintenance of sterile conditions (Koutinas et al. 2014). There are a few additional options for the large-scale production of microbial lipids thanks to the identification of microbes that can produce a lot of lipids when compared to the structural makeup of fats found in the kingdoms of plants and animals, which attracts the financial attention of the industries (Papanikolaou and Aggelis 2011).

The mass production of microbial lipids, which include the most useful fatty acids and may be employed in the medicinal and nutritional domains, is now the top focus. Particularly, the majority of microbial lipids include polyunsaturated acids that are crucial for human health, such as C-linolenic acid (GLA), eicosapentaenoic acid (EPA), arachidonic acid (ARA), and docosahexaenoic acid (DHA). Microbial lipids may be the fundamental feedstock for the creation of biodiesel as they have a similar composition as that of different vegetable oils, which were considered to be the primary fundamental feedstocks for biodiesel production (Papanikolaou et al. 2017).

3.1.5 Oleaginous Microorganisms

Oleaginous microorganisms are those which can store higher than 20% of the extra oil present in their biomass in their cells, pellets, and mycelia. Under the required environmental conditions, i.e., the N_2 limited condition, which stops the subdivision and starts storing polysaccharides, oleaginous microbes like yeast, fungi, bacteria, and microalgae produce microbial lipids, also denoted to as single-cell oils (Diamantopoulou et al. 2014). Whereas those organisms that are unable to store 20% of their biomass are known as non-oleaginous microorganisms (Ratledge and Wynn 2002; Thakur et al. 2023; Muzaffar et al. 2020), secondary metabolites begin producing them in the medium (Diamantopoulou et al. 2014).

The initial trigger for fat accumulation is often attributed to a deficiency in dietary nutrients, particularly nitrogen. When nitrogen supply is limited, the enzyme AMP deaminase becomes more active, breaking down AMP and releasing NH_3 to support cells lacking nitrogen. This disrupts the tricarboxylic acid (TCA) cycle at iso-citrate due to the decrease in the vital cofactor AMP, caused by heightened AMP deaminase activity. Consequently, malate dehydrogenase converts oxaloacetate into malate, which is transported to mitochondria via citrate-malate translocase. Inside the mitochondria, malate is converted into pyruvate through the malic enzyme (ME) pathway, facilitated by the cytosolic transhydrogenase cycle, generating NADPH and CO_2. This synthesized NADPH and acetyl-CoA are then utilized in the production of fatty acids (FA) and triglycerides (TAG) in response to the shortage of nitrogen. It is important to note that different strains exhibit different levels of ME activity. Particular strains are assumed to yield the NADPH necessary to sustenance FA and TAG synthesis through a different FA-allocated enzyme scheme of the cytosol or maybe through the pentose phosphate pathway cycle because they lack this activity (Ratledge 2004).

The process and substrates can change the composition of the lipids that oleaginous microbes produce; the main factors that determine a microorganism's potential for producing lipids are: (1) Lipid quantity: more the lipid produced by a microbial cell, more important it will be considered from an industrial standpoint. (2) Lipid

quality: compared to plants, certain microbes synthesize polyunsaturated fatty acids, although the degree of chain unsaturation varies from microbe to microbe (Papanikolaou et al. 2017).

- **Yeast**: Oleaginous yeast's cells contain lipids, which make up 20% (or up to 70% under nutrient-limiting conditions) of their dry cell weight due to their high oil content. It also grows faster than any other microorganism (Zheng et al. 2012). Different species have different lipid and fatty acid profile contents, and some of the finest oleaginous yeasts are generally originate in the group of *Candida, Cryptococcus, Rhodotorula, Rhizopus, Lipomyces, Rhodosporidium, Trichosporon, and Yarrowia* (Zheng et al. 2012). Unsaturated fatty acids are produced depending on different carbon sources (e.g., glucose, glycerol, xylose, starch, hydrolysate of different industrial and municipal organic wastes), with the highest value of lipid being 53% on a glycerol-based carbon source and the lowest value being 25% on xylose (Easterling et al. 2009).
- **Microalgae**: The cheapest carbon source, i.e., sunlight, is vital for the photoautotrophic culture of microalgae; an organic carbon source other than sunlight is used in the heterotrophic culture; and a mixture of an organic carbon source and light is used in the mixotrophic culture. It is complicated to scale up the autotrophic microalgae because it needs light for its cultivation, so to reduce the expenses of oil production from microalgae, sunlight, and variation in light levels on a daily and seasonal basis are much needed. It is easy to cultivate the heterotrophic culture in a normal fermenter with an organic carbon source, which limits its application. The quantity of lipids in microalgae can reach up to 90% of its cell dry weight (Miao and Wu 2004).
- **Bacteria**: You can cultivate bacteria using simple microbial culture techniques and see rapid cell development. Some bacteria need a unique environment for culture and create lipid profiles that are distinct from those of other microbial oils. Utilizing carbon sources, certain bacteria may produce TAGs, which accumulate during the stationary phase (Ilic-Tomic et al. 2015), which comes after total protein synthesis has stopped. Gram-positive Actinomycetes bacteria, like *Arthrobacter, Dietzia, Rhodococcus, Nocardia* or *Streptomyces* strains, have the capacity to create a significant quantity of TAG as an intracellular product. Table 3.1 displays some examples of these strains. Due to their quick growth and capacity to absorb a range of carbon sources, *Actinomycetes* strains, notably the *R. opcaus* strain, are well recognized for their exceptional TAG quantities of up to 80% of their dry cell weight for industrial tag production (Hetzler and Steinbüchel 2013). Despite the remarkable lipid accumulation capacity of the *Arthrobacter* AK19 strain, this strain is not used in any more research. *Streptomyces* can accumulate up to 50% of its dry cell weight in fatty acids and other similar genus of gram-positive and aerobic bacterium like *Rhodococcus* and *Gordonia* and can also easily convert industrial waste such as molasses, agricultural waste, wastewater from industries or different lignocellulosic hydrolysates into lipids (TAGs).

Table 3.1 Lipid synthesis by several microbial strains cultured on various substrates

Bacterial strain	Substrate	Production mode	Lipid yield (%) (TFA/TAG)	Cell dry weight (g/L)	References
Arthrobacter sp. Ak19	Glycerol	Flask	0.69	10.11	Srinophakun et al. (2017)
Dietzamaris 53	Acetone	Flask	13.4	34.5	Alvarez (2003)
	Hexadecane		19.2		
Gordoniaamarae 106		Flask	61	10	Alvarez (2003)
Gordonia sp. DG	Glucose		72		Voss and Steinbüchel (2001)
	Molasses		95		
	Orange waste		71		
Gordonia sp. KTR9	Ethanol	–	2.3	–	Eberly et al. (2013)
Mycobacterium avium	Palmitic acid	–	5	–	McCarthy (1971)
Nocardia asterides 419	Gluconate	Flask	12.2	64.7	Alvarez (2003)
	Pentadecane		10.8		
	Hexadecane		8.2		
	Gas oil		0.36		
N. corolina	Glucose	–	18.6	–	Alvarez et al. (1997)
	Valerate		23.9		
	Hexadecane		13		
N. globerula 432	Gluconate	Flask	18.6	17.6	Alvarez (2003)
	Hexadecane		18.2		
N. restricta 560	Gluconate	Flask	19.3	30.9	Alvarez (2003)
	Hexadecane		1.7		
Rhodococcus erythropolis	Gluconate	–	21	–	Alvarez et al. (1997)
	Valerate		15.1		
	Hexadecane		17.6		
R. erythropolis 17	Gluconate	Flask	7.7	8.8	Alvarez (2003)
	Pentadecane		56.8		
	Hexadecane		43.4		
	Gas oil		1.8		
R. fascians	Gluconate	Flask	2.8	23.8	Alvarez (2003)
	Pentadecane		4.8		
	Hexadecane		12.9		
	Gas oil		2.9		
R. jostii RHA1	Glucose	–	48.4	–	Hernández et al. (2008)
	Gluconate		56.9		
	Acetate		2102		
	3hb		3203		
	Hexadecane		30.4A		

(continued)

Table 3.1 (continued)

Bacterial strain	Substrate	Production mode	Lipid yield (%) (TFA/TAG)	Cell dry weight (g/L)	References
R. opacus PD630	Gluconate				Alvarez et al. (1997)
	Fructose	Flask	76		Alvarez (2003), Wältermann et al. (2005), Gouda et al. (2008), Voss and Steinbüchel (2001), Kosa and Ragauskas (2012), Wei et al. (2015), Kumar et al. (2015)
			40		
	Acetate		31		
	Citrate		35		
	Succinate		22		
	Propionate		18		
	Phenylacetate		38		
	Valerate		38		
	Pentadecane		39		
	Hexadecane		38		
	Heptadecane		28		
	Octadecane		39		
	Olive oil		87		
	Molasses		92		
	Orange waste		83		
	Sweet whey		83		
	Molasses and sucrose	Flask/30 L	52	37.5	
	4-hydroxy benzoic acid		20.3		
	Vanillic acid		14.6		
	Sweetgum hydrolysate		28.6		
	Dextrose		71		
	Raw dairy wastewater		33		
R. opacus DSM 1069	4-hydroxy benzoic acid	–	16.8	–	Kosa and Ragauskas (2012), Wei et al. (2015)
	Vanillic acid		6.7		
	Softwood lignin		4.1		
	Pine hydrolysate		28.3		
	Kraft lignin hydrolysate		14.2		
R. opacus MR22	Gluconate	–	48	–	Alvarez et al. (1997)
	Valerate		42		
	Hexadecane		43		

(continued)

Table 3.1 (continued)

Bacterial strain	Substrate	Production mode	Lipid yield (%) (TFA/TAG)	Cell dry weight (g/L)	References
R. rhodochrous	Glucose	–	43	–	Shields-Menard et al. (2015)
R. ruber	Glucose		19		Alvarez et al. (1997)
	Valerate		12.		
	Acetate		24		
	Citrate		28		
	Pentadecane		18		
Rhodococcus sp. 20	Gluconate	Flask	7.6	26.2	Alvarez (2003)
	Hexadecane		8.1		
Rhodococcus sp. A27	Fructose	–	33	–	Röttig et al. (2017)
	Sucrose		5		
Streptomyces albus	Complete medium	Flask	56	2.5	Olukoshi and Packter (1994)
S. coelicolor A3(2)	Complete medium	Flask	84	2.2	Olukoshi and Packter (1994)
S. grisezls	Complete medium	Flask	93	3.3	Olukoshi and Packter (1994)
S. lividans	Glucose	Flask	125	2.7	Olukoshi and Packter (1994)
Streptomyces sp. G25	Fructose	–	47	–	Röttig et al. (2017)
	Cellobiose		48		
Streptomyces sp. G49	Fructose	–	31	–	Röttig et al. (2017)
	Cellobiose		29		

3.2 Major Oleaginous Microalgae and Their Significance over Other Oleaginous Microorganisms

In comparison to other oleaginous microbes and fuel crops, oleaginous microalgae have a number of advantages, such as a shorter lifespan (1–10 days) compared to plants, greater light conversion rates, and the need for fewer resources in order to produce the similar quantity of biomass as traditional harvests for biofuel (Schenk et al. 2008; Greenwell et al. 2010). Microalgae may be cultivated on abandoned land, in arid regions, and various other places that are not suitable for producing food, as well as in ponds, tanks, and bioreactors (Greenwell et al. 2010; Yun et al. 1997). Algae may additionally be grown in wastewater. Recently, academia and industry have been more interested in biochemical engineering because of the continued production and harvesting of microalgae. It is possible to extract and create biodiesel from the fatty acids that microalgae generate (Brown and Zeiler 1993). There are certain species of microalgae with oil contents as high as 80%, while ranges from 20% to 50% are more typical (Powell and Hill 2009). Up to 50% of the

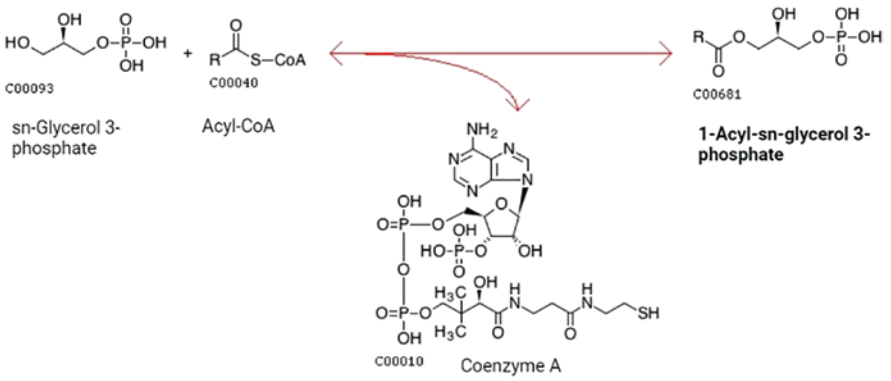
lipids in *Chlorella* may be found, in contrast to as much as 80% in *Botryococcus*. Israel, Taiwan, Japan, Indonesia, the USA, and China often use open-top tanks and cylindrical bioreactors for the industrial manufacturing of triglycerides from microalgae to produce biofuels (Jiménez et al. 2003).

3.3 Enhancing Lipid Synthesis Through Engineered Microalgal Metabolism

3.3.1 Chief Triacylglycerol Biosynthesis Pathways in Microalgae

$$\text{sn Glycerol3 phosphate} + \text{Acyl CoA} \Leftrightarrow \textbf{1 Acyl sn glycerol3 phosphate} + \text{CoA}$$

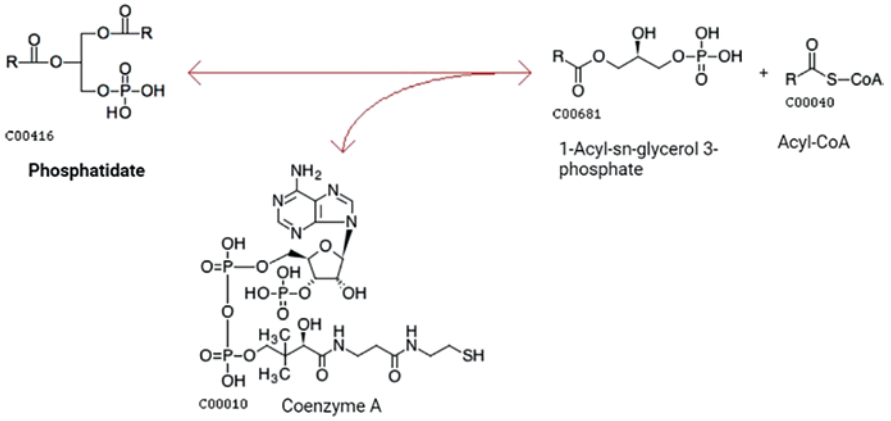
The conversion of sn-Glycerol 3-phosphate and Acyl-CoA into 1-Acyl-sn-glycerol 3-phosphate and CoA is a pivotal reaction in lipid production. It forms the backbone of phospholipids, critical components of cell membranes, regulates energy storage through triglyceride synthesis, and influences membrane fluidity and signalling. This reaction is central to maintaining cellular structure, function, and energy balance.



$$1\ \text{Acyl sn glycerol3 phosphate} + \text{Acyl CoA} \Leftrightarrow \textbf{Phosphatidate} + \text{CoA}$$

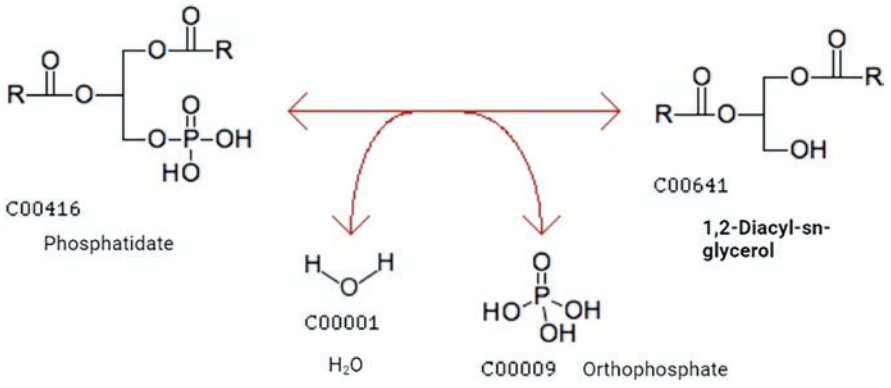
The reaction converting 1-Acyl-sn-glycerol 3-phosphate and Acyl-CoA into Phosphatidate and CoA is a key step in lipid biosynthesis. Phosphatidate is a precursor to various lipids, including triglycerides and phospholipids. These lipids serve vital roles in energy storage, cell membrane structure, and cellular signalling. This

reaction links fatty acid synthesis with lipid production, playing a central role in cellular homeostasis.



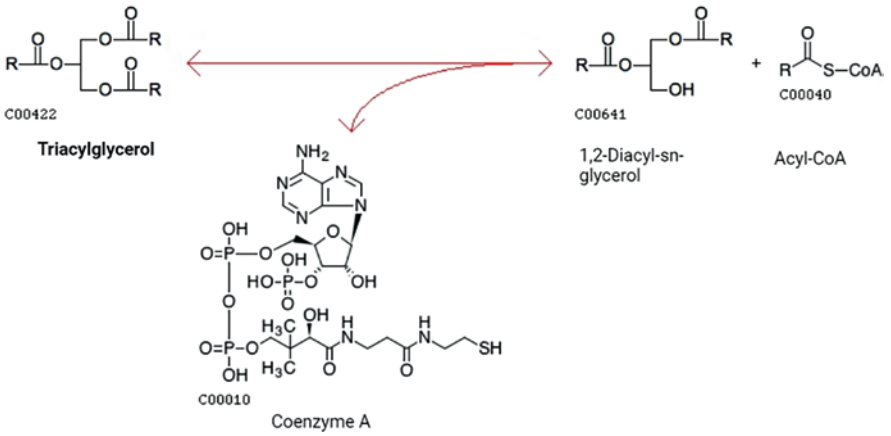
$$\text{Phosphatidate} + H_2O \Leftrightarrow \mathbf{1,2\ \ Diacyl\ \ sn\ \ glycerol} + \text{Orthophosphate}$$

The transformation of Phosphatidate and water into 1,2-Diacyl-sn-glycerol and Orthophosphate holds paramount importance in lipid metabolism. It serves as a critical juncture in the production of phospholipids and triglycerides, which are indispensable for constructing cell membranes, storing energy, and facilitating cell signalling. This chemical process governs the equilibrium between lipid storage and consumption, thus assuming a central role in preserving cellular functionality and internal stability.



$$1,2\ \ \text{Diacyl}\ \ \text{sn}\ \ \text{glycerol} + \text{Acyl}\ \ \text{CoA} \Leftrightarrow \mathbf{Triacylglycerol} + \text{CoA}$$

The final reaction in lipid production, the conversion of 1,2-Diacyl-sn-glycerol and Acyl-CoA into Triacylglycerol and CoA, is of paramount importance in cellular metabolism. Triacylglycerols, commonly known as triglycerides, serve as the primary form of long-term energy storage in organisms. This reaction allows excess energy in the form of fatty acids to be efficiently stored in adipose tissue, which is crucial for maintaining energy balance and providing a readily available energy source during periods of high energy demand. Moreover, triacylglycerols also function as insulators and protect organs. This reaction plays a pivotal role in regulating energy homeostasis and overall metabolic health.



3.3.2 Main Routes of TAG Synthesis

3.3.2.1 Kennedy Pathway

The Kennedy Pathway is a metabolic route that primarily operates in organisms to synthesize phospholipids and glycolipids, the essential components of cell membranes. This pathway plays a crucial role in maintaining membrane integrity and cellular function across various living organisms. Lysophosphatidic acid is acylated first, followed by glycerol-3-phosphate, and finally phosphatidic acid is produced. Phosphatidic acid is dephosphorylated by the enzyme diacylglycerol acyl transferase (DGAT) to produce diacylglycerol, which is then transformed into triacylglycerol (TAG).

3.3.2.2 Acyl-CoA Independent Pathway

Phospholipid: Diacylglycerol acyltransferase uses this pathway to transfer the acyl group from phospholipids (Hildebrand et al. 2013). *Chlamydomonas* species have more DGAT genes than *Arabidopsis*, which increases the complexity of microalgae. Plastids in plants are where fatty acid production takes place. Plant lipid (TAG) manufacture, which is not merely constrained to cells with specific activities, may be triggered by stress in microalgae. Plants do not have certain lipids, such as betaine lipid (Mongrand et al. 1998; Klug and Benning 2001). Isoprenoid compounds are crucial for evaluating biofuel from diatoms because of the prevalence of two unique biosynthetic pathways for isoprene: methylerythritol phosphate (MEP) and mevalonate (MVA), respectively (Lohr et al. 2012).

3.3.3 Investigating the Capacity of Microalgae to Produce Enhanced Bio-Oil Using Metabolic Engineering and Reaction Engineering Methods

Microalgae have recently received more attention because of a shortage of fossil fuels. Microalgal biofuel decreases particulate matter emissions by 50%, sulphur emissions by 98%, and the amount of carbon dioxide outputs by 78% after burning (Xuan et al. 2009). In comparison to other conventional energy sources like hydroelectricity, wind power, or other forms of biomass like plants, household trash, and industrial waste, microalgae are becoming a superior source of biofuel. Microalgae have the added benefit of having a high CO_2 fixation capacity (100 metric tonnes of biomass from microalgal species is capable of fixing 183 metric tonnes of CO_2). Since microalgae produce much less CO_2 compared to other biomass sources, they have a high potential for being turned into biofuel. For instance, microalgal oils can be extracted to produce biodiesel, a biodegradable, reusable, and environmentally beneficial fuel, or microalgal biomass can be fermented to make ethanol and hydrogen. Transesterification might be used to produce microalgae-based biofuel. Using an acid/alkali catalyst and either ethanol or methanol as the solvent, lipids can be transformed directly into ethyl/methyl esters of lipids during the transesterification process (Fukuda et al. 2001; Xuan et al. 2009). The process produces two stages: a lighter stage and the heavy stage, which also includes additional alcohol, water, and contaminants in addition to crude glycerin. Once the light stage is successfully centrifuged and dried off, biodiesel should adhere to ASTM standards (Maa and Hanna 1999). The use of microalgae-based biofuels in CI engines is gaining popularity around the globe.

The metabolic engineering strategy is crucial for attaining the goal of turning microalgae into fuel, along with other techniques that include genetic modification, flux balancing, and finding the desired pathway and its enzymes. An efficient metabolic route must be developed with an appropriate host and other essential

components, such as process aiming and its simulation towards desired product output, in order to make biofuel from microalgae. Finding a suitable pathway that can produce biofuels could be challenging since metabolic processes are, by their very nature, so complicated. The accessibility of sequences of the genome for over 30 microalgae has gained attention, considering the reality that the pathway of metabolism is continually evolving. Without a base or model system, it is very difficult to identify the genes associated with enzymes and integrate them with complicated metabolism. *Chlamydomonas reinhardtii* has become an ideal model for eukaryotic microalgae as a result of this issue. Transcriptomics and proteomics data may be combined with genome-scale metabolic modelling (GSMM) and flux balance analysis (FBA), which are developed and refined following the construction of a base system (Röttig et al. 2017).

Microalgae cannot produce economically viable biofuels owing to (i) poor biomass recovery and (ii) expensive downstream processing and verification. Furthermore, the invention of low-priced procedures for collection of biomasses, its drying, and oil mining in addition to a biorefinery strategy are the only ways of making microalgae-based biofuel realistic.

The application of metabolic pathway engineering and methods of genetic engineering may increase the overall production of this kind of microalgal biofuel. In addition to the methods previously mentioned for increasing the possibility of using microalgae for biofuel, a novel technology called biotic or algal-bacterial relationships is also investigated. Scientists have concentrated on developing high-lipid containing microalgae and employing them for metabolic engineering methods as well as cultivating them in significantly open ponds for the manufacture of biofuels, in addition to the biological discharge reduction technique of capturing the carbon dioxide via coal-fired power plants (Brennan and Owende 2010). In context of microbial production of biofuels and other important molecules, one of the most important problems nowadays is the infections caused in humans by microbes and the by-products of their metabolic processes, which can be tackled through the biochemical engineering techniques making the microbes more and more safe for humans.

3.4 In Silico Metabolic Engineering

In silico metabolic engineering involves the application of computational models and simulations to enhance microalgal lipid production. By integrating biological and biochemical data, these techniques predict how genetic modifications, pathway alterations, and nutrient optimization influence lipid synthesis. Through virtual experimentation, researchers can identify key metabolic nodes, design targeted interventions, and predict outcomes before actual implementation. This accelerates the optimization process, reduces experimental trial and error, and guides the development of high-yield lipid-producing microalgae strains. In silico methods offer a powerful and cost-effective approach to fine-tuning metabolic pathways and

guiding genetic manipulation for efficient lipid production in microorganisms. Large-scale models that use a range of in silico methodologies to examine the roles of many genes, metabolites, transcripts, as well as essential enzymes involved in metabolic fluxes may be used to develop a method of metabolic engineering (Patil et al. 2004; Schmidt et al. 2010). The importance of various computational tools for comprehending important facets of lipid regulation has been shown in a sufficient number of studies, which may be essential for researchers working on biofuels. To establish concepts for improved extraction of lipids from microalgae or to create novel pathways, a number of additional studies also provide descriptions of different flux balance evaluations and metabolic network simulations (Schuhmann et al. 2012). In light of this, the accessibility of metabolic simulations and in silico methods for identifying crucial lipid metabolism may be the distinguishing characteristic and provide sufficient details. Additionally, the development of models that can be used with information from transcriptomics and the fields of proteomics and metabolomics would make it easier to gather the components required for high-quality biofuel. It is certainly necessary to analyse the data on microalgal metabolism of lipids produced from in silico metabolic modification using wet lab tests.

Nowadays, the subject of metabolic engineering uses a wide range of software tools. To assist diverse experimental and analytical procedures, several instruments were developed. All through the metabolic engineering process, computational tools are used to extract and evaluate pertinent data from vast data sets, to present complicated models in an understandable manner, and to suggest effective network building techniques.

The main goal of metabolic engineering is often to maximize the cellular level synthesis of an intriguing chemical. The likelihood that a technique will produce the necessary quantity of output is higher when it diverts cellular resources away from useless directions in an organism. The complexity of metabolic systems makes it challenging to computationally analyse paradoxical knockout tactics adequately, despite the ease with which mutant strains can be created in the lab. In order to optimize production, metabolic flux analysis, or MFA, is used as an investigative tool for the measurement of metabolic fluxes in vivo (Stephanopoulos et al. 1998; Stephanopoulos 1999). Since intracellular fluxes are basically unseen, we may deduce them from stoichiometric relationships with apparent activities. Utilizing quantifiable rates of metabolite absorption and release in the growth medium allows for the calculation of intracellular fluxes for constraint-based methods. Growth mediums that have been 13C-labelled are used to track fluxes across a network while conducting isotopic flow research.

3.4.1 Constraint-Based Analysis

Combining constraint-based research like flux balance analysis (FBA) (Watson 1984; Varma and Palsson 1994; Orth et al. 2010), minimization of metabolic adjustment (MOMA) (Segre et al. 2002), and regulatory on/off minimization (ROOM)

(Shlomi et al. 2005), the metabolic engineer is better equipped to develop a realistic set of focused alterations that lead to the chosen phenotype. Since they are familiar with the underlying concepts and biological importance of these procedures, metabolic engineers have a significant edge. The optimum development and production of an organism may be predicted using FBA with enough time for the microorganism to adapt or respond to desired conditions. Immediately after a knockdown response, MOMA is great at foreseeing negative results. The MOMA hypothesis implicitly assumes that metabolism is still regulated quickly after a knockout by the regulatory mechanism that preserves a wild-type flux pattern. This method spreads the disturbance over the whole system by redirecting flow via a number of channels. Therefore, MOMA outperforms FBA in terms of lethality prediction (Segre et al. 2002). Identification of physiologically plausible flux phenotypes is made easier by the ROOM technique. It figures out how much it would cost to modify the parent strain's regulatory system such that it would be more beneficial to the target strain than it would be in its natural condition. When the ROOM approach gets closer, the best way to speak about the FBA solution is as a conceptual limit. The COBRA 2.0 (Schellenberger et al. 2011) toolbox for MATLAB & OptFlux (Rocha et al. 2010) technologies employ constraint-based methodologies that recommend elimination tactics to optimize metabolite production goals which are connected to cell growth purposes. Both programmes employ the OptKnock algorithm (Burgard et al. 2003) that carefully hunts for the top tactic given a specific sum of knockouts. According to studies by Rocha et al. (2010) and Lun et al. (2009), there are several computational performance difficulties with OptKnock's global search that may be partially resolved by initially decreasing the metabolic model. The two apps (Patil et al. 2004; Lun et al. 2009) give heuristic knockout optimization techniques that enhance computing performance and are not inhibited to a certain sum of knockouts, in contrast to OptKnock. Despite the fact that heuristic techniques cannot offer the best efficient method or a similar strategy across runs, they have been confirmed to deliver outcomes that are equal to OptKnock for application in the actual world (Brochado et al. 2010). The COBRA toolkit gives a tighter interface for working with the core linear programming solvers, in contrast to OptFlux, which has a broad interface for specifying a variety of biological and industrial objectives. Other applications that make use of constraint-based methods consist of CellNetAnalyzer (Klamt et al. 2007), FASIMU (Hoppe et al. 2011), and the Systematic Biology Research Toolbox (Wright and Wagner 2008). Flux balance analysis and associated constraints-based modelling tools have been employed to build several cutting-edge metabolic engineering techniques, such as targeted knockouts and route finding. Constraint-based modelling has been employed in bioprocess applications to optimize the chemical makeup of the medium, pH balance, timing of the switch between aerobic and anaerobic conditions, and microbial co-cultures (Varma and Palsson 1994; Motamedian and Naeimpoor 2011; Hanly and Henson 2011).

3.4.2 Isotopic Flux Analysis

Although constraint-based analysis could reveal potential knockouts, the outcome of a genetic alteration on flux dispersal may range from completely undetectable side effects on regulatory pathways to virtually no change. When combined with the knowledge of global expression, 13C flow investigations are useful in developing an identification of these effects (Sauer 2006). Technologies for 13C flux assessment that employ worldwide or localized inference techniques may be categorized into many categories at the moment. In order to match what is seen in 13C labelling trends to an array of balancing equations established from a network model, 13CFLUX (Wiechert et al. 2001) employs global nonlinear optimization. The initial version of 13CFLUX has a lot of problems about accessibility, availability, and computational efficiency despite being widely used. The most recent version, 13CFLUX2, is much better and offers customizable analysis of spectra for MS, MS/MS, 1H-NMR, and 13C-NMR in addition to using faster optimization methods and switching to a new data file format. 13CFLUX2 features a command-line interface as opposed to Omix, which offers a graphical user interface. A GUI interface and a number of efficient 13C flux analysis tools are also included in the newly released OpenFlux framework for MATLAB (Quek et al. 2009). Additionally, 13C flux may be analysed by means of COBRA and OptFlux. In contrast, FiatFlux's local flux ratios analysis method (Zamboni et al. 2005) is used to calculate the comparative fluxes at branching sites. The mistakes brought on by network model faults are often minimized by this type of reasoning. However, the algebraic formulas that link tagging schemes to flow ratios are often created by hand. As a result, only a few species that consume glucose are currently pre-set in FiatFlux to compute flux ratios.

3.4.3 Techno-Economic Analysis (TEA) of Microbial and Algal Lipid Production

The secret to a successful lignocellulosic microbial oil biorefinery is econometrics. The strain employed to cultivate the lipid and its concentration within the cells and cells generated have a major impact on the price of microbial oil production. The capacity of microbial oil is often expressed as g/L/day or kg/m^3/day, in contrast to the kg/ha/year capacity of vegetable oil plants. The carbon source utilized to make microbial lipids accounts for around 60–75% of overall expenses. To preserve the economics of microbial oils, a low-cost and effective feedstock alternative is adopted.

Few of the technological and economic implications of using oleaginous bacteria to create microbial lipids have been investigated yet. Ratledge and Cohen (2008) provided a supposed cost of the microbial oil, i.e., US$3000/tonne (without the feedstock cost), founded on past work estimations in New Zealand. *Candida curvata* was used to produce cocoa butter from lactose. Not long ago, Koutinas et al.

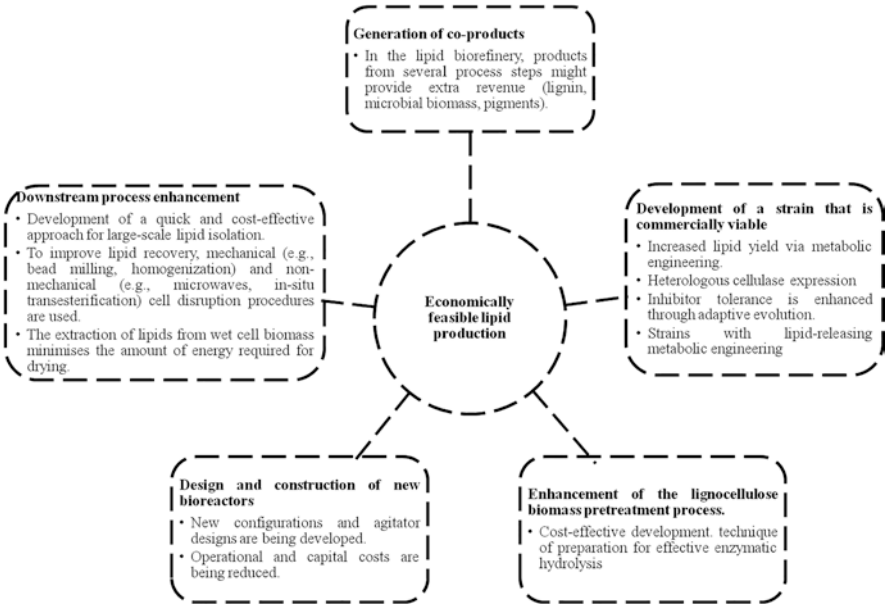


Fig. 3.1 Improvement tactics for the lipid synthesis process

(2014) calculated the presumed cost of production of microbial oil and biodiesel from glucose with a yield of 0.23 g/g lipid as US$5500/tonne oil and US$5900/tonne biodiesel, which is not practicable with vegetable oil prices in the current market, i.e., around US$ 800–900/tonne, and biodiesel is at US$ 4.0/gallon (US$ 1220/tonne) (Koutinas et al. 2014). These tests, which cost between US$400 and US$800 per tonne, do not examine the by-product microbe meal. Improvements in lipid output, productivity, and concentration of lignocellulosic biomass are required to make a microbial biorefinery commercially feasible (Santamauro et al. 2014) (Fig. 3.1).

3.5 Conclusions

Abundant studies have been led to find ways to boost lipid synthesis, lower fermentation expenses, and produce microbial oils via organic and lignocellulosic waste. The approach presents fresh possibilities for waste valorization. To boost lipid daily synthesis, however, microbial concentration must be increased. In this regard, evaluations of fermentation experiments using a fed-batch system with an elevated substrate concentration have been conducted. Operating in a solid-state arrangement would be a wise decision given the characteristics of lignocellulosic biomass, including a number of other by-products. It may be possible to use lignocellulosic substrate directly at high temperatures without the requirement for labour-intensive

preparatory processes. The comparatively low lipid content of fabricating strains, the low susceptibility of microbial organisms to lignocellulosic-derived inhibiting agents, the inadequately substantial biomass concentration found in lignocellulosic hydrolysate, as well as the substantial expenses of product isolation are challenges to the majority of current methods for manufacturing bacterial fatty acids from lignocellulosic biomass. To reduce production costs and improve the viability of bioprocesses, research should focus on the following areas: increasing the concentration of fatty acids and recognizing inhibitors from the pretreatment process, optimizing oleaginous microorganisms via genetic modification techniques and adaptive evolution, developing new effective methods of lignocellulosic material pretreatment and hydrolysis that produce a significant amount of sugars that can be fermented in the fermentation medium, and creative bioprocess design.

Due to the great interest in producing effective biofuel, metabolic engineering of microalgae is a key topic of research. These discoveries involving gene divergence will shed light on a hopeful future for lipid bio-genesis because algae have the highest degree of divergence (67.7% different). The research for biofuel production by microalgae has only been done on the model microalgae *C. reinhardtii,* therefore, a thorough genome-scale metabolic flux-based characterization of TAG accumulation in other microalgal lineages is necessary. This hypothesis has been confirmed by a review of the literature mentioned above. In our opinion, the primary genes involved in lipid production should be our initial points of focus. Since we have mapped the genomes of both plants and algae, it is simple to reproduce the techniques used in plants in microalgae. Indeed, producing the lipid droplets requires using many types of stress responses that microalgae exhibit. Moreover, the availability of nutrients under stressful conditions will play a significant part in lipid output. The storage of energy-rich necessary molecules in the form of triacylglycerol, which might serve as an effective supply of lipid, would change the microalgal metabolism when stressful situations or when the starting substrate concentration is raised.

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Chapter 4
Basic Concepts and Applications of Enzyme-Catalyzed Reactions for Biotechnology



Kongara Hanumantha Rao, Somnath Sahoo, and Jeena Gupta

Abstract Enzyme-catalyzed reactions provide several advantages such as specificity, high catalytic rate, transition state stabilization, etc., over inorganic catalysts and they have brought revolutionary changes in the synthesis of compounds at the industrial level. Enzymes are biological catalysts that are specific to the substrates and they increase the rate of reaction several folds by decreasing the activation energy. Enzyme provides an active site where the substrate/s can take a proper position and orientation to react and further stabilize the transition state favoring the reaction to proceed. Technological advancements like genetic engineering by using fungi, bacteria, and yeast, led to the development of novel enzymes possessing varied applications and specificities and they are still under exploration for new applications. Understanding the basic concepts of enzyme-catalyzed reactions can provide novel insights into the development of new strategies for designing artificial enzymes as well as the new methods for increasing the yields of enzyme-catalyzed reactions.

Keywords Enzymes · Active site · Allosteric site · Synzyme · Abzyme

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4.1 Enzyme-Catalyzed Reactions and General Mechanism of Action of Enzymes

Enzyme-catalyzed reactions are biochemical reactions that can be facilitated or accelerated by enzymes. Catalyst increases the rate of chemical reactions without being consumed or permanently altered in the process. Enzymes are proteins and act as highly efficient catalysts that can accelerate reactions by several folds by lowering the activation energy required for a reaction to occur, thus making it easier for the reactants to reach the transition state and form products. The reactants in an enzyme-catalyzed reaction are known as substrates, and they bind to a specific region on the enzyme called the active site. The active site of enzyme has a definite shape that is complementary to the substrate structure and its chemical properties help to the binding of enzyme to the substrate. This specificity of the enzyme catalyzes the specific reaction by binding to the appropriate substrates and this binding allows the enzyme to undergo conformational changes thus facilitate the reaction. These conformational changes can happen by the rearrangement of bonds within the substrates or the introduction of chemical groups from the enzyme to the substrates and the binding can be achieved through some temporary interactions such as hydrogen bonding, ionic interactions, and hydrophobic interactions (Nelson and Cox 2008). When the substrate binds to the active site, it forms an enzyme-substrate complex. This complex brings the substrate molecules into close proximity and proper orientation, which promotes the occurrence of the chemical reaction. Once the enzyme-substrate complex is formed, the enzyme catalyzes the conversion of the substrate into the desired product(s). Enzymes can accelerate the rate of reaction by lowering the activation energy required for the reaction to proceed. They achieve this by providing an alternative reaction pathway with a lower energy barrier, called the transition state (Fig. 4.1). After the reaction is completed, the enzyme releases the product(s) from its active site. The product(s) may have a different shape from the substrate, and the active site may adjust to accommodate this change

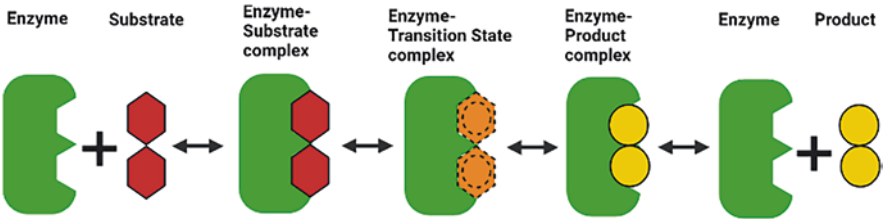


Fig. 4.1 Conversion of substrate to product (or) product to substrate in an enzyme-catalyzed reaction. Enzyme is represented with green. Substrate is represented as paired hexagonal with red color. Transition state is represented as dotted structures of both hexagonal and circles embedded with orange color. The dotted structure indicates transient and unstable nature of transition state which highly energized intermediate state between substrate and product. The product is represented as circular pair with yellow color

during the reaction. Once the product(s) is released, the enzyme is ready to bind to another substrate molecule and initiate another round of catalysis. Enzymes are not consumed or altered during the reaction; they remain unchanged and can be used repeatedly. The specificity of the enzyme is remarkable as a particular enzyme can catalyze a single type of reaction or a particular class of reactions due to the precise arrangement of functional groups on the enzyme's active site. This specificity allows enzymes to carry out specific biochemical reactions in a highly controlled manner. The mechanism of enzyme action can be further influenced by factors such as pH, temperature, substrate concentration, and the presence of inhibitors or activators.

Changes in any of these factors can affect the rate of the enzyme-catalyzed reaction, potentially enhancing or inhibiting the enzyme's activity. Enzyme-catalyzed reactions play essential roles in numerous biological processes, such as digestion, energy production, DNA replication, protein synthesis, etc. They are fundamental to the functioning of living organisms and are targets for many therapeutic innovations and industrial applications.

4.2 How Enzyme Catalysis Is Different from Normal Inorganic Catalysts

Inorganic catalysts are made up of metals and their oxides and almost execute the same function as observed in nature's catalysts. These are also termed as heterogeneous catalysts. The reactants attach to the active sites of the metal surface through adsorption. Therefore, it is crucial to confirm that the metal surface has enough accessible active sites. The number of active sites increases along with the surface area, which thus enhances the catalytic activity. The wide metal surface has supported by several porous materials like carbon, silicone, or alumina. Just after the reaction completion, catalysts are separated into a different phase that is totally different from both initial and final components thus extracting and recycling the catalyst more easily. This characteristic of inorganic catalysts made it a suitable beneficial choice for synthetic processes for bulk isolation and production that are frequently used in both industry and academia.

The presence of Carbon (C), Hydrogen (H), and Oxygen (O) in the chemical structure is the striking feature that distinguishes organic and inorganic catalysts (Védrine 2017). Organic catalyst possesses C, H, and O in their structure while the inorganic ones do not have them. Metals like palladium, copper, and cobalt are examples of inorganic catalysts, whereas enzymes like kinases, invertase, and polymerase are examples of organic catalysts. Though inorganic catalysts cannot be synthesized in live cells and can only be artificially generated, organic catalysts can be made both naturally and artificially. Furthermore, compared to inorganic catalysts, organic catalysts or enzymes are sensitive to variables like pH and temperature.

4.3 Model of the Enzyme-Catalyzed Reaction

More than a century ago, researchers have been trying to understand the process of enzyme catalysis and to find the sites that give enzymes their extraordinary catalytic efficiency. Experimental and computational studies have provided a plethora of knowledge on enzymes, notably highlighting the significance of enzyme structure in catalysis. In both the original and revised versions of the lock-and-key theory, the induced-fit and transition state stabilization theories, it is suggested that enzymes act as structural scaffolds that recognize and bind their substrates in a certain manner (Fischer 1894). After the proper binding of a substrate with an enzyme, enzymes provide a new reaction environment that is quite different from the characteristics of the solvent and enable the reaction to happen more quickly (Bruice and Benkovic 2000). The enzyme (and also cofactor) residues that are bound to the substrate, always present in fixed positions and they have crucial roles in maintaining the reaction (Nagel and Klinman 2009; Nagel et al. 2013). It was observed that mutations of these residues are responsible to alter the rate of catalyzed reaction and thus the fate of reaction (Nagel et al. 2013). It was proved that the electrostatic environment of the active sites of the enzyme are different from the solvent hence it favors the progression of reactions. There was a contradiction that the forms of active sites are complimentary to the transition state of the catalyzed processes (Warshel et al. 2006). Since active sites provide a complimentary structure and electrostatic environment to bind and stabilize the transition state (enzyme-substrate complex), it needs to understand the enzyme activity in terms of structural scaffolds.

Presentation of a biophysical model explaining the operation of enzymes. Enzyme action is significantly influenced by conserved network of residues. The energetic link between the enzyme's active site and the external solvent is made possible by a hierarchy of conformational variations. Movements known as conformational fluctuations enable sampling of remote regions of the protein landscape at intermediate (nanoseconds) to slow time-scales (microseconds and more). The activation energy barrier is broken by the thermodynamical energy from the solvent that is specifically directed into the active site and permits access to conformational substates that are functionally important. A structural change is a motion that modifies the structure of a protein temporarily (or permanently). The dynamical regions of the enzyme that are crucial for its activity, especially those that show modest conformational variations, have not changed throughout time. With the use of this new knowledge of enzymes, hyper-catalytic enzymes may be created by identifying and controlling conformational changes that promote reactions (Agarwal et al. 2012). To develop a comprehensive biophysical model of an enzyme, the information related to various structural components and electrostatic environment at active site and distal residues are required for catalytic activity (Warshel et al. 2006). The environment around the active site region has the best electrostatics to support the catalyzed reaction. There are networks of conserved residues in several enzymes that link surface areas to the active site. These networks provide thermodynamical connection between the catalyzed process and the hydration-shell and bulk solvent.

By improving the energy flow across these networks and optimizing through mutations, new tactics for improving catalytic efficiency have been made possible by the discovery of these enzyme energy networks.

Proteins move internally throughout a time scale of at least 12 orders of magnitude, from femtoseconds (1015 s) to microseconds and longer (>106 s) (Agarwal 2006; Henzler-Wildman et al. 2007). Internal motions of involved molecules occur during the binding and placement of the substrate (and cofactor) and the removal of the product(s) during the operation of an enzyme. The direct contribution of enzyme movements to speed up the rate-limiting phase, which establishes the overall effectiveness of enzyme catalysis, is the more intriguing topic. The chemical (substrate turnover) step, the binding step, or the removal step might all be the rate-limiting step. Understanding catalytic efficiency will be significantly impacted by the discovery of enzyme movements or conformational changes that actively participate and encourage the rate-limiting phase (Agarwal 2005).

The existence of long-range effects and distal regulation of activity in enzymes is a perplexing feature (Lisi and Loria 2016). It is well-known that the part of the enzyme is distant from the active site, there is no clear relationship between them and the functional site, which can significantly alter how the proteins and enzymes work. The modulation of activity by the binding of ligands or biological molecules at a position other than the main site of enzyme action (the active site) is referred to as allostery and cooperative effects (Cook and Koshland 1970; Levitzki and Koshland 1969). Initial research revealed that allosteric modulation was made possible by linked structural interactions and electrostatics (Baldwin and Chothia 1979). The long-term impacts on enzyme performance have recently been explained in terms of energy connection in the enzyme structure (Frauenfelder et al. 2001). For an enzyme-catalyzed reaction to be successful, an activation energy barrier must be overcome. It should be noted that even though enzymes decrease the effective transition state barrier (by raising the ground state and/or reducing the transition state through stabilization), the enzyme-assisted reaction still must overcome obstacles of 5–10 kcal/mol (or more). How do enzymes help us get through these obstacles? There is not enough energy provided by the random thermal fluctuations brought on by enzyme movements (1 kcal/mol at room temperature). The essential energy to get past these obstacles is thought to originate from the solvents thermodynamic energy that is present on the protein's surface (Agarwal 2005, 2006; Ramanathan and Agarwal 2011).

4.4 Applications of Enzymes

Enzymes have the potential to be isolated from cells and employed as catalysts in diverse industrial processes. They are used in several industries ranging from the synthesis of sweetening agents to the modification of antibiotics. They are also applied in laundry detergents, various cleaning products, and contribute

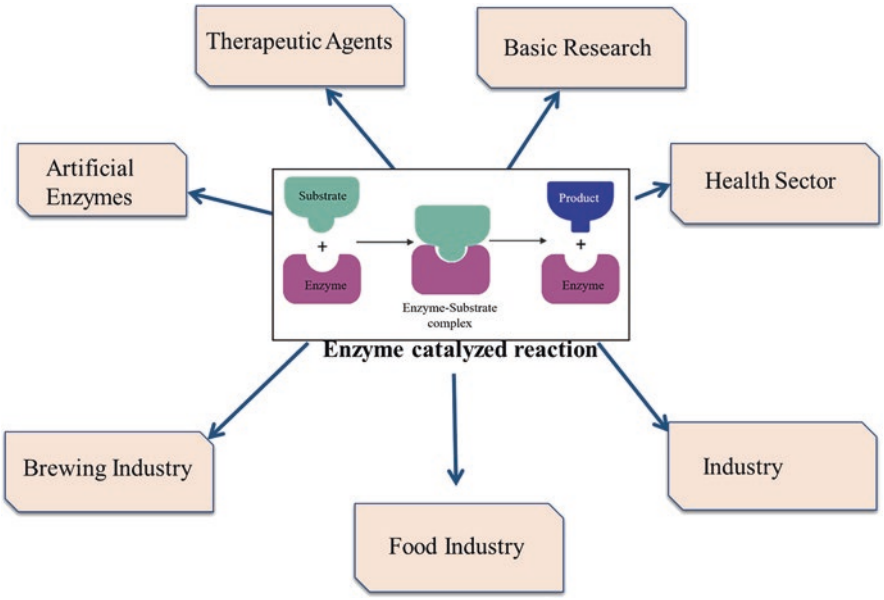


Fig. 4.2 Applications of enzyme-catalyzed reactions in various fields

significantly to the functioning of clinical, forensic, and environmental analytical devices and assays (Robinson 2015) (Fig. 4.2).

4.4.1 Application as Therapeutic Agents

Enzymes are biocatalyst that are broadly used in the medical field to treat several diseases including cancer, heart attack, etc. For example, in acute lymphocytic leukemia, the asparaginase enzyme used to arrest the cancer cell growth by decreasing the level of L-asparagine (Bruno and Njar 2007). Apart from the asparaginase enzyme, there are several enzymes that are used as medicine including collagenase (used to treat skin ulcers), glutaminase (treat leukemia), hyaluronidase (treat myocardial infarction), lysozyme (used to control bacterial infection), etc. (Egler et al. 2016).

4.4.2 Application in the Health Sector

Apart from their function in the treatment of diseases, enzymes have the potential to detect and determine the disease. Their analytical function improves the diagnosis of different malfunctions in blood and serum easily and rapidly. Different

examples provide multifarious roles of enzymes from an analytical viewpoint. Glucose oxidase (clinistix/clinistrip: a biosensor-based glucose detection) is used to detect glucose, cholesterol oxidase determines the serum cholesterol level, Glycerol-3-phosphate oxidase detects the level of serum triglycerides, urease and uricase determines the urea and uric acid respectively (Shingleton et al. 1996; Bauer et al. 2022).

4.4.3 Application in the Food Industry

In our daily life, enzymes play a significant role. In the food production and processing industry, enzymes are used to make food products. Generally, yeast (*Saccharomyces cerevisiae*) produces enzymes that help in the baking of bread. Commercially available glucose, invert sugar, high fructose syrup, cheese, etc. are prepared using different types of enzymes (Konieczna et al. 2012).

4.4.4 Application in the Brewing Industry

Micro-organisms have been used for thousands of years for making products such as wine, beer, vinegar, soy sauce, etc. The micro-organisms (such as yeast) are really used as a good source of enzymes during the manufacture of these products. Fermentation by yeast is the main enzyme-catalyzed reaction and the brewing industry solely relies on this mechanism. Furthermore, papain enzymes are used to prevent chill hazes while brewing (Singh et al. 2019).

4.4.5 Application in Industry

Though enzymes have major functions in health and food sector, it also has an impact to produce other products that improve lifestyle. Washing powder and fabric stain remover are daily used products that mainly consist of different enzymes. Various enzymes are used in leather industry as well as textile industry.

4.5 Artificial Enzymes

(a) **Synzymes**: Synthesizing molecules which mimic the action of natural enzymes (mimic enzyme's function). They mimic a chemical reaction and/or mimic recognition (mimic binding; H-bonding, hydrophobic, ionic, etc.). Artificial

enzymes possess some of the active sites of real enzymes, for example, cyclodextrins operate as ribonuclease and transaminase.

(b) **Abzymes**: Catalytic antibodies. One more approach involves making antibodies with the characteristics of enzymes. An analogue of the supposed transition state of a particular reaction is injected into a mouse and the immune system will make antibodies against it, and some of these may be able to catalyze the reaction in question.

4.6 Conclusions

In this review, we have discussed the function of different enzymes in detail. So, understanding the basic concepts of enzyme and enzyme-catalyzed reactions would definitely provide novel approaches in designing artificial enzymes. Enzymes work very specifically at a certain processing step in the creation of a product, making them very effective catalysts in many different industrial processes. Since the enzymes employed in industrial processes are also derived from natural sources, they can be applied in a similar manner. They assist in providing consumers with environmentally friendly goods that are produced with less waste and less energy, water, and raw materials. Advancement of new technologies will help us to overcome the limitations about enzyme in study of mechanistic insights that further improve the usage of enzyme to various field that may improve the quality of life.

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Chapter 5
Kinetics of Microbial Growth, Substrate Consumption, and Product Formation



Smita Gupta and S. K. Sundar

Abstract Microbial activities could be utilized for the production of secondary metabolites such as antibiotics, growth hormones, etc. The microorganisms can grow on different types of substrates which makes them promising candidates for pharmaceutical, environmental, and other industrial applications. To understand the microbial transformation processes, it is necessary to understand the various kinetic models associated with microbial growth, substrate utilization, and product formation kinetics. Also, various controlling parameters need to be optimized through model development for different types of biological reactors. In this regard, the focus is on topics related to development of kinetics models for microbial growth in batch reactors, chemostats, plug flow fermenters, fed-batch reactors, and immobilized cell systems.

Keywords Growth kinetics · Growth models · Bioreactor · Immobilization · Biofilm

5.1 Introduction

Microbial growth is the process of division of a cell into two daughter cells by binary fission. This cell division results in the growth of cells in a given population. The time it takes for a single cell to double in number is called generation time or doubling time and the growth rate is directly proportional to cell concentration. The growth of a microorganism is dependent on various physical, chemical, and nutritional conditions. The microbial cell utilizes a portion of the supplied nutrients for energy and another portion for product formation. The growth of the microbial cell is to be explored further, as the relation between specific growth of microbe and

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substrate concentration provides valuable information for the development of biotechnological products and contaminant removal from environment. In this chapter, we will discuss various methods of measuring cell number density, cell mass concentration, growth patterns, and kinetics in batch culture. In this regard, various models describing microbial growth kinetics such as the Monod equation, Tessier equation, etc., are discussed. Furthermore, the investigation of growth, substrate consumption, and product formation in various types of reactors are necessary for design and optimization of process conditions. In this context, the performance equation of different types of reactors: batch, continuous, and fed-batch, which are used in microbial systems for substrate utilization and product formation, are highlighted. In addition to the above, inhibition kinetics on microbial growth, and whole cell immobilization systems for continuous culture with an emphasis on biofilm reactors are discussed.

5.2 Microbial Growth Assay

5.2.1 Aerobic and Anaerobic Growth

Bacteria are broadly classified as aerobic, anaerobic, or facultative, based on the reactions that occur to generate energy (Balows et al. 1974). Aerobic growth refers to the growth of an organism in the presence of an oxygenated environment; such organisms are termed 'aerobes' (e.g., *Bacillus, Pseudomonas aeruginosa*). The oxidation of glucose to produce carbon dioxide and water is a well-known example of an aerobic process (Chauhan 2008). Anaerobic growth refers to the growth of an organism in an environment deprived of oxygen, such organisms are called 'anaerobes' (e.g., protozoans). Their metabolism is fermentative, in which organic compounds will be reduced to end products such as organic acids. Some anaerobes even die in an oxygenated environment.

The organisms can be identified by growing them in the media of thioglycolate broth (Todar 2020; Hogg 2005; Kumar 2012). Obligate aerobes migrate towards the top of the test tube as they require oxygen for their survival and oxygen acts as an electron acceptor, e.g., *Cholera vibrio, Pseudomonas aeruginosa* (Fig. 5.1a). Obligate anaerobes can be seen at the bottom of the tube as they migrate away from an oxygen-rich environment, e.g., *Clostridium tetani* (Fig. 5.1b). Facultative anaerobes are aerobic but they can survive in the absence of oxygen and they move mostly towards oxygen-rich environments to generate more ATP. They generate energy both by respiration and fermentation, e.g., *E. coli* (Fig. 5.1c), Microaerophiles require a minimum quantity of oxygen as excess oxygen may hamper their survival and hence, they gather just below the upper part of the tube, e.g., *H. pyroli* (Fig. 5.1d). Aerotolerant bacteria do not require oxygen but they can survive in an oxygenated environment also and hence they can be seen evenly spread throughout the tube, e.g., *Lactobacilli* (Fig. 5.1e) (Kumar 2012).

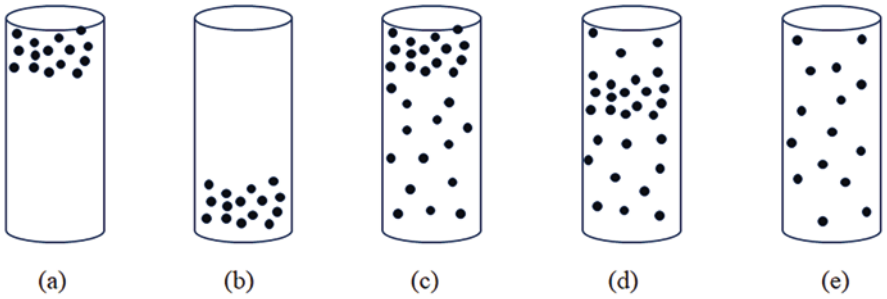


Fig. 5.1 (**a**) Obligate aerobes. (**b**) Obligate anaerobes. (**c**) Facultative anaerobes. (**d**) Microaerophiles. (**e**) Aerotolerant organisms

The culture of anaerobes is challenging because of the need to exclude oxygen and complex growth requirements. Oxygen can be excluded by the production of vacuum, displacement with inert gases like nitrogen or creating an anaerobic environment using Gas Pak jar or McIntosh and Filde's jar, absorption of oxygen by chemical (pyrogallol) or biological methods and by supplementing the culturing media with a reducing agent such as thioglycollate or cysteine. As an alternative, Robertson's Cooked Meat (RCM) broth can also be used, wherein the fatty acids in chopped meat serve as a reducing agent. Reducing agents lowers the oxygen level and brings down the redox potential to less than −300 mV. A redox indicator (resazurin) can be added if required. The indicator will be pink in the presence of oxygen and turns colourless in its absence (Kumar 2012).

While choosing optimal media for culturing anaerobic bacteria, both broth and solid media should be inoculated. The culture media comprises anaerobic blood agar plates with yeast extract, brain heart infusion, amino acids, and vitamin K, a selective medium such as kanamycin vancomycin blood agar, and a broth such as a brain heart infusion broth with added reducing agent. Plates are to be checked after 18–24 h for fast-growing organisms like *B. fragilis*; and after 5–7 days for slow-growing organisms like *Actinomyces* and *fusobacteria*. Aerobic cultures (*Pseudomonas aeruginosa*) should also be grown as a control to identify the growth on anaerobic plates. Anaerobes are generally identified by colonial and microscopic morphology, oxygen tolerance, and biochemical characterization (Kumar 2012). The species are determined based on sugar fermentation and the genus is identified by gram staining and cell morphology.

5.2.2 Microbial Growth Measurements

Bacterial growth is expressed in terms of an increase in either number of individual cells or the amount of biomass (Hogg 2005). Bacterial growth can be determined by counting the cell population at intervals after inoculating a few bacteria into a liquid

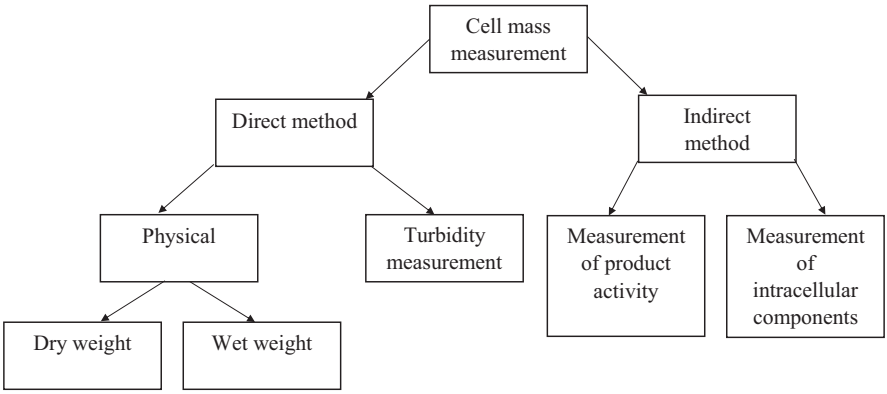


Fig. 5.2 Methods to determine cell mass

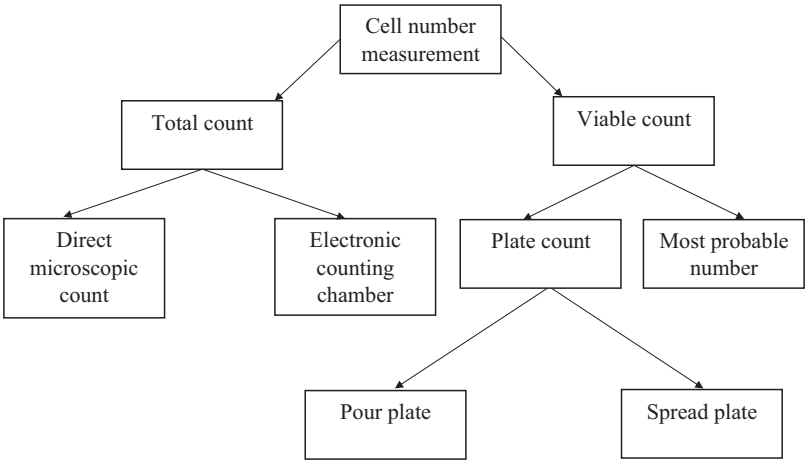


Fig. 5.3 Methods to determine cell number

medium or solid culture media. The growth is detected in the form of colonies on a solid medium whereas by the appearance of turbidity in a liquid medium. The bacterial growth is measured in terms of cell mass or cell number as shown schematically in Figs. 5.2 and 5.3.

5.2.2.1 Measurement of Cell Mass

The direct physical method involves the determination of cellular weight. The limitation is that culturing of cells should be carried out in solids free medium for accurate weight measurement. Usually, the culture broth containing bacterial cells is

subjected to centrifugation or filtration. The collected bacterial cells are washed with a buffer solution or water and their wet weight is measured. The dry weight of the cell is measured after drying the wet cell mass at 80 °C overnight (Shuler and Kargi 2015).

Another rapid and direct method involves the measurement of turbidity of the culture medium by measuring the intensity of transmitted light using a spectrophotometer. The turbidity of the cellular suspension is directly proportional to cell mass (Shuler and Kargi 2015).

The above-discussed direct methods cannot be used in many fermentation processes such as mould fermentation. For such cases, indirect methods which depend mainly on the measurement of chemical activity for product formation or substrate consumption, are used (Shuler and Kargi 2015).

In a batch process, the concentration of intracellular components such as RNA, DNA, and protein varies with time. Measurement of intracellular components also constitutes indirect methods for measurement of cell mass. Biuret, Lowry, and Kjeldahl nitrogen measurements are widely used indirect methods for the determination of cellular protein (Shuler and Kargi 2015).

5.2.2.2 Measurement of Cell Number

Total count requires the use of a microscope or electronic counting chambers and gives a measure of the total number of bacterial cells whether alive or dead (Hogg 2005; Kumar 2012). For observing the cells, a microscope, and special slides known as the Petroff-Hauser counting chamber, Neubauer chamber, and Helber chamber are used.

The methodology involves placing a droplet of bacterial suspension on top of the Petroff-Hauser chamber (Fig. 5.4), followed by placing a coverslip over the chamber (Hogg 2005). By knowing the depth of trapped liquid, volume can be calculated and it can hold a sample volume of $1/20 \times 10^7$ cc (approx.). The number of bacterial cells (N) present in squares of the chamber is counted using a microscope and the number of bacterial cells per cc will be given by $N \times 20 \times 10^7$. The limitation is the requirement of a high concentration of bacterial cells (~10^7 cells/mL).

In an electronic counting chamber (e.g., Coulter coulter), the clean bacterial suspension is forced through an orifice that allows only one cell to pass at a time. The movement of bacterial cell suspension is impacted by electric current. Measurement is very quick and useful for large sample volumes (Kumar 2012).

Viable count gives an estimate of living cells, capable of replication, in a sample. Viable counts can be determined either by the dilution method or the plate count method. In the plate count method, serial dilutions of a sample containing microorganisms are made. The bacterial suspension of known volume is spread onto the surface of agar plates and then incubated at 37 °C overnight. Incubation results in the formation of bacterial colonies that can be seen by the naked eye. Since a single

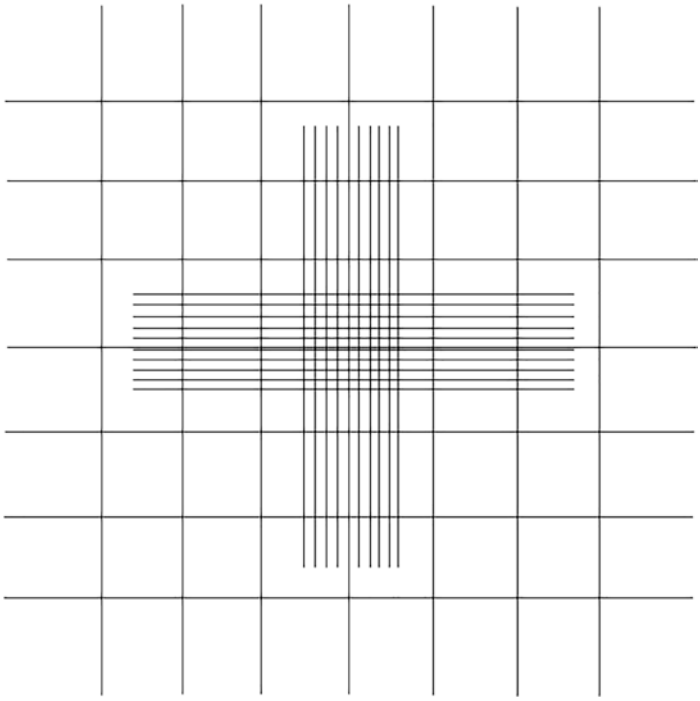


Fig. 5.4 Schematic representation of Petroff-Hauser chamber

bacterial cell grows into a colony by cell division, counting the number of colonies gives the number of bacteria in the original sample, after accounting for the dilution factor. The viable cells are expressed in colony-forming units per ml (CFU/mL) (Hogg 2005).

Most probable number, a common technique to determine microbes in soil samples, waters, and agricultural products is generally employed for quality testing of water. Serial dilutions of water samples are added to MacConkey broth containing bromocresol purple as an indicator. Bacteria in water feed on lactose and produce acid and gas, leading to the colour change of the medium (Kumar 2012).

Apart from the techniques discussed above, other reliable techniques for the evaluation of microbial growth kinetics include the impedance method for detecting the temperature of *Salmonella enteritidis* (Fehlhaber and Krüger 1998) and the colorimetric method for strain identification (Carey et al. 2011). Recently, Loutfi et al. (2020) studied the bacterial growth kinetics of *Bacillus thuringiensis* in a liquid medium under real-time conditions using laser speckle imaging and the experimental results correlated well with parameters obtained from speckle imaging technique. To obtain higher throughput in less time, turbidostats (Takahashi et al. 2014) and time derivative-based microplate readers (Rogers et al. 2022) are used for monitoring the growth of microbes.

5.2.3 *Phases of Microbial Growth*

In a suitable liquid medium and under favourable conditions (nutrients, temperature, etc.) the microorganisms grow rapidly. The growth of a microbe under certain conditions could be controlled for secondary metabolite production, by studying its growth pattern. Bacterial cells need certain conditions for growth, and these conditions vary among different types of bacteria. Bacterial growth is affected by factors such as oxygen, pH, temperature, and light. Since bacteria can be grown easily under laboratory conditions, their growth is analysed using a batch culture. Upon inoculation into a new nutrient medium and under favourable conditions, a bacterium shows a predictable growth pattern termed a growth curve. A typical growth curve (Fig. 5.5) includes four distinct phases: lag phase, exponential or log phase, stationary phase, and death phase. However, only the growth pattern will be similar whereas the details within each phase (cell number, phase length, overall growth time, etc.) will vary between organisms and environmental conditions.

Lag phase: The lag phase refers to the length of time during which a bacterium will tend to adapt itself to the new environment. The bacterial cells will be metabolically active and increase in size without any cell division and the cell number density remains unchanged. The cells will be adapting to the new environmental changes such as changes in temperature, pH, oxygen, nutrients, etc. During this phase synthesis of RNA, enzymes and other growth factors occur. The duration of the lag phase is affected by the concentration of nutrients, growth factors, and age of the inoculum. For instance, when *Enterobacter aerogenes* is grown in a medium comprising glucose and phosphate buffer, the lag phase increases due to a decrease in the concentration of Mg^{2+} (phosphatase activator). Furthermore, to reduce the lag

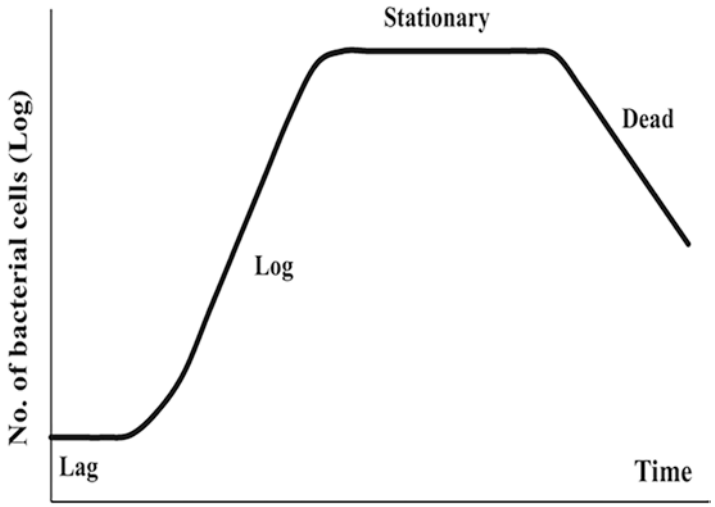


Fig. 5.5 Schematic plot showing various phases of microbial growth (Wang et al. 2015)

phase, cells should be adapted to new conditions before inoculation. Multiple lag phases may occur when more than one carbon source is present in the medium and the phenomenon is termed diauxic growth (Shuler and Kargi 2015).

Log phase: Soon after the generation of necessary growth factors, the bacterial cells enter into the log phase and proceed for cell division by binary fission. The log phase is characterized by a doubling of the cell population (Hogg 2005) where the cell number increases in a logarithmic manner while maintaining the cell constituent. Rapid growth of cells occurs under optimum conditions whereas slow growth occurs under non-optimal conditions. The cells grow at the same rate and the composition within a single cell remains constant. The cells are healthiest and uniform at this stage and bacterial cells at this stage are utilized for experimental research. The cells will double in a constant length of time known as 'generation time' (Hogg 2005). The generation time (g) can be estimated by plotting the logarithm of cell number against time, in a semilogarithmic graph. The slope of the line gives $0.301/g$. Alternatively, by knowing the initial number of cells at the start of the logarithmic phase (N_0) and the number of cells (N) after time 't', the number of generations (n) can be determined by the relation $N = N_0 2^n$. 'n' is the elapsed generation time given by $n = t/g$ from which generation time (g) can be obtained as 't/n'. The log phase continues until there is a depletion of essential nutrients or its growth is inhibited due to the buildup of metabolic waste products that inhibits cell growth and the bacterial cells transit to the stationary phase.

Stationary phase: During this phase, the cells cease to divide and their growth will slow down and try to adapt themselves to starvation conditions. This phase is a kind of equilibrium state in which the number of newly formed cells equals dying cells. However, the cell population remains the same at any time during this phase. The phase is represented by a horizontal line in the growth curve (Kumar 2012). The cells may also produce secondary metabolites, such as antibiotics.

Death phase: In this last phase of bacterial growth, the rate at which cells die dominates the rate of formation of new cells due to a lack of nutrients and other unfavourable environmental conditions. There will be an exponential decrease of viable cells and the viability loss is given by the slope of the curve (Kumar 2012). Also, the cells in this phase won't be able to resume cell division upon transfer to a fresh growth medium.

5.2.4 Factors Affecting Microbial Growth

Microbial growth is affected by many factors such as nutrient composition and various other environmental factors (Hogg 2005; Kumar 2012). The most important factors that influence microbial growth include:

Nutrient composition: The elements that are required for microbial growth are called nutrients. The nutritional requirement also varies among various types of microbes. The nutrient medium must include water, carbon source (molasses,

starch, corn syrup, etc.) nitrogen source (amino acids, proteins, etc.), and mineral salts (Cu, Ca, Ni, etc.). Growth factors such as vitamins, hormones, and amino acids, stimulate growth (Shuler and Kargi 2015).

Temperature: Microbial growth occurs only between a minimum temperature and a maximum temperature. Since solidification and stiffening of lipids in cell membranes occur at low temperatures and cellular protein denature at high temperatures, the growth occurs at a temperature range between these two temperatures. The temperature at which maximum growth occurs within a short period is known as optimum growth temperature (Fig. 5.6). Based on this thermal resistance, bacteria are classified as psychrophiles, mesophiles, and thermophiles (Hogg 2005; Kumar 2012). Psychrophiles grow at 0 °C with an optimum growth temperature of 15 °C (e.g., *Listeria monocytogenes*). Mesophiles grow between 10 and 40 °C (e.g., *E. coli*). Thermophiles grow between 40 and 120 °C (e.g., *Bacillus thermophilus*).

pH: The ionic property of microbial cells is affected by pH. Each microbe requires a definite pH range for its growth and multiplication. While the majority of microbes survive at neutral pH, certain microbes require an acidic or basic environment. Based on the optimum pH requirements, bacteria are classified as acidophiles, neutrophiles, and alkalophiles (Hogg 2005; Kumar 2012). Acidophiles grow in a pH range between 1 and 6.5 (e.g., *Lactobacillus acidophiles*), neutrophiles grow best between pH 6.5 and 7.5 (e.g., *E. coli*), and alkalophiles grow in a pH range between 7.5 and 14 (e.g., *Vibrio cholerae*) (Fig. 5.7).

Ions and salt: The presence of metal ions such as K^+, Zn^+, Ca^{2+}, Cu^{2+}, etc. is mandatory for enzyme production in a bacterial cell. While a majority of bacteria do not require NaCl in the nutrient medium, NaCl is a must for the survival of halophilic microbes such as *Archaebacteria*.

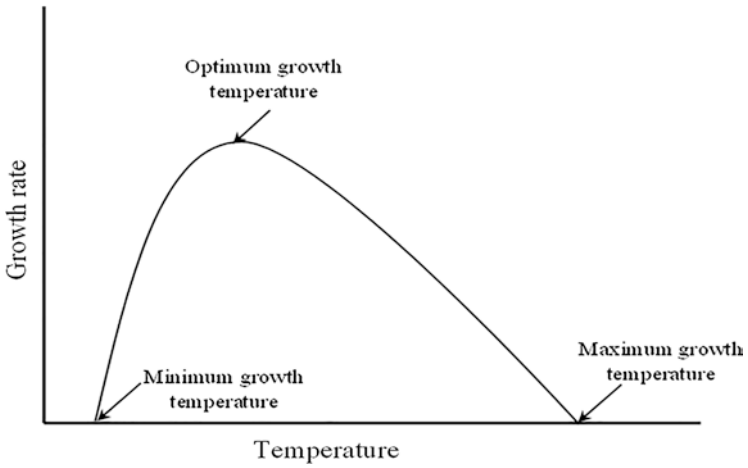


Fig. 5.6 Schematic plot showing effect of temperature on microbial growth rate

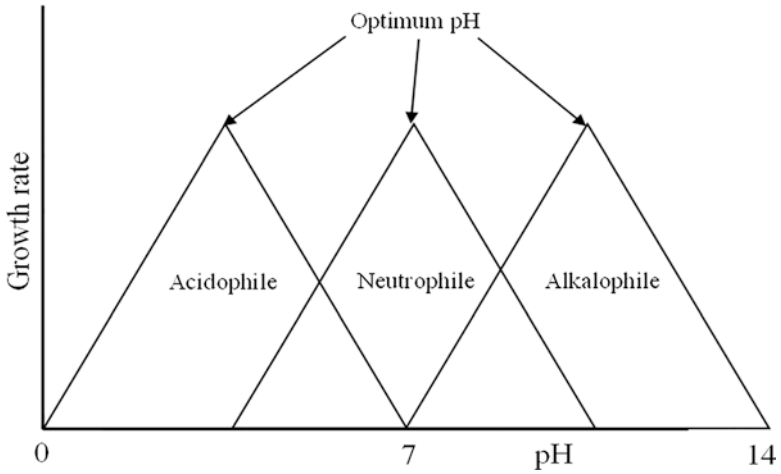


Fig. 5.7 Schematic plot showing effect of pH on microbial growth rate

Gaseous requirement: Microbial growth is influenced by the presence/absence of oxygen and accordingly they are classified as aerobic or anaerobic. Few organisms like *Brucella abortus* grow better in when air contains 5–10% CO_2 (Kumar 2012).

Water: The added nutrients such as sugar, salts, and other chemicals are dissolved in water, making them accessible to the microbes. Further, the metabolic activities are influenced by the amount of water present in the nutrient medium.

5.3 Kinetics of Microbial Growth

When a liquid medium consisting of vital nutrients is inoculated with microbial suspension at an appropriate temperature and pH, the microorganisms will grow. Based on the morphology of cells, the growth of microbial cells occurs in two ways, either an increase in biomass will be accompanied by an increase in the number of cells present as in the case of unicellular organisms such as bacteria, or biomass increases in the size and density instead of number of cells present as in case of multicellular organisms such as moulds. In addition to cell growth, the consumption of some components from the medium and the release of metabolites into the medium also take place. The rates of the latter two processes vary widely with the growth of the cells. In this section, a discussion is focused on the brief introduction of the various models proposed for microbial growth followed by the kinetics of microbial growth for different reactors, product formation, substrate consumption, and inhibition effects on growth kinetics.

5.3.1 Models for Microbial Growth

In microbial cell cultivation, when the cells are inoculated in the medium, it consumes nutrients and substrate available in the medium leading to the formation of products. This microbial growth process is affected by various parameters (like pH, temperature, rheological properties, etc.), phenomena (like heat generation, inhibition, reaction), and interactions (like adaptation, due to change in medium viscosity and heat generation) and should be considered in developing the kinetic model for the microbial growth. But it is practically difficult to develop a kinetic model involving all the parameters, phenomena, and interactions mentioned above. In this regard, a theoretical aspect of different assumptions and representations which are required for the microbial environment of the system was first presented by Arnold Fredrickson and Henry Tsuchiya and is shown in Fig. 5.8 (Bailey and Ollis 2017). Accordingly, based on the complexity involved, the degree of realism, and depending on the description available, models are classified in four ways and they are:

- Structured and segregated,
- Structured and nonsegregated,
- Unstructured and segregated, and
- Unstructured and nonsegregated

Among these models, structured and segregated models are considered to be the most realistic and computationally complex models as they include the structured nature of each cell and the segregation of the cell population into individual cells

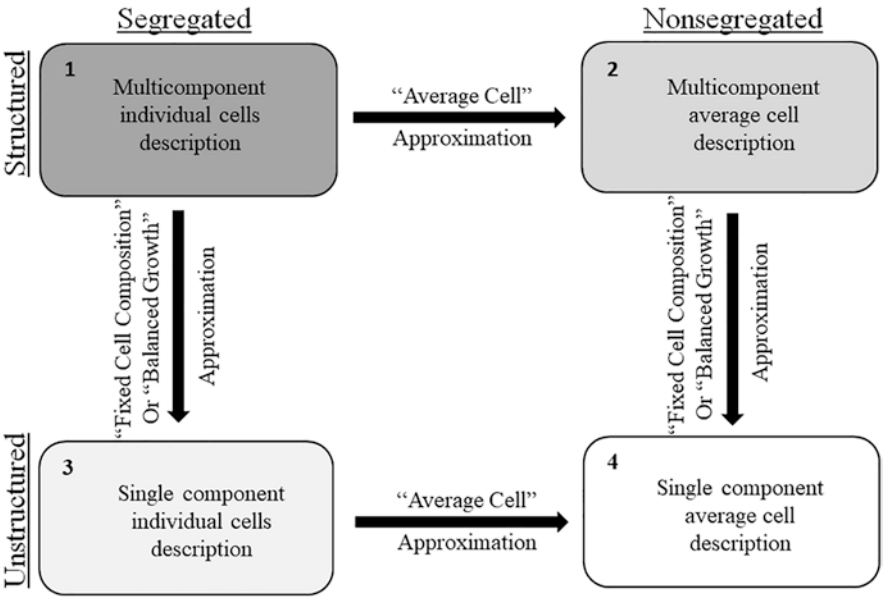


Fig. 5.8 Kinetic models for microbial cell growth

that may differ from each other in the growth kinetics of a bacterial culture. Similarly, unstructured and nonsegregated models are considered to be the most ideal and simplest models which assume fixed cell composition, which is equivalent to the balanced growth of cells. The use of unstructured and unsegregated models can be justified when the response of the cells is fast with respect to the changes in the external environment provided the extent of changes is small (i.e., 10–20% variation from the initial conditions). However, unstructured models cannot provide a satisfactory explanation for the response of culture to rapid changes in the external environment. Also, in most cases, segregation is not an important criterion of cultural response, so nonsegregated models will be satisfactory for many systems. Since, unstructured and unsegregated models are simple and applicable to practical situations, in the further sections, these models are considered for describing the microbial kinetics which is used for the design of ideal bioreactors or fermenters.

5.3.2 Reactors for Microbial Cultivation

In this section, the performance equations for the ideal fermenters are obtained from the unstructured and unsegregated model. For this, the specific growth rate of the microbial cells μ_g, specific growth of limiting substrate consumption rate q_s, and specific product formation rate q_p, in terms of growth limiting substrate concentration (S) are needed to be predicted which can be defined as:

$$\mu_g = \frac{1}{X}\frac{\mathrm{d}X}{\mathrm{d}t} \tag{5.1}$$

$$q_s = \frac{1}{X}\frac{\mathrm{d}S}{\mathrm{d}t} = \frac{1}{X}\frac{\mathrm{d}S}{\mathrm{d}X}\frac{\mathrm{d}X}{\mathrm{d}t} = -\frac{\mu_g}{Y_{X/S}} \tag{5.2}$$

$$q_p = \frac{1}{X}\frac{\mathrm{d}P}{\mathrm{d}t} = \frac{1}{X}\frac{\mathrm{d}P}{\mathrm{d}S}\frac{\mathrm{d}S}{\mathrm{d}X}\frac{\mathrm{d}X}{\mathrm{d}t} = \frac{Y_{P/S}}{Y_{X/S}}\mu_g = Y_{P/X}\mu_g \tag{5.3}$$

where X, P, and S are concentrations of biomass, product, and limiting substrate at time t, respectively. $Y_{X/S}$ and $Y_{P/S}$ are the growth yield and product yield in terms of limiting substrate consumed, respectively, and can be determined as:

$$Y_{X/S} = \frac{X - X_0}{-(S - S_0)};\quad Y_{P/S} = \frac{P - P_0}{-(S - S_0)} \tag{5.4}$$

where X_0, P_0, and S_0 are the initial concentrations of biomass, product, and limiting substrate, respectively.

From Eqs. (5.1)–(5.3), it can be seen that by predicting a specific growth rate, the other two specific rates can be determined if the respective yields are known. In the

case of cellular systems, the Monod equation is one of the most widely used empirical expressions to determine the specific growth rate when a single chemical component is growth limiting and can be described as:

$$\mu_g = \mu_{\max} \frac{S}{K_S + S} \tag{5.5}$$

where μ_g, $\mu_{\max}$, S, and K_s are gross specific growth rate, maximum specific growth rate when $S \gg K_S$, growth limiting substrate concentration, and saturation constant, respectively. The saturation constant K_S is equal to the growth limiting substrate concentration when $\mu_g = \mu_{\max}/2$. The saturation constant K_S is a significant parameter that tells about the affinity of the growth limiting substrate towards the microbial cells. The smaller the K_S value, the larger will be the specific growth rate μ_g, that is $\mu_g = \mu_{\max}$ for $S \gg K_s$ and $\mu_g = (\mu_{\max}/K_S)S$ for $S \ll K_S$. In the case of endogenous metabolism during the stationary phase, net specific growth rate μ_{net} can be described as

$$\mu_{\text{net}} = \mu_g - k_d = \mu_{\max} \frac{S}{K_S + S} - k_d \tag{5.6}$$

where k_d is a first-order rate constant for endogenous metabolism. In general, if the primary product is biomass, fermentation in the fermenters is carried out till the logarithmic growth phase in which $\mu_{\text{net}} = \mu_g$.

The Monod equation empirically fits a wide range of data significantly and is mostly applicable for unstructured and nonsegregated models of microbial growth. However, it fits well for slow microbial growth and low population density. It neither takes care of the death phase nor the inhibition effect in the case of substrate inhibition and product inhibition. Depending on the shape of the μ–S curve, other equations have also been suggested to improve the Monod equation which describes microbial growth in terms of growth limiting substrate. These equations can be the replacement of the Monod equation and are presented as follows (Shuler and Kargi 2015):

$$\mu_g = \mu_{\max}, \text{ if } S \geq 2K_s$$

Blackman equation:

$$\mu_g = \frac{\mu_{\max}}{2K_S} S, \text{ if } S < 2K_s$$

Tessier equation: $\mu_g = \mu_{\max}(1 - e^{KS})$.

Moser equation: $\mu_g = \mu_{\max} \dfrac{S^n}{K_S + S^n} = \mu_{\max} \left(1 + K_S S^{-n}\right)^{-1.}$.

Contois equation: $\mu_g = \mu_{\max} \dfrac{S}{K_{SX} X + S}$.

If more than one chemical species can limit microbial growth, the following equation can be used (Bailey and Ollis 2017)

$$\mu_g = \mu_{\max} \frac{S_1}{K_1 + S_1} \cdot \frac{S_2}{K_2 + S_2} \cdots$$

These alternative equations can give a better empirical fit for the growth of certain microbial cells; however, solving the algebraic solutions are more difficult than the Monod equation. These equations are used for the analysis, interpretation, and design of ideal fermenters. As chemical reactors, depending on the mode of cultivation, microbial cell growth, and fermenters or bioreactors can also be classified into three types that are batch, continuous (continuous stirred tank fermenter, plug flow fermenter and recycled fermenter) and fed-batch.

5.3.2.1 Batch Fermenter

Batch fermenter is used when the mode of cultivation of the microbial cells occurs in batch process. In a batch process, the growth of microbial cells undergoes four major phases which are discussed in the previous section and briefly described in Table 5.1 and graphically presented in Fig. 5.5.

In the batch growth process, all the components required for microbial fermentation or cultivation are added at the initial stage and are taken out at the end of fermentation without any alteration of further addition or removal of nutrients in between. During the fermentation process, no addition and removal of components happens. The reactor used in batch fermentation is known as a batch fermenter (Fig. 5.9). In a batch fermenter, it is assumed that the contents are well mixed and the composition is uniform at all times. However, the batch process is an unsteady state operation in which composition changes with time. In liquid medium upon inoculation with viable cells, the cells will grow exponentially after the lag phase. The change in cell concentration can be determined from the cell mass balance over the batch reactor as

$$\begin{aligned} &\text{Input} + \text{CellGeneration} = \text{Output} + \text{accumulation} + \text{Celldeath} \\ &0 + \mu_g \cdot X \cdot V = 0 + \frac{\mathrm{d}(XV)}{\mathrm{d}t} + k_d \cdot X \cdot V \end{aligned} \tag{5.7}$$

Table 5.1 Phases of microbial growth

Phase	Description	Specific growth rate
Lag	Cell adapts to the new environment, no or very little growth	$\mu_{net} \sim 0$
Log	Growth achieves its maximum rate	$\mu_{net} \sim \mu_{max}$
Stationary	Growth ceases due to starvation	$\mu_{net} = 0$
Death	Cell losses viability and lyse	$\mu_{net} < 0$

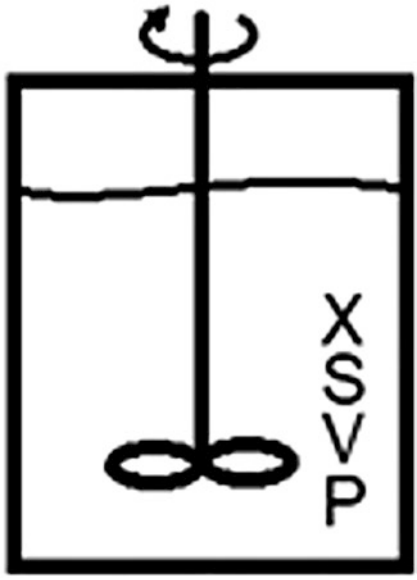


Fig. 5.9 Schematic representation of a batch fermenter

where μ_g and k_d are the rate of specific cell growth and specific cell death, respectively. For constant volume 'V' of fermentation mixture, Eq. (5.7) can be simplified as:

$$\frac{\mathrm{d}X}{\mathrm{d}t} = \left(\mu_g - k_d\right)X = \mu_{\text{net}}X = r_X \tag{5.8}$$

where r_X is the rate of cell mass production. If cell death is negligible in comparison with the cell growth (i.e., $\mu_{\text{net}} = \mu_g = \mu$), then

$$\frac{\mathrm{d}X}{\mathrm{d}t} = \mu X = r_X \tag{5.9}$$

In batch growth, generally μ remains approximately constant and approaches $\mu_{\max}$ as $S \gg K_S$. However, these kinetic constants (i.e., $\mu_{\max}$ and K_S) can be determined graphically after rearranging Monod equation in the various forms of linear equations like Lineweaver–Burk equation, Eadie–Hofstee equation and Hanes-Woolf equation, which are presented in Fig. 5.10. Considering constant m, after rearranging and integrating Eq. (5.9) and applying the limits, the performance equation of a batch fermentation can be derived as:

$$\int_{X}^{X_0} \frac{\mathrm{d}X}{r_X} = \int_{X}^{X_0} \frac{\mathrm{d}X}{\mu X} = \int_{t}^{0} \mathrm{d}t = t \tag{5.10}$$

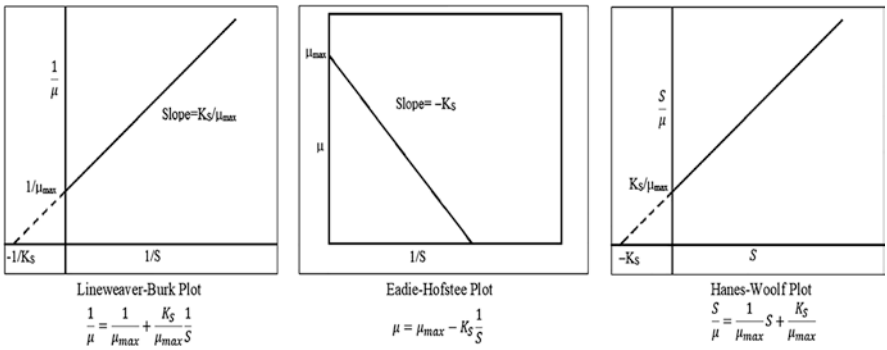


Fig. 5.10 Schematic plot of Monod equation in the various forms of linear equations

$$\ln \frac{X}{X_0} = \mu t \tag{5.11}$$

$$t = \frac{\ln \dfrac{X}{X_0}}{\mu} \tag{5.12}$$

The above equations apply only when r_X is greater than zero. Therefore, the fermentation is considered to be started at the time ($t = 0$) of the beginning of log phase instead of the beginning of the lag phase. From Eq. (5.10), the time for batch growth 't' can be determined graphically from the area under the $1/r_X$ versus X curve between limits X_0 and X. Also, from Eq. (5.10), doubling time and the generation time for the nth generation can be determined.

In general, if Monod kinetics satisfactorily represents the rate of growth during the accelerated growth phase, Eq. (5.10) can be written as

$$\int_X^{X_0} \frac{\mathrm{d}X}{r_X} = \int_X^{X_0} \frac{\left(K_S + S\right)\mathrm{d}X}{\mu_{\max} S \cdot X} = \int_t^0 \mathrm{d}t = t \tag{5.13}$$

Equation (5.13) can be solved if the growth yield (Eq. 5.4) in terms of limiting substrate consumption is known. Substituting Eq. (5.4) in Eq. (5.13) and integrating the final equation will give the following solution:

$$\mu_{\max} t = \left(\frac{K_S Y_{X/S}}{X_0 + S_0 Y_{X/S}} + 1 \right) \ln \frac{X}{X_0} - \frac{K_S Y_{X/S}}{X_0 + S_0 Y_{X/S}} \ln \frac{S}{S_0} \tag{5.14}$$

$$\mu_{\max} t = \left(\frac{K_S Y_{X/S}}{X_0 + S_0 Y_{X/S}} + 1 \right) \ln \frac{X}{X_0} - \frac{K_S Y_{X/S}}{X_0 + S_0 Y_{X/S}} \ln \frac{\left(Y_{X/S} S_0 + X_0 - X\right)}{Y_{X/S} S_0} \tag{5.15}$$

Equations (5.14) and (5.15) designate the sigmoidal-shaped batch growth curve, and accordingly the biomass concentration X maximally reaches to the value of $Y_{X/S}S_0 + X_0$. Here, the predetermined knowledge of the maximum concentration of

the biomass in a particular environment is required. Batch fermentation is simple process and widely used both in the laboratory and industrially to produce biomass, primary metabolites, and secondary metabolites. It is easier to install and maintain and requires lower capital investment. In batch growth process, the risk of contamination is less as the growth period is short in comparison to continuous culture. However, there are certain limitations of batch cultivation. Few of them are lower productivity, high downtime for cleaning and sterilization, difficult to hold the system in log phase for long period, safety issues at the time of filling, emptying and cleaning and variation in quality of the product.

5.3.2.2 Continuous Stirred Tank Fermenter (CSTF)

A continuous stirred tank fermenter (CSTF) is used when the mode of microbial cultivation is continuous in which fresh nutrient medium is continuously added to a well-stirred culture, and at the same time products and biomass are taken out continuously. In continuous culture, microbial growth can be maintained in exponential phase for a long period of time until the additional substrate is exhausted and provided that the medium has been formulated such that the growth is substrate limited, and not toxin limited. This phase of growth will be continued until the fermenter is full. After achieving this stage, further addition of the nutrient medium displaced an equal volume of the culture from the fermenter. In this way, biomass production can be made continuous in CSTF. And after certain period of time and at a suitable feed rate of the medium, the steady state will be achieved when the production of new cells will be balanced by the cells which are withdrawn from the fermenter. At steady state, the concentrations of nutrient, product, and biomass are constant in CSTF and hence, CSTF is also named as chemostat which means constant chemical environment. In an ideal chemostat, fresh nutrient medium is added to the fermenter and the culture is withdrawn at the same rate keeping the volume of the liquid in the fermenter constant. The schematic representation of a simplified chemostat is shown in Fig. 5.11.

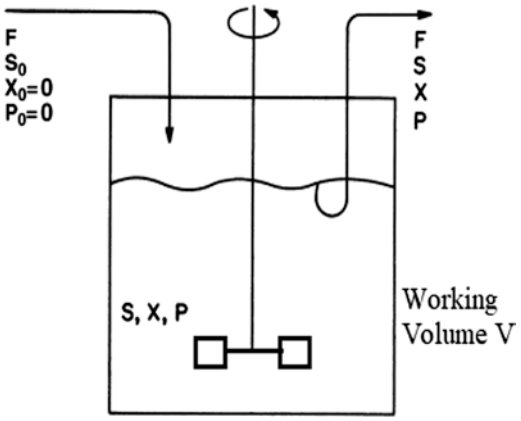


Fig. 5.11 Schematic representation of a chemostat

For this, a material balance on the biomass concentration can be given as

$$\begin{aligned} &\text{Input} + \text{CellGeneration} = \text{Output} + \text{accumulation} + \text{Celldeath} \\ &FX_0 + r_X V = FX + V\frac{\mathrm{d}X}{\mathrm{d}t} + k_d \cdot X \cdot V \end{aligned} \tag{5.16}$$

where F is the volumetric flow rate of nutrient solution, V is the culture volume which is assumed to be constant, X is the biomass (cell) concentration in the exit stream, X_0 is the biomass concentration in the inlet stream, r_X is the rate of biomass production in the CSTF, k_d is the specific endogenous (or death) rate, and dX/dt represents the change of biomass concentration in the CSTF (accumulation) with time.

If cell death is negligible and accumulation = 0 at steady state, then Eq. (5.16) can be simplified as

$$r_X V = F\left(X - X_0\right) \rightarrow \frac{F}{V} = \frac{r_X}{X - X_0} = D = \frac{1}{\tau_{\text{CSTF}}} \tag{5.17}$$

where D is the dilution rate and t_{CSTF} is the residence time. Equation (5.17) represents the performance equation for an ideal chemostat or CSTF. From the given data of r_X and X, the required residence time can be determined as the area of the rectangle of width $X - X_0$ and height $1/r_X$ on the curve of $1/r_X$ versus X. If the initial cell concentration $X_0 = 0$ and $r_X = m_g X$, then Eq. (5.17) becomes

$$D = \frac{1}{\tau_{\text{CSTF}}} = \mu_g \tag{5.18}$$

Equation (5.18) states that in an ideal chemostat, the rate of cell removal will be equal to their growth rate, and the growth rate of cells is equal to the dilution rate. Similarly, a material balance on the growth limiting substrate in the absence of endogenous metabolism can be given as

$$FS_0 - FS - V\mu_g X\frac{1}{Y_{X/S}} - Vq_P X\frac{1}{Y_{P/S}} = V\frac{\mathrm{d}S}{\mathrm{d}t} \tag{5.19}$$

where S_0 is the concentration of growth limiting substrate in feed stream, S is the concentration of growth limiting substrate in effluent stream, q_P is the specific rate of extracellular product formation, and $Y_{X/S}$ and $Y_{P/S}$ are the yield coefficients of biomass and extracellular product, respectively.

If the production of extracellular product is negligible, then at steady state Eq. (5.19) becomes

$$D\left(S_0 - S\right) = \frac{\mu_g X}{Y_{X/S}} \tag{5.20}$$

Since $m_g = D$ at steady state if $k_d = 0$, the above equation can be written as

$$X = Y_{X/S}\left(S_0 - S\right) \tag{5.21}$$

In the same way, taking the same assumption of neglecting endogenous metabolism, the balance on product formation at steady state can be given as

$$FP = q_P XV \rightarrow DP = q_P X \tag{5.22}$$

where q_P is described by Eq. (5.3) in case of growth-associated product which is a function of m_g. For non-growth-associated product formation, q_P is a constant (b).

Now, on substituting the Monod equation for m_g in Eq. 5.18, the growth limiting substrate concentration S at steady state can be determined as:

$$D = \frac{1}{\tau_{\text{CSTF}}} = \mu_g = \frac{\mu_{\max} S}{K_S + S} \tag{5.23}$$

$$S = \frac{DK_S}{\mu_{\max} - D} = \frac{\mu_g K_S}{\mu_{\max} - \mu_g} \tag{5.24}$$

After rearranging Eq. (5.23), the Monod parameters, $m_{\max}$ and K_S, can be evaluated from the graph plotted between $1/D$ versus $1/S$. Equation (5.24) is considered as design criteria of CSTF. Substituting S from Eq. (5.24) in Eq. (5.21), the cell concentration at steady state, if $X_0 = 0$, will be

$$X = Y_{X/S}\left(S_0 - \frac{DK_S}{\mu_{\max} - D}\right) \tag{5.25}$$

Considering the case where $D \ll m_{\max}$, Eq. (5.24) becomes

$$S = \frac{DK_S}{\mu_{\max}} \tag{5.26}$$

In other words, at steady state, the value of growth limiting substrate concentration S in the fermentation medium within the fermenter increases proportionally with the increase of dilution rate D as long as $D \ll m_{\max}$. And from Eq. (5.25) at low values of D, the cell concentration X from the fermenter approximates $Y_{X/S}S_0$.

Considering another case where $D = m_{\max}$, Eq. (5.24) shows that the value of growth limiting substrate concentration S becomes infinite which is not possible as practically the maximum value of S can be the initial concentration of growth limiting substrate S_0. In this case, from Eq. (5.25), the cell concentration X from the fermenter approaches zero. This condition is termed as 'wash out' and under this condition the rate of biomass production can no longer be retained with the dilution rate.

From the above discussion, it is obvious that to get the maximum productivity the dilution rate D needs to be optimized. The dilution rate at which productivity is

maximum can be determined by differentiating DP or DX with respect to D and setting the derivative equal to zero. Considering $k_d = 0$ and $q_P = 0$ and applying Eq. (5.24), the optimal value of D (D_{opt}) for biomass production will be

$$D_{opt} = \mu_{max}\left(1 - \sqrt{\frac{K_S}{K_S + S_0}}\right) \tag{5.27}$$

Since $K_s \ll S_0$, therefore, from Eq. (5.27) D_{opt} may approach the value of m_{max} or the process may reach wash out condition very easily. Hence, fermentation in chemostat at D_{opt} is possible only when the flow rate and liquid volume are exactly constant.

Similarly, in the presence of endogenous metabolism and extracellular product formation, we can also show that

$$S = \frac{K_S\left(D + k_d\right)}{\mu_{max} - D - k_d} \tag{5.28}$$

$$X = Y_{X/S}\left(S_0 - S\right)\frac{D}{D + k_d + q_P\dfrac{Y_{X/S}}{Y_{P/S}}} \tag{5.29}$$

In case the formation of extracellular product is neglected, Eq. (5.28) becomes

$$X = Y_{X/S}\left(S_0 - S\right)\frac{D}{D + k_d} \tag{5.30}$$

5.3.2.3 Chemostat with Biomass Recycle

Chemostat with biomass recycle is used to maintain cell concentration higher than the steady state concentration, in a chemostat. In this way, conversion rate could be increased. It increases the operating flexibility by increasing the critical dilution rate for wash out. It also increases the system stability by minimizing the effects caused by process fluctuation. In this system, the recycling of the cells from the effluent stream is done either by centrifugation, filtration or by settling in a conical tank. Using this system, the major problem of washout in chemostat can be solved. The schematic representation of a chemostat system with cell recycle is shown in Fig. 5.12. Assuming endogenous metabolism as negligible ($k_d = 0$), the material balance on biomass concentration around the fermenter will be

$$FX_0 + \alpha FCX_1 + \mu_g X_1 V = \left(1 + \alpha\right)FX_1 + V\frac{\mathrm{d}X_1}{\mathrm{d}t} \tag{5.31}$$

where 'α' is the recycle ratio based on volumetric flow rates, C is the ratio of cell concentration in the cell recycle stream to the cell concentration in the reactor

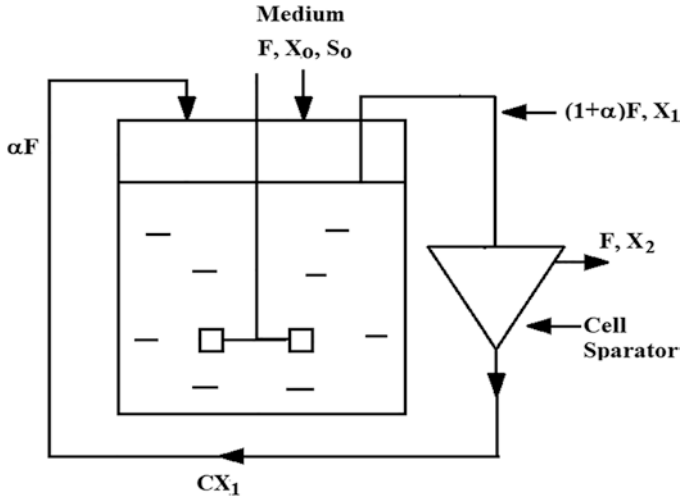


Fig. 5.12 Schematic representation of chemostat with biomass recycle

effluent, F is the nutrient flow rate, V is the culture volume, X_0 and X_1 are the cell concentrations in feed and recycle streams, and X_2 is the cell concentration in effluent from the cell separator. At steady state, if the feed is sterile ($X_0 = 0$), then Eq. (5.31) becomes

$$\mu_g = \left[1+\alpha\left(1-C\right)\right]D \tag{5.32}$$

Equation (5.32) shows that when cell recycle is used, chemostat operation is possible at higher dilution rates than the specific growth rate, i.e., at $D > m_{net}$ as $C > 1$ and $\alpha(1 - C) < 0$. Furthermore, material balance on growth limiting substrate around fermenter gives

$$FS_0 + \alpha FS - \frac{\mu_g X_1}{Y_{X/S}}V = \left(1+\alpha\right)FS + V\frac{\mathrm{d}S}{\mathrm{d}t} \tag{5.33}$$

At the steady state, Eq. (5.33) becomes

$$X_1 = \frac{D}{\mu_g}Y_{X/S}\left(S_0 - S\right) \tag{5.34}$$

Substituting the D from Eq. (5.32) into Eq. 5.34, X_1 will be

$$X_1 = \frac{Y_{X/S}\left(S_0 - S\right)}{\left(1+\alpha-\alpha C\right)} \tag{5.35}$$

Equation (5.35) shows that using chemostat with cell recycle, under steady state conditions, the cell concentration will be improved $1/(1 + \alpha - \alpha C)$ times the cell

concentration of a chemostat without cell recycle. The steady state growth limiting substrate concentration in the effluent is obtained from Eq. (5.32) and Monod equation as

$$\frac{\mu_{\max} S}{K_S + S} = \left[1 + \alpha\left(1 - C\right)\right] D \tag{5.36}$$

Rearranging Eq. (5.36)

$$S = \frac{K_S D\left[1 + \alpha\left(1 - C\right)\right]}{\mu_{\max} - D\left[1 + \alpha\left(1 - C\right)\right]} \tag{5.37}$$

On substituting S, Eq. (5.35) becomes

$$X_1 = \frac{Y_{X/S}}{\left(1 + \alpha - \alpha C\right)}\left[S_0 - \frac{K_S D\left[1 + \alpha\left(1 - C\right)\right]}{\mu_{\max} - D\left[1 + \alpha\left(1 - C\right)\right]}\right] \tag{5.38}$$

From the above derivations, it is clear that in the chemostat with cell recycle at steady state, the cell concentration, its productivity and the rates of substrate consumption are higher in comparison with the chemostat without cell recycle. These systems are used mostly in wastewater treatment methods.

5.3.2.4 Chemostat in Series

Chemostat in series may be advantageous when there is the utilization of more than one growth limiting substrates and for the separation of biomass production from the metabolic product formation. The schematic representation of 'n' number of chemostats in series is shown in Fig. 5.13. Assuming endogenous metabolism as negligible ($k_d = 0$), the material balance on biomass concentration around the nth fermenter will be

$$FX_{n-1} + \mu_{gn} X_n V_n = FX_n + V_n \frac{\mathrm{d}X_n}{\mathrm{d}t} \tag{5.39}$$

$$FX_{n-1} - FX_n + \mu_{gn} X_n V_n = V_n \frac{\mathrm{d}X_n}{\mathrm{d}t} \tag{5.40}$$

At steady state

$$D_n\left(X_n - X_{n-1}\right) = \mu_{gn} X_n \quad \text{where} D_n = \frac{F}{V_n} \tag{5.41}$$

where X_{n-1} is the steady state cell mass concentration of the effluent from $(n - 1)$th fermenter, X_n is the steady state cell mass concentration of the effluent from the nth fermenter, F is the flow rate of medium, D_n is the dilution rate of nth fermenter, and

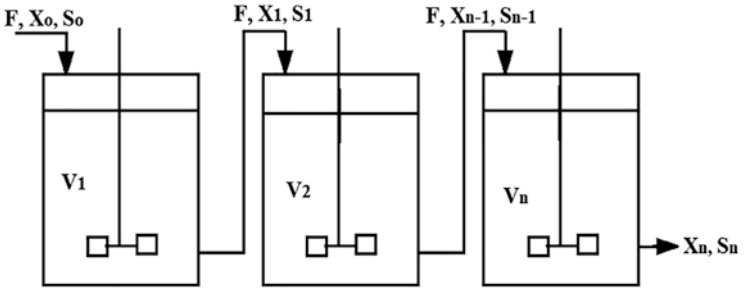


Fig. 5.13 Schematic representation of chemostat in series

m_{gn} is the specific growth rate of microbial cells in nth fermenter. If $n > 1$, then from Eq. (5.41), steady state biomass can be determined as

$$X_n = \frac{D_n X_{n-1}}{D_n - \mu_{gn}} \tag{5.42}$$

Taking the same assumption of endogenous metabolism as negligible ($k_d = 0$), the material balance on growth limiting substrate which is consumed for biomass production around the nth the fermenter will be

$$FS_{n-1} - FS_n - V_n \mu_{gn} X_n \frac{1}{Y_{X/S}} = V_n \frac{\mathrm{d}S_n}{\mathrm{d}t} \tag{5.43}$$

When growth limiting substrate is consumed for the production of extracellular metabolite, the material balance on growth limiting substrate around the nth fermenter will be

$$FS_{n-1} - FS_n - V_n q_{pn} X_n \frac{1}{Y_{P/S}} = V_n \frac{\mathrm{d}S_n}{\mathrm{d}t} \tag{5.44}$$

where q_{pn} is the specific rate of product formation in nth fermenter. Substituting the q_{pn} from Eq. (5.3), Eq. (5.44) will be

$$FS_{n-1} - FS_n - V_n \mu_{gn} X_n \frac{Y_{P/X}}{Y_{P/S}} = V_n \frac{\mathrm{d}S_n}{\mathrm{d}t} \tag{5.45}$$

At steady state, Eqs. (5.43) and (5.45) become, respectively, as

$$S_n = S_{n-1} - \frac{1}{D_n} \mu_{gn} X_n \frac{1}{Y_{X/S}} \tag{5.46}$$

and

$$S_n = S_{n-1} - \frac{1}{D_n} \mu_{gn} X_n \frac{Y_{P/X}}{Y_{P/S}} \tag{5.47}$$

Similarly, the material balance on product formation at steady state will be

$$FP_n - FP_{n-1} = q_{Pn} X_n V_n = Y_{P/X} \mu_{gn} X_n V_n \tag{5.48}$$

$$P_n = P_{n-1} + \frac{1}{D_n} Y_{P/X} \mu_{gn} X_n V_n \tag{5.49}$$

Equations (5.42), (5.46), (5.47), and (5.49) are the general equations to determine the steady state concentrations from any fermenter of the multistage chemostat system.

5.3.2.5 Plug Flow Fermenter

In plug flow fermenter (PFF), the liquid medium moves along a pipe or channel in such a manner that there is no mixing in longitudinal direction, however there is complete lateral mixing. Normally, PFF are not suitable for the growth of suspended cells and rarely used in industrial fermentations. However, they are used frequently for immobilized cells and enzyme reactions. The schematic representation of a PFF is shown in Fig. 5.14. Assuming negligible endogenous metabolism, the material balance on biomass at steady state in the small section (Δz) of volume dV of PFF as shown in Fig. 5.14 can be given as

$$FX_{0z} + \mu_g X \cdot \mathrm{d}V = FX_{z+\Delta z} \tag{5.50}$$

$$FX_{0z} + \mu_g X \cdot A\Delta z = FX_{z+\Delta z} \tag{5.51}$$

$$FX_{z+\Delta z} - FX_{0z} = \mu_g X \cdot A\Delta z \tag{5.52}$$

where 'A' is the cross-sectional area of the reaction and 'F' is the volumetric flow rate of the liquid. Rearranging Eq. (5.52), we get

$$\frac{F}{A} \frac{\left(X_{z+\Delta z} - X_{0z}\right)}{\Delta z} = \mu_g X \tag{5.53}$$

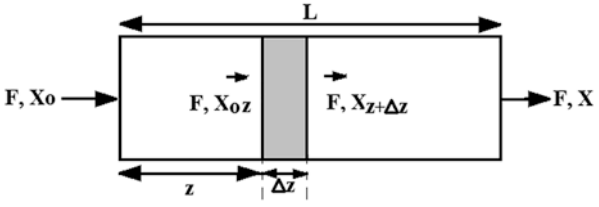


Fig. 5.14 Schematic representation of Plug flow fermenter

Applying limit $z \to 0$ to Eq. (5.53) and putting $u = F/A$ as superficial velocity, we get

$$u \lim_{z \to 0} \frac{\left(X_{z+\Delta z} - X_{0z}\right)}{\Delta z} = \mu_g X$$

or

$$u\left(\frac{\mathrm{d}X}{\mathrm{d}z}\right) = \mu_g X \tag{5.54}$$

Rearranging and integrating Eq. (5.54) we get

$$u \int_{X}^{X_0} \frac{\mathrm{d}X}{X} = \mu_g \int_{L}^{0} \mathrm{d}z \tag{5.55}$$

$$\ln \frac{X}{X_0} = \mu_g \frac{L}{u} = \mu_g \frac{\frac{V}{A}}{\frac{F}{A}} = \mu_g \tau \tag{5.56}$$

where t is the residence time of PFF. Rearranging Eq. (5.56) for t, we get

$$\tau = \frac{1}{\mu_g} \ln \frac{X}{X_0} \tag{5.57}$$

Equation (5.57) is the performance equation of PFF. Equation (5.57) suggests that the performance of PFF will be same as the performance of batch fermenter.

5.3.2.6 Fed-Batch Fermenter

When a batch fermenter is fed with nutrient medium either intermittently or continuously, while culture is withdrawn discontinuously, then the fermenter is considered as fed-batch fermenter and this mode of cultivation is known as fed-batch cultivation or variable-volume continuous culture. This mode of cultivation is mostly used to relieve catabolite repression in case of product formation, avoid inhibition which is caused due to the presence of substrate in culture in excess amount or to overcome toxic effects of some medium components by maintaining low substrate concentration throughout the process. However, productivity is low in fed-batch cultivation owing to high downtime during filling, heating, sterilization, cooling, emptying and cleaning the reactor. In fed-batch reactor, the fermenter volume is increased due to continuous addition of a solution of limiting substrate of the same concentration as that in the initial medium. Consider a batch culture where the biomass concentration at a certain time is given by

$$X = X_0 + Y_{X/S}\left(S_0 - S\right) \tag{5.58}$$

where S_0 is the initial substrate concentration, $Y_{X/S}$ is the yield coefficient, and X_0 is the initial cell mass concentration. At maximum cell mass concentration, substrate concentration S and initial cell mass concentration would be very small with respect to S_0 and X, respectively. At that condition when $X = Y_{X/S}S_0$, if a nutrient feed with the substrate concentration of S_0 is added at a constant flowrate of F, the change in volume of the system can be written as:

$$\frac{\mathrm{d}V}{\mathrm{d}t} = F \tag{5.59}$$

If the initial volume of the fermenter at time $t = 0$ is V_0 and V is the volume of the fermenter at time t, then on rearranging and integrating Eq. (5.59):

$$\int_{V}^{V_0} \mathrm{d}V = \int_{t}^{0} F\mathrm{d}t \Rightarrow V = V_0 + Ft \tag{5.60}$$

At quasi steady state, i.e., $S_{\text{added}} = S_{\text{consumed}}$ and at constant cell mass concentration X, the material balance on cell mass assuming negligible cell death can be given as:

$$FX_0 + \mu XV = \frac{\mathrm{d}(XV)}{\mathrm{d}t} \tag{5.61}$$

$$FX_0 + \mu XV = X\frac{\mathrm{d}V}{\mathrm{d}t} + V\frac{\mathrm{d}X}{\mathrm{d}t} \tag{5.62}$$

At steady state, putting $X_0 = 0$ and $\frac{\mathrm{d}X}{\mathrm{d}t} = 0$, Eq. (5.62) becomes

$$\mu XV = X\frac{\mathrm{d}V}{\mathrm{d}t} \rightarrow \mu = \frac{1}{V}\frac{\mathrm{d}V}{\mathrm{d}t} = \frac{F}{V} = D \tag{5.63}$$

Applying Monod kinetics in Eq. (5.63), we get

$$\mu = D = \frac{\mu_{\max} S}{K_S + S} = \frac{F}{V_0 + Ft} \tag{5.64}$$

Rearranging Eq. (5.64), it becomes

$$S = \frac{K_S D}{\mu_{\max} - D}$$

Now, the cell mass concentration at time 't' in the reactor can be given as $X = \frac{X_{\text{total}}}{V}$; where X_{total} is the total amount of cell mass at time t when the volume is V. Hence, the rate of change in cell mass concentration in terms of X_{total} can be given as

$$\frac{\mathrm{d}X}{\mathrm{d}t} = \frac{V\mathrm{d}X_{\text{total}}/\mathrm{d}t - X_{\text{total}}\mathrm{d}V/\mathrm{d}t}{V^2} = 0 \rightarrow \frac{\mathrm{d}X_{\text{total}}}{\mathrm{d}t} = \frac{X_{\text{total}}}{V}\frac{\mathrm{d}V}{\mathrm{d}t} = XF \tag{5.65}$$

Equation (5.65) can be written as

$$\frac{dX_{total}}{dt} = FY_{X/S}S_0 \tag{5.66}$$

Integrating Eq. (5.66), we get

$$X_{total} = X_0 + FY_{X/S}S_0t \tag{5.67}$$

where X_0 is the amount of cell mass at time $t = 0$. Similarly, the amount of product formation can also be determined using the definitions of $Y_{P/S}$ or q_P and can be given as

$$P_{total} = P_0 + q_P X\left(V_0 + \frac{Ft}{2}\right)t \tag{5.68}$$

where P_{total} and P_0 is the amount of product formed at time $t = t$ and $t = 0$, respectively.

5.3.3 Production Kinetics in Microbial Cultivation

The kinetics of the formation of microbial products can be explained by the following three models (Shuler and Kargi 2015):

5.3.3.1 Growth-Associated Model

This model describes that the rate of microbial product formation is directly proportional to the microbial growth rate. Hence

$$\frac{dP}{dt} = \alpha\frac{dX}{dt} \rightarrow \frac{1}{X}\frac{dP}{dt} = \alpha\frac{1}{X}\frac{dX}{dt} \rightarrow q_P = \alpha\mu \tag{5.69}$$

where α is a proportionality constant. Enzyme production can be considered as an example of a growth-associated product.

5.3.3.2 Non-growth-Associated Model

This model describes that the rate of microbial product formation is directly proportional to the cell mass concentration. According to this model, product formation occurs during the stationary phase when the growth rate is zero. Hence

$$\frac{dP}{dt} = \beta X \rightarrow \frac{1}{X}\frac{dP}{dt} = \beta \rightarrow q_P = \beta \tag{5.70}$$

where α is a proportionality constant. Non-growth-associated products include the production of secondary metabolites, such as antibiotics (for example, penicillin).

5.3.3.3 Mixed Growth-Associated Model

In this regard, Luedeking and Piret proposed that Eqs. (5.69) and (5.70) can be combined to represent the productivity of lactic acid fermentations where product formation occurs during the slow growth and stationary phases. In this case, the rate of product formation is given as

$$\frac{dP}{dt} = \alpha \frac{dX}{dt} + \beta X \rightarrow \frac{1}{X}\frac{dP}{dt} = \alpha \frac{1}{X}\frac{dX}{dt} + \beta \rightarrow q_P = \alpha\mu + \beta \tag{5.71}$$

Equation (5.71) is called Luedeking–Piret equation. Mixed growth-associated products include production of lactic acid, xanthan gum, etc.

Numerous research works are underway to understand the growth kinetics of various microorganisms for synthesis of beneficial biological products in different reactor systems. The experimental data are modelled to achieve optimum conditions for bioreactor operation. The research work by Venkatesh et al. (1993) to study the growth kinetics of *Lactobacillus bulgaricus* for the production of lactic acid from lactose, showed that the productivity is three times higher for CSTR system compared to batch system and three stages were sufficient for complete utilization of the substrate. Similarly, Ariyajaroenwong et al. (2016) performed kinetics study in batch and CSTR system for ethanol fermentation by *Saccharomyces cerevisiae* NP 01 using sweet sorghum juice as substrate. The outcome was that while the Monod and Lineweaver-Burk models were a perfect fit for batch system but not for CSTR system. Ramli et al. (2017) studied the batch fermentation kinetics of *Saccharomyces cerevisiae* for the production of iso-butanol during degradation of valine via Ehrlich pathway. The research work involved estimation of glucose consumption and iso-butanol formation. Monod model was followed for the determination of coefficients of growth kinetics, while Luedeking–Piret model was used for parameter estimation during glucose consumption. The maximum specific growth rate (μ_{max}), substrate saturation constant (K_s), iso-butanol yield and biomass yield were found to be 0.74 h^{-1}, 57 g/L, 4.79 mg/g, and 0.1 g/g, respectively. Another study by Asoka et al. (2021) discusses about the submerged batch reactor for the kinetics of citric acid production by *Aspergillus versicolor*. Growth kinetics, citric acid formation, and glucose consumption were found suitable to represent the experimental data. The conclusion was that the formation of citric acid is growth associated. Unstructured kinetic model describing Monod kinetics was examined for the growth of *Debaryomyces hansenii* in the formation of xylitol from xylose. It was learnt that the maximum growth of yeast cells is the major factor affecting the product formation (Kasbawati et al. 2022). Beck et al. (2022) developed kinetic models for substrate consumption resulting in the production of mannosylerythritol lipids using *Moesziomyces aphidis*. Their research work showed threefold increase in

productivity when fed-batch cultivation compared to batch cultivation. Thus, it can be noted that mathematical modelling and analysis is the best strategy to find optimum solutions involving microbial fermentations with a goal towards maximum productivity of desired bioproduct.

5.3.4 Inhibition Effects on Growth Kinetics

In the presence of growth inhibitors and under high substrate/product concentrations, the microbial growth becomes inhibited and the rate of growth is a function of inhibitor concentration. The mechanism of inhibition is complex and the kinetic constants have any biological significance only if the rate limiting step is single-substrate and enzyme catalysed. The empirical constants are determined by curve fitting. The following are the types of inhibition (Shuler and Kargi 2015).

5.3.4.1 Substrate Inhibition

Microbial growth rate is inhibited under high concentrations of substrate. The inhibition can be competitive or noncompetitive. Substrate inhibition may be prevented by slow and intermittent addition of substrate to nutrient medium.

$$\text{Noncompetitive substrate inhibition}: \mu_g = \frac{\mu_{\max}}{\left(1+\frac{Ks}{S}\right)\left(1+\frac{S}{K_1}\right)} \tag{5.72}$$

$$\text{If } K_1 \gg K_s; \quad \mu_g = \frac{\mu_{\max} S}{K_s + S + \frac{S^2}{K_1}} \tag{5.73}$$

$$\text{Competitive substrate inhibition}: \mu_g = \frac{\mu_{\max} S}{K_s\left(1+\frac{S}{K_1}\right)+S} \tag{5.74}$$

5.3.4.2 Product Inhibition

$$\text{Noncompetitive product inhibition}: \mu_g = \frac{\mu_{\max}}{\left(1+\frac{Ks}{S}\right)\left(1+\frac{P}{K_p}\right)} \tag{5.75}$$

$$\text{Competitive product inhibition} : \mu_g = \frac{\mu_{\max} S}{K_s\left(1+\frac{P}{K_p}\right)+S} \tag{5.76}$$

where K_p is the product inhibition constant.

5.3.4.3 Inhibition by Toxic Compounds

$$\text{Noncompetitive inhibition} : \mu_g = \frac{\mu_{\max}}{\left(1+\frac{Ks}{S}\right)\left(1+\frac{I}{K_I}\right)} \tag{5.77}$$

$$\text{Competitive inhibition} : \mu_g = \frac{\mu_{\max} S}{K_s\left(1+\frac{I}{K_I}\right)+S} \tag{5.78}$$

$$\text{Uncompetitive inhibition} : \mu_g = \frac{\mu_{\max} S}{\left(\frac{Ks}{\left(1+\frac{I}{K_I}\right)}+S\right)\left(1+\frac{I}{K_I}\right)} \tag{5.79}$$

If the toxic compound leads to microbial death, then the expression is

$$\mu_g = \frac{\mu_{\max} S}{K_s + S} - k_d \tag{5.80}$$

'K_d' is the death rate constant.

Uba et al. (2022) demonstrated bioremediation process for the substrate inhibition kinetics using phenol as substrate. It was observed that the growth of *Pseudomonas putida* NAUN-16 was affected at higher phenolic concentrations. Furthermore, the effect of phenol was studied using various substrate inhibition models such as Monod, Teissier, Aiba, etc. and the growth rate was better represented by Teissier model. Das et al. (2020) predicted the inhibitory effects of hydrogen addition to a batch reactor producing hydrogen via acidogenesis of glucose by *Clostridium acetobutylicum*. The inhibitory effect of added hydrogen on microbial growth and hydrogen production was analysed by the modified form of Monod equation. Moderate inhibition was observed till 30% (v/v) of hydrogen and strong inhibition was observed beyond 30% (v/v) of hydrogen. Similar research work was reported by Kalafatakis et al. (2019) for the growth inhibition kinetics of *Clostridium*

pasteurianum under high product (butanol) concentrations using glycerol as a substrate. The fermentation experiments were done by injecting pulses of glycerol under steady state conditions. The yield of butanol started decreasing at higher concentrations of butanol due to inhibitory effect.

5.4 Whole Cell Immobilization

In the formation of secondary metabolites through cell culture, the formation of intracellular products, low product yield, slow growth, and washout of cells are the major contributing factors towards high production cost. These issues can either be eliminated or reduced by immobilized cell culture (Shuler and Kargi 2015). Immobilization refers to confinement of intact cells within a defined space so that cell mobility is restricted with retention of its biological function. By doing this, cells are prevented from entry into the reaction volume containing the substrate and products. The purpose of cell immobilization is to increase the product concentration and decrease the substrate concentration in the product stream, thereby increasing the productivity. Immobilization is an efficient technique to improve cell viability and ensures enhancement of biochemical transformations.

Advantages of Immobilized Cell Culture (Willaert 2009; Shuler and Kargi 2015)

- It enables biomass usage for a prolonged period by providing high cell concentration.
- The entrapped cells, especially shear sensitive cells (mammalian cells) are protected against shear forces and their activity is conserved.
- Immobilized cells can be easily recovered and enables reuse of cells in various types of reactors.
- Washout of cells can be prevented at high dilution rates during continuous fermentation.
- Product recovery and purification via downstream processing is easy due to less contamination by biomass.
- The non-dividing immobilized cells are stable against genetic changes.
- It promotes secondary metabolite production, in some cases.
- It favours microenvironmental conditions (pH gradient, nutrient-product gradient) for cells and increases the product yield.

Disadvantages of Immobilized Cell Culture (Willaert 2009; Shuler and Kargi 2015)

- The efficiency of the reaction is reduced due to mass transfer limitations and accumulation of toxic products.
- The mechanical stability and biocompatibility of the immobilization matrices is a concern.

5.4.1 Active Cell Immobilization

This type of immobilization promotes segregation of cells from aqueous phase thus making their recovery easier after the reaction is completed. Immobilization is induced either by chemical or physical method and is classified in to six types as shown in Fig. 5.15.

Adsorption: This method is simple and inexpensive. It involves the cellular adhesion on the surface of an inert matrix (porous or non-porous) by weak physical forces, such as van der Waals, electrostatic, hydrophobic, and dispersion forces. The support materials used for cell adsorption may be inorganic (alumina, porous silica, glass, ceramics, clay, etc.) or organic (cellulose, starch, activated carbon, etc.) (Shuler and Kargi 2015). Affinity of microbial cells towards the support matrix depends on the chemical nature of the support and cell type, e.g., adsorption of fungus on the wood chips. The choice of support matrix depends on the adsorption capacity and adhesive force. Although biomass loading is limited, the advantages include elimination of cell washout, protection from toxic or harsh environment, easy separation, and cost effective. Porous materials may be used to obtain high biomass loadings, but it may cause pore diffusion limitations.

Cross linking: This is one of the chemical methods of cell immobilization in which glutaraldehyde is used to form insoluble aggregate within which the microbial cells are entrapped. This method does not require any support matrix. However, cross linking will affect the cell metabolism and offer diffusional resistance.

Covalent bonding: The method requires the binding surfaces to be treated with reactive groups such as carbodiimide, cyanogen bromide, S-triazine derivative, glutaraldehyde, etc., for covalent bond formation between cells and surface (Shuler and Kargi 2015). The functional groups could be toxic for the cells and hence covalent bonding is not an ideal method for cell immobilization. However, one may use inorganic carriers such as titanium oxide, cellulose plus cyanuric chloride, carriers with aldehyde, amine, etc., for covalent binding.

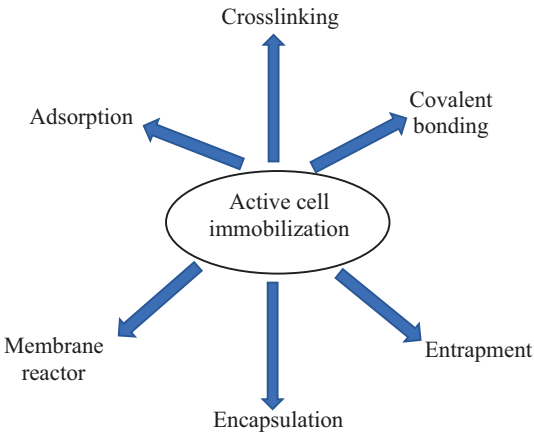


Fig. 5.15 Cell immobilization methods

Entrapment: Physical entrapment using matrices is a common method of immobilization. The polymers used include agar, alginate, carrageenan, chitosan, polyacrylamide, collagen, etc. (Shuler and Kargi 2015). The matrices are formed from polymeric solutions by various methods such as polymeric gelation, polymer precipitation, ion exchange gelation, polymerization, and polycondensation. Cell entrapment in calcium alginate beads is most popular method. Alginate polymer network formation involves gentle gelation technique with tuneable pore sizes, which is advantageous for microbial cells with biological activity.

Encapsulation: Microencapsulation is another type of cell immobilization (Shuler and Kargi 2015; Bailey and Ollis 2017). Microcapsules are composed of core region surrounded by semipermeable membrane. The cells are entrapped within the core region so that transport of nutrients and metabolic products happen though the membrane. Polymers such as cellulose acetate, polystyrene, nylon, etc., are used in the formation of microcapsule. The molecular weight cut off is chosen based on the intended application so that high molecular weights can be retained within the capsule while allowing the low molecular weight nutrients and metabolic products to pass through it.

Membrane reactor: Another form of cell immobilization is membrane-based reactors (Bailey and Ollis 2017). In **flat plate membrane reactor**, semi permeable membrane is present between the cells and circulating medium such that cells are packed in high concentrations. This type of reactor can be designed either for one side flow of nutrients or flow of nutrients in both sides of the immobilized cells. **Hollow fibre reactor** is the simplest form of membrane reactor. The construction is similar to shell and heat exchanger, with tubes fabricated using semipermeable membrane. Cells will be attached to the shell side and nutrient solution flows inside the tubes. Uptake of nutrients by cells take place via semipermeable membrane and metabolic products get excreted in the nutrient stream. The native form of membrane-based reactors suffers from diffusional limitations and hence modifications are required either for gas exchange or product removal. The design of membrane is expensive due to high cost of membrane material. The advantage is they provide simultaneous conversion and product separation with enhanced yields and selectivity.

5.4.2 Passive Cell Immobilization

In this type, cells are naturally attached within solid matrix leading to biofilm formation. Biofilm formation is common phenomena in biological method involving mould fermentation and wastewater treatment. The schematic representation of biofilm is shown in Fig. 5.16. Biofilms are classified as mixed culture microbial films and stagnant biofilms. The former type of biofilm is caused by polymer producing organisms and there is variation of microenvironmental conditions within a biofilm, whereas, in the later type of biofilm, the cellular metabolism is affected by the diffusion of nutrients and products in and out of the biofilm, respectively. Biofilm

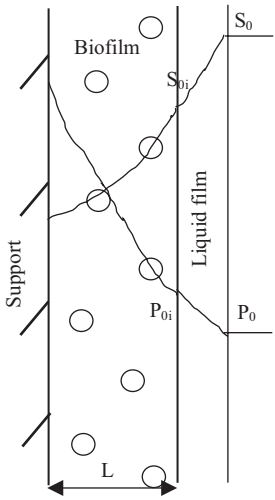


Fig. 5.16 Schematic representation of biofilm formation (Redrawn from Shuler and Kargi 2015)

thickness plays a major role in analysing the performance of biotic phase and hence determination of optimal biofilm thickness is crucial as thin biofilms have low bioconversion rates and thick biofilms may have diffusional limitations (Shuler and Kargi 2015).

5.4.3 Mass Transfer Limitations in Immobilized Cell Systems

Immobilized systems are characterized by extra diffusional limitations compared to suspended cultures. The diffusional limitations are described by Damkohler number based on relative rates of bioconversion and diffusion (Shuler and Kargi 2015).

$$Da = \frac{\text{maximum rate of bioconversion}}{\text{maximum rate of diffusion}}$$
$$Da = \frac{r_{\max}}{\left(D_e / \delta\right) S_0} \tag{5.81}$$

where 'r_{max}' is the maximum rate of bioconversion, 'D_e' is the effective diffusivity of rate limiting substrate, 'δ' is the thickness of liquid film, and 'S_0' is the bulk substrate concentration. For $Da \gg 1$, the rate of bioconversion is diffusion limited; for $Da \ll 1$, the rate is limited by bioconversion rate, and for $Da \sim 1$, diffusion rate is comparable to bioconversion rate. Diffusional limitations can be eliminated by keeping $Da < 1$. When the limiting factor is external mass transfer, the turbulence in liquid phase can be increased by stirring to increase reaction rate, for limitations due intraparticle mass transfer, particle size is to be reduced or void fraction of support material is to be increased to provide increased bioconversion.

When cell growth occurring in biofilms is considered, the thickness of biofilm will increase with time during growth phase. However, since rate of substrate consumption is higher than increase in biofilm thickness, quasi-steady state assumption can be made. Under steady state conditions, performing a differential mass balance within biofilm for rate limiting substrate yields

$$D_e \frac{d^2S}{dy^2} = \frac{1}{Y_{X/S}} \frac{\mu_m S}{K_s + S} X \tag{5.82}$$

With boundary conditions: $S = S_{oi}$ at $y = 0$

$$\frac{ds}{dy} = 0 \text{ at } y = L$$

And if liquid film resistance is made negligible so that $S_0 = S_{oi}$ with the maximum rate of substrate utilization given by $r_m = \mu_m X/Y_{X/S}$, Eq. (5.82) is written as

$$D_e \frac{d^2S}{dy^2} = \frac{r_m S}{K_s + S} \tag{5.83}$$

Non-dimensionalizing of Eq. (5.83) gives

$$\frac{d^2S'}{dy'^2} = \frac{\phi^2 S'}{1 + \beta S'} \tag{5.84}$$

where $S' = S/S_0$, $y' = y/L$, $\beta = S_0/K_s$.

Solving gives

$$\phi = L\sqrt{\frac{\mu_m X}{Y_{X/S} D_e K_s}} = L\sqrt{\frac{r_m}{D_e K_s}} \tag{5.85}$$

Equation (5.85) can be solved numerically. When diffusion limitations are ignored, the maximum rate of substrate flux is given by

$$N_s A_s = -A_s D_e \frac{dS}{dy}\Big|_{y=0} = \frac{r_m S_o}{K_s + S_0}(LA_s) \tag{5.86}$$

where 'A_s' is the surface area of biofilm, 'N_s' is the substrate flux, and 'L' is the biofilm thickness.

Under diffusion limitations, the maximum rate of substrate flux is expressed in terms of effectiveness factor, η

$$N_s = -D_e \frac{dS}{dy}\Big|_{y=0} = \eta\left(\frac{r_m S_0}{K_s + S_0}\right)L \tag{5.87}$$

Effectiveness factor 'η' is the ratio of substrate consumption rate under diffusion limitations to the substrate consumption rate without diffusion limitation (Shuler and Kargi 2015) and is given by

$$\eta = 1 - \frac{\tan h\phi}{\phi}\left(\frac{\omega}{\tan h\omega} - 1\right) \text{for} \omega \le 1 \tag{5.88}$$

$$\eta = \frac{1}{\omega} - \frac{\tan h\phi}{\phi}\left(\frac{1}{\tan h\phi} - 1\right) \text{for} \omega \ge 1 \tag{5.89}$$

where 'β' is the initial substrate concentration (dimensionless) and 'ϕ' is the Thiele modulus. $\eta = 1$ when there is no diffusion limitations, $\eta < 1$ with diffusion limitations. The value of ϕ should be less than 1 to ignore diffusional limitations.

'ω' is the modified Thiele modulus given by

$$\omega = \frac{\phi\left(S_0 / K_s\right)}{\sqrt{2\left(1 + \frac{S_0}{K_s}\right)}}\left[\frac{S_0}{K_s} - \ln\left(1 + \frac{S_0}{K_s}\right)\right]^{-\frac{1}{2}} \tag{5.90}$$

5.4.4 Bioreactors for Immobilized Cell Systems

Bioreactors of varying configurations are available to culture immobilized cells as shown in Fig. 5.17. Due to high fragility of support materials used in immobilization, bioreactors with low hydrodynamic shear are preferred. Commonly used bioreactors include packed bed, fluidized bed or airlift reactors. The choice of bioreactor configuration depends on the method of immobilization (Shuler and Kargi 2015).

Immobilized cell packed bed reactors and fluidized bed reactors can be operated in batch or continuous mode. These reactors approach CFSTR behaviour when the recirculation rate of fluid required is high to maintain stable operating conditions. When the recirculation rate of fluid is low, the system is treated as plug flow reactor. When a mass balance is performed over a differential element of such system (Shuler and Kargi 2015).

$$-F\left(\mathrm{d}S_0\right) = N_s\left(aA\right)\mathrm{d}z \tag{5.91}$$

where 'S_0' is the bulk substrate concentration in liquid phase, 'F' is the flow rate of nutrient, 'N_s' is the substrate flux into the biofilm, 'a' is the surface area of support per unit volume of reactor, 'A' is the cross-sectional area of bed.

Substituting Eq. (5.87) in Eq. (5.91) and integrating gives

$$K_s \ln\frac{S_{0i}}{S_0} + \left(S_{0i} - S_0\right) = \frac{\eta r_m LaA}{F} H \tag{5.92}$$

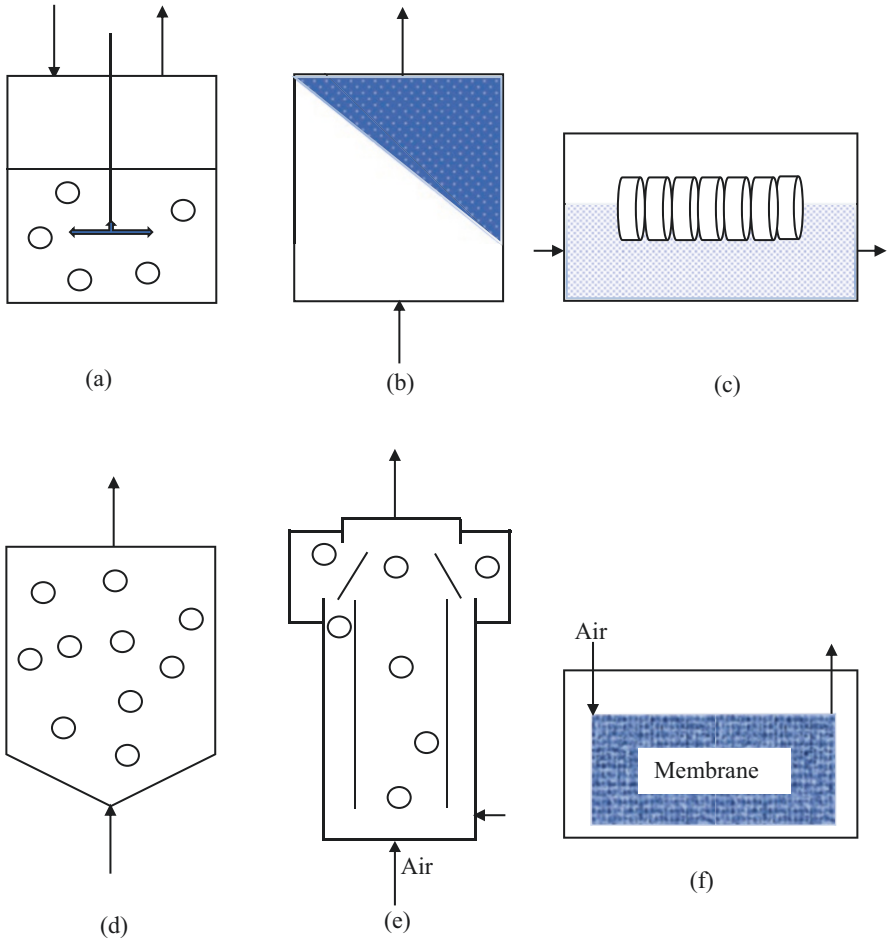


Fig. 5.17 Schematic representation of (**a**) Stirred tank reactor. (**b**) Fixed bed reactor. (**c**) Rotating disc contactor. (**d**) Fluidized bed reactor. (**e**) Airlift reactor. (**f**) Membrane reactor

where 'S_{0i}' is the inlet bulk substrate concentration, 'L' is the thickness of biofilm, 'H' is the height of packed bed. When the substrate concentration in the feed is low, the rate of substrate consumption approaches first order and the above equation reduces to

$$\ln \frac{S_0}{S_{oi}} = -\frac{\eta r_m LaA}{FK_s} H \tag{5.93}$$

Equations (5.92) and (5.93) are used as design equations for immobilized packed bed reactors to determine the height of packed bed to obtain desired level of substrate conversion (Shuler and Kargi 2015).

McCarthy et al. (2017) developed an eco-friendly method to remediate mercury contaminated soils. The work involved immobilization of *Pseudomonas veronii* cells in xanthan gum and coating it over zeolite granules for biofilm formation. The cells remained viable for 4 months even after transportation and the flux rate was as high as 10 μg Hg m^2 h^{-1}. An experimental study by Bandopadhyay (2018) revealed that the activity of α-amylase produced by immobilized cells of *Bacillus thuringenesis* is 3.5 times higher than the activity of α-amylase produced by free cells. Similar work by Ghoshoon et al. (2022) confirmed that the activity of L-asparaginase from immobilized cells of *E. coli* was twofold higher than the activity of L-asparaginase from free cells. While cell immobilization increases the product yield, the effect of interaction between the cells and the material of immobilization also plays a pivotal role. Jiménez-Bonilla et al. (2022) analysed the interaction effects of various polysaccharide-based immobilization materials such as chitosan powder on *Clostridium saccharoperbutylacetonicum* N_{1-4} and *Clostridium beijerinckii* NCIMB 8052, to improve butanol production. The former and the later showed 97% and 21% increase in productivity, respectively. The interaction occurs via adsorption and aggregation and the chemical properties of the immobilization materials affects the productivity.

5.4.5 Advantages and Limitations of Immobilized Cell Reactors

5.4.5.1 Advantages

- Immobilized reactors are stable than chemostats.
- High bioconversion because of high biomass loading.
- In case of high dilution rate, cell wash out is minimized and hence recovery time is quick with less fall in biomass concentration.

5.4.5.2 Limitations

- Intracellular product formation.
- Diffusional limitations.
- System is heterogeneous and hence microenvironmental conditions is difficult (Shuler and Kargi 2015).

5.5 Conclusion and Future Perspectives

This chapter highlighted the microbial growth kinetics resulting in the formation of value-added products such as antibiotics. If the substrate is an undesired one, such as any organic contaminant, then clean environment is created. The chapter also

discussed about various kinetic models and the optimization of controlling parameters. Depending on the intended application, one would be able to choose ideal reactor for maximum product yield. The results obtained from modelling and simulation of biological process could be adapted to design and scale up of processes. Furthermore, by understanding the behaviour of immobilized cell systems, it is quite possible to develop the existing process.

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Chapter 6
Bioreactor Configurations for Fermentation Processes



Pritam Bajirao Patil and Angana Sarkar

Abstract Fermentation technology has gained tremendous importance in a few decades due to its vast applicative fields. One of the main reasons behind this development is the discovery of bioreactors, also called fermenters. These are vessels in which the biotransformation takes place, whether it is a cell-free or cell-mediated transformation. The main purpose of a bioreactor is to provide optimum conditions for the transformation to take place. Bioreactors are generally cylindrical in shape but the design could be modified according to the process requirement. Different types of bioreactors are made depending on the process. For bacterial cultures, regularly stirred tank reactors are used, but for mammalian cultures, bubble column or airlift reactors are preferred due to their low shear force. The material of construction of the bioreactor is also dependent on its application. Commonly, stainless steel of different grades is used but, in some wine-making processes, reactors made of wood are preferred. The basic parts of a bioreactor include an agitator for mixing, a sparger for aeration, and different probes for sensing different parameters like pH, dissolved oxygen, in some cases product concentration. These components also differ from reactor to reactor depending on the process. A bioreactor can be operated in batch, fed-batch, or continuous mode as per the requirement of the process, and these operating modes also define the structure and complexity of the bioreactors. In this chapter, the structure, design, and potential applicative fields of different bioreactors/fermenters are discussed.

Keywords Fermentation technology · Bioreactor · Stirred tank reactor · Photoreactor

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6.1 Introduction

Fermentation technology goes back to decades in the Roman era primarily involving the making of curd, bread, soy sauce, etc. But the word 'fermentation' was coined in the fourteenth century which meant the production of carbon dioxide from an anaerobic fermentation by microbes using energy-rich nutrients (Bourdichon et al. 2021). Now, this term is used commonly for both anaerobic and aerobic cultivation of microbes, algae, plant, or mammalian cells either for the production of a product or the biomass itself. The history of fermentation started with the production of cheese, bread, and curd in 7000 BC and then evolved into the production of alcoholic beverages from fruits, honey, dates, and rice in 6000 BC. The alcohol production process and milk fermentation process are considered pioneers for the new era of production (Ojha and Tiwari 2016). Nowadays, fermentation technology plays a key role in the production of different complex products like industrial chemicals including ethanol, butanol, and acetic acid, pharmaceutical products like antibiotics, monoclonal antibodies, drugs, etc., food products like single-cell proteins, surfactants, curd, dough, enzymes like streptokinase lipases, cellulases, pigments like chlorophyll, anthocyanins, carotenoids, biopolymers like polyhydroxyalkanoates, polylactic acids, and biomass which could be applied as single-cell proteins (Ray and Joshi 2014). These different products can be divided into two basic types, namely, primary products, which are produced during growth like ethanol, acetic acid, enzymes, and pigments, and secondary products which are produced during the stationary phase of growth like antibiotics, surfactants, etc. Primary products are generally considered as growth-related products while secondary products are considered as non-growth-associated products and generally need an induction for their production (Beth 2000).

Although the aim of a fermentation process is to produce a product or the biomass itself, for both, a vessel is required which is called a bioreactor or fermenter. Basically, it is a container that provides optimum conditions required for the efficient production. The first evidence of a fermenter used for a process goes back to the nineteenth century when Louis Pasteur used a fermenter for the production of lactic acid from milk using *Lactobacillus*. Since then, bioreactors have evolved with specific designs for specific processes. The reactor designs are developed according to the process requirements like type of culture, mode of operation, scale of production, properties of media being used, etc. (Ray and Joshi 2014).

The use of different microbes for the production of different products goes long back to the era of alcohol fermentation (Raihofer et al. 2022). But nowadays, numerous products are being produced using different bacteria like organic acids (acetic acid, propionic acid, lactic acid), enzymes (protease, cellulase), biopolymers (polyhydroxyalkanoates, bacterial cellulose), active pharmaceuticals ingredients (API), nutraceuticals, etc. Yeasts are used for the production of alcohol, single-cell proteins, acids, etc. (Raihofer et al. 2022). Antibiotic production using fungi is also a conventional technique since the discovery of penicillin. Nowadays numerous antibacterial compounds are being produced industrially through either solid state or

suspension cultures (Tsivileva et al. 2022). Algae are mainly applied in the wastewater treatment processes along with the production of different industrially important products like nutraceuticals, single-cell proteins (spirulina), and pigments to be used in personal care products (Janssen et al. 2022). Each category of product with its producer has different specificities considering the reactor design, mode of operation, and process parameters. As the bacteria can tolerate a higher amount of shear stress, the fermenters could use different types of impellers for the mixing of the medium in suspension cultures. Although there is a very limited number of bioreactors designed specifically for bacterial culture, stirred tank reactors are used most widely across different industries (Fig. 6.1).

The mammalian cell cultures have been developed for the production of different high-end pharmaceutical and healthcare products like monoclonal antibodies, insulin, APIs, and vaccines (Ritacco et al. 2018). Conventionally some pharmaceutical products including certain enzymes and insulin were produced using genetically modified bacteria. These products required further modifications like post-translational modifications to increase compatibility with the mammalian system. Hence, currently, the production process has been shifted to mammalian cell cultures (Webster and Thomas 2012). But the mammalian cell cultures cannot be treated as microbial cultures as microbes as more robust compared to mammalian cells. Unlike bacteria, mammalian cells cannot tolerate higher shear stress caused during the mixing through agitators, hence for mammalian fermentations, specialized bioreactors are designed which contain different means to achieve proper mixing. The most widely used bioreactor configurations for mammalian cultures include airlift reactor, bubble column reactor, fluidized bed reactor, and fixed bed reactor (Mehdizadeh Allaf et al. 2022).

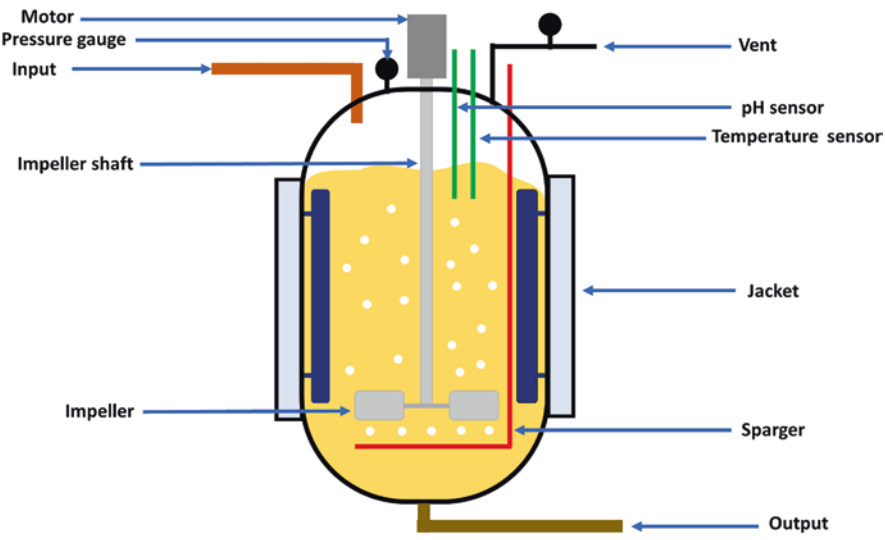


Fig. 6.1 Representation of a basic stirred tank bioreactor

The current chapter focuses on the basic parts of a bioreactor along with different configurations of reactors involved in different types of fermentation processes. Moreover, the basic process of designing a bioreactor along with different control strategies has been discussed.

6.2 Basic Parts of a Bioreactor

There are different processes for the production of different molecules but the basic parts of a bioreactor and their functions remain similar as follows (Singh et al. 2014):

(a) *Vessel:* It is the part where actual reactions take place and is usually made up of stainless steel for medium to large-scale reactors. Small or lab-scale reactors are mainly made up of glass. Both materials have a negligible effect on the growth of both microbes and animal cells hence these are widely used. The size and aspect ratio of the vessel is designed based on the mass and heat transfer characteristics of the liquid being used.

(b) *Impeller:* Also called as an agitator. They are mainly found in the stirred tank reactors for proper mass and heat transfer. They are placed above the sparger in aerobic fermentation to break the air bubbles into smaller spheres to increase the surface area for oxygen transfer. They are placed on an impeller shaft and operated by a gear motor to control the speed of rotation. They could be operated from the bottom or top of the vessel depending on the requirements. Impellers are of different types like Ruston turbine, helical screw type, paddle type, anchor type, etc. and the selection of the best impeller is crucial for a process to be more efficient, for example, the helical screw type is preferred in animal cell culture reactions due to their lower shear stress.

(c) *Jacket:* This is directly welded to the outer wall of the vessel and is crucial in maintaining the temperature inside the vessel. It is attached to both steam and chilled water supply so that the proper valves could be operated when needed and either is passed through the jacket to maintain the temperature inside the vessel. The jackets are constructed in such a way that the efficiency of heat transfer from the jacket to the vessel and then to the liquid medium should be higher to reduce the time required for maintaining the temperature.

(d) *Baffles:* These are rectangular plates fitted on the inner wall of a reactor. The main function of baffles is to avoid the formation of a vortex inside the vessel during mixing. They are not directly welded to the wall of the reactor but are held together with hinges placed on the walls to avoid the deposition of solids and cells as they are difficult to clean.

(e) *Sparger:* This is a crucial part of all reactors used for aerobic fermentations as they provide compressed air through the reactor. These are made up of stainless steel pipes with one end connected to the air compressor from which air is passed through an air filter which sterilizes the air and then this air is passed to the reactor. The design of a typical sparger depends on the process requirement of dissolved oxygen.

(f) *Sensor probes:* These are crucial in every reactor type as most of the processes are dependent on different process parameters. The common sensor probes used in a particular reactor include a pH probe, temperature probe, dissolved oxygen, and specific probes based on the process requirements. These probes are placed from the top of the vessel in such a way that they will be at least 50% immersed in the reaction medium. The probes are connected directly to the controller and the signal is passed continuously to monitor the process. The deviations in the real-time values from the set values will activate the controller system and the proper action would be performed.

(g) *Vent:* This is placed at the top wall or lid of the reactor and is crucial to maintain the positive pressure inside the reactor. The main function of a vent is to remove excess air along with carbon dioxide produced during the growth of cells from the reactor. The vent is always equipped with an air filter to avoid the release of microbes in the environment and contamination of air.

(h) *Pressure gauge:* This is used mainly to check the pressure inside the vessel which should be always maintained at a positive value during the operation.

6.3 Types of Bioreactors for Different Fermentation Processes

A typical bioreactor configuration is dependent on the process requirements but the main driving factor for the choice of a bioreactor is the type of biocatalyst like microbes, animal cells, algae, fungi, or plant cells used. Bacterial cells can withstand higher shear stress produced during mixing by the impeller compared to animal cells. So, for an animal and plant cell culture, bioreactors with different means of mixing are used. Also, for the algae, a reactor with the provision of a light source is required due to its photosynthetic nature. In the case of some fungal cultures, solid state fermentation is used, which further requires a different type of bioreactor.

6.3.1 Stirred Tank Bioreactor

This is the most common and widely used type of bioreactor in bacterial fermentations. As the name suggests, the medium is stirred using impellers to achieve efficient mixing (Kaur and Sharma 2020) (Fig. 6.1). The impellers are mounted on a vertical shaft attached to a geared motor which ensures the adjustment of the rotations per minute (RPM). The position and number of impellers depend on the volume of the reactor along with the extent of mixing required. It is also equipped with baffles of proper dimensions to avoid the formation of a vortex inside the vessel (Kaur and Sharma 2020) as vortex will significantly affect the mixing and aeration. The dimensions and number of baffles depend on the process requirements. The design of the vessel is specifically performed according to the process with a special

focus on the mode of operation, i.e., batch, fed-batch or continuous. Although widely used STRs are cylindrical in nature, the aspect ratio (height/diameter) varies with the production process. A higher aspect ratio is maintained in the processes where the mixing of oxygen from the air to the medium is crucial. Certain advantages of STRs over other reactor configurations are they are easy to manufacture, mixing is more efficient due to the use of impellers, and the oxygen dissolution rate is higher as the impellers break the air bubbles (Tennison-Omovoh et al. 2022). The major disadvantages of the STRs include the higher cost of production and maintenance due to the presence of a number of moving parts and they cannot be used for mammalian cell cultures due to the higher shear stress produced by impellers (Shang et al. 2023). Different products have been produced to date using the STRs and bacterial fermentations as represented in Table 6.1.

6.3.2 Bubble Column Reactor

The bubble column reactors are modified stirred tank reactors without the impeller for mixing and is widely used in chemical, biochemical, and petrochemical processes. It is a pneumatically operated bioreactor where mixing is achieved by passing compressed air through the medium (Fig. 6.2). These are mainly used in animal cell cultures as there is very less shear stress induced during the mixing of the medium as mixing is achieved using air bubbles (Purohit 2012). They can be operated mainly in two operational modes, i.e., semi-batch and continuous. In continuous mode, the liquid leaving the column is fed back to the reactor. The designing of a particular bubble column reactor requires extensive study of heat and mass transfer characteristics, the superficial velocities of gas and liquid, mixing characteristics, and the reaction kinetics, etc. (Thorat et al. 1998). Also, viscous liquids cannot be operated as the mass and heat transfer would be affected significantly. The major advantages include low maintenance cost, high heat, and mass transfer coefficients, compact structure and design hence low capital cost while the major disadvantages include inefficient mixing when not operated at optimum parameters, dead zone formations, and high cost related to the compressed air utilization (Sainz-Mellado et al. 2022). Some of the products produced using the bubble column reactor are mentioned in Table 6.1.

6.3.3 Airlift Reactor

The airlift reactors are also considered as modified bubble column reactors where the whole fluid bed is lifted up due to the differences in the densities. The efficient mixing is achieved through the draft tube principle (Kaur and Sharma 2020). The whole vessel is divided into two compartments, inner and outer (Fig. 6.3). The inner

Table 6.1 Examples of production of different products using different bioreactor configurations

Sr No.	Type of culture	Product	Comments	References
Stirred tank reactor (STR)				
1	*Aspergillus niger* NRRL 330	Cellulase, xylanase	Fungal submerged fermentation in a glass STR of volume 2 L	Iram et al. (2022)
2	*Rhizopus oryzae* NRRL 395	Lactic acid	The effect of oxygen transfer rate was studied on lactic acid production in submerged fermentation in an STR of volume 3 L	Rodríguez-Torres et al. (2022)
3	Mesophilic sewage sludge	Volatile fatty acid	The effect of the C/N ratio on VFA production was studied in a 0.6 L STR	Tennison-Omovoh et al. (2022)
4	*Gluconobacter oxydans*	Xylonic acid	The effect of activated carbon particles on the oxygen transfer rate of production yield was studied	Ding et al. (2022)
5	*Penicillium citrinum*	Chitosan	Continuous submerged fermentation using paper mill wastewater and hemicellulose hydrolysate as low-cost substrates	Namboodiri et al. (2022)
Bubble column reactor (BCR)				
6	*Methylocystis*	Polyhydroxybutyrate	A two-step process for PHB production was employed in BCR	Amabile et al. (2022)
7	*Saccharomyces cerevisiae* PE-2	Ethanol	A BCR of volume 3 L was employed for the simultaneous saccharification and ethanol production to increase yield	González-Gloria et al. (2022)
8	*Pichia fermentans*	Isoamyl acetate	A solid state fermentation in BCR was carried out	Sainz-Mellado et al. (2022)
9	*Pycnoporus sanguineus*	Lignolytic enzyme	Development of a fed-batch strategy in the BCR to produce lignolytic enzymes	de Souza Jr et al. (2022)
10	*Schizophyllum commune*	Schizophyllan	Increase the production yield by preventing oxygen starvation through BCR	Saeedian et al. (2022)

(continued)

Table 6.1 (continued)

Sr No.	Type of culture	Product	Comments	References
Airlift reactor (ALR)				
11	*Clostridium beijerinckii* TISTR 1461	Butanol	The butanol was produced in ALR and production was compared to STRs and BCR	Thanapornsin et al. (2022)
12	*Yarrowia lipolytica*	Citric acid	A new approach of using ALR for citric acid production using glycerol as substrate was developed	Ferreira et al. (2022)
13	*Azotobacter vinelandii*	Alginate	Optimization for the production of alginate was performed along with a reduction in the cost through ALR	Chuacharoen et al. (2022)
14	*Trichoderma reesei* C10 and *Saccharomyces cerevisiae* LGA-1C3S2	Glucaric acid	The highest concentration of glucaric acid was produced in the ALR compared to other reactors with a significant reduction in production cost	Fang et al. (2023)
15	*Mucor indicus*	Biorefinery for ethanol and other value-added products	Different value-added products were produced using ALR	Borujeni et al. (2022)
Fluidized bed reactor (FBR)				
16	*Saccharomyces cerevisiae*	Ethanol	*S. cerevisiae* was immobilized on magnetic beads and ethanol was produced in magnetically stabilized FBR	Liu et al. (2022)
17	*Thermoanaerobacterium, Pseudomonas, Clostridium*	Hydrogen	FBR was used to produce hydrogen using glycerol and cheese whey	de Souza Almeida et al. (2022)
18	*Lacticaseibacillus, Clostridium*	Hydrogen	The effect of increase in the organic loading rate (OLR) on hydrogen production from sugarcane vinasse and glycerol was studied	de Menezes et al. (2022)

(continued)

Table 6.1 (continued)

Sr No.	Type of culture	Product	Comments	References
19	Sludge	Ethanol and hydrogen	Anaerobic FBR was used to study the effect of temperature on the production of ethanol and hydrogen using cassava wastewater as substrate	de Gois Araújo Tavares et al. (2022)
20	*Bacillus subtilis*	Biodiesel	*B. subtilis* was immobilized in the Ferroferric oxide-polyvinyl alcohol composite beads and used as whole cell catalyst in FBR	Liu et al. (2022)
Packed bed reactor (PBR)				
21	*Aspergillus niger*	Invertase	Sugarcane molasses was used as substrate for the invertase production using solid state fermentation in PBR	Martínez-Ruiz et al. (2022)
22	Mixed cultures	Hydrogen	Biohydrogen was produced in packed bed continuous plug flow reactor	Wodołażski and Smoliński (2022)
23	*Oenococcus oeni*	Malolactic	The strain was encapsulated in alginate beads which used as PBR	Ruipérez et al. (2022)
24	*Actinobacillus succinogenes*	Succinic acid	Strain was immobilized in hollow-fiber membrane used as PBR	Vallecilla-Yepez and Wilkins (2022)
25	*Clostridium acetobutylicum*	Butanol	Novel packed bed biofilm reactor was used for butanol production	Raganati et al. (2022)
Photobioreactor (PR)				
26	Mixed cultures	Biohydrogen	Effect of different headspace pressure on biohydrogen production was studied	Xiang et al. (2023)

(continued)

Table 6.1 (continued)

Sr No.	Type of culture	Product	Comments	References
27	*Rhodopseudomonas palustris*	Hydrogen	Hydrogen production was optimized for different parameters using a thermosiphon PR	Bosman et al. (2022)
28	*Spirulina platensis*	Hydrogen	Hydrogen was produced using microalgae in bubbled PR	Saxena et al. (2022)
29	*Halomonas bluephagenesis*	Biofuels, bioplastics	Both products were produced using a PR	Park et al. (2022)
30	*Chlamydomonas reinhardtii*	Bioethanol	Influence of sulfur and light intensity was studied on bioethanol production	Vargas et al. (2022)
Single-use bioreactors (SUB)				
31	*Inonotus hispidus*	Hispidin	Production was carried out in a 10 L wave bag reactor with 4 L culture and the yield was compared with STR	Bergmann et al. (2022)
32	*Metarhizium robertsii*	Conidia	Improving the conidia production by addition of spherical-style packing	Méndez-González et al. (2022)
33	*Trichoderma brev* T069	Biomass as biopesticide	Optimization of production in wave bioreactor	Zhang et al. (2022a)
34	*Saccharomyces cerevisiae*	2G ethanol	Ethanol was produced in simultaneous isomerization and fermentation (SIF) process using xylose	Sandri et al. (2022)
35	*Trichoderma longibrachiatum* and *Trichoderma afroharzianum*	Degradation of lignocellulosic wastes	Solid state fermentation was utilized for the degradation of different lignocellulosic residues	Li et al. (2022)

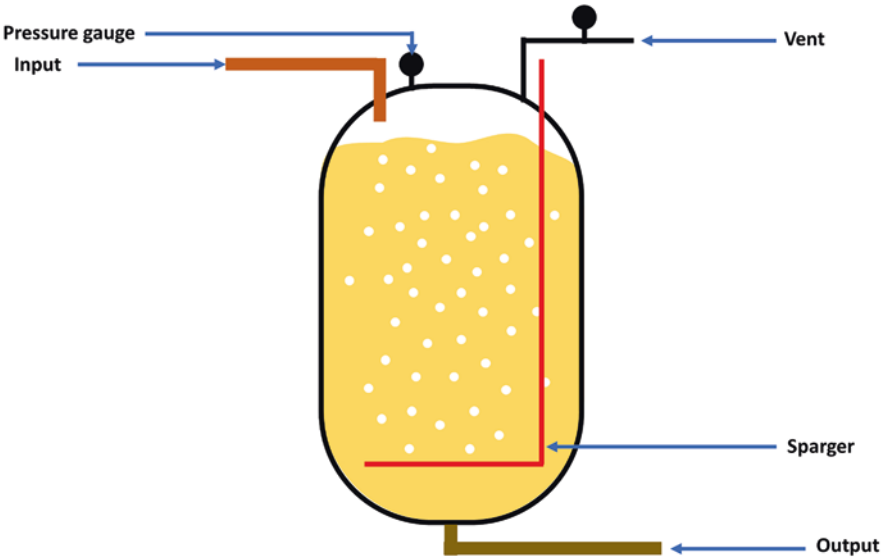


Fig. 6.2 Representation of a basic bubble column bioreactor

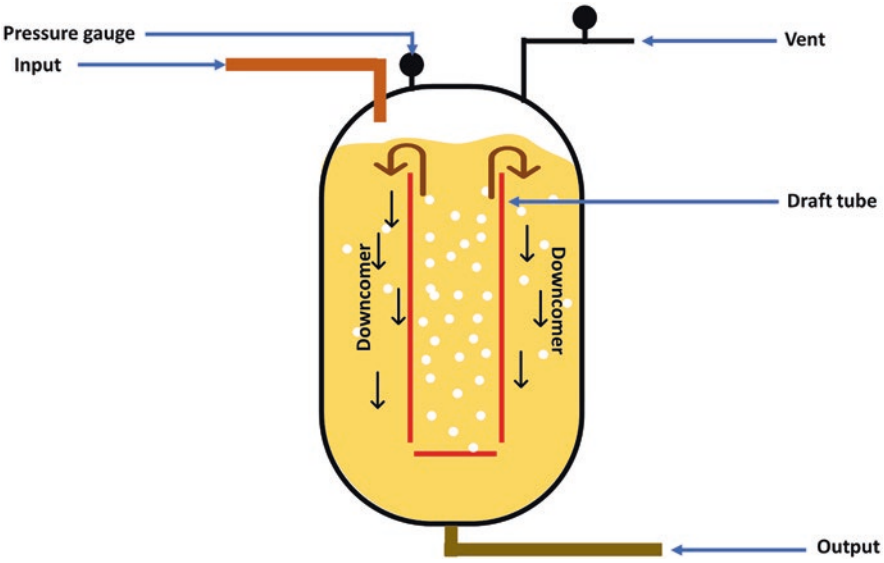


Fig. 6.3 Representation of an airlift bioreactor

compartment comprises the sparger which provides sterilized air at a very high rate, lowering the liquid density in contact. Due to lower density, the fluid is transferred toward the surface, and then again brought down once the air is released (Thanapornsin et al. 2022). The inner compartment is also called the draft tube. The flow of liquid is upside down from the inner compartment to the outer and the cycle continues. It has the advantages like better mixing, lower shear stress compared to other pneumatically operated bioreactor configurations, limited bubble coalescence, higher cyclic movement of fluid, etc. But the main disadvantage of airlift reactors is the higher cost related to the compressed air being used and the formation of dead zones due to improper mixing (Fang et al. 2023). These types of bioreactors are used primarily in mammalian cell culture systems and hairy root culture systems (Table 6.1).

6.3.4 Fluidized Bed Reactor

The fluidized bed reactors involve a carrier/anchorage-based system for the bioconversion reactions using either microbial or animal cell cultures that are anchorage-dependent (Liu et al. 2014). These are characterized uniquely as the solid carrier is suspended using high liquid or gas velocity so as to make the whole solid bed fluidized and behave as a fluid (Fig. 6.4). These are mainly used for operations where either microbes or animal cells cannot grow in suspensions and need a certain anchor to grow (de Gois Araújo Tavares et al. 2022). The fluid with the compounds to be bio-transformed is passed at a high flow rate and when the compounds come

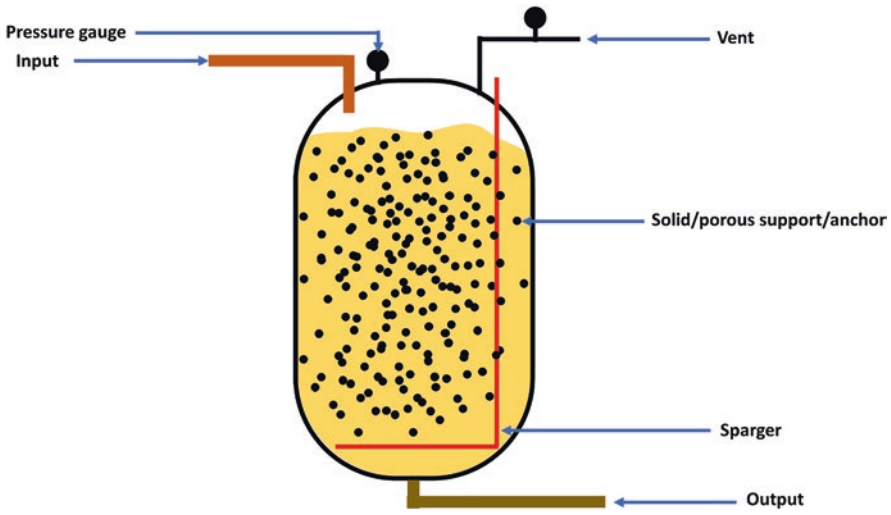


Fig. 6.4 Representation of a fluidized bed bioreactor

in contact with the biofilms on the solid surface, it is converted into the product of interest. These reactors are designed by taking certain parameters into consideration like the type of solid support to be used, the flow rates of gases and liquids, and mass transfer during the operation for higher efficiency of reactions (Liu et al. 2022). The solid carrier to be used is selected based on its biocompatibility with the cell line, density, surface area for the biofilm formation, porosity, and size. The major advantages of such types of reactors include efficient heat transfer and mixing, uniform product in the batch-wise process, and large solid-liquid-gas exchange due to the small grain size of the solids. The disadvantages include the higher cost of operation as the liquid velocity needs to be higher for fluidization, the higher cost of solids, and the difficulty to scale up (Chew et al. 2022). The major applications of fluidized bed reactors are in wastewater treatments, biotransformation reactions where cells are utilized at catalysts, etc. (Table 6.1).

6.3.5 Packed/Fixed Bed Reactor

These reactors differ from the fluidized bed reactor due to the tight packing of the solid support instead of the fluidization (Fig. 6.5). The solid carrier/anchor is packed in a confined space where anchorage-dependent microbes or animal cell cultures are grown in the form of biofilms and biotransformation reactions are carried out (Wodołażski and Smoliński 2022). Also, sometimes the animal cells or microbes could be entrapped inside the solids to increase the viability of cells by providing a controlled environment compared to biofilm formation. Depending on the process

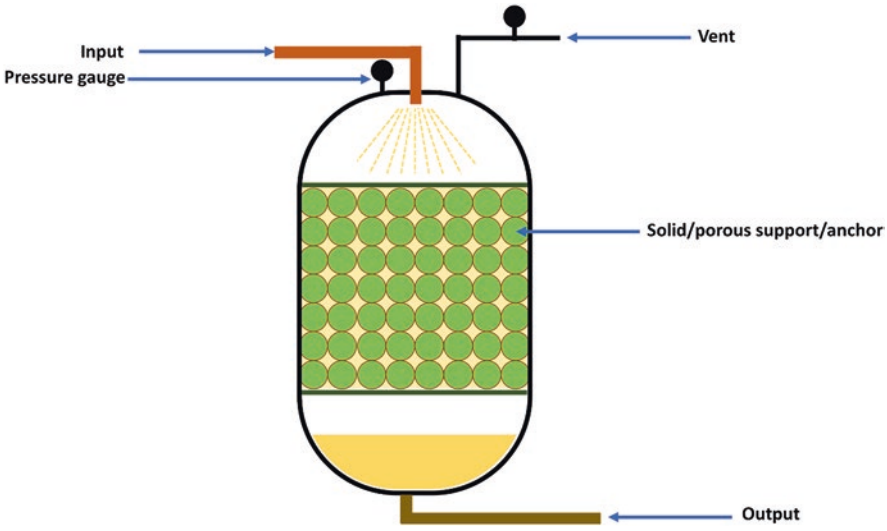


Fig. 6.5 Representation of a packed bed bioreactor

requirements, the fluid can be passed downward or upward. In both cases, the efficiency of the reactor will depend on the mass transfer coefficient between fluid and solid beads. The rate-determining step in the whole process would be the transfer of media components or the compound to be bio-transformed from the outside environment to the inside of the beads. Hence the selection of the perfect solid carrier is the crucial factor to be considered during the design of a packed bed reactor. Also, optimization of the flow rate of the liquid being transferred through the bed is an important factor to be considered as the retention time of the liquid throughout the bed is an important parameter that determines the efficiency of the whole process (Vallecilla-Yepez and Wilkins 2022). Retention time is the minimum time required for a predetermined percentage of conversion of a particular substrate into a product of interest. The rate of reaction and retention time will depend on the mass transfer through the solid support, porosity, and surface area of the solid/bead. The major application of these reactors is in enzymatic conversions and wastewater treatments using bacteria as catalysts. The major advantages of packed bed reactors are the higher viability of microbes or animal cells due to a controlled and protected environment inside the support, higher area for the reactions, economics in the construction compared to the other reactors due to the absence of moving parts, higher product formation rate due to higher contact. The major disadvantages are the higher retention time requirements, the critical procedure for the selection of the best solid carrier, and difficult temperature control with temperature gradient formation (Zhang et al. 2022b). Recent examples of the application of packed bed reactors in fermentation have been mentioned in Table 6.1.

6.3.6 Rotary Cell Culture System (RCCS)/3D Cell Culture Systems

These are newly developed bioreactors for both suspension and anchorage-dependent cultures of animal cells. It is also called a 3-dimensional cell culture system as it facilitates spheroid formations and also provides scaffolds for the growth of different cell lines unlike the conventional cell culture techniques where cells grow in two dimensions (Rodrigues et al. 2011). It also facilitates the growth of co-cultures that resemble parent tissues' growth properties and functions. These types of bioreactors are mostly used to study the effect of different molecules on tissues but can be modified for the production of different bioactive compounds. Structurally these are horizontally rotated bottles with provision for gas exchange through diffusion membranes positioned at the center (Fig. 6.6). The shear stress on the cell cultures during the horizontal rotations is minimized by limiting the rotational speed. The primary operating principle of the RCCS is that when the reactor is filled with the suspension and rotated, the whole liquid medium rotates as a single solid minimizing the shear force (Breslin and O'Driscoll 2013). The cells in the

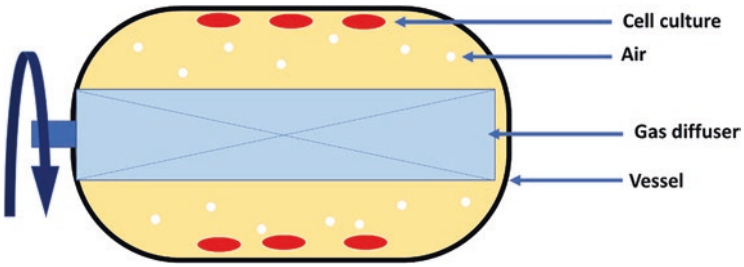


Fig. 6.6 Representation of a rotary cell culture system

suspension retain their position due to the equilibrium in the gravitational, centrifugal, and Coriolis forces. This advantage of low shear stress facilitates the formation of 3D aggregation of large cell masses that may differentiate or resemble the parent tissue. These were primarily developed by NASA at the Johnson space center to study the effect of microgravity on tissues and organoids (Wurm et al. 2007). But nowadays they have been used for the production of different biomolecules as represented in Table 6.1. The major advantages of RCCS over other bioreactors are low shear stress, and 3D cultivation of cells either on scaffold or suspension (Breslin and O'Driscoll 2013).

6.3.7 Photobioreactors

These types of bioreactors include any bioreactor system where photoautotrophic cells are cultured like algae or plant cells. The primary modification includes a setup where the light source is employed to aid the photosynthesis process (Fig. 6.7). The light source is selected to ensure the efficient intensity needed for the growth of either algae or plant cells. The algae are more robust compared to plant cells hence the plant cell cultures need a bioreactor where lower shear stress could be maintained (Bosman et al. 2022). As sterilization conditions are not necessary for the algal cultures, they are mostly cultivated in open ponds at large scale. But plant cell cultures need highly sterile conditions hence the provision for sterilization of medium are provided in the photobioreactor involved in cultures. Stirred tank reactors were used to date with a provision for either sunlight or artificial light source, nowadays glass tube reactors are also used. The tubes are arranged in a linear manner and nutrient medium is passed through them and recycled after every cycle. The light source is provided such that all the tubes would get a similar intensity of light (Saxena et al. 2022). The microalgae or macroalgae could grow in the tube either as suspension or as biofilm inside the tube, but the anchorage-independent algal cultures are preferred as the biofilms are difficult to remove from the reactor if the product is to be extracted from the algal biomass. Such glass tube reactors are used

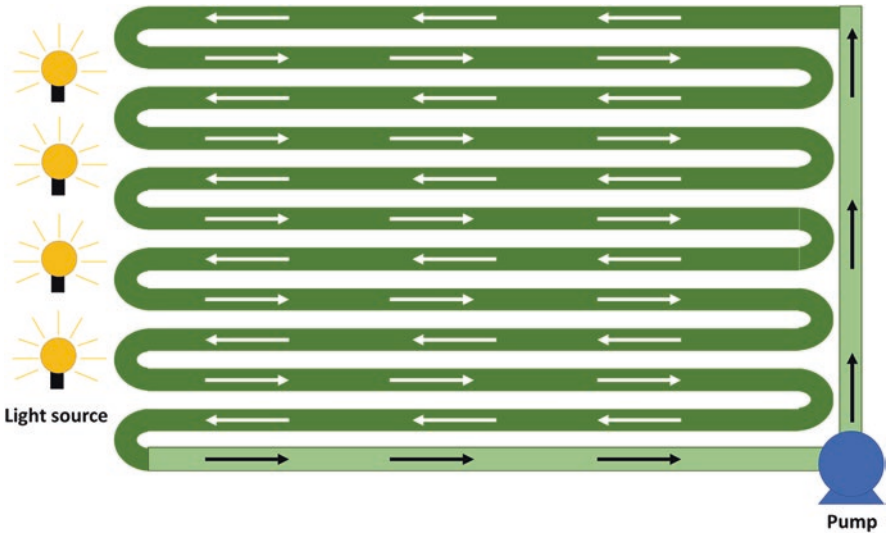


Fig. 6.7 Representation of a photobioreactor

primarily in the wastewater treatment plant where algal biomass is used solely for the treatment and no production of any substances is carried out (Vargas et al. 2022). Many plant tissue cultures utilize photobioreactors as mentioned in Table 6.1.

6.3.8 Single-Use Bioreactors

As the name suggests, these types of bioreactors are used only once. Hence instead of using stainless steel, they are made up of plastic bags so that they can be disposed off easily. Also due to the use of plastic, the cost of their production reduces significantly. They have been developed recently in the 1990s and are primarily used in the animal cell culture industries. The bag and other components used throughout the process are pre-sterilized and used directly (Zhang et al. 2022b). Although they are made up of plastic bags, all the necessary ports at the input and output ends are provided along with different mechanisms for mixing. They can be divided into different types based on the mechanism utilized for the mixing some reactors use impellers while some utilize a rocking motion for mixing. These reactors can be produced for as high as 1000-L culture. The major problem during the production of these reactors is that the whole process should be pre-defined along with the required sensors, input, output, and sampling ports as they have to be incorporated into the bag before sterilization. The whole setup is created and sterilized and then transported to the production facility (Bergmann et al. 2022). Hence at the production facility, only media and inoculum are added and the production is carried out. The major advantages of such types of bioreactors are the low production cost and

easy maintenance of the production process as there is no need for pre-and post-production cleaning and sterilization. But the disadvantages include an absence of provision for the calibration of sensors and the development of new pH and other sensors as conventional probes cannot be used due to the problems of transportation (Galliher et al. 2019). The recent applications of single-use bioreactors are mentioned in Table 6.1.

6.4 Designing a Bioreactor

The different processes have different production process which requires the reactors to be very specific so that the required parameters would be provided. Hence the bioreactors have to be specially designed for every production process which will ensure maximum efficiency of the whole process. There are different parameters and mathematical models that are taken into consideration while designing a reactor (Sharma et al. 2022). The basic process of designing starts with choosing the appropriate reactor, for example, if a process involves the use of some toxic chemicals which might affect the growth of cells, then the packed bed reactors could be used where there will be minimum contact of cells with the toxic chemical. Also, once the reactor is chosen, then the different aspects related to the vessel have to be determined. For example, the aspect ratio (height/diameter) of the vessel is a crucial parameter to be considered as a high aspect ratio will decrease the efficiency of mixing but increase the oxygen transfer rate inside the medium and vice versa. The aspect ratio of the reactor will be determined based on the rheological and thermal properties of the medium and the dissolved oxygen concentration to be maintained. Also, reactors with a high aspect ratio will require more sets of impellers to achieve proper mixing. Once the vessel diameter and height have been defined, the next parameters to be taken into consideration will be the number of input and output ports. In the batch mode of operation, only one input and output port are sufficient but in the case of fed-batch and continuous processes, there have to be different ports for inoculum and medium (Wolf et al. 2019; Catapano et al. 2009; Mersmann et al. 1990). Also, the number of ports for different sensors will depend on the number of sensors to be used which in turn will be determined by the process requirements. The type and number of impellers depend on the rheological properties of the liquid medium. The highly viscous liquid media will require more power for mixing which would in turn require a greater number of impellers (Zhong 2010).

6.5 Bioreactor Control Systems

Most of the fermentation processes operate for 2–3 days. This long process time poses difficulties in continuous monitoring for the optimum conditions. Many factors like dissolved oxygen, carbon dioxide generation, pH, temperature, etc. are

needed to be controlled in real time which is difficult to be performed by a human. Also, the processing time required to sense a parameter and take actions based on its value is higher in humans than a computer, so, different computer-based control systems have been designed to monitor the fermentation process throughout its process time. The choice of parameters to be controlled and the level at which this control is performed depends on the product produced during the respective process (Mitra and Murthy 2021). Factors that are taken into consideration while designing a control system for a process involve functionality meaning the actual need for a control system for a product, regulatory requirements for that particular product, type of product, etc. The basic requirements for a control system are software to interact with the sensors and actuators, a graphics user interface to display the real-time values of different parameters, memory storage to store the datasheets of different production batches, etc. (Liljequist 2017).

Different types of process controls are evolved which are discussed as follows.

6.5.1 Device or Activator Level Control

These are classic controller systems used in the single input and single output processes. Proportional, integrative, and derivative (PID) controls are the most common examples of such control systems. The components that are controlled by this system involves pumps, motor, valves, heaters, etc. which are also called actuators. This controller is suitable for the control of a single variable such as temperature or pH but is not meant for complex processes requiring control of many different parameters at a time (Knospe 2006; Belhaj and Boubaker 2013).

6.5.2 Distributed Control Strategies (DCS)

These are also called microprocessor-based control systems. It incorporates complex control systems with PID control at the instrument level. A host computer is used to perform high-end tasks such as the interpretation of data and a PID controller is used to actually control the actuators. As this involves communications between different control systems, it requires different redundancies, data acquisition, and transmission links, etc. This system also enables remote control when an operator can control the entire plant through a control station. All the real-time data collected by various sensors can either be printed or visualized on the display and the control can be achieved by remotely accessing the PID controller. It can convert data from one format to other like analog to digital or vice versa to gain more control over the system. Smart actuators are developed with their own microprocessor which enables them to perform tasks like auto-ranging, autocalibration, and signal conditioning on site (Tatjewski 2007; Tang and Heider 2009).

6.5.3 Programmable Logic Controllers (PLCs)

These are based on microprocessors with the ability to utilize binary logic and control different electrical instruments. They can be used along with timers, relays, push buttons, etc. for controlling different aspects of a process. Also, they can handle a large number of inputs and outputs at a time and are simple to handle. The important feature of such control is that they are able to time delay a process, can be triggered with time, and can be used with other control systems to make the whole process more robust (Lakerveld et al. 2013).

6.5.4 Supervisory Control and Data Acquisition (SCADA) Strategy

These are used in large-scale industries that need scalable control strategies with multiple redundancies in the process. These are similar to PLCs and DSCs in the microprocessor-based control but are basically software driven. A plant-based software is installed on a server that collects all the data from different inputs throughout the process, interprets it, and controls the whole process. SCADA makes the process easy in terms of process shutdown or startup, maintenance, control tuning, operation interface, etc. (Faedo et al. 2015).

6.5.5 Plant-Based Control Systems

With the digitalization of processes, computer software-based control strategies are widely utilized in industries. Software is hosted in a server or computer which gives the control engineer an integrated view of the whole process through communicating between various electrical instruments, P and ID, and other microprocessor-based control systems. These are very flexible in nature and are important during batch-to-batch transitions or making changes in the process. They also maintain and update datasheets in real time making it easy to troubleshoot the situations that arise due to the changes (Hopkins et al. 2010).

6.6 Future Prospects

There are many configurations of the bioreactors available according to the need of the developed process. Also, the current conventional bioreactors can be modified according to the process requirements in terms of the size of bioreactors, type of impeller to be used, the material of construction, aspect ratio, mass and heat transfer

efficiencies, etc. Still, there is a need for different strategies for designing of bioreactors in more economic terms. The major cost during the designing of a bioreactor lies in the material of construction, hence, different eco-friendly and economical materials could be developed for the construction. The maintenance of bioreactors also contributes to a major portion of the production cost, which could be reduced by either removing or creatively designing the moving parts. Also, the mass and heat transfer efficiency of the rector affects the production efficiency, which needs to be taken into consideration during the design of the bioreactor. Also, the monitoring and controlling of the reactor should be efficient to perform the quick actions in case of deviations from the set value. The overall current process of designing and manufacturing bioreactors could be improved to further decrease the cost and increase the efficiency of production which will ultimately reduce the cost of product.

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Chapter 7
Downstream Processing for Bio-product Recovery and Purification



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Abstract The recovery and purification of various bio-products or biomolecules, notably therapeutics, from sustainable materials, e.g., plant or animal tissue or fermentation broth, together with the repurposing of recoverable materials and the appropriate handling and removal of trash, is referred to as the downstream processing. The downstream concentration and purification program is an important aspect of the manufacturing process that adds significantly to the overall costs. The downstream processing is a multistep process that results in the purified target biomolecule by undergoing several physical and chemical processes and making it separate from other contaminants and impurities present, as the target bio-product is initially trapped in a biological matrix with other non-essential molecules. For the purification and recovery of different biomolecules, several techniques such as chromatography, crystallization, and membrane technology that include filtration (ultrafiltration, microfiltration, and nanofiltration), distillation, and reverse osmosis are available. Ultrafiltration, in particular, has been extensively researched in the industry of pharmaceuticals for downstream processing. Furthermore, the downstream processing of biological products necessitates the employment of specialized unit processes to recover from a complex concoction of molecules, contaminants, and impurities. Each unit operation results in a physical change that affects the concentration and/or purity of the product. Generally, a series of unit operations are needed to affect the product specifications. The prime objective of downstream processing is to efficiently and carefully recuperate the target product with all necessary requirements while also improving recovery yield at the least expensive price. This chapter gives an overview of several unit procedures for recovering, extracting, and purifying pharmaceutical and industrial products.

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Keywords Downstream processing · Biomolecules · Crystallization · Chromatography · Precipitation · Membrane filtration

7.1 Introduction

The understanding of how to obtain biomolecules or metabolites from a mixture of several ingredients after biotechnological/fermentation processes has been critical not only in the chronological advancements of biotechnology from trivial manual techniques to huge commercial technologies, but is also a foremost factor in biotransformation and industrial biotechnology at the moment (Wohlgemuth 2011). The development of novel biopharmaceuticals has benefited significantly in recent years due to advances in molecular biology. Due to the increasing demand for biopharmaceuticals, there is a significant need for highly efficient unit operations or procedures that can be applied on a large scale to purify products derived from biobased resources, e.g., microorganisms, tissues, plants, and animals. With the development of the biotechnological industry, there is an enormous need for novel bio-therapeutics, particularly new medications, antibiotics, glycoproteins, and proteins (Agarwal et al. 2021). To fulfill the growing need, to produce the chosen items in massive amounts, or make them accessible since they are so expensive, the latest efficient technologies must be implemented as soon as possible. Furthermore, a lot of impurities that occasionally have the same chemical and physical properties as the intended molecule, must be removed in order to purify it. Therefore, a sequence of unit operations is required in order to acquire the optimal bioseparation of the desired products from the cells and their environment by applying and optimizing biological systems (Sekhon 2010; Agarwal et al. 2021).

This bioprocess or the therapeutic cell manufacturing technology is divided into two stages, i.e., upstream and downstream processes. The first processing, i.e., upstream one entails a sequence of actions to provide the conditions required for cells to target molecule production, containing the choice of cell line, growth conditions, media components, and process optimization to ensure maximum production of biopharmaceutical. This mainly entails the microbial fermentation, or protein molecules derived from mammalian cell culture, to produce different intended therapeutic molecules (Gronemeyer et al. 2014). These final products, e.g., amino acids, therapeutic proteins, antibiotics, enzymes, hormones, etc., serve varied therapeutic functions, and are known as target proteins. Large-scale bioreactors are used for upstream processing, which is affected by several factors including process type, oxygen supply, pH, temperature, and others that must be taken into consideration (Lee 2003; Petrides 2000; Mutanda et al. 2020). The main focus of downstream processing is the extraction or separation of the intended molecules from the biomass produced during upstream processing. By utilizing specialized unit operations, it involves appropriate procedures for purification, recovery, and characterization from a composite combination of molecules, contaminants, and impurities. For downstream processing, a wide variety of unit operations, including

filtration, centrifugation, and chromatography, may be used (Wohlgemuth 2011). Every unit operation results in a physical and chemical modification, as well as the required grade, which changes the concentration and level of purity of the product. The main goal of downstream processing is to efficiently and safely recuperate the intended molecules with the required specifications while also maximizing recovery yield at the lowest possible cost. Low concentration products need to be purified using methods like chromatography, which raises the original investment and ongoing operational expenses (Agarwal et al. 2021). The goal of a downstream process is to provide the increased yield with highest purity for least amount of money in the shortest time period. As a result, a recovery program must be devised that can satisfy product criteria in the fewest steps possible while also minimizing the operating and capital expenses of downstream processing and the product losses that arise from each step (Kosseva and Rusbandi 2018). It is necessary to set up an appropriate sequence of separation, concentration, purification, and recovery unit operations in order to accomplish this goal. To do this, it is important to have a thorough understanding of the mechanisms underlying each of the unit operations as well as knowledge of the product's physical and chemical properties. This will ensure that the biological activity of the chosen molecule is retained along with its purification and concentration (Gronemeyer et al. 2014). The following are the characteristic unit operations used in downstream processing: product clarity employing methods like filtration, centrifugation, flotation, sedimentation, etc. Cell disruption and extraction of the cell extract using non-mechanical and mechanical methods, purification and concentration of the product using techniques such as precipitation, ultrafiltration, reverse osmosis, chromatography, etc., application of advanced methods for additional purification and refining of the intended product using techniques such as affinity chromatography, gel chromatography, ion-exchange chromatography, etc., and finally stability and purity of the chosen molecule accomplished by approaches like spray drying, freeze drying, crystallization, etc. (Wohlgemuth 2011; Agarwal et al. 2021).

Through downstream processing, numerous significant products have been identified and purified, which has significantly aided in recent advancements, particularly in the biotechnology and pharmaceutical industries. Antibodies, antibiotics, vaccinations, hormones, as well as different industrial enzymes, are all produced using it. The downstream purification process has made it possible to produce a variety of medications with high levels of potency, quality, and purity at affordable prices. Indeed, downstream processing has many different applications (Fig. 7.1, Conner et al. 2014). It produces antibiotics, amino acids, hormones organic acids, vitamins, and vaccines, as the major bio-products. The desired quality of the bio-product is achieved during the downstream process. Important vaccines have been manufactured using a range of quick and simple downstream processing techniques. Since enzymes play a significant part in the food and farming industries as well as the pharmaceutical, paper, and textile industries as well as in scientific research, etc., the development of effective and affordable downstream processing approaches has been applied in the industrial production of enzymes (Roque et al. 2004). The global market for downstream processing, which was estimated to be worth USD

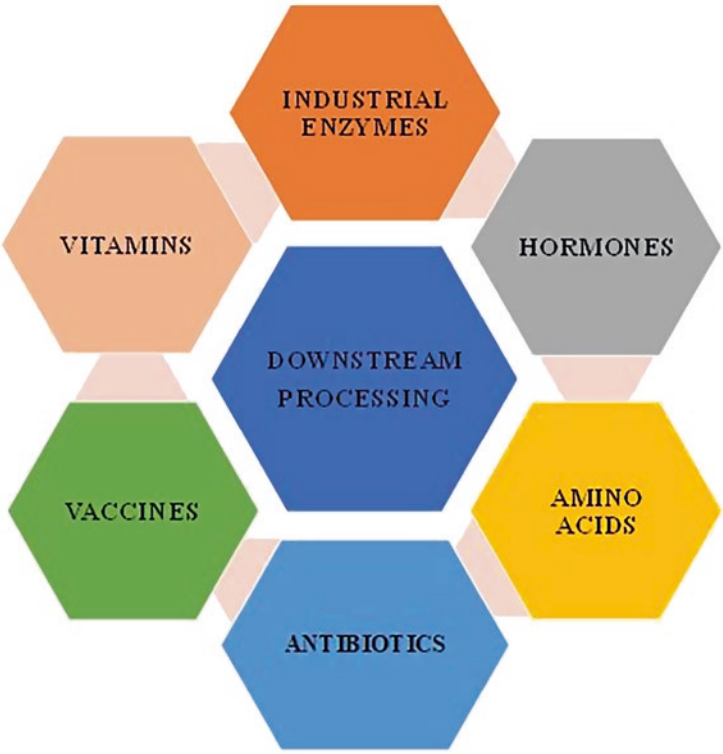


Fig. 7.1 Different applications of downstream processing

22.95 billion in 2021, is anticipated to increase to more than USD 72.30 billion by 2030, growing at a CAGR of 13.6% from 2022 to 2030 (https://www.precedenceresearch.com/downstream-processing-market). In light of these observations, the current chapter outlines a number of unit operations common to bio-product purification and concentration as well as several products recovery programs that require varied purification levels during downstream processing.

7.2 Historical Background

Since the production of wine, vinegar, beer, and many other food products to the current industrialized procedures for producing numerous products, e.g., amino acids, antibiotics, organic acids and other chemicals, proteins, vitamins, solvents, enzymes, and liquid fuels, the operations for recovering products from complicated mixtures produced from bioprocesses have been present throughout human history. As waste has accumulated due to these bioprocesses being scaled up commercially, the need of waste treatment and utilization has increased. The commercial production of these products, which has improved living quality over the previous two

centuries, has been considerably supported by the development of competence in the selection of the optimum product recovery processes (Gudernatsch et al. 1988; Jean-François and Jean 1990; Néel 1995; Petrides 2000; Buchacher and Iberer 2006; Shukla and Thömmes 2010; Carmichael et al. 2018; Bailey and Ollis 2018). The chronological details of the various products developed using downstream processing methods are given below:

4000 bc: Leavened bread was being baked by Egyptians.
2000 bc: Crushed dates were used to make vinegar in ancient Egypt by keeping them for a longer period of time.
1500 bc: In Egypt, Palestine, and Mesopotamia, the art of making wine from crushed grapes and beer from cereals that have germinated, using a lot of yeast were developed.
500 bc: Moldy soybean curds, the first antibiotic, were used to heal boils in China.
100 ad: Chinese people used powdered chrysanthemums to produce the first insecticide.
1761: Edward Jenner, an English surgeon, was credited with developing vaccination by immunizing a child with a viral smallpox vaccine.
1911: The first cancer-causing virus was discovered by American pathologist Peyton Rous.
1928: Scottish scientist Alexander Fleming discovered penicillin.
1942: For the first time, penicillin was mass-produced by microorganisms.
1950s: Production of the first synthetic antibiotic.
1958: For the first time, DNA was produced in a test tube.
1978: Human recombinant insulin was produced.
1979: Synthesis of human growth hormone.
1980: After a 20-year campaign of widespread immunization, smallpox had been eradicated worldwide.
1980: Bio product production by coupling of membrane separator to a conventional fermenter.
1982: The first livestock recombinant DNA vaccination was produced.
1982: The FDA approved the first biotech medication, human insulin made by genetically modified bacteria. The product was developed by Eli Lilly and Genentech.
1986: Hepatitis B vaccine, the first recombinant vaccination for humans, had been approved.
1986: The first anti-cancer medicine developed using biotechnology was interferon.
1990: Establishment of chromatographic techniques for purification of products for commercialization of proteins.
1992: Bovine somatotropin (BST) had FDA approval to improve milk output in dairy cows.
1993: The first of numerous biotech products with significant effects on multiple sclerosis treatment to receive FDA approval was Betaseron®.
1995: In the fight against cancer, recombinantly generated antibodies, immune system regulation, and gene therapy were available in clinics.

1995: Pervaporation, a novel membrane technique introduced to separate liquid mixture from a nonporous perm-selective membrane.
1998: Herceptin®, a pharmacogenomic medicine for patients whose cancer overexpresses the HER2 receptor, received FDA approval.
1999: Creutzfeldt-Jakob Disease (CJD) and Bovine Spongiform Encephalopathy (BSE, popularly known as "mad cow disease") and can be quickly diagnosed by a diagnostic test.
2000: Multimode chromatography and continuous bioseparation processes for the production of biomolecules.
2001: Gleevec® (imatinib), a gene-targeted medication for people with chronic myeloid leukemia, received FDA approval. The first gene-targeted medication to get FDA approval was Gleevec.
2004: The first antiangiogenic medication for cancer, Avastin®, received FDA approval.
2006: Gardasil® as the first vaccine developed against the human papillomavirus (HPV), an infection linked to cervical and throat malignancies, as well as the first cancer prevention vaccine, had been approved by the FDA.
2006: The first regulatory approval for a plant-based vaccination was given to Dow AgroSciences by the U.S. Agriculture Department.
2007: The H5N1 vaccine, the first against avian flu, was approved by the FDA.

7.3 Different Stages of Downstream Processing

The vast majority of individual biotechnological downstream methods may be thought of as a set of steps or unit operations (Fig. 7.2) that share common mechanics and the same underlying sciences as in the chemical engineering unit procedures or operations (McCabe et al. 2018). Each procedure is based on utilizing the special physico-chemical characteristics of the target molecule relative to the other species in the combination. Contrary to chemical engineering, the difficulties in the biotechnological downstream processes are not lessened; rather, they are unique and constitute a significant area of research in biochemical engineering. A separate analysis of molecules with low and high molecular weights is useful due to the wide variety of products. Many biotechnological unit operations that are used to create low-molecular weight products, like phase separation, extraction, crystallization, distillation, or adsorption, are based on similar physico-chemical principles as those used in chemistry. Molecular weight, charge, volatility, hydrophobicity, and solubility behavior are significant physicochemical characteristics that can be used for small-molecular-weight molecules recovery. The underpinning complicated processes' core chemical and biochemical engineering is crucial for industrial production operations. The interaction between chemistry and biotechnology is also advantageous for purification procedures such as the separation of cis/trans-isomers, regioisomers, enantiomers, or diastereomers as well as the protection of labile functional groups and their subsequent deprotection. Unit operations for biotechnological operations include cell harvesting, cell preservation, cell separation, cell

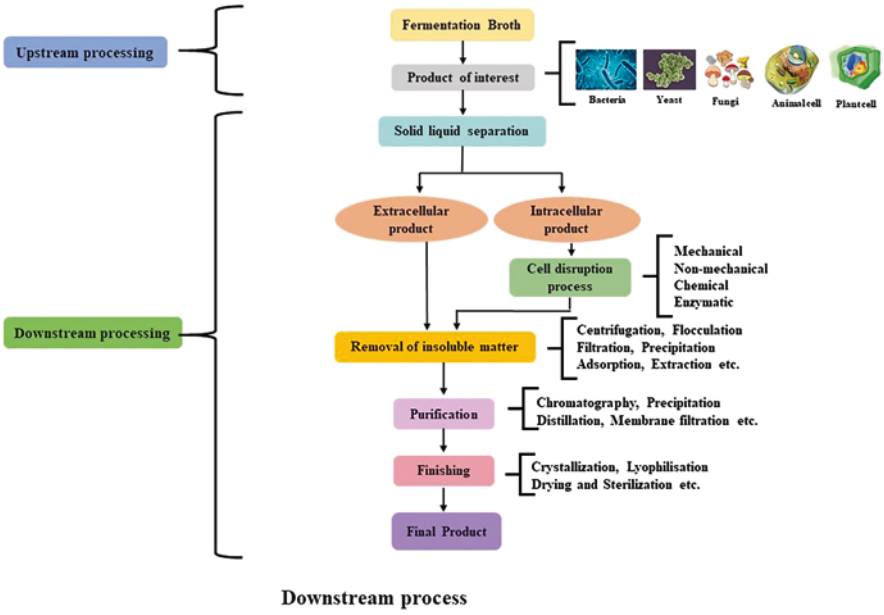


Fig. 7.2 Unit operations involved in downstream processing

homogenization, cell immobilization, product secretion, and cell lysis, which can all benefit from the enormous biochemical experience. This is particularly true for products with high molecular weight, e.g., proteins, polysaccharides, nucleic acids, and other types of biopolymers, where additional considerations like enzyme degradation, process stability, increased viscosity, and folding are factors to consider. The equipment size, cost, and waste generated by this process are typically significantly influenced by the final engineering, and finding the best way to recover the product may require a substantial amount of labor. The utilization of typical unit operations and the employment of recognized and readily accessible methods and materials, such as resins and membranes, are therefore quite interesting. This lowers the risks associated with process robustness, scaling up, Good Manufacturing Practice (GMP) concerns and technology transfer, for tools, processes, and techniques.

7.3.1 Separation of Cells and Extracellular Fluids

The cell or the target molecule is separated from the supernatant during the initial recovery stage of downstream processing. Filtration, centrifugation, flotation, and sedimentation, which involve extracting a solid from a liquid, are a few common solid–liquid separation techniques that can be exploited to separate the extracellular fluid and cells. The solid may consist of salts, undissolved salts, metabolites, live and dead biomass, and intracellular products, etc. The characteristics of the cells and the fluid determine whether to use centrifugation or filtration. Mechanical

strength, density, and particle diameter are examples of solid attributes that have an impact, whereas viscosity and density are examples of fluid properties. In light of this, the first stage is cell separation, and it differs depending on whether the product is extracellular (filtration and cell removal are followed by taking the supernatant after which the product recovery process is initiated) or intracellular (i.e., collection and disruption of the cells, and then removal of the debris by centrifugation before going on to product recovery). Therefore, ultrafiltration can be used to concentrate a target biomolecule that is produced extracellularly, followed by purification (Chisti 1998). Therefore, these two factors determine how the methods of downstream vary: for an intracellular product, the cells are of primary interest, not the extract; and for an extracellular product, the cell extract is of primary interest, not the cells. Hence, separating the cells from the extract is the initial physical stage in the recovery process, which is done to prepare the relevant stream for the product's recovery. If the desired molecule is dissolved in the liquid stream, unit operations or techniques that can purify and concentrate a soluble molecule are used to recover the liquid stream. The easiest way to ensure that the subsequent purifying procedures use the least amount of material is to execute unit activities that allow for some degree of concentration first. When recovering specialized biomolecules, for example, costly technology (such as chromatography) is required to achieve the high-purity specifications. Prior to purification, initial operations to extract the molecule that is linked to the solid stream, either adhering to the cell surface or within the cytoplasm, must be carried out (Agarwal et al. 2021). This requires the breakdown of the cell and the separation of the supernatant from the cell debris or significant quantity of unwanted cells. After that, the supernatant goes through one or more procedures meant to concentrate and purify a soluble molecule.

7.3.1.1 Filtration

The downstream processing industry makes extensive use of filtration, with both continuous and batch filtration finding several uses. When cells are separated on a large scale, as when producing *Saccharomyces* spp., continuous filtration is used, typically with a vacuum drum filter. This entails drawing cells onto the drum surface while maintaining consistent pressure using a rotating drum in a trough of suspended cells (vacuum). Batch filtration is also used, albeit on a smaller scale, and the chamber press is the filter of choice most often. The filter press is made up of a frame-clamped collection of chambers divided by filter plates. The cells in the cell suspension are fed under pressure into the chambers of the filter press as the filtrate is being processed, and they are then deposited as a filter cake on the filter media. When using constant pressure, batch filtration results in a gradual decrease in filtration rate. When using a constant rate, batch filtration results in a gradual increase in pressure. Although filtration is frequently used in the bioprocessing industry as a way to separate cells, this technique tends to produce a compressible filter cake when separating particles with very small dimensions, it would be best to generally avoid it. For the same reason that they are likely not suited for separation by

filtration, low mechanically strong cells also tend to be compressible. Additionally, unless harvesting can be done prior to the layer developing, filtration shouldn't be used to separate cells that have slime layers because it essentially delays filtration by blocking the pores (Jornitz 2019; Moleirinho et al. 2020).

Membrane Filtration

Ultrafiltration and microfiltration are two membrane-based tangential flow filtration processes that can be utilized for purifying viruses for vaccine manufacturing. In particular, the hollow-fiber crossflow membranes' open-flow path encourages the careful handling of shear-sensitive proteins and virus particles, preserving the integrity of the viral structure. The downstream process is therefore particularly appropriate for sensitive molecules (Kalyanpur 2002; Morenweiser 2005). Retroviral particles have also been concentrated and partially purified successfully using membrane filtration. With the use of a two-step microfiltration method via 0.45-m membrane, Braas et al. (1996) have partially purified, concentrated, and salt depleted the retroviral preparations and then ultra/diafiltered through membranes with a 300,000 MWCO, where the virus is recuperated in the dialyzed retentate. Although membrane filtration is a widely accepted technique, there are drawbacks such as fouling of the membrane and, at high pressures, viral infectivity loss that may be caused by shear forces acting on the viral envelope. Additionally, because segregation is dependent on differences in size, the viral particles are coconcentrated with the transduction inhibitors having high molecular weight, which reduces the viral transduction efficiency (Andreadis et al. 1999). Ultrafiltration is used to concentrate high-molecular weight products and separate salts, media components, and other undesirable low-molecular weight by-products, from desired high-molecular weight biopolymers. Microfiltration is used to concentrate cells, crystals, and precipitates (van Reis and Zydney 2007). Reverse osmosis membranes can be utilized for a later concentration step after nanofiltration is used for the concentration and desalination of small-molecular-weight chemicals in downstream processing (Dudziak et al. 1999). Numerous micro-, ultra-, and nanofiltration membranes with various pore sizes and polymers are used in a regular basis in industries. Highly porous membranes made up of ceramic material ranging in size from 1 to more than 1000 nm are among the other interesting membrane materials. There are different sizes of membrane filtration equipment available for obtaining the maximum product recovery yields and least amount of adsorption (Wohlgemuth 2011).

7.3.1.2 Centrifugation

When filtration cannot achieve separation, centrifugation is a high degree of separation technique. The centrifugation technique uses sedimentation to separate cells from liquid using centrifugal force. Centrifugation is still the method of choice for separating cells that would otherwise tend to form a compressible cake, provided

that there is a substantial density difference between the microbe and the fluid. This method is very popular for yeast, particularly when a concentrated yeast cream is needed, and can be applied to particles with a diameter of 0.1–100 μm. The fluid is drawn from the top of a disc stack bowl or a tubular bowl in a centrifuge, and the cells are pelletized close to the bottom of the bowl. Another benefit of centrifugation is that it can be utilized for aseptic separation, which is impossible with separation by filtration. The dynamic viscosity, particle diameter, angular velocity, radius, and the differential in densities between the liquid and the solid all affect how effective centrifugation is (Roush and Lu 2008; Agarwal et al. 2021).

7.3.1.3 Gravity Sedimentation

As microorganisms' terminal settling velocities are too slow to allow for sufficient settling within a tolerable amount of time, gravity sedimentation, which involves gravitational separation rather than centrifugal separation, is only employed in specific situations. The most notable example is when household garbage is treated biologically and bacteria produce microbial masses or flocs. Filtration, centrifugation, coagulation, and flocculation can be combined to increase the potential for an efficient separation. Typically, only the presence of the product in the extracellular fluid would make this approach effective, and it may introduce undesirable molecules into the cells (Agarwal et al. 2021).

7.3.1.4 Flocculation

One of the most efficient and economical techniques for increasing the effectiveness of clarification is flocculation. It is based on the idea that by adding substances (flocculants), which make scattered particles stick together by amplifying the attractive forces between them, it is possible to separate the particles using filtration, sedimentation, or centrifugation. The characteristics of the target products, the nature and concentration of the particles, and other factors can all affect how effective flocculation is in a particular process (Agarwal et al. 2021).

7.3.1.5 Flotation

A separation technique called flotation involves capturing particles smaller than 500 μm. It relies on the capability of the cell to rise to the surface as the cells or particles are adsorbed on gas bubbles, get trapped in a foam layer, and can be collected (buoyancy of the cells). It is a traditional technique for treating wastewater and alcoholic beverages, and it works especially well for unicellular microalgae. There are several techniques used, including electrolytic flotation, dissolved air flotation, dispersion flotation, and ozone flotation (Miettinen et al. 2010; Cramer and Holstein 2011).

7.3.2 Cell Disruption and Separation of Cell Extract

By rupturing the cell and separating the supernatant from the resultant cell fragments, intracellular components are extracted from the cell. A variety of cell disruption techniques are available to rupture the cell wall in order to extract the intracellular products. By using a specific procedure, a better yield and purer version of the intended product can be obtained from inside the cells, although most of the time, these approaches are not always feasible. It might be challenging to sort or separate the unwanted compounds that are released when cells are disrupted in large quantities. Therefore, the method of choice is determined by how well it works with the specific compound. Non-mechanical disruption, mechanical disruption, or a mix of mechanical and non-mechanical disruption are the three different methods used to rupture cells. Biological, physical, and chemical techniques are all used in non-mechanical rupture. Although successful, biological and chemical cell disruption has a narrow range of applications (Chisti 1998; Goldberg 2012; D'Hondt et al. 2017).

7.3.2.1 Mechanical Disruption

Mechanical disruption, achieved through milling or grinding with the aid of mechanically resistant beads, is one of the most popular and efficient approaches of sizable disintegration. The materials used to fabricate the beads are glass, titanium, or alumina. A horizontally spinning drum with a partial suspension of beads and cells revolves at a fast rotational speed (usually 6000 rpm), causing the cells to rupture when they come into contact with the beads. The frictional heating of the cell suspension in this approach would be detrimental to a thermo-labile product like enzymes. To overcome this issue, a jacketed cover is added to the drum through which cooling water is fed to keep the suspension below the point at which the product would be compromised. Another popular technique for mechanical cell rupture is high pressure homogenization, which can typically be scaled up while upholding better efficiency. Here, shear forces cause the cells to disintegrate as the cell suspension is pushed under high pressure via a tiny valve. The degree of cell rupture can be managed through valve pressure control (Günerken et al. 2015; D'Hondt et al. 2017).

7.3.2.2 Non-mechanical Cell Disruption

Biological, chemical, or physical processes can be used to burst cells in non-mechanical ways. The most frequently used physical technique relies on causing a phase transition in the cell suspension. The pressure and temperature variations caused by pushing a frozen cell suspension through an aperture modify the phase and, consequently, the crystal structure and the cell rupture result from changes in

the crystal structure. However, this approach is only appropriate for small-scale applications. Liquid shear force, which is produced by the cell slurry cavitation as a result of a significant pressure decrease, is frequently employed to release intracellular products, e.g., enzymes from microbial cells. Other physical ways include osmotic shock, in which cell walls are broken by sugars, solvents, buffers, or/and thermal shock, in which cells are burst by ice crystals. The osmotic stress is especially relevant for periplasmic enzyme release (D'Hondt et al. 2017).

7.3.2.3 Biological Disruption

Biological disruption caused by lytic enzymes that break down the cell wall polysaccharides or by antibiotics that prevent the formation of the cell wall is efficient but applied for certain products only. The release of enzymes with fewer contaminants, for instance, can be accomplished successfully using this technique. The technique is quite expensive and is combined with other cell disruption techniques. However, there is potential to creatively combine several cell disintegration techniques, such as osmotic stress and antibiotics. Osmotic stress is a very effective technique for bacteria with thick walls since it is used to gently release enzymes from the cells. The cells are disrupted by a change in their internal osmotic equilibrium as a result of the abrupt alteration in salt concentration. After the cells have been broken up, the cell waste still needs to be taken out of the supernatant. Operations for solid–liquid separation are used, especially those that can separate disintegrated debris of the cells with particle sizes smaller than a micron. Consequently, centrifugation is favored over filtration (Lam and Lee 2015; D'Hondt et al. 2017).

7.3.3 Chemical Extraction

Numerous substances including antibiotics, chaotropes, chelating agents, solvents, detergents, oxidizing agents, alkalis, and acids have been used in extensive research on chemical cell disruption. The kind and concentration of chemicals, biomass concentration, time, temperature, and microalgae species are the primary operating parameters. Since chemical techniques frequently depend on particular interactions with cell wall components, they might be more focused than mechanical techniques. Additionally, upscaling is easier, consumption of energy is often lesser, and cell disruption efficacy is higher. However, the price of the chemicals and the value of the goods may decrease their advantages. H_2SO_4 is the most commonly utilized acid and NaOH is the most frequently explored base in acid and alkaline procedures. The use of these pre-treatments for the fermentation and extraction of intercellular molecules including pigments and lipids has been proven to be successful. At temperatures exceeding 100 °C, reaction rates are accelerated and the reactions can be

carried out under low pressure or high pressure. Concentrated acid-based techniques are faster, but they have disadvantages, e.g., the production of inhibitors, challenging chemical recovery, equipment corrosion, and expensive operational and preservation expenses. Diluted pre-treatments appear to be more advantageous for industrial applications while taking more time. Pre-treatments with alkalis and acids exhibit limited selectivity, releasing many components, which makes separation challenging. In alkaline surroundings, proteins can denature, whereas in acidic environments, pigments can degrade (Mendes-Pinto et al. 2001; Nguyen et al. 2009; Miranda et al. 2012; Günerken et al. 2015; Lam and Lee 2015; D'Hondt et al. 2017).

7.3.4 Liquid–Liquid Phase Separation

Aqueous two-phase systems have proven beneficial for the recovery of proteins and enzymes without causing any harm, whereas small molecule product recovery has traditionally been the focus of the extraction of solvents with organic-aqueous two-phase systems. In contrast, the practical usage of phase separations and aqueous two-phase partitioning have generally been constrained in industrial processes due to the expensive phase-forming polymers and the lengthy durations for the development of processes. When, aqueous two-phase separation and ion-exchange chromatography were directly compared, aqueous two-phase separation was found to have a higher overall process yield while being less expensive (Rito-Palomares 2004; Azevedo et al. 2009).

7.3.5 Crystallization and Precipitation

Crystallization and precipitation are two of the most common and preferred downstream processing operations because of the ease of and inexpensive nature of these liquid–solid phase separations. A bioprocess that forms the product at a concentration significantly higher than its solubility in the reaction media is the optimal scenario for this product recovery operation. Changes in temperature, pH, ionic strength, and solvent composition can present opportunities for product crystallization or precipitation if the product is solubilized in the reaction medium even at high concentrations. For both small and large molecules, developing new precipitation and crystallization processes is still difficult, yet rewarding if done well. Even if the ideal circumstances for crystallizing pure molecules have been found, it is not simple to extend these conditions to crystallizing the same compounds in even more complex media. Despite many applications of crystallization technology, it can be difficult to define and enhance crystallization at an industrial scale (Wohlgemuth 2011). For the organic acids recovery, it has frequently been used to precipitate and crystallize materials into insoluble barium or calcium salts. Both proteins and small

molecules have long employed this technique, whereas the former has been done primarily on a small scale. Ammonium sulphate is often used for the precipitation of proteins because of its high solubility, inexpensive price, and extensive information on its saturation concentrations across a wide range of situations. Precipitation and crystallization are both purification and separation techniques that do not require expensive machinery. Consequently, they are enjoying a renaissance as easy and affordable purification techniques for both large and small biomolecules, such as proteins and metabolites.

7.3.6 Adsorption

If organic solvents have negative impacts, distributing the products between an adsorber and the bulk aqueous phase is a useful technique for the recovery of products. It is well known that polymeric adsorbers can be utilized to recover tiny compounds from diluted environments (Garcia 1991). Several commercially available ion exchangers, hydrophobic polymers, and other functionalized polymers can be utilized to provide the finest product distribution between the liquid medium and the adsorber. In order for the product to be recovered, the adsorption parameters, e.g., selectivity, capacity, or kinetics, must typically be evaluated experimentally. Selecting the best adsorbers for small molecule recovery has benefited from the evaluation of adsorbent resins' capacities for product recovery (chiral building blocks and antibiotics). Following the removal of the product-loaded adsorber from the reaction mixture, the product can be readily recovered from the adsorber by solvent elution or extraction, and the adsorber can then be recycled.

7.3.7 Distillation

By gas stripping, evaporation, and then following adsorption or condensation, volatile compounds can be distinguished from nonvolatile products. In addition to conventional alcoholic beverages and biofuels, distillation is preferred for recovering a number of desired molecules such as solvents, terpenes, flavors, and perfumes, and oils from bio-based processes (Moo-Young 2019). The differences in the boiling points of the product and other volatile reaction mixture components determine fractionation according to boiling points. Advanced fine distillation techniques have shown to be effective in the terpenes purification from difficult natural oil mixtures with ingredients having similar boiling points if simple distillations are unable to achieve the desired product purity. Checking the impact of minor contaminants on potential product reactions during distillation at higher temperatures is necessary.

7.3.8 *Chromatography*

An essential standard unit operation and the mainstay of subsequent operations for product recovery is the passage of a product mixture over a chromatography column. High-resolution bioseparations are made possible by variations in how the various components are partitioned between the chromatographic material and the solvent in the column, as well as the optimal conditions for expanding these variations. A flexible yet general downstream process platform must incorporate extremely selective chromatographic capturing steps carried out under efficient product recovery circumstances. For some products with similar biochemical properties, unit operation platforms can be provided via affinity chromatography, adsorption chromatography, gel filtration chromatography (Fig. 7.3) hydrophilic and hydrophobic interaction chromatography, and ion-exchange chromatography (Table 7.1). These important chromatographic techniques separate materials based on charge, polarity, affinity, and size. Ion-exchange chromatography is the most popular chromatographic technique, even though affinity chromatography produces the best purifications. Experimentation is crucial for the optimization of chromatographic separations, and genetic algorithms combined with high-throughput screening have produced potent tools for quick process development (Susanto et al. 2009). Various large and small molecules, e.g., nucleic acids, polysaccharides, and proteins, can be distinguished by an affluence of specialized chromatographic practices that have been developed over the years. Small molecule product class examples include metabolites and lipids. For many mixtures, chromatographic separations provide a high level of resolution, but from an industrial and large-scale standpoint, there are limitations. Membrane chromatography is one alternative that can boost throughput and get over common bottlenecks in column chromatography (Thömmes and Etzel 2007; Moo-Young 2019).

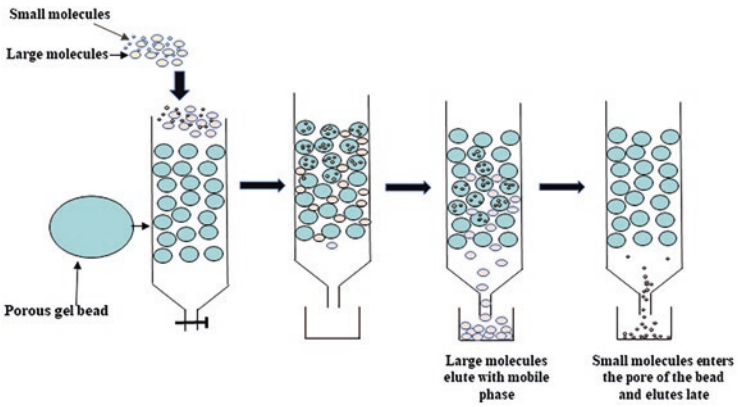


Fig. 7.3 Gel filtration chromatography

Table 7.1 List showing different kinds of chromatography and their industrial applications

S. No.	Kind of chromatography	Mobile phase	Stationary phase	Industrial applications	References
1.	Affinity chromatography	Liquid	Silica gel	Nucleic acid purification from blood, enzymes	Su et al. (2007)
2.	Adsorption chromatography	Liquid or gas	Silica gel	Separation of amino acids, fats, and fatty acids and for the identification of carbohydrates	Martire and Boehm (1987)
3.	Gel permeation chromatography	Liquid	Molecular sieves	Used in molecular weight determination of macromolecules	Weisskopf (1988)
4.	Ion-exchange chromatography	Ion-exchange resins	Liquid	To exchange ions like magnesium, calcium, vitamins, and biological amines	Jungbauer and Hahn (2009)
5.	Reversed phase chromatography	Polar molecules	Non-polar molecule	Isolation of active biomolecules and drugs	Ohmine and Tanaka (1982)
6.	High-performance liquid chromatography	Liquid	Silica gel	Separation of proteins, amino acids, nucleic acids, and drugs	Ambati et al. (2019)

7.3.9 Other Unit Operations

The processes used in electroseparation, such as electrodialysis and electrophoresis, rely on mobility and charge variances in an electric field even though they are complementary to molecular weight-based separation techniques. These processes are utilized in more specialized preparative applications. Over the past few decades, the free-flow electrophoresis apparatus has been used for numerous high-performance separations of cells, proteins, and small compounds. In terms of energy requirements, electrodialysis has proven to be an effective separation technique in the commercial production of amino acids. For process integration and scale-up, ion-exchange membranes are crucial for electrodialysis-based separation technologies (Xu and Huang 2008). Lyophilization can be used to remove volatile buffers from chromatographic separations that included volatile buffers prior to the addition of small molecules. From a range of compounds, the optimum volatile buffer can be chosen based on how thoroughly the removal occurs at the specified pH level, pressure, and temperature.

7.4 Integrated Downstream Processing Unit Operations

The significant aspects of product recovery economics include scalability, quantity, and yield per step of downstream processing unit operations. The use of single-stage processes instead of numerous downstream processing steps can improve overall

operating efficiency by maximizing space-time yield, product quality, timeline, and budget considerations. In situ product removal (ISPR) techniques remove the product from the vicinity of the reaction space thereby preventing its interference (Freeman et al. 1993; Stark and von Stockar 2003). Since Pasteur's biocatalytic tartaric acid resolution procedure in 1858 and current large-scale industrial bioprocesses, integrating product removal unit activities with bio-reactions has been an effective approach of process intensification for enhancing the success of the bioprocess.

7.4.1 Solid–Liquid Separation and Product Recovery

Depending on whether the product is liquid or solid, there are essentially two approaches to integrate these two-unit operations. The solid particles in the first case can either be the product in its purest form, such as crystals, or they can be the product combined with other elements like adsorbers, inclusion bodies, viruses, virus-like particles, and cell organelles. The remaining contaminants, side products, and other reagents are present in the liquid phase and must be segregated from the final product. In the latter case, either the product itself is solubilized in the liquid phase or it is the pure product itself; the remaining impurities are present in the solid phase. The product is often contained in the solid phase of an integrated primary downstream process like expanded-bed adsorption. With this method, product recovery and solid–liquid separation are combined into a single operation (Hubbuch et al. 2005).

7.4.2 Reaction and Solvent Extraction

The organic solvents that are compatible with the bioprocess are required for an organic-aqueous two-phase system for simultaneous reaction and extraction. The production of hydrophobic compounds that are hazardous to cells and biocatalysts and highly soluble in the organic phase, may be increased by these organic-aqueous two-phase systems. A technique that combines a carbon dioxide-induced phase separation with a homogeneous phase made up of organic-aqueous customizable solvents allows for the concurrent recycling of the biocatalyst and product recovery (Broering et al. 2006).

7.4.3 Reaction and Liquid–Liquid Phase Separation

Despite years of promising research exploitations, aqueous two-phase separation applications for extractive bioconversions have not yet yielded widespread applications in various industries (Moo-Young 2019). As more work is done for the

improvement of the aqueous two-phase separations and reduction in the price of phase-forming polymers involved, this study field will experience an increase in industrial applications. An appealing method for obtaining the product in a cell-free phase in extractive fermentation and overcoming low product yield or is the employment of two aqueous phases. By phase-separating the product as it is produced in the reaction, it has been possible to overcome current bottlenecks, e.g., product degradation and inhibition. This technology has been used to recover extracellular products with small and large molecular weights. Since cell disintegration or cell therapy is required in the case of intracellular products, various integration approaches for product recovery must be developed. Taking into account the organic solvent waste contribution of solvent extraction methods, ecologically friendly aqueous two-phase separation holds significant potential for novel integrated bioprocesses, notwithstanding the existing lack of translation of research applications into industrial processes (Rito-Palomares 2004; Moo-Young 2019).

7.4.4 Reaction and Membrane Filtration

A membrane filtering system and a bioreactor in combination can be used to achieve a variety of goals, including recovery of the product from the bioreactor and mass recycling into the bioreactor (Ghisalba et al. 2010). Ultrafiltration membranes are used to recycle large-molecular-weight substances like enzymes or big coenzymes, while microfiltration membranes are used to recycle biological cells. The membranes' symmetrical or asymmetrical shape gives them the option of performing both the dual task of product separation and immobilization of the biocatalyst, or just the single task of product recovery. The biocatalyst can be immobilized at the surface of the membrane or within the porous membrane in the first scenario, where the membrane functions as a separation unit and biocatalyst support, using a variety of techniques such as ionic interactions, covalent attachment, entrapment, adsorption, crosslinking, or gel formation. There are several topologies for the membranes, including spiral-wound, flat-sheet, and tubular constructions. These membrane modules and units can then be put together for standardized couplings to the feed, permeate, and retentate lines. Kula and Wandrey (1987), among others, developed and effectively utilized the enzyme-membrane reactor concept. This integration mode has been produced by the classic work on the simultaneous cofactor regeneration using formate dehydrogenase and conversion of L-leucine from α-ketoisocaproate to L-leucine by enzymatic reductive amination method in a continuously run membrane reactor (Wichmann et al. 2000). At Evonik in Germany, a concurrent regeneration of β-Nicotinamide adenine dinucleotide reduced (NADH) with an industrial version of the synthesis of amino acids enzymatically in membrane reactors was reported (Wandrey 2004). Other companies (such as Sepracor, Sigma-Aldrich, and Tanabe Seiyaku) have successfully used the enzyme-membrane reactor in numerous operational processes, confirming the commercial applicability of this amalgamation of membrane filtering and reaction (Ghisalba et al. 2010). The use of soluble components (enzymes, substrates, and products) gives membrane reactors an

advantage because they can be easily replaced when necessary, like when substrate supply is low or when an additional enzyme is needed to maintain the biological conversion rate due to enzyme deactivation effects. Due consideration must be paid to the production of the enzyme in order to ensure that the free enzyme has sufficient operational stability for usage in the enzyme-membrane reactor. The usage of nanofiltration membranes or charged ultrafiltration membrane enzyme reactors, that are using negatively charged membranes of ultrafiltration to preserve the native cofactor in the reactor, are extensions of the conventional enzyme-membrane reactor concept. For the poorly soluble substrate bioconversion, an emulsion membrane reactor with a separate chamber, an enzyme-membrane reactor loop with a conventional ultrafiltration module, a hydrophilic ultrafiltration membrane for emulsification, and a circulation pump can be used (Ghisalba et al. 2010). Examples of the poor soluble molecules include vitamins, antibiotics, enantiomerically pure intermediates cyclodextrins, anti-inflammatory substances, and anticancer medications (Drioli and Giorno 2009).

7.4.5 Reaction and Precipitation/Crystallization

When a product is impeding a bioprocess or decaying while it is being processed, the combination of the two-unit operations, i.e., precipitation and crystallization can help crystalline products develop. These restrictions of product inhibition and degradation can be overcome by removing the intended material by crystallization quickly after it is produced in the bioprocess. Because the thermodynamics of the integrated bioprocess can be modified in comparison to the nonintegrated process, unfavorable equilibrium states in the non-integrated bioprocess can be resolved by integrating the equilibrium bioprocess with product crystallization. The fermentation and enzymatic deacetylation of adipoyl-7-amino-deacetoxy-cephalosporanic acid in a single reactor is an illustration of such an integrated process (Roa Engel et al. 2009). The integrated procedure has commercial benefits such as reduced industrial costs and cheaper capital expenditures since it uses the liberated adipic acid again in the fermentation, avoids the use of bases and acids for pH shifts, has fewer downstream processing units, and produces fewer waste salts. It has been demonstrated that the integrated case offers considerable advantages over the non-integrated case for product formation and crystallization (Buque-Taboada et al. 2006).

7.4.6 Reaction and Adsorption

For improved product recovery or to get rid of reaction components that are inhibitory at higher concentrations, the reaction part (cell culture, fermentation, or biocatalytic reaction) might be integrated with an adsorption operation. Substrates, products, and by-products can all be the cause of these inhibitions, which stop the

process from operating at higher concentrations. Ethanol production can be increased by preferentially adsorbing fermentation inhibitors from biomass hydrolysates (Ranjan et al. 2009). Adsorbent resins are finding widespread use in the recovery of natural products and antibiotics, and it is crucial to characterize various adsorbers under the relevant conditions (Casey et al. 2007). If both the substrate and the product impede the process, a technique to keep the free concentrations of both below the inhibitory concentrations is to adsorb them both. The technique for substrate feed and product recovery has been effective in raising space-time yields. Therefore, selecting the optimal adsorbers may be done using a simple experimental evaluation of substrate and product adsorption as a function of concentration (Hilker et al. 2004). When a reaction is scaled up and then the product is recovered in a substrate feed and product recovery mode, it is preferable to have a complete conversion and scalable, simple operations at large scale.

7.4.7 Reaction and Distillation

Since ancient times, ethanol-rich alcoholic beverages including whiskey, brandy, rum, gin, and vodka have been conventionally recovered through distillation after being fermented from a variety of fruits, grains, or molasses. Since the nineteenth century, there has been varying interest in producing solvents and biofuels through conventional distillation and fermentation. The development of thermal product recovery methods after fermentation, in terms of energy efficiency, cost savings, and ease of integration, is an essential component of the overall production processes. The volatile product can be condensed, the gas stripped, and the stripped gas circulated back to the fermentation are efficient substitutes for complicated product recovery methods. By lowering product inhibition in the liquid phase, a biocatalytic process or fermentation that eliminates the volatile product via the gas phase might increase conversion yields. It is very intriguing to selectively remove volatile compounds like flavors and aromas, but the specialized fine distillation technology necessitates a partition between the distillation step and the fermentation/reaction process (Wohlgemuth 2011).

7.4.8 Reaction and Chromatography

By feeding unreacted substrate back into the bioreaction and by eliminating products, this coupling is effective for accelerating bioprocesses with unfavorable equilibrium conditions as well as for temperature-sensitive and non-volatile components. Although simulated moving bed reactors are more effective, they can only separate two products at a time. It has been demonstrated that annular reactive chromatography is less effective but more practical for collecting multiple products (Ströhlein et al. 2005). A high-fructose syrup method has been enhanced by mimicking

continuous countercurrent contact of the solid adsorbent with the liquid stream using immobilized glucose isomerase. Adsorption columns were progressed without actually moving the solid adsorbent against the fixed inlets and outputs of liquid streams, while the immobilized enzyme reactors remained stationary (Hashimoto et al. 1983). The coupling imposed additional constraints on the chromatographic separation, necessitating a strong product-substrate separation when other bioreaction components are present. In the biocatalytic condensation of acetaldehyde and glycine to L-allo-threonine, it has been revealed that coupling simulated moving bed separation to an enzyme-membrane reactor that was operating continuously and whose efflux contained, besides the amino acids, the cofactor pyridoxal-5-phosphate and acetaldehyde showed only a slight reduction in separation performance with respect to these two amino acids (Makart et al. 2008).

7.5 Product Purification

Although the direct separation of a highly pure product, for instance, by solid–liquid or similar phase separation would be the ideal purification and downstream processing scheme, the reality at the conclusion of the biocatalytic process is comparable to the work-up in organic chemistry. Numerous purification and downstream procedures are unit operations that are reliable, well-recognized, and scalable. Therefore, when developing purification techniques, available approaches are frequently used. Because of their high selectivity, biocatalytic processes often require fewer side products and auxiliary reagents to be eliminated, however, based on the quantity of intended products produced, the type of auxiliary compounds and educts utilized, advanced separation and purification methods may be beneficial. The simulated moving bed chromatography is a pragmatic purification method that can be used on a large scale when two extremely identical products, such as regioisomers, are produced in an equimolar ratio (Ferreira et al. 2000; Kaiser et al. 2009).

7.5.1 Purification of Metabolites

Because chemical synthesis is frequently either impossible or not practical from an economic standpoint, biological cells are an important source for metabolites. Numerous product purification techniques from biochemistry and chemistry can be applied in biotechnology since metabolites are small-molecular-weight molecules. Due to the extensive structural variability of metabolites, a variety of purification methods are required, varying from simple one-step purifications employing one or several of the previously described unit operations to the most difficult separations of homologues with strikingly similar molecular structures (Schügerl 2013; Wohlgemuth 2009). The variety of very similar metabolites produced by biological cells in biocatalytic pathways might present a difficulty for the purification of a

single product as well as an opportunity for multiproduct bioprocesses. The diversity of metabolites can be reduced by upstream and bioprocess technologies, which also makes it easier to purify such metabolite mixtures into highly enriched and characterized single metabolite products. Large-scale synthesis of commercially significant metabolites including vitamins, amino acids, organic acids, and chiral intermediates requires the selection of high-yield mutant strains and the avoidance of the design of labor-intensive separation procedures. There is a strong demand for quick and easy purification processes such as pH changes, extraction, adsorption, crystallization, and chromatographic separations, with crystallization/precipitation and these processes being the most popular ones. Additionally, by using fine distillation, particular liquid metabolite mixtures can have their purifying potential evaluated.

7.5.2 Purification of Lipids

The huge-scale processing of oils, lipids, and fats involves solvent extraction-based techniques, rendering, expelling, screw-pressing, or other techniques suitable for the uses. Procedures for lipid separation and isolation include distillation, urea fractionation, enzymatic processes, crystallization, and liquid chromatography (Gunstone et al. 1994). Adsorption, partition, supercritical fluid, and ion-exchange chromatography are adaptable and practical methods for preparative scale fractionation of lipid mixtures. For chromatography, sephadex G-25 and sephadex LH-20 have been widely utilized. Other effective lipid purification methods include supercritical fluid and preparative high performance liquid chromatography (Fig. 7.4).

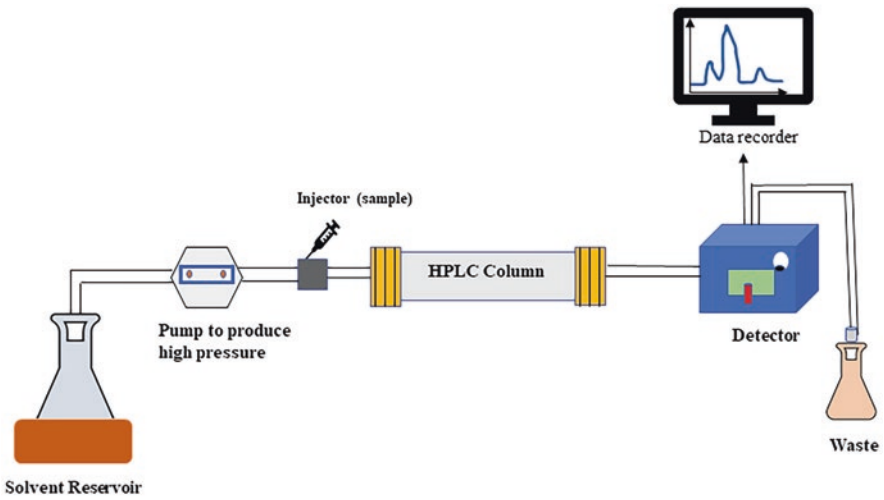


Fig. 7.4 High performance liquid chromatography

7.5.3 *Purification of Proteins*

Although protein isolation and purification have been known for a long time, current difficulties call for better approaches to purify more proteins. There are many different procedures and techniques for protein purification that have been developed over more than a century of experience and expertise (Janson and Rydén 1998), each having specific benefits, drawbacks, and limits. The requirements for the purity and safety of the final product determine how many unit operations are necessary. The establishment of simple and effective protein purification sequences is a significant bottleneck that utilizes the best techniques. Miniature systems have been constructed for the main unit operations and for screening relevant downstream process parameters in an effort to significantly reduce the development time. It is essential to do fundamental research into primary purification techniques since it may result in incredibly new applications that lower this significant portion of production expenses. Since high yields and the conservation of fully functioning proteins throughout protein purification are the objectives, strategies for preventing protein modification or degradation have become crucial. The purification strategy of choice, however, may occasionally involve the separation of inclusion bodies and subsequent in vitro refolding into completely functional proteins.

7.5.4 *Purification of Nucleic Acids*

Traditional molecular biology approaches for RNA and DNA purification include a wide range of well-known laboratory techniques. Several lab-scale techniques for purification of DNA, e.g., density gradient centrifugation with cesium salts, reversed-phase chromatography, and ion-exchange chromatography, can, however, be challenging to transfer to large-scale processes because they require the usage of hazardous and mutagenic reagents, kits, or laborious processes (Ferreira et al. 2000). There are methods for purifying DNA on a large scale that use only generally regarded as safe chemicals and scalable techniques including salting-out, precipitation, and chromatography (size exclusion, hydrophobic interaction, and ion exchange). The quick and easy recovery of plasmid DNA has been made possible by non-chromatographic techniques based on selective precipitation using cetyltrimethylammonium bromide (Murphy et al. 2006) or aqueous two-phase systems containing polyethylene glycol with potassium phosphate or potassium citrate. In a single vessel, the entire process of lysis, precipitation, clarifying, and extraction can be carried out (Frerix et al. 2005). Apposite approaches for large-scale production of RNA employ polyacrylamide-free size-exclusion chromatography (Lukavsky and Puglisi 2004) and increased production and purification of recombinant RNA (Ponchon et al. 2009).

7.5.5 Purification of Carbohydrate Biopolymer

The carbohydrate biopolymers are commonly found in different organisms including plants, animals, fungi, algae, seaweed, mushrooms, bacteria, and yeasts, where they serve a variety of purposes like acting as structural or nutritional reserves. Numerous water-soluble polysaccharides are generated extracellularly in greater quantities, and have a rich history of use in medicine, commercial, and daily life. It is obvious that the purification and analysis present the greatest obstacles for carbohydrate biopolymers because they exhibit the highest structural variety of all biopolymers. Depending upon how the carbohydrate biopolymers are placed in the biological cells, they could need to be deproteinized without affecting the carbohydrate structure. The selective elimination of by-products is necessary for carbohydrate biopolymers with cross-links made of peptide or glycosidic linkages since they are insoluble in aqueous conditions. Polysaccharide production is typically carried out on a very large scale economically. The soluble carbohydrate biopolymers have the advantage of being extractable, but many other similar compounds are generally extracted as well. Although fractionation is based on selective solubilization and precipitation, unlike with low molecular weight compounds, polysaccharides above a certain molecular weight are more difficult to purify and are more difficult to form and isolate due to their higher viscosity, even at low product concentrations. Even at very low concentrations, many physicochemical factors like cations, pH, ionic strength, and viscosity can have a substantial influence on the purification process. It takes the full arsenal of unit operations and case-specific troubleshooting to fractionate structurally different carbohydrate polymers and separate different molecular weight fractions made of the heteropolysaccharide (different building blocks) or homopolysaccharide (same building blocks), only then can purified carbohydrate biopolymers be obtained (Kamerling et al. 2007).

7.6 Product Formulation and Stabilization

The ability of bio-products to be shipped internationally in a biologically active form from the manufacturing place to the application place is essential in our interdependent economy. The characteristics of the formulation and its application, on the one hand, and the minimum shelf-life requirements for products, which are kept permanently under controlled storage or transportation conditions, on the other hand, set the boundary conditions for the formulation and stabilization of the final product. For research, biomedical, and industrial uses, the storage stability of products is crucial, and it is impacted by many of the production process phases that precede before them. However, the final procedures are frequently the most crucial ones (Präve et al. 1984), as they decide the product's form, standardized quality, and contaminants. The ideal product form is a pure powder or liquid, but other preparations, e.g., dried powders as products, immobilized or stabilized solids, and buffered

and stabilized solutions or suspensions are frequently preferred due to factors like production costs, ease of application, or stability problems. The really best standards for stable product formulations have a strong emphasis on inhibiting the molecular mechanisms pertinent to the product degradation pathway. When the product's molecular structure is known, the formulation can be designed to guard against degradation brought on by chemical, microbial, pH, oxidative, and osmotic stress. The addition of protective agents such buffers, anti-oxidants, cryo- or lyoprotectants, and osmolytes can stop changes in the molecular product integrity as well as more physical alterations in solubility, adsorption, and aggregation. Even though there has been a lot of expertise acquired in figuring out the optimal preparation and process to keep and store, stabilize and homogenize, apply and deliver, and keep bio-products in an active state under specific conditions, predictions must be experimentally confirmed for each new preparation. Thus, it is crucial to have process analytical methods that can detect vicissitudes in the molecular integrity and activity of the final product. Biocompatible additives like amino acids, polyethylene glycol, glycerol, and carbohydrates can be helpful for enhancing product production and stability concerns, but it is important to make sure they are compatible with the applications that they are going to be used for. The use of redox buffers or pH is frequently necessary for the quality design for a reliable production process since oxygen or pH can have a significant impact on product stability. Operational stability of products in research, industrial, or biomedical applications is equally crucial. Product stabilization strategies can also focus on creating a unique product design rather than being restricted to those already discussed for storage stability.

7.7 Challenges in Downstream Processing

The manufacturers of monoclonal antibodies are facing a number of difficulties as a result of the current paradigm change in filtration procedures. As manufacturers switch to new processes, keeping prices under control while retaining efficiency requires careful balancing. Due to the fact that some current downstream procedures, including nanofiltration, are incompatible with continuous processing, speed of processing and preventing bottlenecks are two challenges related to efficiency. The production of drugs at higher concentrations has the potential to cause two issues in subsequent manufacturing steps. First, during the separation step, it gets more challenging to reduce the number of contaminants in fluid streams. The processing of bind-elute chromatography and tangential flow filtering is significantly impacted by fluid loss, which is the other issue. Another issue is that processing is significantly impacted by process fluid loss in bind-elute chromatography and tangential flow filtering methods. Because standard resin technology must be regenerated and sanitized after it is used up after a few service cycles, the lack of single-use membranes for chromatography is a problem for manufacturers. This slows down the production process of any desired molecule. Another issue for manufacturers is closed processing. It is more challenging for manufacturers to maintain material

integrity via the connected processes in this method of processing, despite the benefits of higher flexibility, fewer changeover cleaning requirements, and lowered risk of product contamination. Manufacturers also have unique biosafety difficulties with continuous processing. A tighter bioburden management strategy must be implemented since the extended processing time increases the risk of bacterial exposure for processing equipment. Implementing sample collection without endangering the sterility of the industrial environment presents another challenge. Maintaining a sterile environment during this manufacturing process also points to the necessity for more potent techniques of sterilizing devices, e.g., gamma radiation. To verify that filters are virus-retentive, sterile filter certification is necessary (Coffman et al. 2011; Christenson and Sims 2011; González-González and Rito-Palomares 2022).

7.8 Conclusion

Numerous breakthroughs in science and technology, economics, and commerce, as well as in the social, industrial, and environmental spheres, have an impact on the imminent prospects of product recovery technology. But it is undeniably true that technology enhancements in recovery of the intended product with regard to energy conservation, reduction in volatile organic solvent, waste minimization, health, safety, and environmental concerns are advantageous not just for the eco-efficiency of a product recovery scheme, but also for economic aspects e.g., cost reductions for production, since the product recovery costs are significant determinants of the total production costs. Further micro- and macroeconomic advantages on the local and international levels may result from these process enhancements. The reduction of waste during the specific bio-product recovery depends on both technological advancements, such as the replacement of an extraction stage using organic solvents with crystallization, and on discovering new uses for by-products that can transform them from waste into products. In addition to the technical components of waste minimization, the major molecular component of selectivity and the factors controlling it in all product recovery methods are essential for optimizing the recovery scheme to a specific purity of the product of interest. Therefore, novel downstream processing that considers both the engineering and molecular aspects continues to be crucial throughout the entire bioprocess. Due to fluctuating energy prices and the finite supply of some fossil fuels, reducing the amount of input into product recovery bioprocesses becomes not just a cost-saving exercise but also a chance to develop fresh, effective methods for product recovery in order to produce biofuels for cars and aeroplanes profitably. Renewable resources can be used into product recovery processes in a variety of ways, and they also have immense potential as raw materials for bioprocesses. For instance, cellulose and dextran, which are frequently used in chromatography as cross-linked agarose and dextran (such as Sephadex and Sepharose), functionalized cellulose ion exchangers (such as DEAE-cellulose), or as dextran components in aqueous two-phase separation, can be made

from a variety of renewable resources. Product recovery procedures can be improved to be more environmentally friendly, reducing the utilization of volatile organic solvents or reducing the amount of carbon waste burned to carbon dioxide, as it is necessary to reduce the amount of volatile organic carbon compounds and carbon dioxide in the Earth's atmosphere.

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Chapter 8
Principles of Basic and Advanced Techniques Used for the Purification of Enzymes of Industrial Importance



Kongara Hanumantha Rao and S. Haseena

Abstract Recovery and purification are very critical and essential steps in the downstream processing (DSP) of industrially important enzymes and compounds. Purification process involves the enrichment of enzyme proportion by several hundred folds by employing various biophysical–biochemical tools and techniques. The purity and yield depend on the efficacy of suitable methods used in isolation, purification, and packaging of the end product. In the current topic, we have discussed the basic principles and workflow of enzymes isolation and their purification that involves various methods that includes cell disruption, centrifugation, filtration, and precipitation (by inorganic or organic solvents or isoelectric pH) in combination with various types of chromatographic techniques. It is very essential to verify the purity and activity of enzyme at various stages of purification steps by PAGE gel analysis and total activity and specific activity of enzymes. The downstream processing for some of the industrially important enzymes such as amylase, glucose isomerase, lipase, and pharmaceutically important enzymes have been discussed. Thus, understanding the basic principles of enzyme purification will definitely help in designing strategies that can influence the overall economy of enzyme manufacturing industry and eases the handling process.

Keywords Industrial enzyme · Purification techniques · Downstream processing · Enzyme purification · Enzyme source

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8.1 Introduction

Enzymes are found widely in the living systems and are a major factors in the growth and propagation of living systems. The biological reactions in the living organisms are catalyzed by enzymes that generally are known as biocatalysts (Van Beilen and Li 2002). Enzymes catalyze the rate of reaction by lowering the activation energy. They are widely used in various sectors of industry including food and beverage (amylase, invertase, glucose isomerase, pectinase, etc.) (Fasim et al. 2021; Sharma and Beniwal 2014), medicine and health industry to treat various diseases and disorders (example: inborn metabolic disorders/errors Lactase) (Host and Halken 2016), and as therapeutic reagents (streptokinase, protease, lipase, etc.) (Mane and Tale 2015) and as analytical reagents to detect abnormal metabolite levels (biosensor production) (Mehrotra 2016). Realizing the significance of enzymes improving the quality of life, it is imperative to produce enzymes in bulk scale through biotechnological approaches. Bioprocess engineering (Fermentation technology) is a common strategy used in industries in daily manner for large scale production of enzymes. In fermentation technology, downstream processing (DSP) involves all unit operations after industrial fermentation that improve recovery and purity of the final product. This process need to be optimized more to get better production. Due to diverse utilization of enzyme in industrial bioprocessing sector, bulk scale productions of enzymes are performed by the downstream process.

In the 1850s, Louis Pasteur concluded that the fermentation of sugar into alcohol by yeast is catalyzed by "ferments." Later, W. Kühne called these molecules enzymes. The isolation and crystallization of urease by James Sumner in 1926 provided a breakthrough in early enzyme studies. In general, production of enzyme at industrial or laboratory scale, from microbial cells undergoes several steps including sample collection, cell disruption, protein extraction and isolation, enrichment of its concentration by precipitation and ultrafiltration. Further purification of protein is carried out by using different chromatographic techniques followed by drying, formulating, and finally packaging. (Fig. 8.1). Nowadays, microbes are used extensively for the synthesis of enzymes because they are inexpensive, easy to cultivate in fermenters, and availability of gene manipulation methods through genetic engineering. Most importantly, standard quality control checks followed in downstream processing of therapeutic enzymes ensures enzyme product's safety (Gurung et al. 2013; Gervais 2019) for the products derived from microbial factories. Few enzymes have been enlisted in Table 8.1.

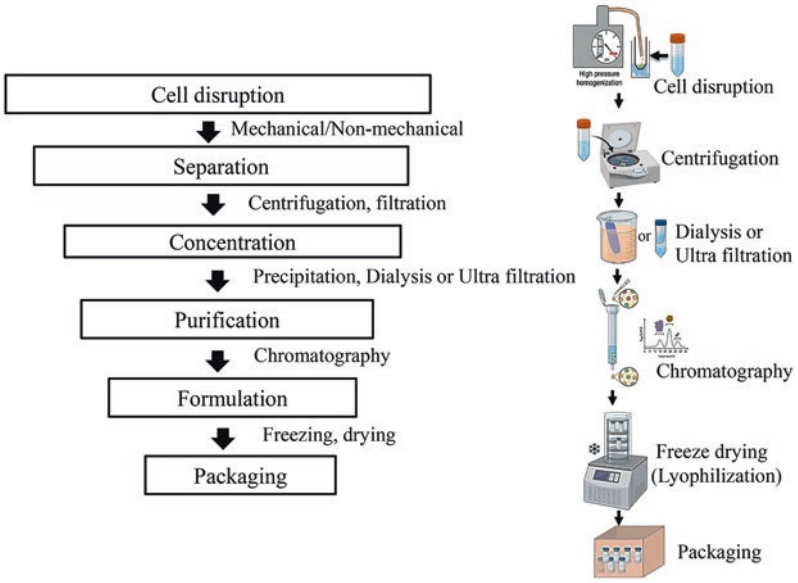


Fig. 8.1 Schematic representation of steps involved in industrial downstream processing of enzymes. Intracellular enzymes extracted by cell disruption through mechanical or non-mechanical means as shown in the flow chart (left panel) and depicted in the pictorial representation (right panel)

8.2 Isolation/Extraction of Enzymes

The isolation procedure depends on the localization and stability of the expressed enzymes. Each proteins have different molecular structure, function, conformation, and isoelectric point due to which it become demanding and require attentive analysis of the quality of the final product (Headon and Walsh 1994). While intracellular enzymes are isolated either by mechanical (solid shear: bead mill, X-press or liquid shear: Ultrasound, French press) or non-mechanical (physical: thermolysis, osmotic shock; chemical: antibiotic, detergents; enzymatic: lytic enzymes) means, extracellular secretary enzymes are isolated by cell separation (Fig. 8.2). The downstream process of cell free extract as well as its extracellular components that are subjected to downstream processing is well described in Figs. 8.1 and 8.2. The current chapter highlighted the fundamental basic principles and workflow of purification that involves various methods including centrifugation, filtration, and precipitation (using inorganic or organic solvents), isoelectric pH in combination with various types of chromatographic techniques. It also illustrates various methodologies, major development in extraction, chromatographic techniques as well as their purification procedure. Various sections of the current chapter provide detailed information regarding basic as well as advanced technique used in the industrial purification of enzymes and their applications.

Table 8.1 Examples of some of the industrially important enzymes and their microbial sources

S. no.	Enzyme name	Microbial source	Enzyme function	Industrial use/reference
1	α-amylase	*Aspergillus niger, Bacillus subtilis*	Converts starch into maltose	Food, fuel alcohol, paper, detergent (de Souza and de Oliveira 2010)
2	Invertase	*Actinomycetales*, *Brevibacterium* sp., *Bacillus cereus* TA-11	Converts sucrose in to glucose and fructose	Beverage, confectionary, bakery, invert sugar, high fructose syrup, etc. (Abolore et al. 2023; Manoochehri et al. 2020)
3	Lactase	*Kluyveromyces lactis*, *aspergillus niger*.	Converts lactose into glucose and galactose	Food industry (Yin et al. 2017)
4	Glucose isomerase	*Saccharomyces cerevisiae, Aspergillus casiellus*	Converts glucose in to fructose	Fructose syrup (Nam 2022)
5	Pectinase	*Aspergillus niger, Saccharomyces, fragilis, Erwinia species*	Polygalacturonases breaks the main chains of pectins and pectin esterases hydrolyses methyl esters	Fruits and vegetable processing, textile processing, coffee and tea fermentation, Prebiotics/ Functional Foods (Haile and Ayele 2022)
6	Lipase	*Bacillus subtilis*, *Penicillium* spp.	Hydrolysis and digestion of triglyceride fats, cholesterol esters, and fat-soluble vitamins	Fat and oil processing, food industry, detergents, pulp and paper industry (Patel et al. 2019)

8.2.1 Purification of Enzyme by Using Centrifugation

The product purification begins with the isolation of desired protein, proceeds and continues until the desired level of purity is reached. The yield in each step is critical because even a small product loss in each purification step adds up to significant losses over the entire downstream process, especially if several purification steps are required to reach a high level of purity. Centrifugation is widely used industrial and laboratory method for taking away of unwanted substance, at particular set of conditions. Some of the widely used centrifugation for isolation of desire enzymes are tubular bowl centrifuge, solid-bowl scroll centrifuge, multi-chamber centrifuge, disc bowl centrifuges, and perforate bowl basket centrifuge. Centrifugation is a process used to spinning up and separation of liquid substance performed by sedimentation of centrifugational force applying to it which is higher the gravitational force (Vitolo 2019). Spinning up and sedimentation of liquid substance on centrifugation depends on particle size, shape, viscosity, density and also rotor speed. Similar substance separated based on the rotor speed applied to it. Tubular bowl centrifuges used for the separation of tiny protein particles, while scroll-type centrifuge for large protein substance.

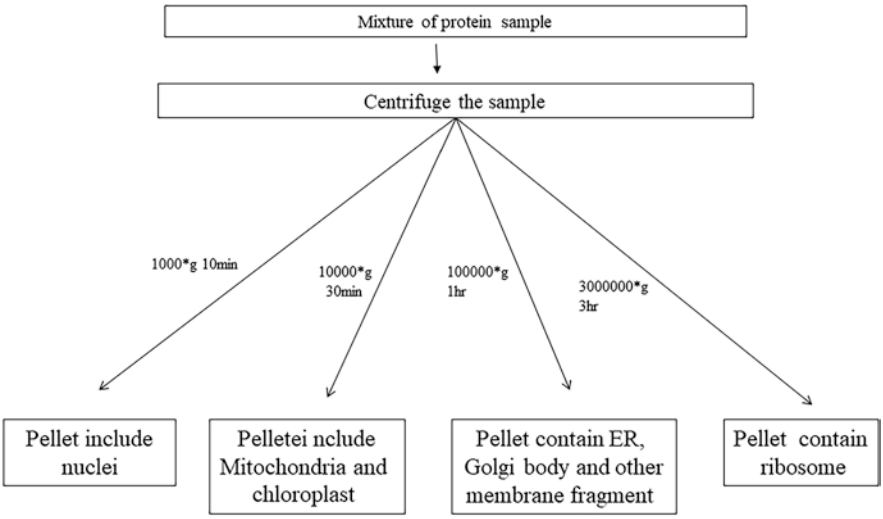


Fig. 8.2 Centrifugation of cell pellet based on the particle size (Adapted from Vitolo 2019). Differential speed centrifugation can be used to isolate organelles

8.2.2 Filtration

Vacuum , pressure and cross-flow filtration are widely used for the industrial purification of enzymes. Diatomaceous earth traps are smaller particles which are useful for the isolation of enzymes though they frequently block the enzyme-containing fluid that might harm the downstream machinery if permitted to get across the filtration.

8.2.3 Precipitation

Insolubular protein precipitation is formed when a reagent is introduced or any circumstances changes. Eventually there are various methods for precipitating a macro-molecule. The ionic strength influences solubility as well. Salting-out is the ionic concentration of the solution increases through the addition of salt makes a decreasing the amount of water quantity therefore allows protein molecules to break down and precipitate (Fig. 8.3). Protein has different stereo-metric and hydrophobic nature, once the amount of ammonium sulfate rises, hydrophobic sites of the protein become more accessible, as a result interaction between hydrophobic regions cause proteins to clump, and perform precipitate. This method precipitates the native proteins without denaturing them. Ammonium sulphate is unusually soluble in cold

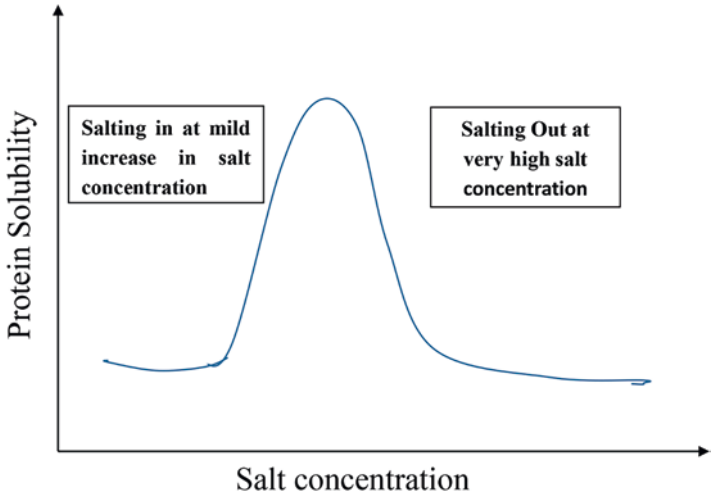


Fig. 8.3 Schematic presentation of salting-in and salting-out of protein. Addition of a small amount of neutral salts like ammonium chloride to a solution can increase the solubility of a protein whereas very high amount of salt result in precipitation of protein by decreasing the availability of water molecules to proteins

buffers. Likewise salting-in of solution participates during ionic concentration decrease via drawing protein to the liquid substance by increasing protein solubility.

8.2.4 pH Variation

It involves the precipitation of protein under its isoelectric point, the pH at which the net overall charge of protein is zero. Proteins frequently precipitate at the point where it becomes isoelectric when their ionic charge is extremely low or high. Proteins have their minimum solubility at their pI. Same amount of positive and negative charges on protein complex, their intracellular repulsion become less, but opposite charge molecules interact and aggregate each other.

8.2.5 Chromatographic Technique

Chromatography involves the constant penetration of a solution across a column loaded via a roughly split material that particularly retards a specific liquid substance (Thömmes and Kula 1995). Downstream process of enzyme purification involves various steps of chromatographic techniques. Chromatography is a strategy that is widely employed because of its capacity to differentiate between complicated mixtures of substances based on tiny changes in physical characteristics

including shape, electrical charge density, hydrophobic along with hydrophilic properties and combinations (Robinson 2015). Enzyme extraction and separation by using chromatographic processes are the most economical, versatile, and high-quality method (Coskun 2016). A standard chromatography instrument established consists of a couple of columns, storage tanks for the phases that move, and the fluids required for recovering, washing, and cleansing the column, as well as the appropriate transports and detectors, all operated by a computing device with appropriate programming.

8.2.5.1 Ion Exchange Chromatography

It is an adaptable, popular, and least expensive column regulation chromatographic procedure with excellent binding potential for the purification of protein, depending on the oppositely charged attraction phenomenon. Each protein possesses both positive and negative charges due to which it has different isoelectric point. The fundamental principle of distinguishing the molecules depends on their electric charge, and efficiency is exceptional at early downstream stages where enormous volumes are handled (Karlsson et al. 1998). Albumin, IgG as well as the products of recombinant DNA technology, such as growth factors and pharmaceutically important enzymes are purified using this technique. For the purification of alkaline protease cationic exchanger are widely used (refer Gupta et al. 2002).

8.2.5.2 Hydrophobic Interaction Chromatography

Depending on the protein hydrophobic nature, different proteins have different degrees of hydrophobic surface following van der Waals interaction. The basic idea of identifying proteins relies on their electrical charge, and efficiency is outstanding during the initial stage of downstream processing where huge amounts are processed. This method is used to control resolution, retention, and efficiency of protein purification (Mahn and Asenjo 2005). Elution of particular enzyme concentration determined via UV light absorbance spectroscopy at a wavelength of 280 nm. Single step purification of an albumin, IgG1, and IgG2a antibodies were well described using hydrophobic interaction chromatography, where different peaks show on detector help to analysis the protein, refer to an excellent review by Manzke et al. (Manzke et al. 1997).

8.2.5.3 Affinity Chromatography

Affinity chromatography is one of the most powerful, highly selective technique used in industrial sector for enzyme production. This technique is used to achieve higher percentage of purity up to 90–98%. The rate of cellular affinity varies according to the sort of ligands utilized, including monoclonal antibodies, particular

inhibitory agents, co-factor, and changed reagent. After removing the contaminating proteins by filtration and thoroughly washing the matrix, the protein of interest is eluded from the matrix by adding high quantities of the unbound ligand in a solution and it can be used at any step of downstream pathway (Firer 2001).

8.2.5.4 Molecular Sieve Chromatography

It is also well known as gel filtration and size exclusion chromatography where separation of molecule takes place according to their size, shape, and molecular weight. In this technique gel beads at stationary phase contains pores that relatively narrow in size and elude smaller molecule out, while the larger molecule completely exclude by space between the column (Walls and Loughran 2011).

8.2.5.5 Thin-Layer Chromatography

In thin-layer chromatographic technique solid absorbent molecules are coated on a glass plate. It is widely used to detect the radioactive, and florescence molecule to visualize the exact position of protein of interest in a mixture of solution (Donald et al. 2006).

8.2.5.6 Paper Chromatography

It is cost efficient and one of the widely used chromatography. In this technique separation of molecule takes place by their differential solubility of mobile and stationary phase. Thick filter paper cellulose used in this technique is a highly saturated bond linked with water.

8.2.6 Total Activity and Specific Activity

The Total activity of enzyme is calculated by measuring the number of moles of product formed or substrate lost per unit time in a given reaction. Enzyme activity is an estimation of the amount of active enzymes present and is dependent on specific circumstances. Enzyme activity gradually increases with increasing concentrations of the substrate until it reaches a plateau and then remains steady. The particular enzyme has an ideal temperature and pH level at which it performs best. As a result, the activity of an enzyme is conditional. Specific activity indicates the number of substrates transformed by enzymatic processes per unit of time per unit amount of protein. Enzyme activity carried out per total enzyme is referred as overall enzyme activity. The specific activity of the enzyme at certain substrates in the case of a pure enzyme will be constant.

8.2.7 Purity of Enzymes

Purification is the critical step in downstream processing of enzyme which involves identification of specific protein of interest from a *crude extract* from the mixture of cells containing many other unwanted components in order to get the most specific activity with the desirable outcome as compared to its initial activity. For the enzyme purification, many procedures are used that perform as main success in the selection of the most appropriate treatment widely. The degree of purification of the enzymes in industry involves various methods as described above for the *purification and isolation* of enzymes production in large scale.

8.2.7.1 Sodium Dodecyl Sulfate Polyacrylamide Gel Electrophoresis (SDS-PAGE)

Different techniques are used for the separation of desired proteins and further to determine their purity at subsequent steps in the process. SDS-PAGE is one of the well-established, simple, rapid, inexpensive, and analytical techniques used to separate the mixture of proteins and estimate the purity and integrity of enzyme at various stages of downstream processing during enzyme purification. Both native and denaturing PAGE gels are used to analyze the integrity and purity of enzyme. Under denaturing conditions of protein, samples are boiled before subjecting them onto SDS-PAGE electrophoresis. Reducing as well as non-reducing phase with a difference in their pH were run on SDS-PAGE (Singh et al. 1991). For observation and analysis, each of the bands is much sharper and adequately separated.

8.2.8 Immobilization of Enzymes

Enzyme immobilization is one of the most promising approaches for exploiting enzyme based processes in biotransformation, diagnostics, pharmaceutical, and food industries. Several hundreds of enzymes have been immobilized in a variety of forms including amylase, lipases, proteases, invertase, etc. Utilization of immobilized enzymes has a great potential in several food industries due to their excellent functional properties, simple processing, and cost effectiveness. Compared to the free enzymes, immobilized forms are more robust and resistant to environmental changes. In this method, the mobility of enzymes is artificially restricted to changing their structure and properties (Bhavaniramya et al. 2019). Various types of nanomaterials such as nanofibers, nanoflowers, and magnetic nanoparticles are significantly used as a support material in the immobilization methods.

8.2.9 Industrial Application of Enzyme

The huge amount of enzymes are utilizing for the development of pharmaceutical, starch conversion, food, detergent, paper, textile, and also in biofuel production (Kirk et al. 2002). Applications of enzymes in the food industry are many and diverse, ranging from texturing to flavoring. Common to more or less all food applications, the enzymes are applied to processed food products as processing agents upstream from the final product. Among all enzymes protease covers nearly 60% of entire worldwide utilization, which participate in pharmaceutical drug designing, food, cleaning, and washing sector by contributing nearly 25% of overall marketing. Several new and advanced technologies were improving for the optimization of enzymes applications, for the use of recombination protein production to provide better efficiency.

8.2.9.1 Role of Enzymes in Food Industry

Various sources of enzyme ancient generation is utilized in food fermentation processing, and new technologies are emerging to improve taste, quality, and durability. Enzymes plays excellent role in food processing industry. For example, enzyme peptidase is utilized for milk coagulation as well as cheese processing which contains additional antioxidant property. Some of the enzymes used in food industry are shown Table 8.1.

8.2.9.2 Role of Enzymes in Detergent Industry

Currently, detergent industry is a rapidly growing sector which utilizes the enzymes such as protease, amylase, cellulases, peroxidase along with detergent as additives in industrial manufacturing of detergent via combination with different hydrolases for the excellent removal of critical stains. Constant new techniques are developing for further improvement in detergent composition to overcome the problem related with efficient cleaning and stain removal.

8.3 Conclusion

The expeditious growth of enzyme used in industrial sector made it possible for versatile application such as in textile, pharmaceutical drug designing, food processing, fermentable alcohol production, starch conversion, research purpose, agricultural sector, and many more. The production of desired enzymes on a large scale in a short span of time may be possible by designing more industrially standardized procedures, and making the entire process of producing enzymes based on their

features. Further, a critical analysis and research are required to decrease expenses incurred in each step. Nearly 500 industrial based applications directly or indirectly depend on enzymes and their by-products (Adrio and Demain 2014). The utilization of bioreactor using downstream processing of enzymes plays vital strategy in industrial sector of enzyme production because expenditures incurred for collecting sample, cell disruption, filtration, precipitation, concentration and formulation of end product, as every step in industrial bioprocessing involves costs for sample collection, filtration, precipitation, concentration, and purification of the end product in the commercial sector of enzyme production. The total efficiencies of enzyme processing benefit from excellent efficiency for output. Purification of proteins may involve from basic ingredients to extremely refined kinds, depending upon their utilization. The purity of the enzyme is increased numerous times while the end product decreases ,and it is often below ten percent of the amount of activity of the initial substance. The normal downstream approach remains advantageous in getting enzymes via acceptable purification and avoiding eliminating undesirable substances because every further step becomes expensive with regard to machinery, personnel, and overall enzyme activity reduction. Commercial enzyme production mainly consists of concentrated broths of fermentation with additions to maintain the activity of the enzyme as it involves a very low transitional purity, while pharmaceutical and analytical proteins are required to be exceedingly purified (Kalyanpur 2002). Excellent purity is capable of being produced by processing the solution of enzymes acquired from earlier crude purification operations through centrifugation, filtering, and precipitating using various chromatographic techniques. Intracellular enzymes, regardless of their source, always require cell disruption as an initial stage. Removal of debris from cells, the macerate is passed through filters, and centrifugation. The resulting solution is processed with a variety of downstream processes based on the level of purity required by the enzyme utilization. As a result, with a tiny filtrate quantity, downstream processes tend to be simpler to handle machines and operate (Belter et al. 1987). The early steps of the downstream procedure involve filtration of cell suspensions, and followed by product precipitation using various chromatographic techniques. Finally, formulation and packing of the end product are carried out. Processing activities such as drying, formulating, and packing remain critical stages for stabilizing an enzyme preparation with a shelf-life of at least 1 year. Enzymes employed in drugs must have a long shelf-life. The goal of industrial downstream processing is to regain the desired product with the requisite criteria in the safest and most effective manner possible while also maximizing recovery output at the lowest possible price and to produce desire proteins in a huge amount.

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Chapter 9
Bioprocess Strategies for the Production of Short-Chain Organic Acids from Microbial Cells



Pratiksha Pundir, Harsha Shabhanani, and Amarchand Chordia Murugan

Abstract The utilization of biotechnological processes to extract valuable compounds from renewable feedstocks is becoming increasingly popular. It is a cost-effective solution for producing fuels, building blocks, and solvents from biomass. Organic acids are naturally occurring metabolites that have a broad range of applications in various industries, including food, pharmaceuticals, and bio-based materials. Short-chain organic acids, specifically those within the C_2 to C_6 range, play a vital role as intermediate metabolites in cells and can also be used as substitutes for certain chemical products in the market. Organic acids can serve as precursors for various bulk chemicals and commercially significant polymers. However, for bio-based production to be cost-effective, microbial production systems must achieve exceptional results in terms of yield, productivity, purity, and substrate consumption flexibility. Although microbial organic acids have been predominantly used in food-based industries, they offer great potential as precursors for various chemicals that could enhance their marketability. Our book chapter delves into the latest advances in metabolic engineering that have facilitated the development of microbial cell factories capable of producing crucial organic acids for industrial use. We also provide an overview of recent progress and challenges in producing short-chain organic acids using microbial cell factories.

Keywords Short-chain organic acids · Microbial cells · Metabolic engineering · Metabolic flux · Catalytic conversion

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9.1 Introduction

Throughout history, the process of fermentation has been a valuable technique utilized to create a diverse range of food and animal feed. Organic acids, found naturally in abundance, have played a pivotal role in this process. However, in the twentieth century, the industrial production of organic acids through chemical processes had a detrimental environmental impact. While the chemical industry has provided numerous useful products, its reliance on non-renewable fossil fuels, environmentally destructive production methods, and the generation of toxic by-products, waste, and nonrecyclable or degradable products has raised concerns. As a result, there is now mounting pressure on the industry to become eco-friendly and more sustainable (Hatti-Kaul et al. 2007). Thus, it is crucial for the chemical industry to adopt a business strategy that prioritizes social, safety, health, and environmental benefits along with its technological and economic objectives to ensure long-term sustainability. Microbial fermentation is a promising method for producing organic acids that are sustainable and eco-friendly. The technique has been successfully used to produce citric acid on a large scale since the 1920s, paving the way for other microbially fermented chemicals like Penicillin in the 1970s. Organic acids are low-molecular-weight compounds that are generated as by-products of microbial fermentation of carbohydrates and substrates similar to them. These acids include carboxyl, sulfonic, alcohol, and thiol, and their functional properties are determined by their acidic groups. Due to their versatility as chemical building blocks, organic acids have become increasingly prevalent and can be produced by various organisms, including bacteria, fungi, and yeast.

Organic acids that are short-chained are categorized into five different groups based on the number of carbons they contain, namely two-carbon (C_2), three-carbon (C_3), four-carbon (C_4), five-carbon (C_5), and six-carbon (C_6) organic acids. Becker and Wittmann (2015) provided this classification. Acetic, glycolic, glyoxylic, and oxalic are some examples of two-carbon (C_2) acids. Meanwhile, three-carbon (C_3) compounds include propionic acid, lactic acid, ethylene carboxylic acid, 3-hydroxy propanoic acid, pyruvic acid, and propanedioic acid. Four-carbon (C_4) acids, on the other hand, include butanoic acid, isobutyric acid, α-ketobutyric acid, succinic acid, tartaric acid, fumaric acid, maleic acid, and oxaloacetic acid. Five-carbon (C_5) acids are butane-1-carboxylic acid, delphinic acid, 2,2-dimethyl propanoic acid, glutaric acid, α-ketoglutaric acid, 2-furoic acid, and itaconic acid. Finally, six-carbon (C_6) acids include caproic acid, adipic acid, citric acid, aconitic acid, and muconic acid, as stated by Sun et al. (2020).

Short-chain organic acids are highly versatile and valuable substances with various applications across various industries. These substances are utilized to produce many essential products, including food, pharmaceuticals, soaps, detergents, polymers, and textiles. For example, calcium and sodium salts of propionic acid are highly effective food preservatives used in human and animal products. Meanwhile, itaconic acid is widely utilized in the production of synthetic fibers, resins, and plastics. In the food processing industry, malic and citric acids are in great demand due to their numerous applications.

In recent years, the development of biotechnological tools has significantly improved the production of organic acids. Microorganisms are being engineered to produce short-chain organic acids as crucial intermediates, and *E. coli*, a model strain, has become a common choice for this purpose. *Aspergillus oryzae*, *Aspergillus terreus*, *Aspergillus niger*, *Yarrowia lipolytica*, *Propionibacterium acidipropionici*, *Propionibacterium jensenii*, and *Saccharomyces cerevisiae* are commonly utilized as host microbes to produce short-chain organic acids, in addition to *E. coli*. Fermentation utilizes various cofactors, including affordable raw materials, as substrates. With these advancements, the production of organic acids has become more efficient and sustainable, providing a reliable supply of these valuable substances for various industries.

Several studies conducted by esteemed researchers, including Hong et al. (2012), Li et al. (2020), Nikaido (2003), and Wu et al. (2015), have shown that the production of microbial organic acids can have a significant impact on various aspects of cellular functioning, like the structure of cellular walls, intercellular organic acid levels, and metabolic fluxes. However, the inherent complexity of the process and intercellular signaling for metabolic regulation can pose challenges that hinder the large-scale production of organic acids (Lee and Kim 2015; Liu and Nielsen 2019; Pérez-García et al. 2018).

Scientists have been working hard to find ways to produce microbial organic acids more efficiently and cost-effectively while also improving their overall performance. Several studies by leading researchers, including Lee and Kim (2015), Liu and Nielsen (2019), and Pérez-García et al. (2018), have highlighted the challenges involved in commercial production. However, significant progress has been made in recent years through the use of innovative approaches that combine systems and synthetic biology. Microbial cells can be modified through genetic engineering to inhibit their natural competitive pathways and incorporate synthetic heterogeneous pathways. This results in a significant boost in productivity and a higher yield of the desired compounds. Notably, experts such as Becker and Wittmann (2015), Chen et al. (2018), Choi et al. (2019), De Paula et al. (2019), Westbrook et al. (2019), and Xu et al. (2020) have made important contributions to the field by optimizing various cellular functions and capabilities. Numerous techniques have been extensively studied by scientists with the aim of enhancing the tolerance of microorganisms to low pH of organic acids. These techniques include blocking by-product biosynthesis pathways, adaptive evolution, cellular and intercellular transporter bioengineering, fine-tuning gene expression, and protein expression profiles. As a result of these approaches, *E. coli*, *Saccharomyces cerevisiae*, and *Corynebacterium glutamicum* have been transformed into microbial cell factories capable of large-scale production of a diverse range of biochemicals. This transformation has been documented in various research studies, including those conducted by Alonso et al. (2015), Cheng et al. (2019), Joung et al. (2011), and Song et al. (2016).

This book chapter provides a comprehensive overview of the latest developments in the biosynthesis of short-chain organic acids. It also discusses various metabolic engineering techniques that can be used to enhance acid production in microbes. Additionally, the chapter examines the challenges and opportunities involved in establishing efficient microbial cell factories for producing short-chain organic acids.

9.2 Biosynthesis of Short-Chain Organic Acids

9.2.1 Organic Acids

Organic acids, particularly short-chain organic acids and short-chain fatty acids, have proven to be valuable natural metabolites in several industries, including food, pharmaceuticals, and bio-based materials. The most commonly found short-chain fatty acids are acetic (C_2), propionic (C_3), and butyric (C_4) acids. These organic acids have a low molecular weight and one or more acidic groups, which makes them versatile and useful in various applications. The research by Annunziata et al. (2020) and Sun et al. (2020) highlights the importance of organic acids and their potential benefits in various industries.

9.2.1.1 C_2 Organic Acids

Acetic acid, also known as ethanoic acid, is a versatile organic compound that has a wide range of applications. It is used to produce various products, including vinyl acetate, acetic acid ester, acetic anhydride, cellulose acetate, and monochloroacetic acid (Vidra and Németh 2017). While it is present in trace amounts naturally in fruits like pears (Naraian and Kumari 2017), it is primarily manufactured through chemical processes. Acetic acid plays a crucial role in various industries, and its production and uses continue to evolve with ongoing research and technological advancements.

Acetic acid production is primarily determined by the species and productivity of microorganisms, according to Raspor and Goranovič (2008). Various microorganisms, including bacteria, yeasts, and a few filamentous fungi, produce this organic acid. The most important bacteria are gram-negative, obligate aerobic, ellipsoidal or cylindrical, and they can be found alone, in pairs, clusters, or chains. They have the ability to oxidize ethanol into acetic acid. Natural acetic acid producers include the aerobic bacteria *Acetobacter*, *Gluconacetobacter*, and *Gluconobacter species*, as well as the anaerobic bacteria Acetobacterium and Clostridium species (Raspor and Goranovič 2008; Vidra and Németh 2017). Acetic acid bacteria under aerobic conditions convert ethanol into acetic acid in neutral and acidic media. This makes them crucial for the vinegar industry (Li et al. 2015; Sengun and Karabiyikli 2011).

Studies have shown that acetic acid has a significant impact on the bacteria responsible for food spoilage, beyond just altering pH levels. The acid can quickly enter cells, lower pH levels, and affect metabolism through the acetate anion (Trček et al. 2015). Acetic acid is a naturally occurring by-product produced by acetic acid bacteria during food processing. These bacteria are commonly used in the food industry and play a vital role in determining the flavor, appearance, and shelf life of food. They convert carbohydrates and alcohol to organic acids during the fermentation process (Sengun and Karabiyikli 2011). The acetic acid production pathway has three distinct forms.

(a) Direct oxidation pathway: Alcohol dehydrogenase and aldehyde dehydrogenase convert ethanol into acetic acid (Sakurai et al. 2012; Vidra and Németh 2017).
(b) The Wood-Ljungdahl pathway: Under anaerobic conditions, CO_2 produced by the conversion of glucose to acetyl-CoA can be transformed to acetic acid (Ragsdale and Pierce 2008; Vidra and Németh 2017).
(c) The glycine synthase pathway: Glycine is generated from CO_2 and then converted to acetic acid through the actions of glycine reductase and acetate kinase (Ljungdhal 1986; Vidra and Németh 2017).

Recent research on acetic acid manufacturing has primarily focused on two key areas.

(a) Coculture technology (Wang et al. 2013): The microbiological coculture technology has been used successfully to produce acetic and lactic acid due to its high efficiency, simple process, and low cost (Park et al. 2013; Tang et al. 2011).
(b) In order to enhance cellular tolerance to acetic acid, it is essential to increase the expression of transport proteins. A range of transporters, such as syn aconitase and ATP-binding cassette transporters, have been identified and developed, as an excess of acetic acid can impede cell growth (Nakano et al. 2004, 2006).

9.2.1.2 C_3 Organic Acids

Propionic Acid

Propionic acid is a clear and scentless liquid that boasts the chemical formula $C_3H_6O_2$. This C_3 carboxylic acid plays a pivotal role in the creation of animal feed, preservatives, drugs, herbicides, and fragrances. In the CO_2 fixation process, C_3 plants generate three-carbon compounds, which account for roughly 85% of all plant species worldwide (Liu et al. 2012). Propionic acid serves as a vital intermediary between formic acid and acetic acid. It can dissolve in water but can be isolated by adding salt (Abdel-Azeem et al. 2023). Additionally, short-chain fatty acids, which can be linear or branched and contain 1–6 aliphatic carbon atoms, are created as well (Rios-Covian et al. 2021). Recently, a comprehensive evaluation of progress in strain engineering, process optimization, and new metabolic pathways, tools, and methodologies for enhancing microbial synthesis of C_3-C_5 diols has been conducted (Vivek et al. 2021).

To produce propionic acid, metabolic engineering techniques mainly involve coculture technology, adaptive evolution, and eliminating unwanted by-products. There are three potential areas for further research: Firstly, due to its strong inhibition of cell growth, propionic acid could be better tolerated by using an effective transporter or an autonomous evolution mutation (Goswami and Srivastava 2001; Zhang and Yang 2009). *Selenomonas ruminantium, Anaerovibrio lipolytica, Propionibacterium shermanii, and P. acidipropionici* are some microbial species that can naturally produce propionic acid by fermenting lactate (Ahmadi et al. 2017;

Liu et al. 2012). Propionic acid was produced at a large scale using genetically modified *Propionibacterium jensenii* ATCC 4868 from glycerol (Zhuge et al. 2013).

To optimize the production of propionic acid, it is imperative to develop innovative genetic manipulation tools for the natural producer *P. acidipropionici*. Among these tools are glycosylase base editors. By utilizing adaptive evolution and inhibiting the production of acetic acid, a novel strain of *P. acidipropionici* was able to generate an impressive 97.0 g/L of propionic acid, as detailed in a study conducted by Zhang and Yang (2009).

There are a couple of limitations to using *P. acidipropionici* for propionic acid production. Firstly, the process of adaptive evolution in *P. acidipropionici* is both time-consuming and costly, leading to higher operational expenses. Secondly, due to the absence of synthetic biology tools for *P. acidipropionici*, redirecting metabolic flux towards the production of propionic acid is challenging. To overcome these obstacles, a 3-hydroxypropionic acid/4-hydroxybutyric acid pathway has been developed that utilizes propionyl-CoA as a metabolic intermediary to produce propionic acid (Berg et al. 2007). However, this pathway requires thorough and systematic analysis, and specific strategies must be employed to improve propionic acid production, such as increasing the intracellular precursor pool of succinic acid (Li et al. 2021). Two methods for producing propionic acid are chemical and microbiological. While chemical methods cause environmental pollution and generate numerous by-products that are difficult to separate, microbiological methods represent a more sustainable option. In recent microbial research, the utilization of glycerol as a substrate for the production of 3-HP has garnered significant attention. *E.coli* (Chu et al. 2015; Kim et al. 2014) and *Klebsiella pneumoniae* (Huang et al. 2013) are the primary strains used for microbial fermentation to generate 3-HP. By enhancing the central carbon metabolism of these strains, their capacity to produce and tolerate 3-HP has increased by 60 times compared to the wild-type strains (Liu et al. 2018b). This has resulted in higher yield, titer, and productivity.

1,3-Propanediol (1,3-PDO) is a three-carbon glycol that has many effective qualities, such as usage as a plasticizer in paints and adhesives, as well as a monomer in the production of polyesters, polyurethane, and plastics. The conventional chemical production of 1,3-PDO involves the hydration of acrolein and the hydro-formylation of ethylene oxide, both of which use expensive catalysts and require challenging environmental factors, including high temperatures and pressure. However, the yield and titers of 1,3-PDO are still lower. As a result, research has shifted towards microbial biosynthesis (Kaur and Sridhar 2023; Maervoet et al. 2011; Pflügl et al. 2014; Vivek et al. 2019b) to address these challenges.

Several types of microbes, such as *Klebsiella*, *Clostridium*, *Lactobacillus*, and *Enterobacter*, have been found to be capable of producing 1,3-PDO. By introducing the necessary genes for 1,3-PDO metabolism, native strains can be genetically modified to create the desirable compound. This process has numerous benefits, especially for downstream processing (Kaur and Sridhar 2023; Maervoet et al. 2011; Vivek et al. 2019b). 1,3-Propanediol (1,3-PDO) is synthesized using glycerol as its native substrate. The process of metabolizing glycerol involves two distinct linked oxidative and reductive processes, according to Gungormusler et al. (2010). The

genes gdrA and gdrB encode glycerol dehydratase enzyme reactivating factors, as noted by Ma et al. (2010). The yield of 1,3-PDO can be increased by introducing these genes into suitable hosts through genetic engineering, as demonstrated by Raynaud et al. (2011), Vivek et al. (2019a), and Vivek et al. (2019b).

Pyruvate

Pyruvate is a molecule that plays a vital role in bacterial processes. It possesses properties of both carboxylic acids and ketones (Wieschalka et al. 2013) and is extensively utilized in the production of food, cosmetics, and pharmaceuticals. Additionally, it serves as a precursor for several other compounds and polymers (Li et al. 2001; Zhu et al. 2008). Bacteria like *E.coli* and *C. glutamicum* can be engineered to produce pyruvate, primarily through the glycolysis pathway (Van Maris et al. 2003). However, the major hurdle in synthesizing pyruvate from glucose is the breakdown of pyruvate by the pyruvate decarboxylase (Pdc) enzyme. The pdc mutant prevented acetyl-CoA production, causing two fermentation problems:

1. Moderate amounts of ethanol or acetate were necessary for aerobic growth.
2. Strains of the Pdc mutant could not grow in synthetic media containing glucose (Van Maris et al. 2004).

Researchers addressed the issue of glucose-limited growth in the Pdc mutant by overexpressing the gene glY1, which encodes threonine aldolase, in *S. cerevisiae*. This enzyme can convert threonine into glycine and acetaldehyde. The generated acetaldehyde successfully addressed the problem of acetyl-CoA supply in the cytoplasm, as it served as the precursor of acetyl-CoA (Van Maris et al. 2003).

The process of producing pyruvate involves utilizing eukaryotic microorganisms, specifically multi-auxotrophic yeasts, that have been extensively researched (Li et al. 2001). Pyruvate is a vital element in the carbon and energy metabolism of all organisms. Through the conversion of pyruvate to acetyl-CoA by pyruvate dehydrogenase, which is then fully oxidized to carbon dioxide and water by the TCA cycle, energy is supplied for cellular growth and upkeep (Zhu et al. 2008).

Schreiner et al. (2005) deleted the aceE gene, coding for the E1 p subunit of the pyruvate dehydrogenase complex (PDHC), in *C. glutamicum* during their investigation of PDHC. The *C. glutamicum* ΔaceE strain obtained from previous studies required acetate or ethanol for glucose growth as an extra carbon source (Blombach et al. 2009; Schreiner et al. 2005). This strain proved to be an excellent starting point for engineering *C. glutamicum*, as it allowed for the effective production of L-valine (Blombach et al. 2007, 2009; Krause et al. 2010b), isobutanol (Blombach et al. 2011), 2-ketoisovalerate (Krause et al. 2010a), succinate, and pyruvate, among others.

Scientists have explored various alternative sources of pyruvates instead of glucose. For instance, lactic acid (Gao et al. 2010), alginate (Kawai et al. 2014), mannitol (Yoshida et al. 2015), and D/L-alanine (Hossain et al. 2016) have been investigated as potential substrates.

Lactic Acid

Lactic acid, also referred to as 2-hydroxypropanoic acid, is an organic compound with the chemical formula $C_3H_6O_3$ and the structural formula $CH_3CHOHCOOH$ (Senthuran et al. 1997). It can be synthesized chemically or produced through carbohydrate fermentation (Senthuran et al. 1997). This hydroxycarboxylic acid participates in various processes, including oxidation, reduction, condensation, and hydroxyl substitution, which lead to a range of lactic acid derivatives (Joshi et al. 2010).

The versatility of lactic acid has made it a valuable asset in a multitude of industries, such as chemical, food, pharmaceutical, textile, and leather. For decades, lactic acid bacteria have been utilized in the food industry to produce this compound through microbial techniques that have been continuously refined since the early 1900s. As a result, lactic acid has attracted an immense amount of interest, as demonstrated by numerous studies (Datta and Henry 2006). During the process of fermenting sugars, lactic acid bacteria require a complex set of nutrients that can result in homo-, hetero-, or mixed acid fermentation, as noted by Hofvendahl and Hahn-Hägerdal (2000). However, the acidity of the environment during fermentation makes lactic acid highly vulnerable, which often requires the use of neutralizing chemicals such as $CaCO_3$, NaOH, and NH_4OH, resulting in additional expenses (Datta and Henry 2006; Hofvendahl and Hahn-Hägerdal 2000). Furthermore, it is worth noting that *Saccharomyces cerevisiae* produces lactic acid in a medium containing glucose at a low pH, according to Dequin and Barre (1994). Fermentation of the *Lactobacillus* and *Sporolactobacillus species* can result in the production of lactic acid. Recently, lactic acid-producing strains of *E. coli* (Chen et al. 2014; Wang et al. 2019, 2015a; Zhou et al. 2012), *S. cerevisiae* (Nguyen et al. 2013; Zhao et al. 2010), and *K. pneumonia*e (Chen et al. 2014; Feng et al. 2014, 2017; Wang et al. 2015a) have been developed. Lactic acid is a main metabolite that is naturally produced by lactic acid bacteria (LAB) (Datta et al. 1995).

It is generally known that the bacteria *Carnobacterium funditum* and *Carnobacterium alterfunditum* may produce lactic acid in microorganisms (Franzmann et al. 1991). Since lactate dehydrogenase can convert pyruvic acid into lactic acid, numerous techniques have been used to increase the production of lactic acid, including preventing undesirable by-products (Chen et al. 2014; Feng et al. 2014, 2017; Wang et al. 2015a; Zhou et al. 2012), optimizing the expression of lactate dehydrogenase (Chen et al. 2014; Wang et al. 2015a; Zhou et al. 2012), dynamically controlling cell growth (Zhou et al. 2012) and product production, modular engineering (Chatzifragkou and Papanikolaou 2012; Wang et al. 2019), transporter engineering (Wang et al. 2015a), and increasing the uptake rate of substrates (Chatzifragkou and Papanikolaou 2012; Wang et al. 2019). System biology and metabolic engineering could be used to further enhance lactic acid generation. Its global metabolic network is analyzed using systems biology, and after that, metabolic engineering strategies are created to control its metabolic flux (Li et al. 2021).

9.2.1.3 C_4 Organic Acids

Butyric Acid

Butyric acid is a type of carboxylic acid with a linear chain structure and a molecular formula of $C_4H_8O_2$. It is characterized by an unpleasant odor and appears as a colorless, oily liquid (Li et al. 2021). Due to its extensive use in various industries, including industrial chemicals, meals, medicines, animal feed supplements, and fragrances, butyric acid has gained significant attention from researchers (Luo et al. 2018). According to IMARC Group, the global butyric acid market size was US$ 177.1 million in 2021 and is expected to increase to US$ 401 million by 2027, with a compound annual growth rate (CAGR) of 14.4% from 2022 to 2027.

Currently, there is a growing interest in using butyric acid as a precursor to biofuels (Dwidar et al. 2012). However, butyric acid is mainly produced chemically on an industrial scale by synthesizing it from petroleum-based feedstocks through chemosynthesis. One common method involves using butyraldehyde, produced through oxo synthesis from propylene, to manufacture butyric acid (Dwidar et al. 2012; Jiang et al. 2018; Zigová and Šturdík 2000). While butter processing can also extract butyric acid, the chemical option is more cost-effective and practical (Dwidar et al. 2012; Zigová and Šturdík 2000). Alternatively, butyric acid can be produced through the fermentation process, making it a more natural and organic option that meets the increasing demand for such products. Additionally, the rising price of crude oil required for chemical synthesis has made the fermentation process a more viable alternative (Zigová and Šturdík 2000).

Many bacterial species, particularly Gram-positive, anaerobic, chemoorganotrophic, spore-forming bacteria, are responsible for generating butyric acid primarily in anaerobic settings. Strains capable of producing butyric acid include *Eubacterium*, *Fusobacterium*, *Butyrivibrio*, *Clostridium*, Butyric *Bacterium*, *Sarcina*, and *Megasphera* (Zigová and Šturdík 2000). However, when it comes to industrial use, *Clostridial strains* are the preferred option because of their higher productivity. *Clostridium* sp. strains offer a more economically viable solution for producing butyric acid or butanol. There are three strains of *Clostridium—C. butyricum, C. tyrobutyricum,* and *C. thermobutyricum*—that have been extensively studied for their ability to produce butyric acid on a commercial scale. Out of the three, *C. tyrobutyricum* is considered the most promising microbe for the bioproduction of butyric acid. It has been proven that this particular strain is capable of producing high selectivity butyric acid, and has the ability to tolerate high concentrations of butyric acid (Dwidar et al. 2012; Sheehan 2022; Zigová and Šturdík 2000). The natural producers of butyrate, *Clostridium species*, employ the buk, hbd, bcd, thl, and crt genes in their butyrate synthesis pathway (Saini et al. 2014).

To improve the butyrate/acetate ratio during butyric acid production by *Clostridium* spp., genetic engineering is employed in multiple approaches. Recent studies are briefly described below:

Joung et al. (2011) conducted a study on metabolic engineering strategies to produce butyrate in recombinant *Escherichia coli* strains. The researchers utilized

gene cloning and expression techniques from *Clostridium acetobutylicum* in *E. coli*, which were then transformed with pTT and pHCB, and named SMJ01 and SMJ02. The study also explored the impact of deleting the pta and ptsG genes in *E. coli B* and *K* strains on butyrate productivity, but no significant improvement was observed. Additionally, the study compared the effectiveness of two different carbon sources, glucose and glycerol, and found that using glycerol as a carbon source resulted in a significant increase in butyrate synthesis. Specifically, the LB medium with 1% (v/v) glycerol produced 0.56 g/L of butyrate.

Clostridium tyrobutyricum is capable of fermenting glucose, xylose, and other carbon sources to primarily produce butyrate, acetate, H_2, and CO_2. However, the presence of acetic acid in the fermentation process reduces the amount of butyric acid produced and increases the cost of separating and purifying butyric acid. Therefore, it is essential to enhance the butyrate/acetate ratio to produce butyric acid economically. A study by Suo et al. (2018) found that *C. tyrobutyricum* can significantly decrease its acetic acid content by increasing the flow of acetyl-CoA to butyrate. This is done by overexpressing the genes for butyryl-CoA/acetate CoA transferase (cat1) and crotonase (crt). The overexpression of butyrate-producing genes helps in controlling the process that produces acetic acid by inactivating the gene that detects acetate formation. This enhances the synthesis of butyric acid from glucose.

In another study by Zhu et al. (2005), the gene linked to the pathway that produces acetate was disrupted in *C. tyrobutyricum* using integrational mutagenesis, resulting in the PTA-deleted mutant (PTA-Em). This increased butyrate synthesis while decreasing acetate production. Although the mutant grew more slowly, it generated 15% more butyrate and 14% less acetate than the wild-type strain in batch fermentation using glucose as the substrate. It was discovered that a mutant strain of *C. tyrobutyricum* exhibited slower growth compared to the wild-type strain. However, it yielded a 15% increase in butyrate production and a 14% decrease in acetate. Further research focused on the wild-type strain's acetate kinase gene (ack) segment. Upon deleting the ack gene, the mutant strain demonstrated a butyrate production rate that was roughly twice that of the wild-type strain (Liu et al. 2008). The mutant strain displayed increased tolerance to butyrate, as proven by growth kinetics analysis conducted with varying initial butyrate concentrations in the medium. This resulted in a 23.5% increase in butyrate production (0.42 g/g glucose) compared to the wild-type strain (0.34 g/g glucose), reaching a higher final concentration of 41.7 g/L (compared to 19.98 g/L for the wild type).

Succinic Acid

Succinic acid is a naturally occurring dicarboxylic acid that belongs to the C_4 family, with a chemical formula of $C_4H_6O_4$. It is produced from liquefied petroleum gas and used in various industries, including food and beverage, medicine, leather and

paint, biodegradable polymers, cosmetics, and inks. Succinic acid is also used as a precursor in the manufacturing of bulk chemicals and resins. However, its current production process involves hydrolyzing petroleum products, which is not cost-effective.

The world market for succinic acid is estimated to be USD 222.9 million in 2021 and is expected to grow at a compound annual growth rate (CAGR) of 9.7% from 2022 to 2030. To make the production of succinic acid more cost-effective and sustainable, a cheaper method needs to be developed.

Succinic acid is a common intermediate in the metabolic pathways of several anaerobic microbes. It plays a crucial role in the production of propionate by anaerobic rumen bacteria (Agarwal et al. 2007; Chotani et al. 2000). Therefore, it is frequently produced as a metabolic intermediate by several anaerobic bacteria.

Researchers have investigated and examined the production of succinic acid from various carbon sources in a wide range of microorganisms. Among these microorganisms, *Actinobacillus succinogenes* and *Anaerobiospirillum succiniciproducens* have garnered the most attention due to their ability to produce a relatively significant amount of succinic acid. While the former is strictly anaerobic, the latter is facultatively anaerobic (Guettler et al. 1999; Nghiem et al. 1997). Lee et al. (2003) discovered a new bacterium called *Mannheimia succiniciproducens* MBEL55E, which produces succinic acid and can be found in the rumen of cattle. Researchers have attempted to establish a biological method of producing succinic acid using fungi such as *Aspergillus fumigatus, Aspergillus niger, Byssochlamys nivea, Paecilomyces variotii, Lentinus degener, Penicillium viniferum,* and yeast *Saccharomyces cerevisiae* under aerobic and/or anaerobic conditions. Additionally, some have engineered *E. coli* to produce succinic acid, likely by overexpressing malate dehydrogenase and nicotinic acid phosphoribosyl transferase (Lee et al. 2003; Liang et al. 2011; Lin et al. 2005; Liu et al. 2011).

Fumaric Acid

Fumaric acid is an important chemical that has numerous commercial applications. It is used as an acidifier in foods and medicines and as a raw material for the production of polymeric resins. Fumaric acid is estimated to be 1.5 times more acidic than citric acid. Studies have shown that fumaric acid can be effective in preventing food-borne illnesses and extending the shelf life of products such as fresh beef (Joye 2019; Yang et al. 2011). In 2020, the market for fumaric acid reached a value of around USD 660.2 million and is projected to grow at a CAGR of 4.8% by 2026, reaching a value of approximately USD 874.7 million. Fumaric acid can be produced by two main methods. The first method involves using various hydrocarbons such as benzene, n-butane, or a mixture of n-butane and n-butene as raw materials, and is known as petrochemical production. The second method involves the conversion of biomass by microorganisms, as detailed in a study by Felthouse et al. (2001).

As fossil fuels continue to deplete and environmental pollution becomes more of a concern, businesses are turning to eco-friendly alternatives. One promising method is the biotechnological production of fumaric acid, which uses inexpensive raw materials like apple juice, lignocellulose, food waste, and glycerol as a source of carbon. Researchers are currently exploring this biochemical route to fumaric acid production with the aim of implementing it on a larger scale. Due to the depletion of fossil fuels and the increasing public concern about environmental pollution, many companies are now exploring eco-friendly alternatives. One such method involves the biological production of fumaric acid using low-cost raw materials such as apple juice, lignocellulose, food scraps, and glycerol. Rigorous research is being conducted to establish the biochemical pathway for the production of fumaric acid, as reported by Ilica et al. (2019) and Straathof and Van Gulik (2012). Producing fumaric acid from naturally occurring *Rhizopus species* can be challenging due to their formation. However, genetic engineering techniques have been developed to produce fumaric acid in non-fumaric acid-producing organisms (Li et al. 2014; Song et al. 2013; Xu et al. 2012a).

Malic Acid

Malic acid is a dicarboxylic acid consisting of four carbons. It is used extensively in the chemical, food, cosmetic, and pharmaceutical industries. Based on the year 2022, the market is expected to be worth $USD 227 million in 2023, with revenues projected to reach $USD 324 million in 2030, exhibiting a CAGR of 5.2% from 2023 to 2030. In the food sector, it is primarily used to enhance the sweetness or sourness of fruit and vegetable juices, jams, candies, carbonated soft drinks, and wines (Zelle et al. 2008). It is also used as a preservative in animal feed and as a flavoring agent and acidity regulator in food (Goldberg and Rokem 2009; Lee et al. 2019). Did you know that malic acid is a natural organic dicarboxylic acid commonly found in vegetables and fruits such as cranberries, guava, grapes, papaya, lingonberries, peaches, passion fruit, pears, plums, pineapple, and raspberries? It gives these foods a pleasant sour taste and is often used in cosmetics like self-tanner, cleanser, and face cream to adjust the pH level. Interestingly, the name “malic” comes from the Latin word for apple, as it is closely associated with this fruit (Gurtler and Mai 2014; Lee et al. 2019).

Malic acid can be extracted from apple juice, but the process is expensive and impractical due to the low L-malic acid content, which typically ranges from 0.4% to 0.7% (Wang et al. 1998). Nowadays, commercial malic acid is produced by hydrating either maleic acid or fumaric acid derived from petrochemicals (Iyyappan et al. 2019; Wang et al. 1998). Alternatively, fumarase enzyme-mediated enzymatic conversion of fumaric acid generated from petrochemicals can also result in malic acid. There are three metabolic pathways that can generate malic acid: the glyoxylate cycle pathway, the tricarboxylic acid pathway, and the reductive tricarboxylic acid pathway (Iyyappan et al. 2019). The production of fermentative fumaric acid from renewable resources may be more economically feasible than

petrochemical production (Straathof and Van Gulik 2012). The most recent technique involves the fermentation of a carbon source such as glucose to produce L-malic acid (Naude and Nicol 2018). Some filamentous fungi can produce high levels of organic acids and exceptional yields close to the theoretical maximum without the use of metabolic engineering techniques (Ochsenreither et al. 2014). The reversible hydration of fumarate to L-malate is catalyzed by fumarase in the fungus *Rhizopus oryzae* (Cao et al. 1996; Song et al. 2011). By using a two-stage dissolved oxygen management technique, *Rhizopus oryzae'*s ability to produce fumaric acid is improved. It has been observed that although some strains of *Aspergillus niger* and *Aspergillus flavus* can produce fumaric acid as a small by-product of fermentation, their potential for fumaric acid production has not yet been explored (Fu et al. 2010). On the other hand, various microorganisms, such as *Aspergillus species, Penicillium species, Zygosaccharomyces rouxii,* and *Ustilago trichophora*, have been identified, characterized, and utilized for the production of L-malic acid (Dai et al. 2018).

9.2.1.4 C_5 Organic Acids

Glutaric Acid

In the production of polyesters and polyamides such as nylon-4,5 and nylon-5,5, the dicarboxylic acid glutarate is widely used due to its linear chain structure, as stated by Zhao et al. (2018). However, the bacterium *P. putida* can only produce a small amount of glutaric acid due to its conversion to acetyl-CoA, according to Revelles et al. (2004, 2005). The davDT gene products are responsible for the conversion of lysine to δ-aminovaleric acid, which is further degraded to glutaric acid (Revelles et al. 2004). Glutaric acid production involves several pathways. The L-lysine catabolism pathway produces the highest level of glutaric acid (Revelles et al. 2004; Rohles et al. 2018). Other pathways include the 2-ketoglutaric acid reduction pathway (Pérez-García et al. 2018), the reverse β-oxidation pathway (Adkins et al. 2013), the carbon chain extension pathway, the 2-ketoglutaric acid decarboxylation biosynthesis process (Wang et al. 2017), and the reverse adipate-degradation pathway (Zhao et al. 2018). Glutaric acid has been produced by modifying the metabolism of *E. coli* (Shin et al. 2016; Wang et al. 2017) and *C. glutamicum* (Kim et al. 2019).

Itaconic Acid

Itaconic acid, also known as methylene succinic acid, is a hygroscopic, crystalline powder that dissolves in water, acetone, and ethanol. The usage of itaconic acid is widespread across various industries such as plastics, paint, and pharmaceuticals. Itaconic acid is known for its painkilling and anti-inflammatory properties. Additionally, its ethyl, methyl, and vinyl esters are polymerized and used to produce

various materials like plastic and elastomers. These materials are then utilized for adhesives and coatings, as explained in a study by Kirimura et al. (2011b). It is used in the manufacture of bioactive chemicals, lattices, textiles, detergents, and cleansers, among other products (Okabe et al. 2009; Willke and Vorlop 2001). The market for itaconic acid is expected to grow at a CAGR of 7.8% from USD 101.4 million in 2022 to USD 110.4 million in 2023–2028.

Aspergillus terreus and *Ustilago maydis* are the most commonly used microorganisms for synthesizing a valuable acid used in industrial processes. Besides these, *Candida* sp.*, Pseudozyma antarctica,* and several *Aspergillus species* can also produce this valuable acid (Vandenberghe et al. 2018). Nowadays, *Aspergillus terreus* is the primary organism used in the commercial synthesis of itaconic acid. It often uses glucose or molasses as a key carbon source, which has been previously processed. The synthesis of itaconic acid by *A. terreus* is influenced by four specific environmental factors, which are the carbon source, pH value, trace element content (e.g., Mn), and dissolved oxygen (DO) concentration. Li et al. (2011) have reported on these factors. Previously, only fungi that consumed sugar, such as *A. terreus* and *U. maydis*, were able to produce large quantities of itaconic acid. However, in recent years, bacteria strains have been genetically modified to produce itaconic acid from glucose (Harder et al. 2018), glycerol (Jeon et al. 2016), and acetate (Noh et al. 2018).

5-Amino Valeric Acid

Oxaloacetic acid is a chemical compound that serves as a building block for the production of 5-aminovaleric acid and glutaric acid. These two chemicals are commonly used in the manufacture of plastics and are important precursors for various industrial products. Furthermore, 5-aminovaleric acid, which is a five-carbon platform chemical, is a major component in the production of bio-based polymers such as nylon 5 and nylon 6,5,5, and is also a precursor to Valero lactam. Glutaric acid and 1,5-pentanediol are manufactured using this C_5 platform (Park et al. 2013; Wang et al. 2021; Wendisch 2020).

The synthesis of 5AVA from L-lysine has been produced using three distinct roots.

The process of lysine metabolism involves different routes and enzymes. Route A is mediated by two enzymes: L-lysine monooxygenase (davB) and 5-amino valeric amidase (davA), as reported by Park et al. (2013). Route B, which involves the reaction catalyzed by L-lysine oxidase (raiP), was described by Cheng et al. (2018). On the other hand, the three-step pathway includes L-lysine decarboxylase (LdcC), putrescine transaminase (PatA), and γ-aminobutyraldehyde dehydrogenase (PatD) according to Jorge et al. (2017).

5-amino valerate is a compound that can be produced from L-lysine using three different pathways. One of these pathways is the 5-amino valerate pathway. Enzymatic techniques are also used to produce this compound, which offers the benefits of easy recovery, high substrate conversion rates, and simple reaction mixture composition (Liu et al. 2014). Research by Park et al. (2013) created

recombinant *Escherichia coli* strains to produce 5-amino valerate from L-lysine using davB and davA genes from *Pseudomonas putida* KT2440. It has been reported that *P. putida* and *P. aeruginosa* naturally produce 5-aminovaleric acid (Jorge et al. 2017; Revelles et al. 2005). Recently, various microorganisms, including *E. coli* (Adkins et al. 2013; Cress et al. 2015) and *C. glutamicum* (Rohles et al. 2016; Shin et al. 2016), have been developed for the conversion of glucose into 5-aminovaleric acid.

9.2.1.5 C_6 Organic Acids

Muconic Acid

Muconic acid is a highly valuable bio-product that is being utilized in various industries. It is used to produce special functional resins, food additives, bio-plastics such as nylon-6,6 and polyurethane (via an adipic acid intermediate), polyethylene terephthalate (PET) (via a terephthalic acid intermediate), textiles, carpets, cans, agrochemicals, and medicines (Eudes et al. 2018; Xenios et al. 2022). Unsaturated carboxylic acid muconic acid, also known as 2,4-hexadecenoic acid, has three isomers of MA labeled as cis, cis-MA, cis, trans-MA, and trans, trans-MA. These isomers are used for manufacturing industrially important compounds, including caprolactam, terephthalic acid, and adipic acid (Eudes et al. 2018). The muconic acid market had a value of US$ 97.3 million in 2021. According to IMARC Group, the market is expected to expand at a CAGR of 6.88% between 2022 and 2027 and will reach a value of US$ 146.6 million by 2027.

Glucaric Acid

Glucose oxidation produces various acids such as glucaric acid, gluconic acid, and glucuronic acid under different conditions (Shendurse and Khedkar 2016). A reaction between glucose and concentrated HNO_3 converts aldehyde and primary alcohol groups to -COOH groups, leading to the formation of saccharic or glucaric acid, a dibasic sugar acid. D-glucaric acid is a naturally occurring aldaric acid that is usually present in trace amounts in fruits and vegetables. Its concentration varies from 0.1 g/kg in grapes and lettuce to 3.5 g/kg in apples and broccoli. It has various applications in the food, medicinal products, and chemical sectors (Shendurse and Khedkar 2016; Walaszek et al. 1996). The global market for glucaric acid was valued at over USD 0.9 billion in 2020 and is expected to grow at a CAGR of 9.9%, reaching approximately USD 1.6 billion by 2026. Metabolic engineering of microorganisms, including *E. coli* (Moon et al. 2009, 2010; Shiue and Prather 2014), *S. cerevisiae* (Chen et al. 2018; Gupta et al. 2017), and *Pichia pastoris* (Liu et al. 2016), is necessary to generate D-glucaric acid from renewable biomass materials for creating a sustainable D-glucaric acid biosynthesis process.

Glucaric acid production has been achieved through various metabolic engineering techniques. These include screening enzymes with high catalytic activity (Chen et al. 2018; Liu et al. 2016), preventing the formation of undesirable by-products (Moon et al. 2010; Qu et al. 2018), controlling cell growth and product biosynthesis dynamically (Moon et al. 2010; Qu et al. 2018), colocalizing pathway enzymes (Moon et al. 2010), and enhancing the soluble expression of myo-inositol oxidase (Reizman et al. 2015).

Adipic Acid

Adipic acid is a highly valuable organic diacid widely used in the production of an array of commercial products, ranging from polyurethanes, nylon, food, to perfume. Adhesives, thermoplastic polyurethane resins, plasticizers, and synthetic lubricants are among the products that rely heavily on adipic acid (Deng et al. 2016). However, the traditional approach to producing adipic acid involves using nitric acid to oxidize a combination of cyclohexanone and cyclohexanol-KA oil, which leads to the production of nitrous oxide, a potent greenhouse gas that causes the depletion of the ozone layer and acid rain. To mitigate these detrimental environmental effects, researchers have developed an eco-friendlier method for producing adipic acid. Instead of using nitric acid, the production of adipic acid using H_2O_2 and sodium tungstate is considered a superior option (Draths and Frost 1994; Isac-García et al. 2016). This method involves the oxidation of cyclohexene and has fewer negative environmental effects than the conventional technique. In conclusion, adipic acid plays a pivotal role in producing a wide range of commercially valuable goods, and it is crucial to develop more sustainable production methods to minimize its environmental impact.

Researchers are showing great interest in the biological synthesis of adipic acid, as they aim to develop methods that are more efficient, cost-effective, and sustainable. Adipate, which is a precursor to adipic acid, can be produced biologically in two ways. The first method involves biological accumulations of cis,cis-muconic acid and d-glucaric acid, which can be further catalyzed to form adipate. The second method involves direct synthesis of adipate from a carbon source (Deng et al. 2016). Researchers have successfully produced adipic acid using biologically derived precursors such as cis,cis-muconic acid or D-glucaric acid, followed by a chemocatalytic conversion process (Niu et al. 2002). Additionally, adipic acid can now be directly synthesized by engineered *Escherichia coli* using a non-natural synthetic pathway (by reversing beta-oxidation) (Kallscheuer et al. 2017; Yu et al. 2014).

Citric Acid

Citric acid, a typical metabolite of plants and animals, is found in citrus fruits (e.g., lime, lemon, and raspberry) and pineapple (Angumeenal and Venkappayya 2013). Citric acid is a colorless substance that easily dissolves in water. Its molecular weight is 210.14 g/mol. Citric acid is a significant metabolic intermediate, initiating the tricarboxylic acid (TCA) cycle. It has various uses in different industries such as

pharmaceuticals/chemicals, food/beverages, electroplating, textiles, and cosmetics. According to research by Angumeenal and Venkappayya (2013) and Moresi and Parente (1999), pure citric acid has a molecular weight of 210.14 g/mol, and Kirimura et al. (2011a) found that it initiates the TCA cycle. Adeoye et al. (2015) and Kirimura et al. (2011a) also reported that it has a variety of uses in the aforementioned industries. The global market for citric acid is expected to grow at a CAGR of 4.1% between 2023 and 2028, with an estimated volume of 3.29 million tons (Adeoye et al. 2015).

As of 2022, the global sales volume of citric acid was recorded at 2.59 million tons. It is expected that the market will grow at a Compound Annual Growth Rate (CAGR) of 4.1% from 2023 to 2028, and reach a volume of approximately 3.29 million tons by the end of 2028.

Three processes were formerly used to manufacture citric acid: citrus fruit extraction, chemical synthesis, and fermentation. Citric acid can be synthesized chemically by using one of the following methods: (a) adding lime (CaO) to citrus, (b) adding limestone ($CaCO_3$) to citrus juice, or (c) adding calcium chloride to citrus juice after the addition of caustic soda (NaOH). In all of these methods, the most important step in the acidification process is the addition of dilute sulfuric acid (Naveen et al. 2022).

A significant amount of the global demand for citric acid is fulfilled through biotechnological sources. This is because research has shown that the production of citric acid using synthetic or chemical methods is not cost-effective in comparison to biotechnological techniques (Adeoye et al. 2015). *A. niger*, a common black mold, is widely used in the industry as a citric acid-producing cell factor for the synthesis of citric acid (Behera 2020; Papagianni 2007).

Selected strains' fermentation capabilities are sometimes enhanced through the use of mutagens. To reduce production costs, several strategies, such as metabolic engineering and optimal fermentation conditions, have been developed. Consequently, CA has become a crucial organic acid (Kaur and Sridhar 2023). Metabolic engineering is an effective method for improving the synthesis of citric acid, while also providing a better understanding of how citric acid metabolism is controlled. This approach has largely replaced traditional mutagenesis methods. In *A. niger*, more advanced metabolic engineering techniques are employed, such as engineering carbon utilization by deleting the α-glucosidase-encoding gene and overexpressing glucoamylase (Wang et al. 2016), enhancing the citric acid biosynthesis pathway through aox1 overexpression (Hou et al. 2018), inhibiting oxidative phosphorylation (Wang et al. 2015b), modifying the mycelium morphology through RNA silencing of the chitin synthase gene (X. Sun et al. 2018), and studying the effects of Mn^{2+} concentration on citric acid production (Dai et al. 2004).

6-Aminocaproic Acid

Epsilon amino caproic acid (EACA), also known as 6-aminohexanoic acid, is a substance that has a similarity to the amino acid lysine. The generic term for aminocaproic acid is AMICAR. EACA has been approved by the FDA to treat bleeding caused by enhanced fibrinolytic activity (fibrinolytic hemorrhage) and to help

patients with traumatic hyphemia reduce constant blood loss. It has been granted orphan drug status due to its effectiveness in treating these conditions (Castillo et al. 2018; Michalets and Harris 2018). The first human trials of EACA, which is an antifibrinolytic agent, were conducted by Japanese scientist Okamoto in the 1950s (Michalets and Harris 2018). Caprolactam, the precursor of nylon-6, can be produced by polymerizing 6-aminocaproic acid (6-ACA). This process opens the ring structure of the monomer (Dahlhoff et al. 2001; Turk et al. 2016), which is a key step in the production of nylon-6. The Beckmann rearrangement is a significant industrial process worldwide, resulting in the production of 1-caprolactam, the precursor to nylon (Dahlhoff et al. 2001). The primary methods of metabolic engineering for synthesis of 6-aminocaproic acid are developing rate-limited enzymes (Cheng et al. 2019), enzyme screening for important pathways (Turk et al. 2016), and combinatorically enhancing the expression of different genes (Zhou et al. 2015).

9.3 Strategies to Improve the Production of Short-Chain Organic Acids

9.3.1 Metabolic Engineering for Improved Microbial Strain

It has been found that microbes are capable of producing chemicals by developing and enhancing their metabolic pathways. As the world becomes more energy and environmentally conscious, biomass has been gaining attention as a renewable energy source (Atsumi et al. 2008). However, there are several variables that can affect the function and overall conversion efficiency of the biosynthetic pathway, such as the availability of precursors, the demand for cofactors, and the optimal expression of the enzymes involved in the process. In some cases, a single strain may not be able to provide the required pathway due to the complexity and length of the recombinant pathway, known as metabolic burden (Wu et al. 2016). According to Chotani et al. (2000), achieving high yields and volumetric productivities requires a focus on the overall metabolic environment within the production host, establishing metabolic pathways that optimize carbon flow from substrate to final product.

Metabolic pathway engineering involves four essential components:

1. Improving the completion of the primary metabolic pathways leading to the desired product by removing rate-limiting, transcriptional, and allosteric regulations;
2. Blocking competing pathways;
3. Increasing carbon commitment from central metabolism to the primary metabolic pathway; and.
4. Modifying secondary metabolic pathways to enhance energy metabolism and increase the availability of critical enzyme cofactors.

Engineering microbial catalysts for fuel and chemical synthesis is a promising and sustainable alternative to traditional petroleum-based technologies (Gallezot 2007; Larson 2006). Scientists have developed various genetically modified microorganisms that can produce biofuels, bulk chemicals, and other valuable compounds (Atsumi et al. 2008; McKenna and Nielsen 2011; J. Park et al. 2012; Thakker et al. 2012; Zhu et al. 2014). Metabolic engineering involves introducing non-native DNA sequences to alter the behavior of microorganisms and create desired enzymes. By mimicking the flavonoid biosynthesis pathway in plants, this bioengineered enzyme system can replicate the process with the help of precursors already present in the host microbe (Cress et al. 2013; Milke et al. 2018). Microbial production of flavonoids has gained significant attention recently due to its low energy requirements, high product purity, and minimal waste emissions such as sulfates, nitrates, or nitrites (Du et al. 2020).

Researchers have made significant progress in developing microbial cocultures that are engineered rationally. The aim is to disperse the metabolic burden of complex and lengthy biosynthetic pathways into multiple strains/species. This approach is expected to improve bioproduction performance by overcoming the limits imposed by metabolic load. Several studies have been conducted in this regard, such as those by Jawed et al. (2019), Jones et al. (2016, 2017, p. 201), Liu et al. (2018a), Saini et al. (2015), Tsoi et al. (2018), and Zhang and Wang (2016). Citric acid, also known as 2-hydroxy-propane-1,2,3-tricarboxylic acid, is an intermediate in the tricarboxylic acid cycle. It plays a vital role in extracting energy from carbohydrates, lipids, and proteins through acetyl-CoA oxidation. In addition, citric acid is an important product in the organic acid industry due to its versatility in various applications such as in food and beverages, pharmaceuticals, cosmetics, detergents, and organic chemical production (Karaffa and Kubicek 2003; Legiša and Mattey 2007). Microbes such as *Absidia* sp.*, Acremonium, Botrytis, Eupenicillium, Penicillium,* and various *Aspergillus* sp. (e.g., *Aspergillus niger, Aspergillus awamori, Aspergillus nidulans, Aspergillus luchuensis,* and *Aspergillus flavus*) have been found to produce citric acid (Tong et al. 2019). Citric acid is produced by bacteria, yeast, and filamentous fungus. Bacteria strains such as *Bacillus* sp.*, Brevibacterium* sp.*, Corynebacterium* sp.*, Klebsiella* sp.*, and Pseudomonas* sp. are known to produce citric acid (Anastassiadis et al. 2008).

A study conducted by Wang et al. (2016) identified several ways to simplify the primary metabolic pathway of *A. niger* for the production of citric acid. This included designing targets to improve carbon consumption, enhance biosynthesis and precursor production, remove by-products, reduce feedback inhibition, and improve the respiratory chain. The techniques used in *A. niger* for metabolic engineering are extensive, including improved carbon source utilization, citric acid production, precursor supplements, alternate respiratory chains, feedback inhibition relief, and by-product elimination. In their previous review, Yin et al. (2015) reported on situations involving universal techniques such as increasing citric acid production and eliminating by-product creation, which were previously discussed by

Ruijter et al. (1999). During the production of citric acid in *A. niger*, one molecule of pyruvate is decarboxylated by the mitochondrial pyruvate dehydrogenase (PDH) complex to produce acetyl-CoA. The other molecule of pyruvate is carboxylated to oxaloacetate by pyruvate carboxylase (PC) in the cytosol with CO_2 fixation. The TCA cycle then uses citrate synthase (CS) to condense oxaloacetate and acetyl-CoA to produce citric acid. Cytoplasm is the last location for the transportation of citric acid, which is then collected in the mitochondria before being secreted out of the cell. Citric acid has a potential molar yield of 1 mol/mol glucose (Wright et al. 1996; Zhang et al. 2013).

Succinic acid is a metabolic fermentation product that is both anaerobic and a tricarboxylic acid cycle intermediate, as stated by Zeikus et al. (1999). It has a variety of applications, including as a chemical precursor, food additive, ion chelator, and medicinal supplement. Maleic anhydride is the primary source of succinic acid production through chemical means. Additionally, the naturally isolated obligate anaerobe *Anaerobiospirillum succiniciproducens* (Davis et al. 1976), as well as facultative anaerobes from the *Pasteurellaceae* family, including *Actinobacillus succinogenes* (Guettler et al. 1999) and *Mannheimia succiniciproducens* (Lee et al. 2002), have been found to efficiently produce succinic acid.

During anaerobic fermentation, *Escherichia coli* produces succinic acid as a minor fermentation product. However, the bacterium prefers to generate acetic acid, formic acid, lactic acid, and ethanol instead of succinic acid. To increase the production of succinic acid and decrease the production of other metabolites, metabolic fluxes must be redirected. In contrast, *M. succiniciproducens*, a bacterium found in the rumen, may produce succinic acid as the primary fermentation product. Therefore, it is believed that separate fermentation routes in two bacteria could induce various fermentation patterns. Lee et al. (2005) reported these findings. By combining genome and pathway comparison, conducting in silico metabolic characterization, and performing knockout tests, it is possible to successfully engineer the metabolism of *E. coli* for increased production of succinic acid. This approach can be applied in general to enhance the metabolic synthesis of other microbial strains as well.

Formic acid is a chemical that finds its application in multiple industries, such as leather and textile processing, cleaning and descaling, pH maintenance, and as a deicer. Salts, on the other hand, are commonly used in various oil field operations to remove carbonates, and in the flue gas desulfurization process (Crable et al. 2011).

Formate is produced in organisms like *Escherichia coli* that metabolize carbohydrates via glycolysis, through the pyruvate formate-lyase enzyme, which is a glycyl radical enzyme (Selmer et al. 2005; Shisler and Broderick 2014). This enzyme has activation and inactivation mechanisms (Sawers and Watson 1998). Many uses of formate and NADH-coupled formate dehydrogenase have been documented. For instance, *Candida boidinii* or other species' NAD^+-dependent formate dehydrogenase can convert 1 mole of formate and NAD^+ into 1 mole of NADH and CO_2 (Berríos-Rivera 2002a, b; Sun et al. 2018). These uses include adding extra redox to improve succinate yield (Balzer et al. 2013; Litsanov et al. 2012), utilizing electrochemically formed formate in chemical production (Li et al. 2012), using formate

as a redox driving force (Shen et al. 2011), and in mannitol formation (Kaup et al. 2004).

Zelle et al. (2008) aimed to investigate a method for metabolic engineering of Pdc- *S. cerevisiae* to produce malate. The process involves decarboxylating pyruvate to oxaloacetate and then reducing it to malate. The method has a theoretical maximum yield of 2 mol malate per mol glucose. The goal of this study is to investigate the effects of three genetic modifications on malate production. These modifications include overexpression of pyruvate carboxylase, highlevel expression of cytosolic malate dehydrogenase activity, and the functional expression of a heterologous malate transporter. Malate transport rates via diffusion across the plasma membrane of *S. cerevisiae* will also be examined (Volschenk et al. 2003). The study will be conducted using a glucose-tolerant, C_2-independent Pdc strain.

There are now three intracellular mechanisms in microorganisms that have been identified; the first is oxidative TCA in which acetyl-CoA and oxaloacetate combine in mitochondria to synthesize one molecule of citrate. Then, via a series of enzymatic reactions, the citrate is further oxidized to L-malate, resulting in the emission of two molecules of CO_2, and due to the emission of CO_2, the total molar yield of the TCA pathway is just 1 mol^{-1} (Kovilein et al. 2020). There are different TCA enzyme isoforms found in either mitochondria or in the cytosol of some *Aspergillus* species (Bercovitz et al. 1990; Ma et al. 1981; Niesche and Haase 2012), which indicates the existence of an extra systolic L-malate synthesis pathway. The second method is reductive TCA (rTCA), which is also known as the cytosolic route that begins with the pyruvate carboxylase (pyc)-catalyzed carboxylation of pyruvate to oxaloacetate and the subsequent conversion of oxaloacetate to L-malate by malate dehydrogenase (mdh), and glyoxylate. A maximum theoretical L-malate of 2 mol mol-1 glucose is created if pyruvate is produced through glycolysis, a non-oxidative process that is ATP-neutral. Three intracellular mechanisms have been identified in microorganisms by Kovilein et al. (2020). The process of the TCA cycle involves various mechanisms for the production of malic acid. The first mechanism is oxidative TCA, which occurs in mitochondria where one molecule of citrate is synthesized by combining acetyl-CoA and oxaloacetate. Citrate is then oxidized to L-malate through a series of enzymatic reactions, resulting in the emission of two molecules of CO_2. Due to this CO_2 emission, the total molar yield of the TCA pathway is only 1 mol^{-1}. Different TCA enzyme isoforms have been found in either mitochondria or in the cytosol of some *Aspergillus* species, indicating the existence of an extra systolic L-malate synthesis pathway, which involves a net fixation of CO_2 (Zelle et al. 2008).

The second method for malic acid production is the glyoxylate pathway, which also involves citrate produced during the oxidative TCA process. In this pathway, citrate is converted into succinate and glyoxylate, which is then combined with acetyl-CoA to produce L-malate. The maximum yield of this glyoxylate pathway is limited to 1 mol mol^{-1}. *Clostridium tyrobutyricum* can increase the yield of butyric acid synthesis through metabolic engineering. When the carboxylation of pyruvate

replaces the oxaloacetate used in oxidative TCA, the highest yield for this noncyclic variant of the glyoxylate cycle is increased to 1.33 mol/mol^{-1}.

C. tyrobutyricum ATCC 25755 is known for producing various end products, with the most important being butyric acid. It also generates acetic acid, H_2, and CO_2 as by-products (Lee et al. 2016; Michel-Savin et al. 1990; Zhou et al. 2014). In *C. tyrobutyricum* ATCC 25755, the EMP (Embden-Meyerhof-Parnas) pathway is responsible for converting glucose into pyruvate, which is further converted into acetyl-CoA by pyruvate: ferredoxin (flavodoxin) oxidoreductase (PFOR) (Lee et al. 2016). Two different pathways are involved in the production of acetic acid and butyric acid in *C. tyrobutyricum* ATCC 25755. The first pathway involves the conversion of certain acetyl-CoA to acetic acid, which is catalyzed by phosphotransacetylase (PTA) and acetate kinase (AK) as mentioned in the studies conducted by Liu et al. (2008) and Zhu et al. (2005). The second pathway involves the conversion of the remaining acetyl-CoA to butyryl-CoA, which is then converted to butyric acid by butyryl-CoA/acetate CoA transferase (CAT). This conversion process involves several enzymes such as thiolase (THL), 3-hydroxybutyryl-CoA dehydrogenase (HBD), crotonase (CRT), and butyryl-CoA dehydrogenase (BCD), as described in the study conducted by Lee et al. (2016) and Liu et al. (2008). During this process, the by-product of acetic acid is reabsorbed into cells by converting to acetyl-CoA, which in turn leads to a relatively high butyrate/acetate ratio and an increased yield of butyric acid.

9.3.2 Improving Catalytic Conversion Rate

While biodiesel has been extensively researched in recent years (Chen et al. 2020; Deneyer et al. 2015; Hermida et al. 2015), higher calorific value hydrocarbons like those found in gasoline (C_5-C_{12}) or volatile alkanes (C_1-C_3) are still preferred for biomass energy (Hussain et al. 2017; Singh et al. 2020). Traditionally, hydrocarbons were produced through the decarboxylation of fatty acids, but this method required high temperatures and resulted in unwanted by-products, making it both costly and hazardous (Santillan-Jimenez and Crocker 2012; Verduyckt and De Vos 2017; Wu et al. 2016).

In a study conducted by Da Silva et al. (2013), it was found that *R. oryzae* lipase (ROL) can be used to convert caryophyllene into a mono-epoxide product through chemoenzymatic oxidation. However, only a 16% conversion rate was achieved, indicating that ROL may play a catalyzing role in the pre-hydrolysis reaction. As a result, *R. oryzae* CGMCC 3.5040 was selected as the enzyme production strain for alkene epoxy catalysis. Although *R. oryzae* CGMCC 3.5040 whole cells can catalyze the α-pinene epoxidation process, they exhibit only moderate catalytic activity and operational stability. To address this issue, the study attempted to immobilize the whole cells using glutaraldehyde (GA) crosslinking to provide structural support and fixation (Monsan 1978).

Sena et al. (2021) produced biodiesel from agro-industrial fatty acids by expressing lipase from *R. miehei* (RML) using a recombinant *Pichia pastoris* strain. They found that treating GA-immobilized RML-PIR1_ increased the strength and durability of the biocatalyst. In addition, treating *Rhizopus oryzae* with GA-crosslinking improved the catalytic activity and operational stability of the whole catalyst in chemoenzymatic epoxidation, in contrast to unprocessed cells (Xu et al. 2023). Scientists from Beisson and colleagues in 2017 discovered a new form of fatty acid photo decarboxylase called CvFAP in *Chlorella variabilis* NC64A. This discovery has opened up a new possibility for the generation of bio-alkanes by directly converting long-chain fatty acids into alkanes. The catalyzed reactions of CvFAP can be used for the production of value-added fine chemicals or fuel alkanes, as stated by Lin et al. (2021) and Zhang et al. (2020).

According to recent research (Lakavath et al. 2020; Rauch et al. 2020), CvFAP, like other flavin-containing enzymes, is vulnerable to photochemical deactivation. To prevent this from happening, it has been recommended by Scrutton and colleagues to keep it in the dark or under red light for as long as possible (Lakavath et al. 2020). However, Wu et al. (2021) have found that the main cause of CvFAP photoinactivation is the photoexcitation of the protein in the absence of a substrate, such as carboxylic acid, which it can convert. They have also proven that by introducing caprylic acid, the photostability of CvFAP can be significantly improved, particularly under illumination conditions. With its one carboxylic group and five hydroxyl groups, gluconic acid is utilized in various industries including paper, food, and pharmaceuticals. Its production through the oxidation of glucose is essential to the current economy. To achieve more control over synthesis, a chemocatalytic system is preferred over the traditional enzyme-based oxidation system as it is more resistant to acidic and basic environments (Wu et al. 2021).

A sustainable and innovative approach to producing gluconic acid is by directly utilizing cellulose. Through a simple reaction, glucose can be transformed into gluconic acid by selectively oxidizing the aldehyde group into the carboxylic group. Noble metals such as Pt, Pd, Au, and their alloys, which are supported by heterogeneous catalysts, are often used for this purpose (Basheer et al. 2005; Cao et al. 2016; Liu et al. 2017; Onda et al. 2008; Zhang et al. 2013). Onda et al. (2008) conducted a study on the catalytic capabilities of Pt nanoparticles loaded on different supports, including activated carbon (AC), MgO, Al2O3, and SiO2, for the oxidation of glucose in a basic medium. The Pt/AC catalyst demonstrated the highest output of gluconic acid at 45% at 353 K under airflow. It was found that Pt nanoparticle dispersion on AC is higher than on other supports, which may be a factor in the improved activity of Pt/AC.

Lactic acid is commonly produced through the fermentation of carbohydrates like fructose and glucose. However, this method has some issues, such as high enzyme costs and poor productivity. To overcome these challenges, many chemocatalysis systems using trioses like glycerol, dihydroxyacetone (DHA), and glyceraldehyde (GLY) have been studied (Dusselier et al. 2013). It is more attractive, but also more difficult, to convert lignocellulosic biomass, specifically cellulose, to lactic acid. This is because the glycosidic connections linking the glucose units need to

be broken, as well as a specific C-C bond within each glucose unit. Unfortunately, there are not many effective catalytic systems for the direct conversion of cellulose to lactic acid (Coman et al. 2015; Wang et al. 2013). However, Sn-based catalysts have been used with some success in the transformation of mono- or disaccharides like glucose and sucrose (De Clippel et al. 2012; Holm et al. 2010; Nemoto et al. 2016; Tolborg et al. 2015).

Formic acid can be produced through oxidative conversion of cellulose-derived carbohydrates. However, it is currently produced using high alkaline concentration and expensive H_2O_2 (Deng et al. 2014). Recently, vanadium-based catalysts, such as H5PV2Mo10O40, have been developed as a less expensive and more efficient alternative for this process (Li et al. 2012; Niu et al. 2015; Wolfel et al. 2011).

In their 2016 study, Deng et al. reported that in recent years, renewable biomass has been successfully converted into various organic acids, ranging from C_1 to C_6, under either aerobic or anaerobic conditions. Using catalytic systems, cellulose is transformed into lactic acid or levulinic acid through a series of tandem transformations involving hydrolysis, isomerization, retro-aldol reaction, dehydration, rehydration, and rearrangement processes (Deng et al. 2016).

There are three main ways to improve enzyme activity. Firstly, we can create new reactions by searching for or designing new enzymes through enzyme screening or de novo enzyme design. Secondly, we can increase the catalytic activity of existing enzymes through protein engineering or directed evolution to produce high concentrations of target chemicals. Thirdly, we can enhance bioconversion to make chemical synthesis more efficient and reduce production costs (Li et al. 2021) (Figs. 9.1 and 9.2).

9.3.3 Enhancing Metabolic Flux

Microbiologists who focus on sustainable production of organic acids through microbial fermentation processes are particularly concerned with regulating metabolic pathways in microbial cells, especially for large-scale production. Over the last decade, experts have utilized molecular biology techniques to gain a deeper understanding of cellular and metabolic functions, as well as various aspects of biochemical processes (Stephanopoulos and Sinskey 1993). Producing bioproducts through cell metabolism engineering requires not only building blocks and ATP energy molecules, but it may also lead to inefficiencies in the cell. These inefficiencies can hurt microbe productivity, restrict their potential, and cause physiological changes. To address these problems, various strategies can be used such as metabolic balancing, enhancing respiration, using dynamic regulatory systems (Cui et al. 2021, p. 202; He et al. 2016), genome engineering (Choi et al. 2019; De Paula et al. 2019; Li et al. 2014; Tran et al. 2022), decoupling cell growth (Lo et al. 2016) from production phases, and co-utilizing nutrient resources. To successfully implement these strategies in industrial settings, innovative approaches such as genome-scale models, carbon metabolic flux analysis, and computational techniques

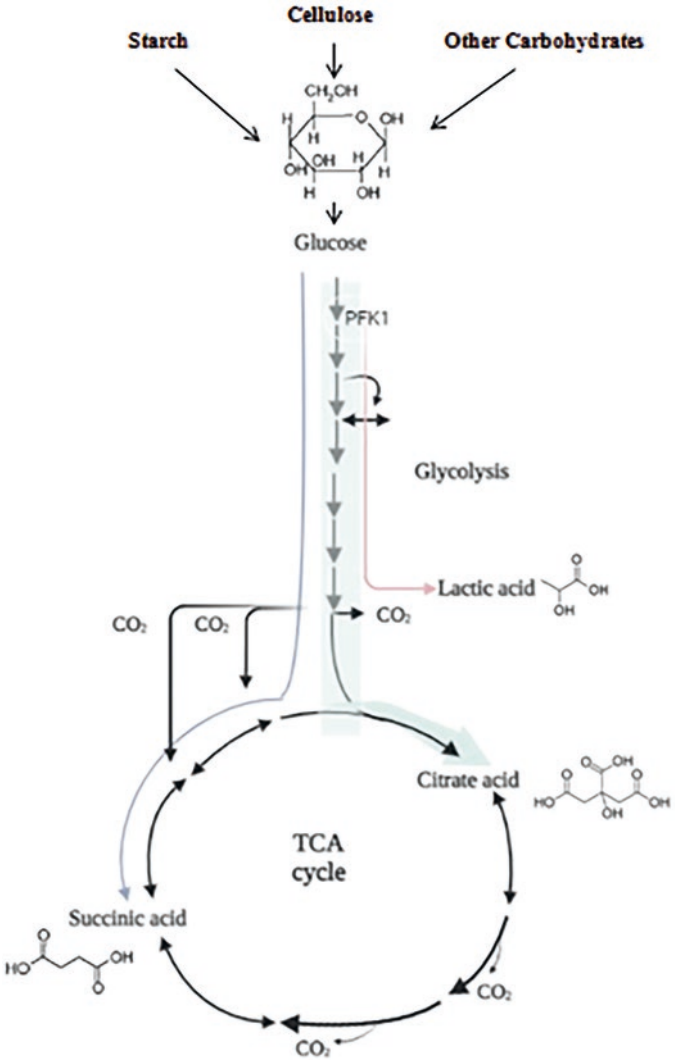


Fig. 9.1 This schematic illustrates the conversion of glucose to citric, lactic, and succinic acid. Only the main pathways are shown for clarity, with one enzymatic conversion per black arrow. The thickness of each colored arrow represents the annual production of each acid. Abbreviations such as pfk I and TCA (phosphofructokinase I and tricarboxylic acid cycle) are used (Papagianni 2007)

should be used to accurately predict and standardize metabolic costs (Wu et al. 2016).

Cells rely on ATP to fuel all their functions, including biosynthesis and maintenance. However, certain biological and physical factors can limit their energy metabolism capacity (Phillips and Milo 2009). For instance, the ratio of cell membrane surface area to volume can impact the efficiency of mass exchange and the

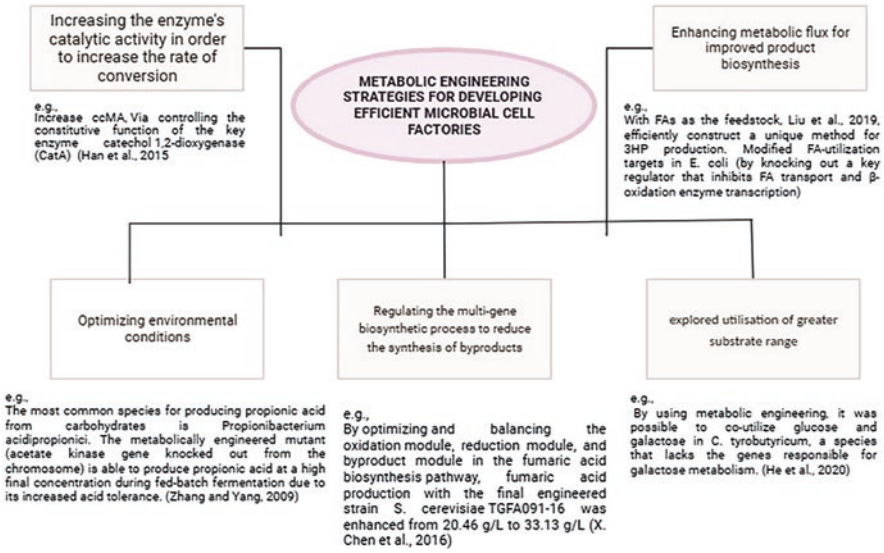


Fig. 9.2 Here is a simplified overview of bioprocess strategies for producing short-chain organic acids from microbial cells

maximum nutrient and oxygen uptake rates (Zhuang et al. 2011). In the case of *E. coli* cultures, glucose uptake rates are capped at around 18 mmol/gDW/h in anaerobic conditions and 11 mmol/gDW/h in aerobic conditions, with 5.3 ATP molecules produced per glucose molecule consumed in aerobic conditions compared to 1.96 ATP per glucose in anaerobic conditions (Chen et al. 2011). This puts a hard limit on cell catabolic rates. Furthermore, ATP synthesis is an inefficient process that generates heat as a by-product, which can further limit cell productivity. For engineered strains, proton gradients may be dissipated through the cell membrane instead of charging ATP synthase, resulting in poor P/O ratios (Wu et al. 2015). As such, ATP consumption can significantly impact cell physiology and bioproduction (Sauer and Bailey 1999; Varma and Palsson 1994; Wu et al. 2016).

A study was conducted using flux balance analysis (FBA) to demonstrate the significant impact of energy burden on biosynthesis (Varman et al. 2011). The results showed that the native metabolism of cells can withstand a certain amount of ATP loss without affecting biosynthesis. However, if the demand for ATP increases, cells can use respiration to maintain their well-being. When the catabolism and respiration mechanisms cannot accommodate further ATP expenditure, the biosynthesis yield suddenly drops, leading to minimal production yields. When a strain's metabolic status declines, it becomes highly sensitive to suboptimal growth conditions, such as low oxygen levels, resulting in unstable performance. Additionally, the study found that the yield plateau phenomenon differs for the three products. Fatty acid synthesis requires both NADPH and ATP and is sensitive to oxygen supply, maintenance loss, and P/O ratios. Isobutanol synthesis only uses NAD(P)H and can be produced during the non-growth phase, making it more likely to achieve

stable industrial production. Acetate synthesis generates net ATP and has the largest yield plateau, making it more likely to achieve robust production. In a recent study involving flux balance analysis (FBA), the impact of energy burden on biosynthesis was investigated. By utilizing a constraint-based model, researchers were able to simulate the effects of ATP metabolism on the yields of *E. coli* fatty acid, isobutanol, and acetate. The findings revealed that the native metabolism of cells can tolerate a certain level of ATP loss without affecting biosynthesis. However, when the demand for ATP increases, cells can rely on respiration to maintain their optimal function. But when the catabolism and respiration mechanisms cannot accommodate further ATP expenditure, the biosynthesis yield suddenly drops, leading to minimal production yields. When a strain's metabolic status declines, it becomes highly sensitive to suboptimal growth conditions like low oxygen levels, resulting in unstable performance. Additionally, the study found that the yield plateau phenomenon differs for the three products. Fatty acid synthesis requires both NADPH and ATP and is sensitive to oxygen supply, maintenance loss, and P/O ratios. On the other hand, isobutanol synthesis only uses NAD(P)H and can be produced during the non-growth phase, making it more likely to achieve stable industrial production. Lastly, acetate synthesis generates net ATP and has the largest yield plateau, making it more likely to achieve robust production. These results can help guide future research on biosynthesis and provide valuable insights for industries seeking to optimize their production processes.

9.3.4 Flux Analysis to Measure Metabolism and its Constrains

Fluxomic measurements in systems biology are a valuable tool for understanding how carbon sources are converted into products. These measurements help determine how cell resources are allocated during biosynthesis (Varman et al. 2011). By analyzing the stoichiometry of biochemical reactions and measuring inflow fluxes, metabolic phenotypes and biosynthesis yields can be predicted using FBA. FBA can also categorize cell energy metabolism into non-growth associated maintenance (NGAM) by resting cells and growth associated maintenance (GAM). While FBA requires constraints to analyze solution spaces and has a high degree of uncertainty, it remains a useful tool. Complementing FBA with C tracing, or 13C-metabolic flux analysis (C-MFA), provides a more comprehensive understanding of functional pathways throughout the metabolic network (Celik et al. 2010; Heyland et al. 2011). C-MFA can validate gene function, identify metabolic engineering targets, and quantify energy flows. When combined with transcriptomics/proteomics and genome-scale modeling, C-MFA can provide a comprehensive understanding of cell responses to metabolic burdens at different cellular levels (He et al. 2014; Sauer 2006). While C-MFA has limitations in synthetic biology, there have been innovative techniques developed to enhance strain performance. Genome-scale MFA and isotopically nonstationary MFA offer quick snapshots of cell metabolism, which can be particularly useful when analyzing samples (Buescher et al. 2015). However,

the process of calculating fluxes through complex metabolism can take a long time. Additionally, this method may not be suitable for fast turnover cycles of individual enzymes. Despite these challenges, C-MFA remains a valuable tool in improving the activity of heterologous enzymes in the production pathway, especially when strain performance is close to industrial levels (García Martín et al. 2015; Gopalakrishnan and Maranas 2015).

9.3.5 Metabolic Engineering Strategy for Addressing Metabolic Limitations

In the realm of traditional fermentation engineering, managing the conditions of the bioreactor—such as pH and oxygen levels—and utilizing various media, like yeast extract, can help reduce the burden on the metabolic process. In the past, metabolic engineers would rely on natural selection through random mutations or adaptive evolution to enhance the synthesis of growth-related products. For example, *E. coli* strains could be "trained" to increase their growth rate by using lactate. Through 250 generations of evolution, certain metabolic nodes—including phosphoenolpyruvate, oxaloacetate, and acetyl-coenzyme A—could be improved (Hua et al. 2007). A potential approach to address biosynthesis pathways involves splitting them into two distinct parts and leveraging two different species to share the metabolic load. The emergence of synthetic biology tools has opened up refined and sophisticated strategies to regulate cell metabolism at various levels, minimizing metabolic load. Transcriptional regulators have recently garnered significant attention as modifying certain regulators can relieve metabolic restrictions and enable the flow of cell carbon and energy (Perrenoud and Sauer 2005). Many microorganisms exhibit catabolite repression, which impacts their carbon, energy, and cofactor metabolism in response to various carbon sources. *E. coli* is a prime example, where glucose is the favored carbon source. By eliminating catabolite repression regulators and upregulating genes essential for pentose metabolism, the cell can rapidly grow while utilizing both glucose and xylose simultaneously (Kim et al. 2015). According to Smolke and Keasling's research, the correlation between DNA, mRNA, and protein levels is significantly nonlinear. Moreover, they observed that mRNA stability and DNA copy number have a notable effect on protein production. Therefore, to construct a pathway efficiently, it is crucial to optimize both plasmid copy number and mRNA processing, as per their findings (Smolke and Keasling 2002). Synthetic biology provides numerous approaches to decrease energy consumption. A prominent area of interest is regulating cell production via transcriptional and protein levels. Dynamic sensors and regulatory systems (DSRSs) are included in these strategies, which implement a biosensor to detect metabolite intermediate levels and regulate enzyme expressions. This is advantageous in avoiding the biosynthesis of redundant RNAs, proteins, and metabolites while enhancing energy and carbon utilization efficiency (Xu et al. 2012b, 2014; Zhang et al. 2012). DSRSs, or Dynamic Pathway

Regulation Systems, are unique in that they have the ability to either encourage or inhibit pathways depending on the specific conditions of cell growth or intermediate metabolite concentrations. This allows for the separation of cell biomass growth and production phases, such as those involved in quorum sensing, so that resources can be allocated towards one primary task at a time. Recently, scientists have been utilizing CRISPR systems to carefully control gene expression and manipulate chromosomes in a methodical manner (Cress et al. 2015). While plasmid engineering has been utilized, it is not without its limitations, such as instability and increased burden due to DNA and enzyme synthesis. Synthetic biology, however, has the potential to reprogram the entire cell genome over time, yielding a highly efficient and minimal cellular structure (Forster and Church 2006; Gibson et al. 2010; Trinh et al. 2008).

9.3.6 Metabolic Modeling for Flux Metabolism Engineering

In order to optimize biochemical productivity when creating engineered strains, it is important to keep in mind the potential impact of metabolic burden. Fortunately, there are a number of helpful review papers available that cover various approaches to model-driven metabolic engineering design. For example, Medema et al. (2012) provide a comprehensive review that outlines a six-step de novo design process. This process involves predicting possible biosynthetic routes, ranking pathways, modeling biochemical function in a network context, selecting pathways, refactoring and integrating pathways using molecular-level design specifications of necessary DNA constructs, and ultimately synthesizing the desired product. While these approaches are effective at prioritizing the specification of biochemical pathway usage needed to achieve theoretical yields of the desired product, it is important to remember that experimental strains may not always meet these theoretical yields due to unaccounted in vivo factors such as overall metabolic burden.

9.3.7 Genome-Scale Modeling in Metabolic Pathways

Genome-scale modeling is a powerful tool for understanding the metabolic capabilities of an organism based on its genetic makeup. By establishing relationships between genes, proteins, and biochemical reactions, GSMs provide valuable insights into an organism's metabolic processes. However, it is important to consider the molecular details of gene expression when using basic GSMs, as this can impact the accuracy of predictions. Recent advancements in GSMs have addressed this issue by incorporating expression data to better account for metabolic costs. For example, an E-Matrix was developed to incorporate synthetic reactions for DNA, mRNA, proteins, and protein complexes in a sequence-specific manner. This approach has led to improved predictive power in ME models of *Thermotoga*

maritima and *E. coli.* Additionally, modularized subsystems have been used to create a whole-cell model of *Mycoplasma genitalium*, allowing for dynamic modeling of each process. The WholeCellKB database has been established to support the development of additional whole-cell models using this approach. These advancements in GSMs provide valuable tools for understanding the complex metabolic processes of organisms at a molecular level.

9.4 Conclusion and Prospects

This chapter introduces microbial engineering and its application in producing C_2-C_6 organic acids, including monocarboxylic, hydroxy carboxylic, amino carboxylic, dicarboxylic acids, and monomeric units for polymers. To optimize biosynthesis, metabolic engineering techniques have been developed, involving reprogramming microbial cell factories like *E. coli*, *S. cerevisiae*, and *C. glutamicum*. However, further research is necessary to establish efficient microbial cell factories for producing C_2-C_6 platform chemicals and improve titer, yield, and productivity.

The evidence suggests that the most efficient way to establish an industrial process for microbial organic acid production is by utilizing natural producers with comprehensive bioprocess engineering. It is important to consider the diversity of species since a natural producer may already exist for most small organic acids (Kern et al. 2007). Although metabolic engineering can address particular issues, like limiting by-product formation or expanding carbon sources used, our focus should be on influencing intricate cellular processes such as transport and morphology. These processes are crucial to the performance of the process, yet we still lack sufficient knowledge to modify them rationally. Therefore, it is crucial to adopt a holistic approach when setting up an industrial process for microbial organic acid production.

One crucial aspect of advancing microbial organic acid production processes for industrial use is to enhance strain robustness and stress resistance. At present, the primary objective is to lower costs, as the current production expenses hinder large-scale manufacturing. Enhancing the stress resistance of microorganisms involved in the process could significantly decrease production costs. Nonetheless, the existing scientific literature lacks a comprehensive overview of this process. Academic research often focuses on isolated issues, disregarding the broader context of industrial production. Fortunately, synthetic biology has leveraged cutting-edge tools to expedite the development of cell factories, including genome editing, expression control, and high-throughput screening. These innovative technologies have enabled metabolic engineering to play a pivotal role in hastening the process of strain development for organic acid production, thereby promoting a more sustainable world.

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Chapter 10
Metabolic Burden Analysis in In Vitro Culture of Plants: A Bioprocess Strategy for Efficient Process Design

Utkarsha Srivastava and Ashwani Mathur

Abstract Plants serve as the source of a myriad of products owing to the therapeutic properties of their secondary metabolites. The increasing commercial importance but low yield of secondary metabolites has now raised interest in research focusing on increasing their yield. Alternatively, studies are conducted for mass propagation of plants. Although bioreactors are used extensively for microbial systems, they are now being considered as suitable alternatives for the production of secondary metabolites and mass propagation of plants. The successful design of a bioprocess strategy to culture different morphological forms of plants, viz. whole plant, callus, friable cell, and suspension culture, largely depend on the successful design of bioprocess parameters. The production of metabolites always imparts metabolic burden on the cell, a bioprocess parameter, scarcely explored for the production of metabolites in higher eukaryotes, especially plants. The metabolic burden is indicative of the ability of a cell to assimilate the resources (nutrients) and use the energy, thus derived, for growth and metabolite production. It is evident that metabolic burden is a parameter to assess the additional load on cells (or organism) in production, and understanding of the parameter will pave the way for designing an efficient bioprocess strategy. Previous studies have explored oxygen uptake rate, as an important parameter, indicative of metabolic burden. The importance of oxygen assimilation rate on plant cell culture and its possible impact on cell metabolism is being presented.

Keywords Plant · Metabolic burden · Oxygen · Assimilation · Bioprocess

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10.1 Introduction

Plant cell and tissue cultures are being extensively used for more than 45 years now for the production of secondary metabolites (Hibino and Ushiyama 1999). These products have found applications largely in pharmaceuticals (Wink et al. 2005), food industry, and cosmetics. Bioreactors are used for the large-scale production of these metabolites although this system is now also used for the expression of recombinant proteins by growing plant cells and tissue cultures. Various factors are known to influence the performance of bioreactors and the manufacturing of desired product. These factors include both the basic characteristics of bioreactors and the limitations presented by the plant cells to growth conditions. The characteristics of bioreactors under consideration encompass the shear stress, aeration, oxygen transfer, and flow. During the growth of plants in bioreactors, they are prone to various stresses which cannot be measured directly but the effect of which can be seen as a decline in the production of biomass or a reduced specific growth rate. The specific growth rate is a measure of the metabolism and the energy requirements of the system which is directly dependent on the availability and uptake of oxygen. This is measured by oxygen uptake rate (OUR), which is indicative of the metabolic burden on the cell under these conditions. A study conducted by Taticek et al. (1990) demonstrated that during initial phases of growth oxygen uptake rate in plants increased and later it decreased which corresponded to a decline in productivity (Taticek et al. 1990). Thus, we can use OUR as a parameter to determine metabolic burden on the cell indicative of the stress faced by plants during growth in a bioreactor.

10.2 Cell Growth and Maintenance

Culturing of plant cells is undertaken for various purposes, the most important being the extraction of secondary metabolites. This requires the production of these compounds encompassing upstream and downstream processing. Owing to the different morphology of plant cells, these organisms require unique approaches for the designing of bioreactors and its operation. Various factors influence the growth of plant cell cultures in bioreactors which need to be considered for the appropriate growth and maintenance of these cultures in bioreactors. The most relevant factors to be considered are (1) morphology of the cell, (2) degree of cellular accumulation and rheology of the culture, (3) wall growth with foaming, (4) sensitivity to shear stress, (5) rate of growth, demand for oxygen, and evolution of metabolic heat; some of which are discussed below.

1. Morphology of the cell and degree of cellular aggregation—The morphology of the cell is an important parameter for defining the culture conditions required for culturing in a bioreactor. For plant cells in suspension cultures the size of the cell

ranges from 50 to 100 μm. The degree of cell aggregation is determined by the morphological features of the species under consideration and the stage of growth coupled with the conditions of culturing. Cell aggregation may or may not be desired depending on the purpose of the bioreactor and the end product; for example cell aggregation is extremely unwanted for plant cell culture bioreactors which aim to produce recombinant proteins. In general, aggregation of cells is undesired majorly because it hinders with the proper aeration and agitation of the system and proper availability of nutrients to individual cells. It also leads to sedimentation of the large aggregates of cells. Large clumps of cells also complicate the handling of culture broth for downstream processing. The morphology of the cultured plant cells also influences the amount of end product obtained as has been reported by Curtis and Emery (1993). Their study focussed on obtaining recombinant proteins from tobacco suspension cultures, the results of which reported that the elongated filamentous cells were more difficult to handle and gave inferior results as compared to spherical cells. This may be attributed to the fact that elongated filamentous cells form a cellular network by entangling, which affects the packed cell volume leading to higher viscosity in the system (Su and Arias 2003).

2. Rheological properties—The rheological properties of media strongly influence the activity of the bioreactor by impacting mixing, transfer of heat and oxygen, and the concentrations of the cells. Various elements influence the rheological properties of plant cell culture. These include cellular congregation, volume of biotic phase in comparison with the cell number, amount of water at cellular level, dimensions of the cell, morphology of cell, and the degree of aggregation. Suspension cultures of plant cells which have high density also have high viscosity generally. It can be attributed to the property of these cultures to reach an exorbitant volume fraction in biotic phase (PCV greater than 50%) also in batch cultures. The media remaining after the culturing of cells and their growth is not generally viscous and follows the properties of Newtonian fluid. Cultures consisting of huge lumps of cells are not as viscous as those containing cells having elongated structure that are intertwined to form a filamentous cellular network (Su and Arias 2003). Shear thinning or pseudoplastic characteristics are exhibited mostly by those plant cell suspension cultures that are viscous with greater density (Doran 1993; Kieran et al. 1997). Under this condition, the apparent viscosity of the culture turns out to be lower under high shear which makes dispersion of bubbles and mixing more effectual in the region of the impeller, which is under extreme shear compared to the regions away from the impeller that experience greater viscosity. This ultimately leads to inefficient mixing and improper mass transfer.
3. Aeration and foaming—Aeration is important for maintaining the proper growth conditions of the cell. Foaming is required to eliminate the adverse after-effects of foam developed in the bioreactors. The cells entrapped in the layers of foam reduce the concentration of volumetric biomass in the culture broth. These cells aggregate together to form a thick layer like a meringue, which adheres to the

vessel wall and the probes. This accumulated layer will become necrotic and secrete chemicals inhibitory in nature which may include proteases or superannuated cell organelles. Under conditions of excessive foaming, foam can overflow and clog the filter of the air vent, thereby raising the chances of contamination. A study by Abdullah et al. (2000) focussed on the examination of strategies used to overcome foaming and accumulation of foam on the wall leading to wall growth in bioreactors. This study identified bubble-free aeration with the use of silicone membrane tubing having thin walls as the only strategy which was capable of eliminating the growth on vessel walls due to foaming completely (Abdullah et al. 2000). A major drawback of bubble-free aeration is that it is not suitable for bioreactors of large volume since the membrane surface to volume ratio is reduced leading to reduced transfer of oxygen on scaling up (Qi et al. 2003; Su et al. 1992). Although the problem of foaming needs to be resolved, the basic design for solving the issue lies in the fact that as the reactor is scaled up geometrically, its cross-section area per volume ratio decreases and the wall growth can be expected to be low.

4. Shear sensitivity—Cells of plant origin are highly susceptive to hydrodynamic shear due to the large size of their vacuoles and the primary cell wall structure although the age of the species also greatly influences the sensitivity to shear stress. The intensity of exposure of cells to hydrodynamic shear coupled with the duration of exposure affects the cellular response of the species to this condition. The basic line of comparison of data for shear studies of a range of plant species lies in their cumulative energy dissipation, hydrodynamic conditions, and physiological indicators (Doran 1999; Dunlop et al. 1994; Kieran 2001), the most prominent index being the cumulative energy dissipation.

10.3 Metabolic Burden: A Kinetic Parameter for Determining Cell Stress

Aerobic organisms require oxygen for proper metabolism, which is directly linked to their respiratory cycle. Information on the uptake of oxygen is indicative of the metabolism in the plant system. Under conditions of controlled growth as in bioreactors, this uptake may become a limiting factor for propagation of plants and the production of secondary metabolites from the same. This is mainly because of the requirement of the organisms for appropriate aeration and agitation to circulate the gases in the system and prevent the development of anaerobic culture conditions. The uptake of oxygen is indicative of the energy requirements of the plant system and is measured in terms of Specific Oxygen Uptake Rate (SOUR), which is associated with the metabolic burden and an increase in the requirement of energy (Chisti 1999b). The metabolic burden is influenced by various parameters both biotic and abiotic.

10.3.1 *Effect of k_La on Metabolic Burden*

Plants are aerobic organisms and therefore have specific requirements for oxygen. Aeration of floral cell cultures in bioreactors serves three major functions of maintenance of aerobic conditions in the culture vessel, obliteration of metabolic heat, and desorption of volatile products. The specific oxygen uptake rate is a measure of the respiration of these floral cells. It depends on various parameters including conditions for cultivation and the phase of growth of the plant. It is found to be of the order of 6×10^{-4} g O_2/g of dry cell weight/min in general (Chisti 1999b; Kieran et al. 1994). The oxygen transfer rate (OTR) gives an overview of the amount of oxygen that is retained by the system after a fixed amount of oxygen is passed through it. A high OTR gives the best conditions for sustaining the aerobic cells since it can efficiently fulfil the respiratory demands and hence enhance the maturation of the plants and the manufacturing of desired metabolites. Excessively high OTR may be a hindrance since incompetence of the system to regulate aeration in the bioreactor will affect flow of both oxygen and carbon dioxide. This coupled with insufficient supply of oxygen will pose a major hindrance in the cell growth and the metabolism of the plants, leading to cell death. In order to prevent oxygen limitation, it is recommended to keep the level of dissolved oxygen greater than the critical oxygen concentration, which has been reported to be 15–20% of air saturation (Kieran et al. 1994).

A study conducted by Pavlov et al. (2005) on the impact of concentration of dissolved oxygen on cell growth over a range of air saturation (10–50%) supports the conclusions made by Kieran (2001) (Pavlov et al. 2005; Kieran et al. 1994). In his research, Pavlov reviewed the effect of dissolved oxygen by varying the constitution of air:nitrogen ratio while maintaining constant speed of agitation. He considered *Lavandula vera* cells for manufacturing rosmarinic acid and reported that the maximal biomass for this metabolite was found at 50% air saturation while 30% air saturation gave the highest volumetric yield of this metabolite. A prominent reduction was observed at 10% air saturation for both the cell growth and the production of secondary metabolites.

Control of limits for dissolved oxygen by regulating aeration at fixed speeds of agitation proves advantageous for cell lines that are sensitive to shear stress (James et al. 2000). Conversely the parameter for dissolved oxygen is not ideal for upgrading a process; instead, parameters to be considered include volumetric mass transfer coefficient (K_La) and the air flow rate, which is related by Eq. 10.1.

$$\frac{dC_L}{dt} = K_L a.\left(C^* - C_L\right) \tag{10.1}$$

where C^* is the concentration of dissolved oxygen at saturation,
C_L is the concentration of dissolved oxygen at time t,
K_L is the mass transfer coefficient,
a is the area of gas-liquid interface per unit volume of liquid (Chisti 1999a).

The coefficient of mass transfer (K_La) is dependent on both agitation and aeration. Therefore, the values of K_La at the start are measured for various reactors. A study by Zhang and Zhong (2004) reported that the initial K_La had a major impact in governing the growth of cell and the production of secondary metabolite ginseng, polysaccharides, and saponins for a bioreactor system employed for the cultivation of *P. notoginseng* cells of high density in 3 L Centrifugal Impeller Bioreactor (CIB), which achieved the highest productivity at 30.2/h. The information provided by K_La was further used to upscale the process from 3 L centrifugal impeller bioreactor to 30 L (Zhang and Zhong 2004). The ideal rate of air flow is presumed to fall in the scale from 0.1 to 0.5 L/min for cell suspension cultures for plant systems although increase in the rate of flow during the process of cultivation to be equivalent to the physiological state of the cell could be a major advantage (Eibl and Eibl 2008). Haas et al. (2008) utilized this scheme of aeration having a rate of flow of 0.5 L/min till the second day of cultivation and 1.0 L/min for the consecutive days to successfully establish increased growth of *H. annuus* cells, which also allowed the cumulation of biomass up till 15 g dry biomass per litre (Haas et al. 2008).

The availableness of oxygen dynamically influences the performance of aerobic processes. Additionally, these bioprocesses are generally carried out in liquid media for plant tissue culture. Gases are known to be less soluble in liquid since they also contain ionic salts besides nutrients in case of bioreactors, affecting the rate of dissolution of oxygen in liquid media of bioreactors and the dissolved oxygen concentration in the system. The rate of uptake and utilization of oxygen by plants is high which demands a greater availability of dissolved oxygen and better agitation and aeration. This makes oxygen transfer the crucial and rate-limiting step in the bioprocess system of aerobic organisms (Gupta and Rao 2003; Gogate and Pandit 1999). Change in the availability of oxygen drastically influences the kinetics of metabolism while an increased availability of dissolved oxygen enhances the yield of secondary metabolites (Vardar and Lilly 1982).

The oxygen absorption rate (OAR) was employed previously to study the effect of dissolved oxygen on the growth of organisms in bioreactors. It is described as the milli-mole of oxygen which is absorbed for every 1 L of solution during a period of 1 min (Tautorus et al. 1994). Presently OTR is extensively used for the rate of transfer of oxygen and the numerous resistances encountered during the transfer of oxygen from a pure gaseous medium to liquid medium in bioreactors (Garcia-Ochoaa et al. 2010). In gas-liquid system, which is fully agitated, OTR is deduced by observing the balance of oxygen across the liquid-gas interface. The driving force, which is represented as $C_G/H - C_{o_2 \cdot L}$, is proportional to the OTR and can be represented by Eq. 10.2.

$$\mathrm{OTR} = k_L a\left\{\left(\frac{C_G}{H}\right) - C_{O_2}\right\} = k_L\, a\left(C^*_{O_2} - C_{O_2}\right) = k_L a\left((p_{O_2 G} - p_{O_2 L}\right) \qquad (10.2)$$

k_L gives a measure of the opposition for the transfer of oxygen across the liquid and gas interface for a bioreactor system with *a* as the specific area of the surface.

C_G is the concentration of oxygen in the gaseous phase while $C_{o_2,L}$ is the concentration of oxygen in the liquid phase, and H is Henry's constant (Van't Riet 1975).

In the liquid phase the concentration of oxygen at the liquid-gas interface is given by the term $C_G/H = C^*_{O_2,G}$ (Suijdam et al. 1978). L_{O2} is the solubility of oxygen while $pO_{2.G}$-$pO_{2.L}$ is the difference in the driving pressure (Maier et al. 2004). k_La greatly affects the OTR. The maximum oxygen transfer rate (OTR_{max}) can be obtained by Eq. (10.3) considering that the resistance offered by liquid phase is negligible. In this condition:

$$OTR_{\max} = k_L a C^*_{o_2} = k_L a\, p_{o_2 G} \tag{10.3}$$

10.3.2 Oxygen Uptake Rate (OUR) and Its Effect

The rate of oxygen uptake by the cells (plant system) is related to the concentration of dissolved oxygen (C) by Eq. (10.4).

$$\text{OUR} = Q_{o_2} \text{X} = k_L a\left(\text{C}^* - \text{C}\right) \tag{10.4}$$

where X denotes the concentration of cells on the basis of dry weight and C^* is the concentration of dissolved oxygen at saturation at 25°C.

If the growth of the culture is restricted only by C but it is much smaller than saturation constant (K_m), the specific rate of growth (μ) can be obtained by Eq. (10.5).

$$\mu = \frac{1}{X} \cdot \frac{\text{dX}}{\text{dt}} = \frac{\mu_m C}{K_m + C} = \frac{\mu_m C}{K_m} \tag{10.5}$$

From Eqs. (10.4) and (10.5) above, the specific rate of growth can be rephrased into Eq. (10.6).

$$\mu = \frac{\mu_m}{K_m}\left(C^* - Q_{o_2}\frac{X}{k_L a}\right) \tag{10.6}$$

From this equation it can be concluded that μ is directly related to the function of X/k_La if the specific rate of oxygen uptake (Q_{o_2}) is constant.

A study by Kondo et al. (1989) on the hairy roots of carrots in various bioreactors concluded that the growth of hairy roots was affected by variations in OUR with the maximum growth rate of 0.63 g/L (Kondo et al. 1989). The type of bioreactor used also influences the OUR. Studies conducted on rotating drum bioreactors containing hairy root cells showed intermittent adherence of the cells to the vessel wall as the cells were lifted up and dropped back in the medium as the drum rotated (Astarita 1967). Cells were disrupted due to the intermittent uplifting and dropping periods, which slowed the rate of growth of cells. The cells, mostly, remained attached to the walls and

regained growth upon dropping down. This reactor is found to be associated characteristically to slow the growth rate. To overcome this condition, a foam sheet of polyurethane is used which serves as a support for the cells that are attached to the bioreactor wall (Linek et al. 1970). This study therefore demonstrated that specific growth rate and OUR are related while OUR is directly related to an increase in biomass. The cultures demonstrate a decrease in the accumulation of biomass at highest levels of oxygen which is relative to the saturation specific growth rate. Reduction in growth rate is an indication of oxygen toxicity which requires statistically substantial data to overcome its influence, which can be achieved by including the inhibition term, C_{o_2}/k_i, in the denominator for saturation kinetics; the equation for which is mentioned below.

$$v = \frac{v_{max} + S}{K_m + [S]} \tag{10.7}$$

where S is the concentration of substrate.

K_m is the Michaelis constant,

v_{max} is the maximum velocity of the system, and

v is the reaction velocity.

10.3.3 Relationship Between OUR and Cell Biomass

Respiratory activity is used as a measure for determining the metabolic conditions and, therefore, the biomass concentration. Growth rate and biomass concentration are associated with the OUR by Eq. (10.8) mentioned below.

$$\text{OUR} = \frac{1}{Y_{X/O}} \frac{dX}{dt} + m_0 X \tag{10.8}$$

In this equation, $Y_{X/O}$ gives the yield of biomass from O_2, and m_0 is the coefficient of maintenance for O_2.

The above equations can be differentiated to signify sigmoidal growth (dX/dt), which is then substituted in Eq. (10.8) and is related to the OUR and X as shown in Eq. (10.9) below.

$$\frac{dX}{dt} = \mu \cdot X \left(1 - \frac{X}{X_m}\right) \tag{10.9}$$

This equation can also be represented as Eq. (10.8), which is related to the OUR and X by Eq. (10.10). As given below.

$$\text{OUR} = \frac{-1}{Y_{X/o}} \mu \frac{X^2}{X_m} + \left(\frac{1}{Y_{X/o}} \mu + m_o\right) X \tag{10.10}$$

The above equation can also be used to experimentally obtain the concentration of biomass and the OUR (Kocabas et al. 2006).

10.4 OTR and its Influence on OUR

A continuous supply of oxygen is required to achieve the productivity limit for aerobic cultures, which requires proper aeration and agitation of the system. This is important to promote efficient transfer of mass from liquid medium and thorough mixing in the bioreactor vessel, both of which are necessary to accomplish optimum growth and product formation. The study of oxygen mass transfer depends on several parameters, which include transport parameters as k_La and the consumption of oxygen by the organism.

The values of both OTR and OUR vary depending on biomass and conditions for transport of oxygen, which are defined by the hydrodynamic conditions of the bioreactors. OUR is also dependent on the oxygen mass transfer (Calik et al. 2004; Gomez et al. 2006). OUR is also influenced by the speed of the stirrer and the phase of growth (Garcia-Ochoa and Gomez 2005). OUR increases in the lag phase of growth specifically during the exponential stage after which it decreased slowly at the stationary stage due to decline or inhibition of metabolic activity of the cells while the least values of OUR were obtained for oxygen transport limitation conditions, although higher oxygen transport conditions may also negatively influence OUR due to the inhibition of growth by oxygen. qO_2 increases linearly with an increase in biomass concentration (CX) upon increasing the speed of the stirrer. Specific OUR reaches a minimum value when the CX attains its maximum value (in stationary phase) and is associated with the maintenance of the culture. The coefficient for oxygen consumption for maintenance (mO_2) and the yield of oxygen (Y_{XO}) are dependent on the conditions of transport of oxygen prevalent in the broth, which is, in turn, dependent on the speed of the stirrer and thereby the hydrodynamic conditions.

10.5 Methods to Measure OTR

10.5.1 Sulphite Oxidation Method

The basic principle of this method lies in the oxidation of sodium sulphite to sulphate in the presence of oxygen. Various metal ions can act as catalysts for this reaction including Co^{+2}, Fe^{+}, Cu^{+}, and Mn^{+} (Linek and Vacek 1981; Harris and Roper 1964). Eq. (10.11) below depicts the reaction when metal ions are employed as catalysts (Hermann et al. 2001).

$$SO_3^{2-} + 0.5O_2 \xrightarrow{\text{Catalyst}} SO_4^{2-} \tag{10.11}$$

When the reaction in the boundary layer of liquid side is slow, the OTR is measured in the non-accelerated system. Under this condition the Hatta number (Ha) is less than 0.3 where the Hatta number is calculated by the formula given in Eqs. (10.12) and (10.13).

$$\text{Hatta number}\left(\text{Ha}\right) = \frac{\text{reaction rate}}{\text{mass transfer rate}} \tag{10.12}$$

$$\text{Hatta number} = \frac{\sqrt{\left\{\frac{2}{n+1}\left(C_{0_2}^{*}\right)^{n-1} D_{O_2}\right\}}}{k_L} < 0.3 \tag{10.13}$$

Here *n* is the order of reaction with respect to oxygen,
K_L is the reaction constant for n^{th} order, and
D_{o_2} is the coefficient of diffusion for oxygen.

For conditions where Ha > 0.3, the acceleration of reaction must also be considered (Linek and Vacek 1981; Sharma and Danckwerts 1970; Robinson and Wilke 1974). Prior knowledge on kinetics for homogenous systems of chemical reaction is required for applying this method. In spite of the fact that this method of determination of OTR has been extensively studied (Linek 1966), there is not much agreement for:

1. Order of reaction with respect to oxygen.
2. The concentration of sulphite at which the order of reaction becomes 0 in sulphite.
3. The region where the concentration of cobalt (metal ion) catalyst is linearly related to the reaction rate constant.

The parameters of the reaction need to be adjusted as per the regimen of reaction. The major factors that influence the k_L in this method are the concentration of catalyst, temperature, pH, and light irradiation (Linek and Vacek 1981). An equation to study the influence of concentration of catalyst was proposed by Linek and Tvrdik (1971) and Reith and Beek (1973). During this oxidation, the pH of the reaction initially drops and reaches to a pH of 5 at the end, which can be attributed to the buffering capacity of the weak acid, sulphite ($pK_S = 7.2$). The sulphite is converted to sulphate, which is a strong acid and has no buffering properties (de Waal and Okeson 1966). The determination of k_La for this method has been defined separately by Ruchti et al. (1985) and Hermann et al. (2001) for different conditions and regimens. This method of sulphite oxidation for the determination of OTR has also been used to deduce the area juncture between liquid and gas in devices for absorption like contractors that are agitated mechanically, bubble column reactors and packed column reactors (Arora et al. 2018; Suresh et al. 2009).

10.5.2 Dynamic Method

This method of determination was proposed by Bandyopadhyay et al. (1967). It is established on the rationale of following the concentration of dissolved oxygen during the short period when aeration is interrupted in the fermenter (Ziv 2000). A sterilizable probe for dissolved oxygen capable of quick response, similar to the laboratory model of Beckman, is needed to acquire the requisite information via this technique. This is a direct procedure which works on the basis for employing the bioreactor system. For a basic system of fermenter which is sufficiently aerated, the rate of oxygen uptake under steady-state conditions is given by:

$$O_2 \text{ uptake} = r.X = \text{OTR} = k_L \text{a}\left(C^*_{O_2} - C_{O_2}\right) \tag{10.14}$$

Here r is the specific rate of oxygen uptake per unit mass of fermenting organism, and X is the concentration of biomass in the bioreactor.

A mean driving force is necessarily used as the transfer occurs over an integral volume, although the use of ($C^*_{O_2}$—C_{o_2}) is under considerable debate. The change in concentration of dissolved oxygen under non-gassing situation is given by:

$$\frac{dC}{dt} = -\text{r} \cdot \text{X} \tag{10.15}$$

This equation is invalid for the instantaneous period when the air for aeration is switched off since the air bubbles held up in the system require some time to escape the fermenter. Most of the surface aeration which can be easily measured occurs at high speeds of agitation which can be reduced by either lowering the speed of agitation or by sparging dinitrogen over the surface, both of which may influence the OTR. Hence, the change in dissolved oxygen concentration after non-gassing for the period of aeration is given by:

$$\frac{dC}{dt} = k_L \text{a}\left(C^*_{O_2} - C_{O_2}\right)_{mean} - \text{r} \cdot \text{X} \tag{10.16}$$

A plot of $C_{o_2 \cdot L}$ and dC/dt is used to determine the value of $k_L a$, which implies that the only measurement necessary is the concentration of dissolved oxygen at different intervals of time. This method is found to be internally consistent since it requires a single measuring device which is independent of a reference measurement. This method has, therefore, been employed in several studies (Kilian et al. 1983; Ducros et al. 2009). A major drawback of this method is the inconsistency of the results when compared with the data obtained by the steady-state method of measurement for the rate of oxygen uptake. This discrepancy can be attributed to the effect imparted by extremely small bubbles in viscous broth, which necessitates the correction in the $k_L a$ value to take into account the hold-up of small bubbles (Amanullah et al. 1998). In spite of this drawback, this method is largely

recommended and used by modifying the equation of k_La as has been reported by Ducros et al. (2009) in a study where the equation used was:

$$C = Z_1 + \frac{Z_2}{1 - k_L a \tau_E} e^{-k_L at} + Z_3 e^{-t/\tau_E} \tag{10.17}$$

where C is the concentration of oxygen as shown by the probe (g/L).

$$Z_1 = C^*_{O2} - \left(Q_{O2} X / k_L a\right)$$

Z_2, Z_3 = integration constant.
$C^*_{O_2}$ is the concentration of dissolved oxygen for saturation (g/L).
Q_{o_2} is the specific rate of oxygen uptake.
τ_E is the time constant of probe (h).

10.5.3 Gassing out Method

Suijdam et al. (1978) first described this method. The vessel to be used for culturing is filled to the working volume with a reference medium which is later absolved of oxygen by passing nitrogen gas through the vessel till the medium becomes devoid of oxygen (Suijdam et al. 1978). To prevent changes in temperature of the liquid phase, the nitrogen gas is humified with water at 50°C prior to releasing it in the shake flask, which ascertains that a constant temperature is maintained during the gassing out process. An oxygen probe is used to determine the rate of dissolution of oxygen for which the dynamic time constant must be less than 1 s in order to establish that the measurements for concentration of dissolved oxygen are not interfered by time lag. Determination of k_La is done directly by measuring the rate of increase in the concentration of dissolved oxygen after this concentration was decreased by sparging nitrogen through the reactor (Puthli et al. 2005). In case of shake flask cultures, Eq. 10.18 is used to determine the response of the process.

$$k_L a = \frac{-\ln\left(\dfrac{C^*_{O_2} - C_{O_2}}{C^*_{O_2} - \left(C_{O_2}\right)_{mean}}\right)}{t} \tag{10.18}$$

Here $C^*_{O_2}$ is the concentration of oxygen in the liquid phase at time $t = 0$.

This process involves no chemical reaction which may adversely affect the precision of measurement and also influence the resistance of the liquid film besides being suitable for application to various media making it advantageous for application. A major drawback of this system is that it does not work efficiently for systems which involve culturing of non-respiring systems. A polarographic digital meter for measuring dissolved oxygen is used to measure its concentration after sparging with

nitrogen to evaluate C_{o_2L} to zero, in which case the change in the concentration of oxygen in the liquid with time is extremely slow and the delays in response of the cell can be neglected. Another major limitation of this method is the obligatory use of membrane type electrodes, which may not have an appropriate response time over a short period of time affecting the real change in the rate of oxygenation (Veglio et al. 1998).

10.6 Importance of Abiotic Parameters in Plant Culture

A regulation on the growth of biomass in bioreactors requires the control of various conditions, which include the bioreactor functioning and its environment and the abiotic parameters that influence the plant growth in general (Heyerdahl et al. 1995). These abiotic parameters include pH, nutrient and mineral availability, supply of oxygen and removal of carbon dioxide, addition of growth regulators, agitation of the liquid culture medium, and the cell density (Ilan et al. 1995). Some of these parameters are discussed below.

10.6.1 Oxygen

The amount of oxygen present in the gas phase on the top portion of the bioreactor vessel and the air bubbles entrapped inside the medium constitute the total amount of oxygen present in the bioreactor apart from the dissolved oxygen present in the medium. A sparger is present at the base of the culture vessel, which releases air into the culture medium. The oxygen transfer coefficient (k_La) determines the amount of dissolved oxygen present in the reactor vessel that is available for utilization to the plants. The metabolic activity of the plants affects the level of dissolved oxygen, which depletes with the increase in cell biomass affecting the culture yield. Plant cells have a metabolic rate and doubling time lower than that of microbes and, therefore, require a less and more controlled supply of oxygen in general. Aeration rate is generally found to be inversely related to the biomass growth with a high aeration rate leading to reduction in growth of biomass (Cazzulino et al. 1991). Although the requirements of oxygen depend on the plant species under consideration, oxygen is crucial and must be provided continuously to the plant system in the culture vessel with proper aeration since it affects the metabolic activity and supply of energy besides anaerobic conditions in the reactor vessel. Agitation, stirring, aeration, flow of gas, and the size of bubbles influence the levels of dissolved oxygen in the culture broth. When porous irrigation tube was used as a sparger, it was found to generate fine bubbles and high values of k_La while lowering the mechanical stress and providing a high rate of growth. A study on the bioreactor cultures of carrot embryogenic tissue has denoted an affected cell differentiation process on reducing the

levels of oxygen to 10% (v/v) while increasing the levels of oxygen from 21% to 80% in Boston fern cultures demonstrated an improved growth rate from 0.61 s^{-1} to 0.92 s^{-1} (Cragg 1992).

10.6.2 Carbon Dioxide

Carbon dioxide is known to affect both the agar cultures and the suspension cultures used for the production of secondary metabolites although the impact of carbon dioxide supplied during proliferation and multiplication phase of plants in media rich in sucrose present in bioreactors remains unclear. Enrichment of CO_2 beyond 0.36% is not required in the absence of photoautotrophic conditions. A high rate of aeration, not the presence of excessive oxygen, is also known to inhibit growth due to a deficiency of CO_2 or removal of several culture volatiles, including CO_2 (Hegarty et al. 1986). The requirement of CO_2 is not essentially linked to photosynthesis but to other metabolic pathways of amino acid biosynthesis.

10.6.3 Ethylene

Ethylene gas is generally produced by suspension cultures and accumulates in the head space of the flasks of liquid cultures, which is different from that of continuously aerated cultures in bioreactors. “Stripping” of volatiles from the culture media is a consequence of high aeration rate, which is a frequent requirement at high densities of biomass.

10.6.4 Mineral Nutrients

Murashige and Skoog (MS) medium is generally used for plant cultures with various modifications in its constituents. The type of culture, size, and type of biomass of plants and the physical properties of culture influence the availability of mineral nutrients. The abiotic factors including pH, aeration, and mineral concentration and availability affect the viscosity of the medium and further determine the rate of absorption of nutrients (Camara 2007). The liquid cultures have better availability and uptake of nutrients since the plants are better exposed to the culture medium, which enhances the rate of uptake. In case of bioreactors with humidified air or condensers, the availability of nutrients is largely influenced by the rate of absorption of nutrients and the lysis of cells (Murashige and Skoog 1962). Ammonia has been reported massively as a limiting factor for the growth of biomass. Availability of other minerals as phosphate and nutrients as carbohydrate is also known to affect the growth rate but to a lesser extent.

10.6.5 pH

The initial pH in most cultures of plant is between 5.5 and 5.9. A change in pH of media occurs mostly during autoclaving or biomass growth since most of the media are not buffered. Studies have reported a massive drop in the pH to 4.0–4.5 in various suspension, embryogenic, and organ cultures within 24–48 h (Westerterp et al. 1963), which has been found to be related to the uptake of ammonium initially and the acidification of media caused by cell lysis. The pH was found to raise to 5.0–5.5 after a few days, which was studied to be influenced by the uptake of nitrates as has also been seen in culture of spruce species which reported an increase in pH to 6.5–6.8 after 14 days of culturing.

10.6.6 Temperature

Temperature is known to influence various abiotic parameters, and the response of plants to these conditions is variable. Regulation of temperature inside the bioreactor is easier in liquid medium either by manipulating the temperature using a heating element or by jacketing the vessel in an envelope of circulating water. Mostly the temperature of bioreactors is kept constant at 25 °C (Ruffoni et al. 2010).

10.7 Conclusion

Several factors, both biotic and abiotic, are known to influence the efficiency and productivity of a bioreactor, and the plant cells and tissue cultures used. A major parameter for consideration is the oxygen uptake rate (OUR), which is affected by the oxygen transfer rate and the aeration and agitation in the bioreactor system influencing the growth of cell biomass and the production of the desired end product. Although proper aeration coupled with appropriate agitation is crucial to maintain the flow of gases in the system (important for aerobic cultures), excess of these conditions will also negatively influence the productivity of the bioreactor by decreasing the rate of cell biomass growth. It is, therefore, important to understand the conditions of oxygen uptake and its transfer in the bioreactor, which is indicative of the metabolic burden on the cells during the process. OUR and OTR are affected by both biotic and abiotic parameters. While the requirement of oxygen varies depending on the plant species, the abiotic factors as temperature and pH affect the dissolution of gases in the culture medium, which may be either liquid or agar based. Hence, the knowledge of metabolic burden, with respect to specific plant species under consideration, is important for defining the type of bioreactor, its properties, and conditions for maximum productivity.

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Chapter 11
Advances in Bioreactor Design for Plant Cell Culture



Tripti Singh and Ashwani Mathur

Abstract The growing interest in disposable bioreactors has taken the world with stride. Bioreactors are well-controlled engineered systems that hold the capability of supporting a biologically active situation for conducting aerobic or anaerobic biochemical processes. Bioreactors have been used for growth and metabolite production from various eukaryotic and prokaryotic hosts. The plant system attracts special scientific interest due to varied morphological forms and design of bioprocess operations accordingly. One of the major challenges in the mass scale cultivation of myriad of morphological forms of plants is the validation of the process that necessitates the need to explore alternative solutions. The choice of the bioreactor and its optimization has been proven a great tool towards the successful production of desired metabolites. The recent advancement in the design of the bioreactor is important for developing strategic approaches for growth and metabolite production from plant systems. Plant cells are also used as an alternative host for recombinant protein production. Recent advances in bioreactor design for specific applications and the introduction of disposable bioreactors are a breakthrough solution to the regulatory requirements and validation guidelines. Various types of bioreactor designs used for plant culturing and metabolite yield thereof are presented.

Keywords Secondary metabolites · Callus · Plant propagation · In vitro cultures

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11.1 Introduction

Plants are pivotal for human existence from time immemorial. They are the primary producers and the source for harnessing solar energy for the production of biomolecules, which is required for their sustenance and for supporting complex food webs. Plants are also a rich source of diverse phytocompounds that have already been explored for their application in food, feed, and pharmaceutical sectors and as sustainability solutions. Previous studies have reported the well-defined circadian regulation in living systems, including plants, is largely responsible for regulating physiological and plant behaviour to abiotic conditions. First demonstrated in 1729 by de Mairan, this process enabled regulated or controlled division of plant cells, associated with UV exposure from the sun (Bouget et al. 2008). The plants have been well-explored systems to produce growth-associated metabolites, called primary metabolites, and non-growth-associated metabolites referred to as secondary metabolites. The primary metabolites are required for the growth and viability of an organism and include metabolites such as carbohydrates, protein, and lipids, which are significant for the growth and development of plants (Bosco and Butnariu 2022). The existing study (and reference therein) has reported that most of the secondary metabolites are biosynthetic modifications of primary metabolites, produced via alternative biochemical reactions including glycosylation, hydroxylation, and methylation, to name a few (Twaij and Hasan 2022). Previous studies have elucidated the synchronized role of abiotic components including media components on the regulation of primary and secondary metabolites (Liu et al. 2010; Lloyd and Zakhleniuk 2004. In a significant study using mutants of *Arabidopsis* sp., the role of primary metabolism in regulating secondary metabolism was explored. In another similar study, Liu et al. (2010) explored the nitrogen supplement-mediated regulation of primary and secondary metabolites in *Chrysanthemum morifolium*. The plant is well utilized for its primary and secondary metabolites, with latter accounting for a large pool of therapeutically and commercially important metabolites. The plant-derived secondary metabolites have been categorized into four major groups based on their nature: (1) Terpenes, (2) Phenolics, (3) Nitrogen (N) containing secondary metabolites, and Sulphur (S) containing secondary metabolites (Bosco and Butnariu 2022; Twaij and Hasan 2022. These compounds are commonly biosynthesized via three biosynthetic pathways, viz., (a) Phenolic biosynthesis from the malonic acid pathway, phenylpropanoid pathway (Adeboye et al. 2014), and shikimic acid pathway; (b) Terpene biosynthesis from mevalonic acid and MEP pathway; (c) Nitrogen compound from amino acids (Twaij and Hasan 2022) (Fig. 11.1).

The biological functions associated with most of the plant-derived metabolites, especially secondary metabolites that were considered as metabolic wastes and toxins for human cells, were largely deciphered and elucidated in the last four to five decades (Twaij and Hasan 2022). The production of such metabolites is largely regulated by the response of cells to abiotic or environmental factors, specifically

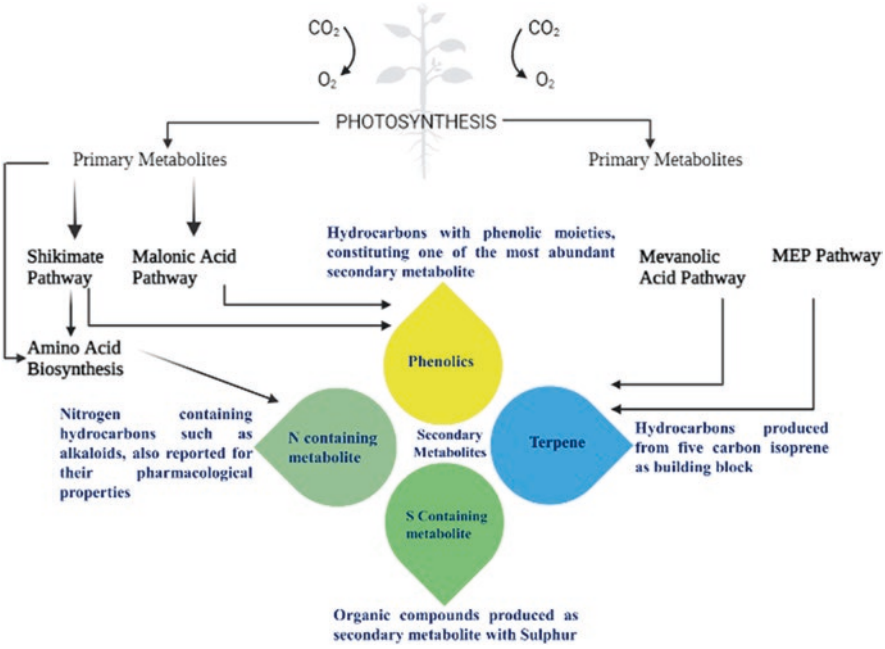


Fig. 11.1 Secondary metabolite categorization and biosynthetic pathways reported for their biosynthesis

for plants cultivated in different habitats or in in vitro culture environments (Oksman-Caldentey and Sevón 1995). The low yield of many metabolites in the cultivated plant system is another challenge that fascinates scientific exploration for isolation and characterization of unexplored metabolites via plants bioprospecting (Kewalanand and Prajapati 2018).

One of the biggest challenges faced by the world is the increasing human population but depleting cultivable land. According to World Bank data, with around 38% of global land, the global cultivable land accounts for 5 billion hectares (https://www.fao.org/sustainability/news/detail/en/c/1274219/). Further, the global agricultural land has been reduced from 38% to 36.5% in 2020 (https://data.worldbank.org/indicator/AG.LND.AGRI.ZS). Moreover, the role of environmental factors and demography in regulating plant growth is well documented.

An alternative strategy is the cultivation of plants under controlled environmental conditions in a bioreactor. Advances in plant biotechnology had harnessed the totipotency of plant cells as the significant driving force for the propagation of sterile plants using different plant tissues and explants. This property obtrudes as a versatile approach for cultivation of plant cells, under controlled environmental conditions to friable cells, suspension cells, or differentiated plants (Feher 2019). Many such methods such as protoplast fusion and anther culture have also been used for the propagation of plants with specified genetic traits.

11.2 In Vitro Culture of Plants: Role of Different Morphological Forms

Previous studies have reported different morphological forms of plants, being used for plant propagation, associated with diverse applications. Some of the key morphological forms, which were designed for the propagation of plants of similar germplasm, were further extended for mass cultivation of plant cells and metabolite production, thereof.

11.2.1 Callus Culture

The calli are observed during cultivation and in vitro propagation of plants under regulated abiotic conditions. In nature, the callus induction in plants was reported more than 200 years ago (Neely 1979 and reference therein).

The word is derived from the Latin word '*Callum*' meaning hard (Ikeuchi et al. 2013). During micropropagation of plants, callus induction has been observed due to varying concentrations of phytohormones, such as auxin and cytokinin (Srivastava et al. 2014). The morphology of the callus varies significantly in its form and texture from hard mass to soft tissues. Also, the shape of the cells arranged in the calli ranges from spherical to irregular mass (Bhatia 2015). Studies have shown that variations in the type of the callus lead to diverse metabolite profiles in *Arabidopsis thaliana* (Iwase et al. 2011). The bioactive compounds from the callus are collected at a specific stage generally during the stationary phase due to the higher production of secondary metabolites during this phase of their growth cycle (Srivastava et al. 2014). Due to diversity in the size, shape, and nature of cells in a callus, bioreactor design modifications have been reported. In one of the studies on exploring the effect of reactor design and agitation mechanism on the propagation of embryonic rice callus, Moon et al. (1999) reported better propagation of embryonic rice calli in 'turbine blade reactor' (TBR®) over air lift reactor and jar reactor with turbine impeller. Moreover, the TBR imparts less hydrodynamic stress and better nutrient availability to the immobilized calli. In another significant study, Goojgi et al. (2022) explored bioreactor design modification to overcome nutrition availability and mass transfer limitation during calli culture of *Arnebia pulchra* and *Hyoscyamus niger*. They suggested a novel open-flow spray bioreactor (OFSB) for the propagation of the calli of the two plants and feeding strategy (Goojgi et al. 2022). However, the mass transfer limitation with the 3D morphology of plant calli often restricts their use for mass scale production of their metabolites.

11.2.2 Friable Cells

The diverse nature of calli enables their sub-categorization based on morphological attributes. Depending on the ability to regenerate different plant parts, the calli can be categorized as rooty, embryonic, and shooty (to name a few), depending on the organ they regenerate. Calli without any apparent organ regeneration depending upon their tissue characteristics are often referred to as friable or compact calli (Ikeuchi et al. 2013). The colour of such calli often differs from other types of calli. Friable calli are also reported as an intermediate step in the development of suspension plant cells from calli due to loose attachment of cells. One of the major advantages of this type of morphology is since cells are loosely bound it is easy for them to fall apart and thus are useful for the establishment of suspension cultures (Evans et al. 2003; Chourey and Sharpe 1985). A comparative study by Jamil et al. (2018) on metabolite profile, after induction with elicitor jasmonate, showed variation in the phytoconstituent in the suspension culture and friable callus suggesting possible variation in molecular responses of the two morphological forms to similar abiotic factors.

11.2.3 Suspension Culture

Suspension culture, also called as plant cell line, is one of the most versatile morphological forms of plant with homogeneous undifferentiated cells used for the production of plant-specific metabolites (Kieran et al. 1997). In suspension cell cultures, callus cultures consisting of an aggregated mass of cells are inoculated in liquid media and agitated until refined cell suspension is achieved. Suspension cell cultures are preferable due to easy maintenance via subculturing into new and fresh media, short life cycle, and increased rate of cell multiplication (Cai et al. 1987). The plant cells differ from other prokaryotic and eukaryotic cells in terms of morphology, physico-chemical properties, and requirement of abiotic conditions. This necessitates an in-depth understanding of various abiotic factors that play a pivotal role in plant growth and metabolite production. One of the key attributes affecting plant cell growth is liquid rheology and hydrodynamic properties. In a classical review, Kieran et al. (1997) collated information on different plants suspension cells, grown in Newtonian and non-Newtonian fluids, and the effect of flow regime on cell growth and biomass. In one of the recent reviews, Motolinía-Alcántara et al. (2021) had highlighted the role of engineering parameters including temperature, pH, nutrient composition, and liquid rheology on the choice of bioreactors and diversity in reactor design and operation attributed to diversity in the properties of plant suspension system.

11.3 Basic Requirements for Plant Cultivation in Bioreactors

11.3.1 Historical Perspective of Plant Propagation in Liquid Medium

The dynamic changes in the growth rate of plants and plant-derived metabolites' yield with changes in abiotic (demographic conditions, season, and other associated environmental factors) and biotic conditions restrict the cultivation and/or collection of plants from diverse habitat and use of the alternative strategy of propagating them in bioreactors (Ruffoni et al. 2010).

The formation of an unorganized mass of cells, also called callus (*plural:* calli), was first observed in late eighteenth century by Duhamel du Monceau; however the concept of totipotency and plant tissue culture techniques were the outcome of scientific achievements of the German scientist, Gottlieb Haberlandt, who in 1902 proposed the techniques for aseptic culture of cell, tissue, and other explants from the plants for scientific and commercial applications (Thorpe 2007). The twentieth century witnessed the used of plant tissue culture techniques for the study of plant cytology, propagation of disease-free plants, germplasm preservation, and production of plant-derived metabolites (Thorpe 2012). The initial experimentation of Haberlandt in 1902, on culturing of the isolated cells on *Lamium purpureum*, was performed using liquid, sucrose-supplemented Knop's media. However, based on other similar studies using gelled agar media, agar-gelled media were a recommended choice for plant tissue culture as they mimic the natural environment (Preil 2005). However, further studies using liquid media for plant propagation find their wings as liquid media was also relevant for the scalability of the processes under controlled environmental conditions.

11.3.2 Bioreactors for Plant Propagation

There are diverse bioreactor design modifications reported for the propagation of plantlets, friable cells, and suspension cultures from plants. The bioreactor systems can be categorized based on the liquid mixing mechanisms into (1) mechanically driven, (2) hydraulically driven, and (3) pneumatically driven (Lehmann et al. 2014). The design modification in the types of reactors has been reported for different plant systems and different morphological forms. One of the pivotal parameters of importance during fermentation of plant cells and tissues is mass transfer. The design of bioreactors must overcome mass transfer limitations, which is often difficult to estimate in complex media (Riet 1983). The existing kinetic correlations have shown the role of oxygen uptake rate (OUR), maintenance coefficient, yield coefficient, and biomass on the growth rate of the organism (Eq. 11.1) (Riet 1983).

$$\left(r_s\right)=\frac{\text{Biomass production rate}\left(r_x\right)}{\text{Yield coefficient}\left(Y_{\frac{X}{S}}\right)}+\left(m_x\right)\times\left(M_x\right) \tag{11.1}$$

where r_x is substrate consumption rate, m_x is maintenance coefficient, and M_x is biomass in the reactor (Riet 1983). Further, the heat production rate was observed to be the function of the OUR (Heijnen and Roels 1981). This may be explained by the role of oxygen as an electron acceptor in most of the aerobic processes, involving energy dynamics. However, the dissipation of heat from the fermentation system is equally important for the successful growth of organisms. The heat transfer rate (HTR) from the bioreactor is a function of the heat transfer coefficient (α), temperature gradient (ΔT), and cooling surface area (A) (Eq. 11.2).

$$HTR=\alpha.A.\Delta T \tag{11.2}$$

This further suggests that the choice of bioreactor is largely regulated by optimized growth conditions and the design (including material of construction) of the bioreactor (Henzler 1982). The advances in reactor design have been done keeping in mind the engineering aspects associated with high density and metabolite yield, which is often a challenge in plant cell culture. Studies compared in Table 11.1 showed the diversity in the types of bioreactors vis-à-vis plant morphology. The plants and their morphological forms (hairy roots, plantlets) are shear sensitive, and the impact due to high mechanical shear can be paramount on their growth and metabolite yield. Previous studies have shown that different morphological forms of plants have been cultured in stirred tank, air lift, bubble column, trickle bed, and mist bioreactors, which are the design modifications in mechanically and pneumatically driven reactors (Mishra and Ranjan 2008). The shear stress is evident in the pilot scale/large scale hairy root culture. Furthermore, different plants have shown different responses when cultured under similar types of bioreactors, suggesting diversity in response to shear stress is largely regulated by plant species (Weathers et al. 2008). Apart from design modifications in the type of bioreactors, scientific efforts have also been accentuated towards reducing production costs. The disposable bioreactors have obtruded as a well-accepted technological advancement to reduce the overall cost of manufacturing. The bags that can be used and disposed of after a successful harvest is a technology that was in its infancy in the late twentieth century but is now finding a suitable place in most commercial manufacturing units. In one such seminal studies by Ducos et al. (2008), disposable bioreactors have been successfully used for the production of three million embryos of *Coffea canephora*. The dynamic amalgamation of advances in reactor designs has played a pivotal role in reducing the overall cost of plant propagation and production of plant-derived metabolites thereof at large scale.

Table 11.1 Diverse catalogues of plants cultured in different types of bioreactors

Type of reactor	Plant	Morphological form	Design/ operation	Metabolite/ Product	References
Mechanically driven	*Withania coagulans*	Suspension culture	Culture flasks	Establishment of suspension culture; withanolides production	Mirjalili and Esmaeili (2022)
Mechanically driven	*Ageratina pichinchensis*	Suspension culture	Erlenmeyer flasks with MS media on shaking at 110 rpm at 25 °C with 16 h white light illumination	(−)-Artemesinol, (−)-Artemesinol glucoside, encecalin, and 3,5-diprenyl-acetophenone	Sánchez-Ramos et al. (2020)
Mechanically driven	*Lactuca sativa*	Suspension culture (transformed and untransformed)	Culture flasks	Establishment of suspension culture; total phenolic and flavonoid content	Ismail et al. (2019)
Mechanically driven	*Azadirachta indica*	Hairy root culture	Modified stirred tank reactor with polyurethane foam (PUF); disc for root support	Biomass production and azadirachtin accumulation	Srivastava and Srivastava (2012)
Pneumatically driven	*Catharanthus roseus*	Hairy root culture	Air sparged bioreactor with no other mixing	Air sparging with no other mixing was reported to be more growth supportive than stirred tank or media recycling	Nuutila et al. (1994)
Mechanically driven	*Atropa belladonna*	Hairy root culture	Stirred tank bioreactor with stainless steel net for hairy roots	Atropine production	Lee et al. (1999)
Pneumatically driven	*Beta vulgaris L.*	Hairy root culture	Different airlift bioreactors (cone, balloon, bulb, drum and column bioreactors of 5 l capacity) were compared	Betacyanin production	Shin et al. (2002)

11.4 Conclusion

The growing concern over depleting cultivable land and increasing demand for plant and their metabolites had created an imbalance that needs immediate attention. Plant-based metabolites are an integral component for human survival. The diverse metabolite pool, concomitant with diversity in the morphological forms of plants, has taken the world with stride. The plant system is well accepted for the production of commercially important metabolite with their diverse applications in food, feed, and pharmaceutical sectors. The unique properties of every plant system make it even more challenging to develop bioreactors for a common morphological form. The common bioreactors used for plant (and their diverse morphological forms) culture are categorized based on mixing behaviour to mechanically, hydraulically, and pneumatically driven bioreactors. However, the differences in the responses of plant cells to diverse cultivation conditions had provided an impetus for bioreactor design modifications. The engineering attributes have been significant in successful process design and scale-up. The joint effort of academics and industries in process optimization and validation at pilot scale is the need of the hour.

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Chapter 12
Role of Nanotechnology and Artificial Intelligence (AI) in Waste Management



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Abstract With a growing population, an improvement in the quality of life, advancements in the industrialization of medicine, food, agriculture, energy, electronics, etc., and an increase in dependency on high-tech tools and accessories, there is also a substantial rise in the waste generated. This exponential growth of by-products/waste has become a hazard to the environment. Traditional ways of handling this huge mass of waste like collection, treatment, transport, storage, disposal, etc. are becoming a major challenge at times. It is widely accepted that no single method or process can effectively be used in managing the vast quantity of waste that is being generated on a daily basis. So, a new and more customized approach that amalgamates both the traditional and advanced technologies is the need of the hour. One such promising field is nanotechnology. Imparting nanotechnology to the physical, chemical, and biological management methods needs further exploration. Nanomaterials due to their unique characteristics such as high surface-to-volume ratio, enhanced activity/sensitivity, better dissociation, etc., can work wonders, if applied strategically after careful studies. Nanomaterials like zero-valent iron (ZVI), zeolites, carbon nanotubes, etc. have been under investigation for their potential in waste management and treatment for a while. Furthermore, waste can be valorized by employing bioprocess technology for the production of bioproducts such as hydrolytic enzymes, primarily carbohydrases for bioethanol production, and, to a lesser extent, fragrances, biosurfactants, biopesticides, bioplastics, organic acids, phenolic compounds, etc. This chapter's emphasis will be on the role of nanotech-

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nology and bioprocess technologies in waste management and treatment, and ways to monitor the process of waste management using artificial intelligence.

Keywords Nanotechnology · Artificial Intelligence · Waste Management · Industrialization · Nanomaterials

12.1 Introduction

With the exponential rise in living standards, rapid urbanization, and advancement of technology in various fields like agriculture, food processing, medicine, animal husbandry, energy, electronics, transportation, etc. there has been a rampant increase in waste generation (Chand and Chand 2022). According to the UN Environment Programme, it is estimated that a whopping 11.2 billion tons of solid waste is generated globally, which contributes more than 5% of the greenhouse gases emission. It is reported that only 60–70% of this waste is collected, of which only a mere 20–24% is treated while the rest is dumped into landfills to decay or dumped into streams (Soni et al. 2022). By the year 2050, it is anticipated that solid waste generation will reach roughly 3 billion metric tons worldwide. However, these conventional methods do not address the key hurdles as they are labor-intensive, unregulated largely, but not effective in preventing contamination of groundwater bodies and air in the vicinity (Nanda and Berruti 2021). According to the UN Environmental Programme, engineered landfills facilitate a relatively safer domestic solid waste disposal, as the permeation of waste into groundwater is prevented to a greater extent. Unfortunately, engineered landfills still face issues with emission of greenhouse gases, foul odor, windblown debris, pest infestation, and a significant fire hazard (David et al. 2020).

Other treatment techniques like thermal recovery or waste-to-energy deal with thermal disintegration of inert and moisture containing waste to produce electricity (Mukherjee et al. 2020). These techniques, though efficient, have a huge impact on the environment if proper care in segregating the waste is not followed. Segregation of waste is extensive, prolonged, tedious, labor-intensive, and requires substantial training and skilled labor. On top of these, the composition of the waste generated also has a major role in the effectiveness of the procedure (Maddikunta et al. 2022). Despite its potential, waste-to-energy techniques also rely on the calorific outcome of the waste, location, climate, and related factors. Another aspect of these techniques is the considerable amount of residual waste that is generated.

It is estimated that domestic waste generation is rising more than 90% every decade. The rapid increase in utility of materials made of plastics, dyes, synthetic leathers, polymers, etc. adds more challenges to the existing technologies in waste treatment (Li et al. 2023). Along with domestic waste, commercial, industrial, and infrastructure construction activities are adding more waste, and this type of waste cannot be treated by conventional domestic waste treatment methods. In addition to

this, all this waste ends up contaminating groundwater bodies significantly (Mor and Ravindra 2023).

Considering all the above-mentioned aspects, it is imperative to have an integrated treatment method that can address majority of the issues raised in conventional treatment methods. Even a mode of real-time monitoring is also required to follow-up and assess the treatment process. This real-time monitoring should gather information related to parameters like moisture, composition, emissions, temperature, etc. This requires a large workforce, investment in infrastructure, integration of multiple treatment methods, and policy changes at socio-economic and government levels. This chapter puts an emphasis on integrating advanced treatment techniques like catalytic nanomaterials and artificial intelligence (AI) based monitoring. Further management of waste treatment can be achieved by employing bioprocess techniques that are efficient, economical, and less labor intensive. Synergizing "Nanotechnology" and "Artificial Intelligence" will stand as a cutting-edge approach to handle waste management efficiently. Researchers are working in this direction and realizing the concepts of nanotechnology, and AI plays a significant role in achieving the amalgamation.

12.2 A Brief Insight into Nanomaterials

Nanomaterials, as per International Organization for Standardization (ISO), are materials with any of the external dimensions or surface in nanoscale, i.e., in the size range of 1–100 nm (Chintagunta and Pallavi 2022; Chintagunta et al. 2021). In short nanomaterials are those with at least one of its dimensions in a three-dimensional space under 100 nm (Lane 2001). Nanomaterials have drawn huge attention in recent years, and the term "nanoscale science and technology" is attributed to the development of methods that have the ability to manipulate matter atom by atom (Service 2000). Due to their distinctive physical and chemical properties, nanomaterials are found to have promising applications in the design and development of catalytic, mechanical, optical, electronic, and diagnostic devices with enhanced activities (Rosi et al. 2006; Chan et al. 2002). Due to their huge surface to volume ratio, nanomaterials exhibit quantum size effect (Li et al. 2006). As the size of the particle decreases, the surface atoms have a much stronger tendency to interact or react with the neighboring atoms to attain surface stabilization (Service 1998).

Tailoring the characteristics of materials by controlling the size of the particles has been of great interest for material science (Thess et al. 1996). The change in characteristics of nanomaterials from their bulk counterparts can be attributed to two major effects. Firstly, the number of surface atoms is in a large fraction compared to the total atoms. Surface atoms have a distinctive contribution to the total free energy, thermodynamic properties, surface plasmon resonance, etc. (Murray et al. 1993). Secondly, the intrinsic properties of the nanomaterials exhibit quantum size effects (Rossetti et al. 1983). When the size of the material falls below 100 nm,

electronic and optical properties exhibit variation by a great extent (Schmid and Simon 2005).

Nanomaterials, due to their unique and/or enhanced properties, have found potential applications in various fields from electronics to health care and so on (Guntur et al. 2018). It is estimated that nano-based commercial products currently on the market worldwide can be valued at around USD 64 billion, and more products are being introduced daily (Linsebigler et al. 1995). It is vital to have a clear understanding on the key aspects of nanomaterials such as their origin, nature, classification, application, impact on the environment, etc. to better utilize them. There are a wide range of naturally occurring nanomaterials including organic and/or inorganic moieties like ultrafine clay particles, certain proteins, materials like hydroxyapatite, etc. that are safe for the environment. However, nanomaterials generated or released into the environment by anthropogenic activities are of concern and should be carefully monitored (Hong et al. 2006). These nanomaterials are classified into different groups based on their dimensionality as zero dimensional (0D), one-dimensional (1D), two-dimensional (2D), and three-dimensional (3D) nanomaterials. Zero-dimensional (0D) nanomaterials have all their external dimensions in the nano scale. One-dimensional (1D) nanomaterials have at least two external dimensions in the nano scale whereas one dimension in the micro scale (Burdett 1984). Two-dimensional (2D) nanomaterials have one of their external dimensions in the nano scale while the other dimensions are not in the same size range. Three-dimensional (3D) nanomaterials are made up of materials that have internal features in the nano scale whereas the external dimensions are not in the nano scale.

Furthermore, nanomaterials are classified into different groups based on their composition or their constituent elements (Zhang et al. 1999). Nanomaterials made up of metals like silver, zero-valent iron, etc., or metal oxides like ZnO, TiO_2, Fe_3O_4, etc., semiconductors, or ceramics are included in inorganic nanomaterials. Nanomaterials comprising polymers, resins, micelles, biological molecules, etc. are included in organic nanomaterials (Zhan et al. 2003). Fine nanomaterials such as activated charcoal, carbon nanotubes (CNTs), fullerenes, graphite, graphene, diamond-like carbon (DLC), etc. are included in carbon-based nanomaterials. Finally, nanomaterials that are made by a combination of different materials including organic, inorganic, carbon-based, metal-ligand complexes, metal-organic frameworks, etc. are included in nanocomposites.

There is significant potential in reusing large quantities of waste and by-products as energy sources or beneficial materials. On the one hand, submerged fermentation is the most common method for producing valuable bioproducts. This technology, which is generally used to valorize liquid wastes, has many benefits, including the ability to develop a highly controlled bioprocess, as the design and implementation of bioreactors have been extensively reported. In order to treat substantial organic waste, however, rigorous pretreatments are required. These pretreatments are frequently environmentally hazardous and energy- and water-intensive, resulting in a highly diluted, low-productivity product stream.

12.2.1 Essentials for Nanomaterials in Waste Management

As discussed earlier, nanomaterials due to their enhanced/superior properties pave way to an alternate method of waste treatment. The application of nanomaterials as potential alternative for conventional treatment of selective contaminants can be promising as nanomaterials exhibit increased reactivity and can be implemented in a more targeted way (Thulasiraman and Ganesapillai 2023). On the other hand, nanomaterials can also be used in combination with traditional waste treatment procedures like chemical treatment, physical separation, biological treatment, membrane-based filtration, etc. Different nanomaterials with different modes of operations and mechanisms can be applied in waste treatment either individually or in combinations based on the nature of waste. The role of nanomaterials in waste treatment can be for disinfection, adsorption, decomposition, degradation, conversion into less hazardous or environment friendly compounds, etc.

12.2.2 Nanomaterials as Catalysts

Nanomaterials due to their high surface-to-volume ratio have a greater active surface area availability and hence have higher catalytic output when applied in waste treatment.

12.2.2.1 Inorganic Nanomaterials

Due to their relative ease in synthesis and characterization, these nanomaterials are highly sought after for large-scale applications in catalysis. Various inorganic nanomaterials like metals and metal oxides are studied extensively for their catalytic properties (Deepak et al. 2011). Extensive studies have been reported over the past few decades on the application of silver nanomaterials for disinfecting. Due to their unique chemical, optical, electrical, and biological properties, zero-dimensional and one-dimensional silver (Ag) nanomaterials are being used in various consumer fields like health care, food, medicine as well as domestic and industrial effluent treatment (Shankar and Rhim 2015). Silver nanomaterials can be easily synthesized through chemical route via chemical reduction of silver salts using reducing agents like sodium citrate or borohydride. However, the environmental implications raised concerns due to the hazardous and toxic nature of the reagents and solvent used in chemical synthesis methods along with the toxic by-products (Gurunathan et al. 2015). To negate these issues and to make the nanomaterial synthesis more sustainable and cost-effective, green or biological approach towards nanomaterial synthesis is more desirable. Numerous studies reported on green synthesis of silver nanomaterials using plant extracts, bacteria, and biological molecules like enzymes and amino acids (Kumar et al. 2014; Leung et al. 2010). Similarly, synthetic approach can also

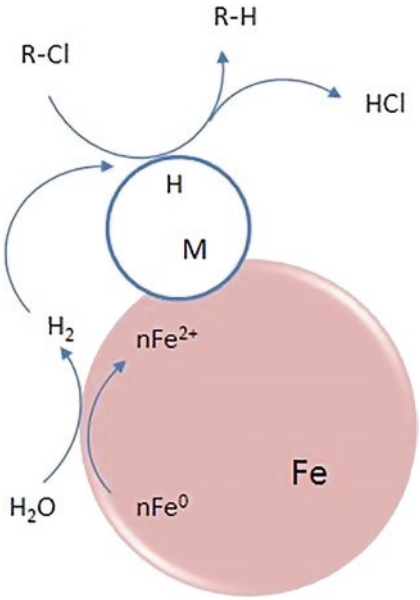


Fig. 12.1 Hydro-dehalogenation mechanism of zero-valent iron-based bimetallic nanomaterials, where M = Pt/Pd/Ni/Zn/Cu

be applied to the synthesis of a wide range of nanomaterials including gold, zero-valent iron, nickel, etc. Extensive studies are also reported on the application of zero-valent iron nanomaterials of varying shape morphologies in catalytic reduction of organic moieties in waste (Kalishwaralal et al. 2010). Zero-valent iron nanomaterials are reported to be highly efficient in the reduction of hydrocarbons and halogen derivatives of hydrocarbons. Zero-valent iron nanoparticles exhibit high efficiency towards dehalogenation, are cost-effective, and have benign environmental implications. However, zero-valent iron nanomaterials have one key disadvantage, as the reduction of the effluents is driven through electron transfer from the surface of iron, and iron due to its relatively high electropositivity is prone to oxidation, making them difficult for long-term application. Extensive studies to increase the half-life of zero-valent iron have found out that introducing a secondary metal like platinum, palladium, nickel, zinc, copper, etc. can increase the half-life of iron nanomaterials. In bimetallic nanomaterials, the secondary metal acts as a catalyst and controls the electron transfer from iron surface. Bimetallic nanomaterials based on zero-valent iron are more efficient in hydro-dehalogenation of halogenated organic moieties. Figure 12.1 shows the mechanism of hydro-dehalogenation.

12.2.2.2 Metal Oxide Nanomaterials

These are another type of inorganic nanomaterials that are noticeably well explored for their catalytic properties and have a potential in waste management (Janotti and Van de Walle 2009; Gomez-Solís et al. 2015). Metal oxide nanomaterials are relatively more adaptable in waste management compared to metal nanomaterials, as

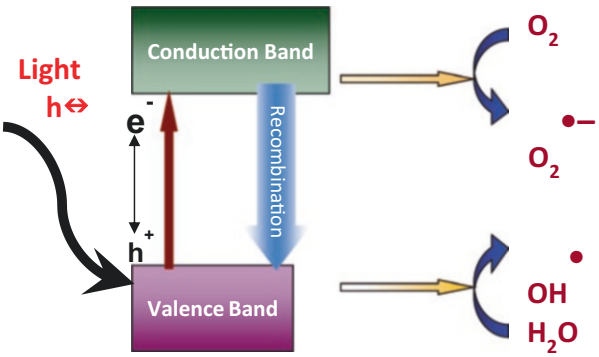


Fig. 12.2 Schematic diagram of photo excitation of TiO_2

they are robust and exhibit insulator/semi-conductor properties, stable structural geometry, and metallic characteristics in certain cases (Zhang et al. 2008; Rao et al. 2002). Owing to their high surface-to-volume ratio and their ability in forming highly porous structures metal oxide nanomaterials like titania (TiO_2) (Ohno et al. 2001), zinc oxide (ZnO) (Daneshvar et al. 2004), etc. are efficient sorbents and can adsorb a wide range of pollutants like heavy metals, dyes, volatile organic compounds, oils, etc. While on the other hand reactive metal oxide nanomaterials like iron oxide (Fe_2O_3 & Fe_3O_4) (Herrera et al. 2001; Yavuz et al. 2006), manganese oxide (MnO_2) (Li et al. 2007; Zheng et al. 2006), magnesium oxide (MgO) (Stoimenov et al. 2002; Makkhluf et al. 2005), etc. can be used to treat biological and organic effluents. Titania-based nanomaterials in particular are of great interest due to their adsorbent as well as photocatalytic properties. Titania, a semiconductor with a large band gap in the range of 3.05–3.2 eV, based on the purity and phase, is photoactive in UV radiation and is a potent photocatalyst forming electron-hole pair that initiates a chain of redox reactions (Zhang et al. 2008) (Fig. 12.2). It has been reported by multiple research groups that higher electron-hole separation can be attained in TiO_2 by reducing the particle size or by introducing crystal defects by doping it with different materials like nitrogen, carbon, sulfur, metal, and metal oxide impurities.

TiO_2 nanomaterials (rutile, anatase, or brookite) can be synthesized in few simple steps through solvo-thermal or hydrothermal route followed by high temperature treatment (Egerton et al. 2006). Doped TiO_2 nanomaterials also exhibit photocatalytic degradation of large organic molecules in visible light as the dopant improves the photo-excited charge separation, thereby reducing the band gap (Kumar 2019; Kumar and Ravi Kumar 2021). Furthermore, nanocrystalline TiO_2 is also known to adsorb heavy metal impurities such as arsenic Ar(V) and Ar(III) at pH ranges of 7.5–8 (Pena et al. 2005) and Cr(VI) and Cr(III) at neutral pH conditions (Jiang et al. 2006). Studies have also reported that nanocrystalline TiO_2 exhibited significant light-induced antimicrobial effect on both gram-positive and gram-negative bacteria and few fungi as well (Maness et al. 1999; Yu et al. 2005). Due to its physical and chemical stability, high resistance towards

corrosion and high temperature tolerance TiO_2 nanomaterials can be made into different forms with varying dimensionality such as nanorods, nanotubes, nanowires, nanoparticles, nanosheets, and nanofilms and even form composites with other nano and macro materials (Bavykin et al. 2006; Cao et al. 2004; Shieh et al. 2006).

Iron oxide nanomaterials, which are formed in two different forms hematite (Fe_2O_3) and magnetite (Fe_3O_4), have exhibited a wide range of applications in waste and effluent treatment with their selective adsorption and oxidative properties (Herrera et al. 2001). Hematite nanomaterials exhibit anti-ferromagnetic nature and are stable in different chemical and environmental conditions, making them highly adaptable for removal of anions and few heavy metals species from wastewater (Hsia et al. 1994; Munoz et al. 2002). Studies have also found out that when fused with photoactive species like TiO_2, hematite nanomaterials exhibited high efficiency in the removal of total organic carbon at neutral pH condition (Sun et al. 2004). Magnetite nanomaterials, on the other hand, exhibit unique physical and chemical properties due to their ferromagnetic nature as well as having a combination of both bivalent (Fe^{2+}) and trivalent (Fe^{3+}) iron (Yavuz et al. 2006; Takafuji et al. 2004). Magnetite nanomaterials can be magnetized, and this makes them an ideal candidate for separating magnetic heavy metals from effluents (Cornell and Schwertmann 1996). Moreover, modified magnetite nanomaterials have also shown higher efficiency in the detection and recovery of heavy metal toxins like As(V), Cr(VI), and radioactive metals toxins (Wang et al. 2006). Magnetite nanomaterials readily undergo oxidation under suitable conditions turning into more stable hematite nanomaterials; this property enables them to be effective in decomposition and dissociation of effluents. Magnetite nanomaterials, when combined with other nanomaterials like silver and titania, exhibit superparamagnetic nature and also act as a potent antibacterial property against both gram-positive and gram-negative bacteria (Gong et al. 2007; Chen et al. 2008).

Polymorphic manganese oxide nanomaterials have superior structural flexibility along with innate chemical properties and are well explored for their potential in molecular adsorption and catalytic degradation of industrial effluents including heavy metals, radioactive metals, dyes, and organic pollutants (Li et al. 2007; Zhu et al. 2007). Due to their polymorphic nature and relatively high density and robustness, MnO_2 nanomaterials can be synthesized in various structures like nanorods, nanowires, nanosheets, nanosieves, and other 3-D hierarchical structures through different synthetic routes (Zheng et al. 2006; Li et al. 2005). β-MnO_2 nanomaterials in particular exhibit selective oxidation properties against alcoholic and phenolic groups, as shown in Fig. 12.3. This selective oxidation ability can be applied in VOC abatement (Lamaita et al. 2005). Furthermore, mesostructured MnO_2 nanomaterials when combined with gold or iron oxide nanoparticles excelled in oxidative decomposition of a wide range of organic moieties, with better efficiency at temperatures around 60 °C (Sinha et al. 2007). MnO_2 nanomaterials in the size range of 20–30 nm exhibit great affinity towards adsorption, decoloration, and

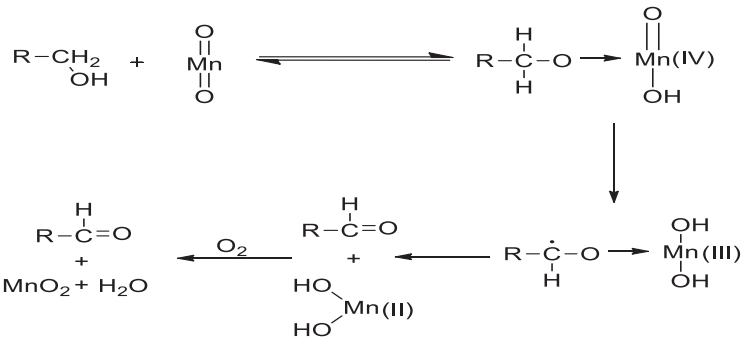


Fig. 12.3 Mechanism of oxidation of alcohol group by MnO_2 nanomaterials

mineralization of dyes and solid-liquid separation, which can be implemented in the separation of effluents from industrial waste (Espinal et al. 2004) (Fig. 12.3).

12.2.2.3 Carbon-Based Nanomaterials

They are vastly studied over decades for their versatile nature, chemical stability, superior flexibility while retaining mechanical strength, extremely high surface areas, and availability of a very large functionalized area (Chatterjee and Deopura 2002). Various functionalized carbon-based nanomaterials like activated charcoal, carbon nanotubes (CNTs), graphene, graphene oxide, etc. can be produced in large scale with few optimization steps (Navrotskaya et al. 2020). Among the carbon-based nanomaterials, carbon nanotubes are most extensively studied for their unique mechanical, optical, electrical, high specific surface area, superior adsorption efficiencies, and adjustable surface functionalization properties (Aqel et al. 2012; Baby et al. 2019). CNTs can be simply explained as rolled-up graphene sheets retaining the hybridization of graphene, but with very high mechanical strength and flexibility compared to graphene sheet (Volder et al. 2013). Both single-walled (SWCNTs) and multi-walled (MWCNTs) carbon nanotubes exhibit great potency in adsorption of heavy metals, polar/non-polar organic molecules, volatile organic molecules, oil and other nanomaterials (Goh et al. 2013; Tessonnier and Su 2011). CNTs can be fine-tuned to adsorb or separate molecules with targeted functional groups by applying specific functionalization protocols (Azelee et al. 2017). Functionalized SWCNTs even exhibited very high adsorption performance against highly volatile organic contaminants like benzene, toluene, hexane, etc. (Choi et al. 2013; Gao et al. 2014). Similarly, graphene oxide (GO) showed great potential in waste management. Graphene oxide exhibits ample activity and mechanical strength and can be functionalized with relative ease. Functionalized graphene oxide mainly comprised hydroxyl groups with a small quantity of epoxide or carboxylic acid groups, based on the functionalization protocol (Adeyemi et al. 2016).

12.2.3 Nanofilters and Membranes

Quite a good quantum of studies has been reported on filters based on ceramics, polymers, metal-organic frameworks (MOFs), and composites immobilized or functionalized with nanomaterials. Traditionally various ceramic and polymer-based membranes/filters have been employed in domestic and industrial wastewater treatment (Pei et al. 2008). These filters rely mainly on physical phenomenon like sieving, adsorption/absorption, electrostatic interactions, etc. for effluent filtration (Ebenezer and George 2014). However, certain parameters like membrane fouling, surface sedimentation, varying levels of effluent rejection, etc. limit the lifetime of these traditional filters (Pei et al. 2008). Studies reported on nanocomposite membranes established them to have relatively higher selectivity, better anti-fouling/permeability, rigidity, and longer lifetime (Jafarinejad 2017). Nanomaterial immobilized polymer matrixes like polysulfone (PSf), cellulose acetate (CA), polyvinylidene derivative (PVDF), chitosan, etc. have shown great anti-fouling and selective separation of effluents (Vijayakumar et al. 2013). Furthermore, composite and hybrid nanofillers supported by porous polymer or ceramic substrates demonstrated superior performance towards effluent separation or desalination (Ahmad et al. 2020; Salehi et al. 2018).

12.3 Role of Artificial Intelligence (AI) in Waste Management

Artificial intelligence (AI) is a branch of computer science in which machines are built with intelligence similar to humans, allowing different tasks such as problem solving and decision making to be completed in less time (Abbasi and El Hanandeh 2016). Recent advancements in technology has led to the application of AI in different fields such as health care, automobiles, education, solid waste management, and many more (Abdallah et al. 2020). Artificial intelligence techniques such as artificial neural network (ANN), support vector machine (SVM), genetic algorithm (GA), linear regression (LR), and decision tress (DT) have been found to be useful in all stages of solid waste management (SWM). Using AI in SWM has several advantages, including increased efficiency, safety, quality, and lower labor costs. However, there are some limitations, such as a scarcity of empirical data and a lack of public awareness (Andeobu et al. 2022). SWM has five different steps such as waste generation, storage and collection, transport, treatment, and disposal (Fig. 12.4).



Fig. 12.4 Various steps involved in solid waste management

12.3.1 Waste Generation

Predicting the amount of waste that will be produced accurately is a crucial prerequisite for effective solid waste management. One can design and run an efficient waste collection system by predicting the volume of Municipal Solid Waste generated (Bircanoğlu and Arıca 2018). AI techniques (ANN, SVM, GA, DT, LR) can be used at this step. Among these techniques, ANN has found effective in predicting the accurate amount of waste generation. The concept of the neural network found in animal brains serves as the basis for the creation of artificial neural networks, a key AI method. An artificial neural network is made up of many tiny components called neurons that are organized into three layers. Input, hidden, and output layers are the three distinct layers (Guo et al. 2021). The input layer receives the input whereas the output layer predicts the output. In between these two layers lies the hidden layer that performs the major computational part of the neural network.

Firstly, the neurons of primary layer will be fed with the input value. Neurons of one layer are connected to neurons of another layer through weighted connections. Weighted connections are nothing but a pre-assigned numerical value, which is referred to as weight. Now, the input value will be multiplied with this weight and the resultant value will be transmitted as input to the hidden layer. The neurons in hidden layers are associated with a value called bias. The bias value will be added to the input of hidden layer and then this final value will be placed into an activation function (Gupta et al. 2019). A mathematical operation known as an activation function determines whether or not a neuron will become activated. Data is sent to the output layer's neurons if it becomes active. To ascertain how many neurons will activate in the hidden layer, a trial-and-error approach is used. The amount of waste that will be generated will be precisely determined by the hidden layer's trial and error method. This entire process is called forward propagation.

However, in order to acquire accurate prediction with ANN we need to train the network. Backward propagation, a method, is used to provide this training. We give the network input and output during the training phase (Low et al. 2016). The output that was predicted and what actually happened was then compared. The magnitude of weights and the direction of connections will be changed based on the inaccuracy in order to make the anticipated output resemble the actual output. The three layers of artificial neural network have been shown in Fig. 12.5.

12.3.2 Collection and Storage

The second step in solid waste management is collection and storage. This involves collecting all sorts of solid waste from different sectors and storing it in intelligent waste bins before it is transported. An intelligent waste bin is a smart bin powered

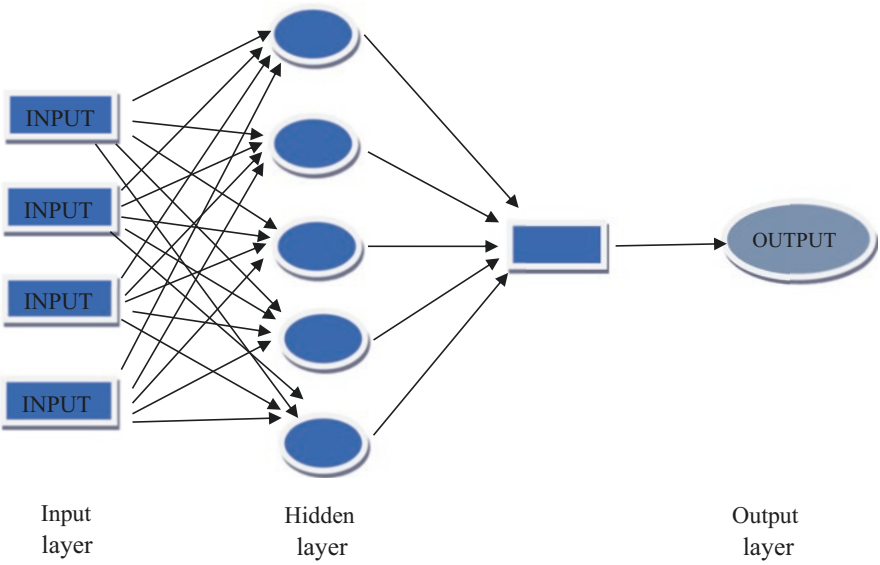


Fig. 12.5 Layers of artificial neural network

by artificial intelligence (AI) that uses genetic algorithms or ANN to recognize, classify, and organize waste (Majchrowska et al. 2022). However, the most widely used technique for the construction of these bins is artificial neural networks. Serious environmental and health risks result from improper trash disposal. Researchers created these bins to easily recycle and classify different kinds of waste in order to solve this problem. These smart bins will have AI-based sensors and cameras, allowing a central monitoring station to keep an eye on the bin's level. The benefit of this monitoring is that it lowers labor and transportation costs (Masebinu et al. 2017).

An intelligent waste bin segregates all the solid wastes into glass, paper, plastic, and metal. These bins compress waste, make recycling easier, and keep track of the contents' levels. One of the most well-known smart trash cans is called Bin.e, and it is used in a number of Asian, European, and Australian nations. However, additional research is still needed in this area of solid waste management.

12.3.3 Waste Transportation and Sorting

The use of advanced modern technology in smart solid waste management is to ultimately minimize the transportation costs and ensure an overall efficient work flow. In general, a large number of small-sized vehicles are employed in waste transportation, in order to make sure smooth travel on busy and congested roads. Hence, there is a need to reduce fuel consumption (Neelakandan et al. 2022). Cloud

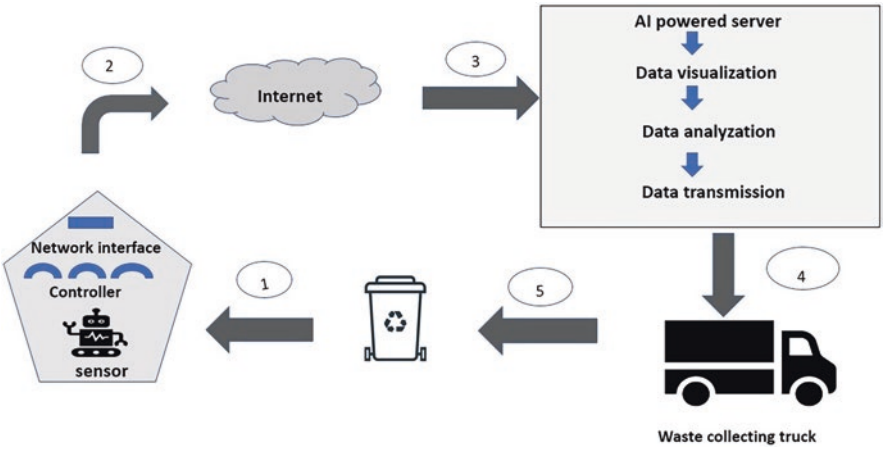


Fig. 12.6 Use of artificial intelligence in waste transportation

computing and IoT (internet of things) play a crucial role here. Basically, the smart dustbins are fitted with IoT devices having embedded sensors that have the capability to measure its environmental changes like overall waste content, temperature, pressure, and moisture thrown into them (Fig. 12.6). This data can be stored and exchanged in real time with the central waste management system, so that they are well aware of which dustbins are overflowing/completely filled and need to be taken care of immediately. This IoT technology does not require any man-man or man-computer interaction. The IoT is connected to an IoT gateway which is like a central hub that enables information exchange between the IoT sensors and the connected devices via the cloud which helps to store the extensive amount of data, which can be accessed through the internet (Roo et al. 2011).

Deep learning algorithms play a vital role in waste sorting by recognition of different kinds of waste components like plastic, organic, electronic, and hazardous clinical wastes. Convolutional neural network (CNN) is the best type of artificial intelligence preferred for waste classification. Deep learning algorithms have multiple layers, and each layer performs a complex operation like data acquisition, data visualization, and data analyzation (Wei et al. 1997). The CNN algorithm initially needs to be provided with millions of images in different viewpoints, backgrounds, color, size, etc.; the algorithm learns from the pixels (smallest unit in a digital display) of an image, thus mimicking the human process of learning and recognizing. Later this image information is converted into data, weights are assigned by various nodes to each element of an image, and finally an output would be generated by the neural network in the form of series of binary true/false questions, which involves highly complex mathematical and statistical calculations (Yurtsever and Yurtsever 2019). Now this generated result would be compared with that of human generated label and confirmed. If there is no match the CNN marks it as an error and tries to adjust the weights again and again; thus it gains accuracy over a period of time.

12.3.4 Waste Treatment and Disposal

The traditional methods of waste treatment and disposal techniques are not very efficient and have many built-in flaws, which could pose serious effects on the environment. Machine learning (ML) is a sub-field of artificial intelligence, which plays a crucial role in waste treatment and disposal (Abdallah et al. 2020). The basic principle of machine learning is that it tries to establish a relationship between the input and output variables of a particular data, for example plastic, learns from the parameters involved, and tries to predict the new data automatically based upon the decisions taken in the past by looping mechanisms. Basically, ML involves three steps, i.e., training, wherein the algorithm is trained with data that involves various parameters, cross-validation, the stage where the data is tuned into a best fit model optimized with all hyper parameters, and testing step where the data is validated using a data set (Andeobu et al. 2022). ML can predict the possible materials that could be recovered from waste and provide the best suitable technique to dispose of the waste safely (Fig. 12.7). Therefore, the intervention of AI in waste management makes the overall process automated and efficient.

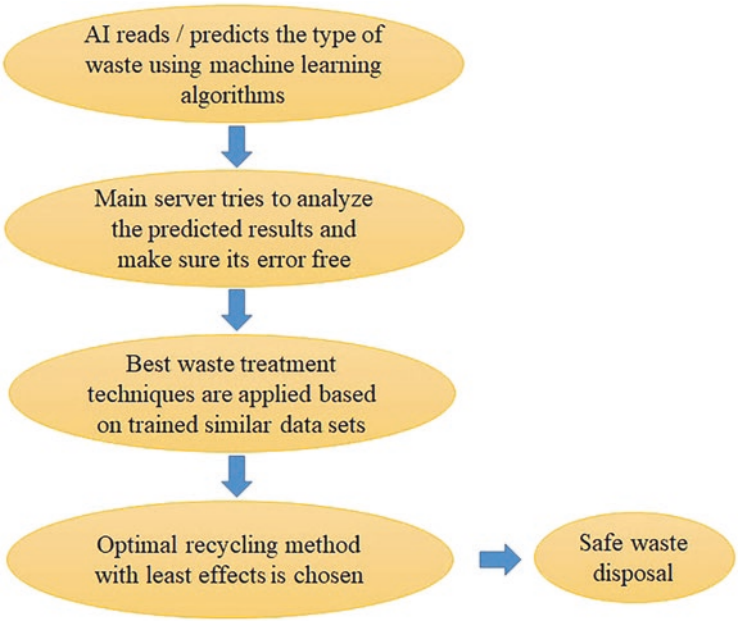


Fig. 12.7 Application of artificial intelligence in waste treatment and smart recycling

12.4 Bioprocess Technology

In bioprocessing, solid-state fermentation and submerged fermentation are two distinct approaches for producing various products, including enzymes, pharmaceuticals, biofuels, and organic acids. Both techniques entail microorganism cultivation. Solid-state fermentation (SSF) is a more intriguing technology that has gained traction in recent years. SSF is a fermentation that takes place in the lack or near absence of free water. In contrast, submerged fermentation (SmF) is the process by which microorganisms grow and decompose substrates in the presence of an abundance of free water. The SSF approach enables the use of solid organic wastes as substrates without the need for preparation, simulating a natural environment for the growth of microorganisms. Its advantages include low energy and water requirements, concentrated bioproducts, and being an environmentally benign process. SSF is an appealing technology, but there are significant constraints that prevent it from being implemented on a large scale, the most important of which being reactor design and the influence of upscaling on process productivity (Farinas 2015). SSF is a competent process for the effective utilization of waste biomass for the production of value added products. Through this bioprocess technology, several products such as hydrolyzing enzymes, antibiotics, bioplastics, biopesticides, organic acids, fragrances, and biofuels are being produced (Table 12.1).

Table 12.1 Bioproducts obtained from waste biomass through solid-state fermentation process

Waste biomass	Bioproduct	Inoculum	Production
Coffee husk, compost, biowaste digestate, agricultural waste	Cellulases	*Trichoderma* or *Aspergillus* strains, specialized consortia	2–10 FPU/gDM
Biowaste digestate, wheat bran, sorghum Stover, corn cob, soybean meal, wheat bran, coffee husk	Xylanases	*Trichoderma reesei, Aspergillus Niger*	58–2919 U/gDM
Soy fiber	Amylases	*Thermomyces lanuginosus*	35,000–228,000 U/gDM
Soy fiber, biowaste, Brewers's spent grain	Biopesticides		2.1×10^7–8.5×10^9 CFU spores/gDM
Sugar beet molasses and winterization oil cake	Sophorolipids	*Starmerella bombicola*	0.58 g/L*h
Sugarcane bagasse and sunflower seed meal	Rhamnolipids	*Pseudomonas aeruginosa*	0.19 g/L*h
Sugarcane bagasse and sugar beet molasses	Aromas	*Kluyveromyces Marxianus*	47.6 mg ester/gDM
Fig residues, black rice	Phenolic compounds	*Aspergillus niger, Aspergillus awamori*	1.7 mg–10.19 mg gallic acid/gDM
Food waste, agroindustrial waste	Bioplastics: PHB	*Bacillus* sp., *Lactobacillus* sp.	0.1–0.91 g/gDM
Cassava bagasse	Pollullan	*Aspergillus pullulans*	19–32 g/gDM
Agro-industrial waste	Cordycepin	*Cordyceps* sp.	8–25 mg/gDM

12.4.1 Hydrolyzing Enzymes

The hydrolytic enzymes that have protuberant application in biofuel production are cellulases and xylanases. Cellulase and xylanase production from wastes has traditionally relied on many *Trichoderma* and *Aspergillus* genus species. Using *Trichoderma* as inoculum and agroindustrial wastes as substrates has been found to yield the maximum cellulase production, with activity production ranging from one to hundreds depending on the operational parameters. Cellulase is substantially stimulated by cellulose; for example, low cellulase productivities (below 3FPU g^{-1}DM) are produced when the cellulose concentration of the substrate is less than 30% (Mejias et al. 2018).

12.4.2 Biosurfactants

Due to their potential to replace chemically manufactured surfactants, lower toxicity, higher biodegradability, and temperature endurance, the production of biosurfactants has recently developed interest among various researchers (Singh et al. 2019). According to Krieger et al. (2010), biosurfactants have numerous uses in cosmetics, soil bioremediation, and creation of novel polymers. A class of extracellular biosurfactants known as sophorolipids are produced by non-pathogenic strains. In commercial submerged fermentation, *Starmerella bombicola* synthesized 3.7 g L^{-1} h^{-1} of sophorolipids. Jiménez-Peñalver et al. (2018) designed an unconventional system that used molasses and winterization residues as substrates by *S. bombicola* for sophorolipids production. The yield and productivity obtained through unconventional system was 0.21 g g^{-1} and 0.58 g L^{-1} h^{-1} respectively, which was found to be poor in comparison to the submerged process. As the type and yield of sophorolipid are influenced by the substrates, Wang et al. (2019) employed agroindustrial sugar and obtained the yield of 0.06–1.07 g g^{-1}DM.

12.4.3 Aroma Compounds

Aroma compounds are another intriguing bioproduct produced by SSF. Rose-scented 2-phenethyl alcohol (2-PE) and floral-scented 2-phenethyl acetate (2-PEA) are among the most widely used aroma compounds (Guo et al. 2017). Martínez et al. (2017) reported a residue-based productive process for obtaining fruit-like fragrances via SSF using *Kluyveromyces marxianus* and sugarcane bagasse with a yield up to 12.1 mg 2-PE per gram of dry substrate (gTS) and 3.9 mg 2-PEA per gram of dry substrate (gTS). In addition, the production yield of 2-PE and 2-PEA increased by 33.6% and 23.8%, respectively, when a readily

metabolizable sugar source was added to the system. According to the published literature, aroma production via SSF allowed for higher sugar concentrations, less reagent use, and comparable volumetric yields compared to submerged fermentations.

12.4.4 Biopesticides

Biopesticides, particularly those derived from *Bacillus thuringiensis*, are an additional bioproduct that has been researched. Soy fiber and biowaste are just a few examples of the organic residues that have been used as substrates in the many studies (Ballardo et al. 2017; Cerda et al. 2019). An inexpensive method of biowaste valorization was reported by Ballardo et al. (2017); they used non-sterile biowaste to create a compost-like material enriched with biopesticide activity via the action of *Bacillus thuringiensis*. Other reported biopesticides are those derived from agro-industrial residues and *Beauveria bassiana*. Specifically, Qiu et al. (2019) reported the production of biopesticides from waste grain from breweries. These findings are consistent with those reported for bacterial biopesticides, yielding productive results and providing a strong basis for the development of a representative production process.

12.4.5 Phenolic Compounds

The generation of phenolic compounds by SSF has been discussed by other researchers (Buenrostro-Figueroa et al. 2017; Shin et al. 2019). A SSF process was created by Shin et al. (2019) using black rice bran as the substrate and *Aspergillus awamori* as the inoculum for the synthesis of phenolic chemicals. Approximately, 1660 μg pherulic acid g^{-1} of substrate produced by the described process in a 3-day operation demonstrated its effectiveness. Additionally, these authors emphasized the need for a pretreatment to prepare the phenolic chemicals for extraction. This is consistent with the findings of Lee et al. (2019), who showed that enzyme components might be extracted using water to prevent the synthesis of phenolic compounds.

12.4.6 Bioplastics

Bioplastics, such as polyhydroxybutyrate (PHB), whose manufacture using SSF has been garnering interest, are another area of interest in bioproducts. Industrial waste, wastewater sludge, agricultural waste, and food waste can all be used as a feedstock to make PHB (Rivero et al. 2017).

12.5 Conclusion and Future Studies

As we have already established, modern problems need modern solutions. A +culmination of different wastes (domestic, industrial, commercial, construction, etc.) needs a more holistic and multi-level treatment method. Over the past few decades, efforts have been made to integrate nanotechnology in bench-level waste management, with promising results. Nanotechnology, a conglomeration of different branches of science and engineering, has great potential as it deals with materials custom built with very specific applications. Similarly, for nearly a decade, artificial intelligence (AI) has gained prominence, due to its ever-developing deep learning, perception, reasoning, probabilistic methods, problem-solving, artificial neural networking (ANN), etc. making AI pertinent in highly intellectual tasks. AI has become pervasive and very dominant in mainstream utilities like internet search engines, transport assistance (traffic control, autonomous vehicles), facial recognition/labeling, automatic language translation, virtual assistance, etc. Hence, convergence of conventional treatment methods with nanotechnology and artificial intelligence (AI) holds a higher prospect of success.

Nanotechnology, despite its great potential, suffers from physical constraints in its workability as building devices, testing, and integrating them into the macroscopic world is a meticulous process. It needs great control and optimization to make workable scale models based on nanotechnology, and perhaps AI-assisted machine learning can bridge this gap largely. Similarly, AI can also assist in aspects like designing, optimization, assessment, failure prediction, troubleshooting, etc., of the devices/working models made of nanotechnology by developing paradigms based on artificial neural networking, genetic, or evolutionary algorithms. Therefore, synergic development of nanotechnology and AI in the future can develop efficient devices and infrastructure that can address the current and upcoming challenges in waste treatment and management.

In conclusion, essentially any bioproduct can be produced using SSF while solid organic waste is also being utilized. SSF serves as a crucial instrument to close the gaps in the transition to a circular bioeconomy in this way. However, in order for SSF to compete with commercial production systems based on SmF and consolidated end-of-pipe valorization technologies, efficient procedures must be developed.

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Chapter 13
Economics of Fermentation Processes



Swasti Dhagat and Satya Eswari Jujjavarapu

Abstract The fermentation process plays a pivotal role in the production of biopharmaceuticals, food processing, beverage and biofuels. With the advancement in biotechnology, understanding the economics of the fermentation process becomes indispensable in assessing the feasibility of the production process, which ultimately affects the selling price of the final product. There are several factors such as cost of raw material, consumables and equipment, downstream processing, human resources involved, etc. that are responsible for the overall cost of a fermentation process. This chapter provides a comprehensive overview of the fermentation economics and discusses all these factors in detail. It also talks about different software tools which are essential to estimate the cost of a bioprocess. The chapter also highlights the role of bioprocess optimization in increasing the efficiency of the process while reducing the production cost. The advantages of the use of computational tools for process optimization over the traditional one factor at a time approach to significantly reduce the time and cost required for experimentation are also discussed. Finally, the strategies of strain improvement for enhanced production of the desired product are also provided in detail in this chapter.

Keywords Fermentation economics · Process optimization · Strain improvement · Maximum product yield

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13.1 Introduction

Fermentation technology converts raw materials into useful and valuable end products, which are beneficial to the end users. The manufacture of these valuable products only becomes sustainable when the production processes are not only economically viable but profitable. The cost of a fermentation process plays a very vital role in determining the feasibility of any bioprocess at industrial scale and the cost of the product or the market value of the product. Hence process economics becomes an important element of biotechnology and chemical engineering. The unit production cost (cost per kilogram) is calculated by dividing the annual operating cost by the annual rate of production. The unit cost of product as well as the selling price varies inversely with the market size of the product (Petrides 2000).

The cost of a fermentation process or the economics of a fermentation process are affected, directly or indirectly, by a number of factors. These factors include the cost of raw materials and consumables, cell cultures involved in the process, fermentation duration, steps involved in the purification and recovery of the products, treatment of effluent, investment in the equipment or equipment costs, labour costs, land requirements, etc. All of these factors along with the analysis of profit are used to estimate the value of a fermentation product. Among these, raw materials, labour, utility and fixed costs are the major contributors of total fermentation cost. The production of biopharmaceuticals also includes an additional cost required for new drug development, which takes a lot of time for research and development, and the price for which ranges from millions to billions of dollars. The approval of such biological products by various organizations also adds up to the cost of production (DiMasi et al. 2003, Gilbert et al. 2003).

The rise in the production cost encourages the need for effective process design strategies and tools that can evaluate the process development cost at an early stage and eliminate the non-viable processes. The utilization of optimization techniques for designing processes can also assist in the reduction of overall cost of the bioprocess. In order to compete with the existing products, the fermentation process must utilize the microbial, plant and animal cultures, which produce high yield end products at a reasonable price. Hence, research on designing of suitable media and improvement of microbial, plant and animal cells must be encouraged as a part of development programme in an organization. Before starting any fermentation process, it becomes important to have knowledge about the economics of the process, the parameters that play a significant role in the bioprocess and the ways in which the cost of process can be reduced without reducing the quality of the bioproducts. The main aim of this chapter is to enlighten the readers on various factors contributing to the cost of the fermentation process and the estimation of the cost of a bioprocess. It also deals with ways to reduce the overall cost of a bioprocess and thereby reducing the cost of the end product in the market.

13.2 Factors Affecting the Cost of Bioprocess

A fermentation process consists of three primary objectives: efficient utilization of substrate, process productivity and maximization of product yield. All these objectives majorly contribute to the cost of the fermentation process. As mentioned previously, the major factors that contribute to the total cost of a bioprocess are raw materials, consumables, cell cultures involved in the process, fermentation duration, steps involved in the purification and recovery of the products, treatment of effluent, investment in the equipment or equipment costs, labour costs and land requirements.

13.2.1 Raw Materials

In a fermentation process, raw materials include fermentation media, recovery solutions as well as chemicals used as cleaning agents. Carbon source and nitrogen source being the primary part of fermentation media consist of a major portion of the cost of raw materials. The most common carbon sources used in a fermentation process are carbohydrates such as glucose, fructose, galactose, mannose and xylose, fats/oils such as vegetable oils, soybean oil and corn oil, starch from food and food waste, molasses, lignocellulosic biomass, and chemicals such as ethanol, methanol and glycerol. Nitrogen sources are classified as inorganic and organic sources. Inorganic sources are ammonia, ammonium sulphate, ammonium nitrate and urea while the organic sources are cottonseed, soybean and peanut meal, corn pulp, meat extract, yeast extract, yeast peptone, peptone, etc. Trace elements, namely calcium, copper, cobalt, iron, manganese, molybdenum and zinc, and macronutrients such as potassium, phosphorus, magnesium and sulphur are also supplemented in the fermentation medium as raw materials. Even purified water (water for injection) is an important determinant of the cost of raw materials (Petrides 2000).

Certain commodities, such as biofuels, are produced from agricultural residues and lignocellulosic feedstock, which are generally waste products. Hence, the cost of raw materials for such products will be lesser. The production and purification of pharmaceuticals need raw materials of the highest grade. The purification process includes chemicals associated with product recovery and equipment cleaning which needs to be pure and adds up to the cost of the process. Therefore, the cost of raw materials is a major factor determining the economics of the fermentation of pharmaceutical products.

13.2.2 Consumables

Consumables determine a significant portion of the cost of a bioprocess. They are non-durable goods that need to be used up quickly and replaced often. They are also damaged during processing. Single-use disposable bags and containers are

gaining popularity in biopharmaceutical industries as they reduce the time and effort spent in cleaning and sterilization (Petrides et al. 2008). Single-use bioreactors also reduce the time between different cycles and increase process flexibility. Separation membranes and chromatographic resins contribute majorly to the cost of the biopharmaceutical industry. The cost of resins of hydrophobic interaction chromatography and ion exchange chromatography ranges from $500 to $2000 per litre of resin whereas affinity resins for purification of monoclonal antibodies range from $5000 to $15,000 per litre. These resins need to be replaced periodically. The cost of polymeric chromatographic resins for the purification of small biomolecules is cheaper than the previous ones with longer lives. Similarly, the cost of membranes used for filtration ranges from $50/m^2 to $800/m^2 depending upon their application. Polymeric membranes are cheaper than ceramic membranes (Petrides 2000).

13.2.3 Investment in the Equipment or Equipment Costs

Equipment is a prime necessity to run a biopharmaceutical industry, and equipment costs fall under direct fixed cost of an industry. The cost of equipment is not limited to the purchase of equipment but also includes costs associated with its installation and maintenance. Equipment costs fall under the category of non-recurring cost, which means that the majority of the cost under this head is incurred only once during the start-up of the industry. For any organization, the periodic maintenance of equipment becomes a mandate for smooth running of the unit operations, which forms a part of recurring cost to be incurred by the organization at regular intervals.

13.2.4 Cell Cultures Involved in the Process

The development of microorganisms involved in the fermentation process also adds up to the cost of the fermentation process. Microorganisms can either be procured from a cell bank or the pure culture can be isolated in the laboratory. Large volumes of fermentation medium require a significant amount of culture for the production of suitable product. Hence, inoculum development, a part of upstream processing, forms a significant part and one of the most important steps of bioprocess technology as any undetected microbial contamination at this stage can result in the failure of the entire fermentation process. The upstream processing also includes experimental design, cloning and expression of recombinant proteins in cells and genetic modification of cells to obtain high yield. Hence, the economics of a fermentation process must also consider this factor.

13.2.5 Fermentation Duration

Even though a less significant one, fermentation duration also indirectly determines the cost of a bioprocess especially in product-oriented industries. The products which require a short span of time for culture can be purified quickly and the next batch of production follows whereas in case of products which require long fermentation duration, the number of cycles will be reduced.

13.2.6 Steps Involved in Purification and Recovery of the Products

The number of steps needed to purify a product also regulates the cost of the product. The compounds that require high purity will have to pass through multiple steps of purification while crude products do not need to undergo extensive purification process. Hence, the cost of production of products of high purity will be high. But the cost associated with purification cannot be substituted as it directly impacts the safety of consumers.

13.2.7 Human Capital

An industry needs a lot of skilled and knowledgeable manpower for it to run successfully, which includes employees from all the departments of the industry. Human capital consists of all the employees in an organization that solve business problems and provide competitive solutions for the progress of the organization. The cost associated with labour depends on the scale of industry and the stage of operation. An industry focussed on only research and development requires experts from multiple fields while a production unit requires operators skilled to handle a particular facility. A small-scale industry dedicated to a single product is generally less automated and hence requires more operators whereas in automated industries, a single operator can handle multiple steps remotely. Moreover, employees in industries with production units operate on a shift basis, with the number of shifts determined by the demand for the product. The cost (salaries and incentives) associated with each employee depends on their skill and varies across the globe (Pollak et al. 2012).

13.2.8 Land Requirements

Land is one of the primary requirements for establishing an industry. The cost of land is not just limited to the procurement of land but includes the cost associated with the construction of building and its maintenance as well. The land for an

industry can either be owned, rented or leased by the organization for a fixed tenure. In some countries, governments also support the establishment of biotechnological facilities by providing them loans at suitable rates. This factor also falls under direct fixed cost and is one of the major non-recurring costs of a biotechnology industry.

13.2.9 Utilities

This includes the cost associated with electricity, water and heating and cooling agents. The continuous running of reactors generates a lot of heat and hence refrigerants or chilled water is required to remove the heat and maintain the temperature of the reaction. On the other hand, sterilization of equipment needs steam of high purity; the cost of which depends on the type of fuel used and its scale of production. The cost of distilled and MilliQ water also falls under this head.

Treatment of wastewater and the safe disposal of sewage and hazardous waste constitute another important operating cost. A biotechnology industry uses different types of solvents, which are ultimately washed away in the waste stream. The disposal of solvents becomes expensive and challenging. Single-use systems are disposed off by incineration whereas solid waste must be collected separately in polybags. The wastewater needs to be treated before it is released into the environment. The exhaust gas too must be treated before it is discharged in the atmosphere.

13.3 Estimation of Costs of Bioprocesses

As discussed in the previous section, the cost of a bioprocess depends on various factors. It also depends on the bioproduct produced by the fermentation process. Low-value high-volume products such as biofuels can be produced from cheap raw materials in large quantities and do not require many purification steps. Hence, the cost of bioprocess for the production of such products is less but the overall yield of the product is high. High specialty biochemicals that are used in therapeutic applications are subjected to multiple purification processes. Also, they require expensive raw materials and are produced in smaller volumes. The production of such high-value low-volume product becomes costly, thereby increasing the selling price of the final bioproduct. But care should be taken at each and every step as failure in any step of the production process will result in loss of money, time and efforts. Therefore, it is advisable to perform techno-economic analysis (TEA) at every step to assess economic feasibility of production.

Techno-economic analysis (TEA) is used to check the potential of fermentation process for many chemicals and proteins. Currently, many online tools are available that help in evaluating the feasibility of a production process. A bioprocess TEA calculator was developed by Lynch to evaluate the commercial competitiveness of potential bioprocesses (Lynch 2021). The calculator helps the user in making

relevant decisions about the development of technology by estimating the production cost from process parameters and financial metrics. Along with the research and development team of any organization, this tool will also be beneficial to start-ups, funding agencies and investors to analyse the development of an organization from beginning (Kennedy 2021). Another tool which helps to model, evaluate and optimize batch and continuous process in biotechnology, biopharmaceutical and food processing industries is SuperPro Designer by Intelligen, Inc. (https://www.intelligen.com/products/superpro-overview/#) (Viswanathan et al. 2020). It also models and evaluates wastewater treatment and water purification processes. In this tool, the user needs to provide the materials utilized in the process and initialize a process. It enables the user to design and optimize manufacturing processes simultaneously along with treatment process to prevent and/or control pollution. It also has a databank with physical and thermodynamic properties of more than 1200 materials and 60 examples from various industries.

BioPET (Biorenewables Process Evaluation Tool) has been developed by Claypool and Raman. This tool helps in the estimation of preliminary cost and minimum selling price of a product or process. It gives feedback and capital costs similar to commercial process modelling software and therefore has a potential to estimate the cost of bioproducts at an early stage. It enables users to provide seven unit operations, namely, fermentation, separation, purification and three stages of catalysis (Claypool and Raman 2013; Viswanathan et al. 2020). The Process Economic Analyzer by AspenTech is also a model-based approach to estimate the generation of project capital cost and operating costs. This tool includes equipment to determine operating costs of an industry and investment analysis and results in automatic generation of process flow diagrams. The Process Economic Analyzer estimates the capital and operating costs to compare and screen multiple processes (Inc 2023).

13.4 Optimization of Bioprocess Design

Due to the complexity of biological systems, industries utilizing bioprocess technology face several challenges in bioprocess design. These challenges may be due to variations in biological system used in the process (microorganisms, plants or animal cells), which can be difficult to elucidate due to various metabolic pathways, optimization of product yield, limitations of oxygen and nutrients transfer impacting cell growth and productivity, overexpression of genes and/or pathways for high product yield, maintenance of aseptic conditions compromising purity of product, complex and expensive purification and separation steps, the time for bioprocess development and issues in scale-up such as shear stress, mass transfer limitations and temperature gradients. Hence, monitoring and controlling all factors involved in the bioprocess using a reliable methodology are important. These challenges also encourage the need to optimize various process parameters to achieve desired outcomes which helps in saving time and money, both being prime aspects of any bioprocess industry (Toms et al. 2017). It is also necessary to identify the optimal

conditions of a bioprocess which play a significant role in maximizing product quantity and quality while minimizing the cost and variation among these factors. To overcome these challenges, a multidisciplinary approach requiring expertise in biotechnology, bioengineering and process optimization along with computational modelling and process control is required.

The optimization of a bioprocess includes different aspects of the process in order to achieve higher yields and better efficiency at a cheaper cost, which depends on the specific bioproducts being produced. The optimization of a bioprocess requires careful monitoring of processes and measuring key process parameters to identify the prospects of improvement. It also ensures that the process meets the desired objectives in an efficient and sustainable way. The optimization of a bioprocess involves experimental design, statistical analysis as well as process modelling.

Conventionally, a bioprocess is optimized by varying one factor at a time (OFAT) wherein all factors are kept constant except one. This is a major challenge in a bioprocess as there are many critical parameters which need to be controlled and monitored to achieve an effective bioprocess, and testing one factor at a time will result in numerous experiments consuming a lot of time and money. Also, the one factor at a time approach does not consider the interaction among critical process parameters which may lead to ambiguous conclusions (Schmitz et al. 2016; Dhagat and Jujjavarapu 2021a, b).

Design of experiments helps in planning, conducting and analysing tests to assess the critical process parameters and their interaction influencing the process (Kasemiire et al. 2021). The design of experiments reduces the time and cost of experimentation by providing critical process parameters and optimizing the bioprocess to achieve maximum yield. It also provides a set of experiments required to achieve optimal parameters. The very first objective of the design of experiments is to define the problem to be addressed. Generally, in a bioprocess optimization the problem statement is to increase productivity, minimize production time or improve cell growth. Design of experiments requires the identification of critical process parameters (commonly termed as factors) that affect the bioprocess. Some of the main critical process parameters are substrate concentration, pH of the medium, temperature, time of incubation, agitation speed, etc. The range of all the process parameters must also be established. The selection of a suitable experimental design is also an important criterion for the optimization of a bioprocess design. The most common experimental designs used in any bioprocess is Central Composite Design, full factorial design, fractional factorial design, Box-Behnken Design and response surface methodology.

Once the experimental design is obtained, experiments are performed as per the experimental design plan and the responses are recorded for each run. The interactions between various factors can be studied by fitting a response surface model. This also enables to identify an optimal operating condition for the bioprocess. Even though the optimization process predicts the best parameters for the operation of a bioprocess, the results from the optimization process need to be validated experimentally to ensure their consistency and reproducibility. The optimized conditions can be further used to scale-up bioprocess for large-scale production after considering the mass and heat transfer and agitation rates.

13.5 Strain Improvement to Maximize Production

Microorganisms produce a wide variety of primary and secondary metabolites of significant industrial importance in food, pharmaceutical, chemical, agriculture and cosmetic industries. Even though the products arising from the fermentation process have a very high market demand their productivity is too low to be commercialized in a cost-effective manner. The yield of product can be increased by optimizing the fermentation medium and growth conditions but these conditions are restricted by the ability of microorganisms to synthesize the products. Hence, strain improvement becomes significantly important for the commercialization of any fermentation process and determines the overall cost of a bioprocess. Strain improvement mainly aims at the improvement of microbial strains to produce compounds of interest at a faster rate and high yield and is an important area of research in the industry. This involves selection, development and optimization of cell lines or microorganisms to increase their ability to produce the desired product. Apart from improving the microbial strain, search for strains which produce high yield of products over a wide range of reaction conditions (temperature and pH) is also advantageous as they will be able to tolerate variations in reaction conditions during a fermentation process. Along with this, the growth of microorganisms on a simple cheap medium and strains which are resistant to contamination will be advantageous.

There are three strategies by which microbial strains can be improved, namely, selection of high-yielding mutants, genetic engineering and metabolic engineering. A population of variant microorganisms that produce more product than the parental strains is termed high-yielding mutant strains. High-yielding mutant strains are selected from the parental strains by growing them in a selection agent that is toxic to the parental strain but non-toxic to the mutant. Genetic engineering deals with manipulating or altering the genetic sequence of the microorganism to increase the product yield. This is performed by insertion, deletion or modification of genes of interest of the organism. Metabolic engineering is altering the metabolic pathways of microorganisms towards the production of the desired product. This is generally performed by knocking out genes not required for product formation and replacing them with genes essential for encoding enzymes for desired product formation.

On the other hand, improvement of plant cells is faced with certain challenges due to the complexity in growth requirements and their biology. Nevertheless, plant cells in bioprocessing can be improved by genetic engineering, gene silencing, transcription factors, organelle engineering, metabolic pathway engineering, use of promoters, marker genes, tissue-specific expression, optimization of cell culture conditions and selection of cell lines. This can be done by introducing or overexpressing genes related to the biosynthesis of product of interest, reducing the competitive pathways or increasing the efficiency of metabolic pathways. Some other methods are by suppressing the genes which negatively impact the production of the compound of interest, by manipulating the activity of transcription factors regulating gene expression or by engineering the metabolic pathways to increase the flux towards the production of the desired compound. The plant cells can also be

improved by optimizing the functionality of their organelles, such as chloroplasts or mitochondria. Screening and selection of cell lines that naturally produce the desired compound and creation of hybrid lines with increased productivity also result in improvement of strains of plant cells. Different omics approaches and metabolic profiling help to identify suitable targets for improvement. Another approach is to select plant cell lines producing high quantities of desired products. In order to select high-producing cell lines for increased production of secondary metabolites, it is important to understand the physical and chemical factors that regulate the production of secondary metabolites. Hormone optimization during culture of plant cells, environmental conditions, media composition and supplementation of media with nitrogen and phosphorus sources also affect the production of secondary metabolites in plant cells directly or indirectly. Some chemical agents, such as biotin, 5-methyltryptophan and glyphosate, are also used to select high-yielding plant cell lines (Bhatia et al. 2015).

Similar to plant cells, strain improvement in mammalian cells is an important strategy for the production of robust and high-producing cell lines. Mammalian cells are used for the production of vaccines, biopharmaceuticals and bioproducts by metabolic, genetic and physiological modifications. The first step in strain improvement is the selection of a suitable cell line for the bioprocess. The mammalian cell lines can be genetically engineered by transfection, gene editing, gene overexpression and gene knockdown. The modified cell lines are screened for increased growth rates and product quality and higher productivity. The modification of metabolic pathways within the cells is also used to enhance productivity and product quality. Optimization of bioprocess conditions (culture medium, pH, temperature of bioprocess, bioreactor design, etc.) is also responsible for the improvement of mammalian cell lines and the production of the suitable product. As the mammalian cell cultures are used for the production of biopharmaceuticals and vaccines, they require stringent regulatory compliance of the production of bioproducts with high quality and purity.

The improvement of any cell line (microbial, plants or mammalian) is a complex and iterative process, which requires continuous monitoring and assessment of the performance of the improved cell lines. It needs the collaboration between microbiologists, cell biologists, genetic engineers, bioprocess engineers, quality control experts and compliance officers for the increased production of bioproduct with high purity and at a reasonable cost.

Many computational tools have also been developed which utilize methodologies for strain design and improvement. They provide researchers a valuable insight into the various metabolic and genetic pathways of microbial, plant and animal cells and enable them to design efficient and effective strategies for strain improvement. Computational tools have also been used to predict different genetic modifications for high productivity, high yield and substrate utilization (Long et al. 2015). These tools analyse genomes of cells and identify genes and promoters linked with the production of the target product. This helps in selecting suitable genes for knockout or overexpression for improved production of the desired product. The metabolic interactions and fluxes in the cell are simulated by metabolic network models. These

computational models along with experimental data help in the identification of key metabolic pathways and engineering the pathway for optimal product synthesis. Many different databases are available with different enzyme-catalysed reactions for pathways with known reactions. These databases are utilized to develop metabolic network models. Examples of such databases are KEGG, SEED, BRENDA, MetRxn, etc. (Sreenath and Polasa 1985; Henry et al. 2010; Kumar et al. 2012; Schomburg et al. 2012; Kanehisa et al. 2014). For those compounds with no knowledge of biological synthesis, de novo methods have been developed to identify the reaction pathways and enzymes required for their catalysis. BINCE can identify enzymes which catalyse novel reactions based on molecular simulations and similarity of substrates (Brunk et al. 2012). RetroPath (Carbonell et al. 2011), DESHARKY (Rodrigo et al. 2008), SimPheny Biopathway Predictor (Yim et al. 2011), and GEM-Path (Campodonico et al. 2014) are some of the de novo pathway prediction algorithms based on substrate similarity, pathway length, cellular growth rate, etc. The analysis of those genes that are upregulated or downregulated under certain conditions can also result in the identification of targets for genetic manipulation.

Computational tools can also be employed to design synthetic promoters for enhanced expression of target products ultimately resulting in high product titres. The prediction of protein structures helps in designing proteins with better catalytic activity. Considering the large amount of genomics data available they provide an easier and faster approach for analysis and comparison of data and identification of genetic variations in genomes of different strains. The optimal combination of genetic modifications for increased production can also be obtained from optimization studies. The prediction of the strain design before experimentation in laboratory also reduces the number of experimental runs saving time and resources.

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